Conditional existence of four-dimensional Yang–Mills on a finite torus along Bałaban's route

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Abstract

Bałaban's renormalization-group construction of four-dimensional SU(2) lattice Yang–Mills theory depends on control of the running coupling. This control is his Theorem 2, whose proof was never published. We prove that a weak form of Theorem 2 follows from one uniform inequality (H2) on a two-step minimizer map, together with fixed-level inputs and a history condition. The weak form gives two-sided bounds on $1/g_k^2$ with an additive constant, and it suffices for Bałaban's Theorem 1. We prove the fixed-level inputs and the history condition relative to named inputs: Bałaban's published proofs, his inductive assumption (H0), stated conventions, and one external result on the constant-field effective action. At $L = 13$, inside Bałaban's range $L > 11$, we prove (H2) by a computer-assisted argument. An explicit majorant and an outward-rounded interval branch-and-bound give $\rho_1(j, K)^2 \leq 10 < 13$ for every level j, including $j = \infty$, and every momentum K, so $\rho_2 \leq 10 < 13$. The same argument gives (H2) at $L = 15, 17, 21$. This proof uses a stated convention for Bałaban's averaging operator. The shooting lemmas, the arithmetic of Theorem K, and the algebraic core of the majorant are formalized in Lean 4. Second, on a fixed torus, we prove that a uniform moment bound (H-U) implies compactness of the lattice Schwinger functions of the centred plaquette density, and that every limit point satisfies the torus forms of the Osterwalder–Schrader axioms OS0–OS3. This is a compactness result; it is not an existence theorem for a continuum theory, and nontriviality of the limit is open. We reduce (H-U), but we do not prove it. It rests on sourced versions of Bałaban's bounds; on a hypothesis (FX), which we reduce to a holomorphy bound (H-hol) for Bałaban's increments, history bounds and three named inputs of the large-field step; and on a bridge from the source form to (H-U). A dependency table lists the hypotheses of each block. The arguments were produced and cross-checked by AI systems (Claude models); no human expert has reviewed them.

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1 Introduction

1.1 The problem

The Clay problem on quantum Yang–Mills theory asks for a four-dimensional quantum gauge theory with compact simple gauge group that satisfies axioms at least as strong as those of Osterwalder and Schrader [19, 20], and that has a mass gap [15]. One constructive approach starts from Wilson's lattice action on a torus with lattice spacing ε and takes $\varepsilon \to 0$. The bare coupling $g_0(\varepsilon)$ must then go to zero at the rate that asymptotic freedom prescribes, so that the effective coupling at a fixed physical scale stays fixed. This paper treats the first limit only: lattice spacing to zero on a fixed torus. It does not treat the infinite-volume limit or the mass gap (Section 4.7).

Identifiers. Each object of the paper has an identifier of the form P-nnn. The prefix gives the kind: R for an input statement (most of them Bałaban-level), H for a hypothesis, condition or reading, L for a result proved here, K for a convention, Q for a closure item, X for an external result. At its first mention in a subsection, an object is named and its identifier follows in parentheses, for example "the inequality (H2) (R-025)". Later mentions use the name only. The registry `IDS.md`[1] in the repository [14] lists every identifier. The dependency table (Section 4.6), the table of inputs (Section 7.7) and the appendices use the identifiers as their index.

1.2 Bałaban's programme and the missing Theorem 2

Between 1984 and 1989 Bałaban built a renormalization-group (RG) analysis of lattice gauge theories [1, 2, 3, 5, 4, 6, 7, 8, 10, 9, 11, 12]. His main output for $d = 4$ is Theorem 1 of CMP 122 II: uniform bounds on the effective densities at every RG scale, under the hypothesis that the sequence of effective couplings stays small. His Theorem 2, which controls the running coupling, was to discharge this hypothesis. The two statements read as follows.

Theorem 2 [8, p. 259]: "Let $d = 4$, $G = SU(2)$, and let γ be a sufficiently small positive constant, then for a sufficiently small positive g there exists a bare coupling constant $g_0 = g_0(\varepsilon, g)$ such that the sequence of the effective coupling constants g_k is contained in the interval $]0, \gamma]$, and $g_K = g$. Moreover, there exist constants β, β', $0 < \beta \leq \beta'$, such that g_k satisfy the inequality $1/g^2 + \beta \log(L^k\varepsilon)^{-1} \leq 1/g_k^2 \leq 1/g^2 + \beta' \log(L^k\varepsilon)^{-1}$. (0.31)" Bałaban adds that the proof "will be given in a separate paper". It was never published. In [12, p. 355] he writes that it "has not been published yet, so we have Theorem 1 with the assumption".

Theorem 1 [12, p. 355]: "If the sequence of the effective coupling constants is contained in an interval $]0, \gamma]$ with a sufficiently small positive γ, then the effective densities ρ_k have the form, and satisfy all the conditions and bounds, described in Sect. 2 [III]." The bound (0.1) there reads $\chi_k \exp[-(1/g_k^2)A(U_k) - E_-|T_\eta|] \leq \rho_k \leq \exp E_+|T_\eta|$.

Without Theorem 2, the hypothesis of Theorem 1 is an assumption on the unknown coupling flow. Every continuum statement built on Bałaban's bounds inherits this assumption.

[1] Project files named in this paper live in the `notes/` directory of the `yang-mills` repository [14].

1.3 The two targets and the main results

The paper has two targets. The first target is a proof of enough of Theorem 2 to remove the coupling hypothesis from Theorem 1. The second target is a compactness theorem for the lattice Schwinger functions on a fixed torus, with the Osterwalder–Schrader properties proved for every limit point. Both results are conditional: each holds under named hypotheses, stated in the theorem.

Theorem A (weak Theorem 2)**.** *Let $G = SU(2)$, $d = 4$, and L odd, $L > 11$. (For odd $L \geq 3$ the statement holds if Bałaban's inductive assumption (H0) (H-202) and his construction of $\beta_j(g)$ at this L are added as hypotheses.)*

(i) Assume the following.

- *The uniform inequality (H2) (R-025) of Section 3.7.3.*
- *The fixed-level inputs (PI) (R-026), (F) (R-027), (M_j) (R-028), V1 (R-029), and (G2) (R-030) of Section 3.6.1.*
- *The history condition L5 (R-031) of Section 3.4.5.*
- *The bound $|\beta_{k+1}(h, g) - \beta_{k+1}(0)| \leq C_\beta g$ on every past h, including non-monotone pasts (item 7 of Section 7).*

Then the conclusions (b), (c-pair), and K-018 (S) of Theorem 2w (Theorem 3.2) hold.

(ii) Assume the conclusions of Theorem 2w, the condition L5, and the volume hypothesis (Vol) (K-032) of Section 2.1. Then Bałaban's Theorem 1 holds for the bare coupling $g_0(K, g)$, for every K and every $g \in]0, g_1]$, without his coupling assumption.

Part (i) follows from Theorem K (Theorem 3.14) and Lemma DT (Lemma 3.6). Part (ii) uses Bałaban's published proof of Theorem 1, the cross-scale sweep of Section 3.4.3, the list of later constants (Proposition 5.85), and the finite-volume identification of Proposition 5.95 (Section 3.8). A second route (Route B, Theorem $\mathrm{B_{PW}}$ in Section 3.6.2) also gives (i), under its own hypotheses (items B1–B7 of Section 7.2). Both routes rest on (H2).

The hypotheses of (i) hold in the following sense.

- (H2) holds at $L = 13$ (Theorem 6.2) and at $L = 15, 17, 21$ (Proposition 6.4 in Section 6.8). At other L it is a hypothesis.
- (F) holds by Theorem 5.134, (M_j) by Theorem 5.102, and L5 by Theorem 5.104, each relative to the inputs listed there.
- (PI), (G2) and V1 hold at each fixed level, relative to the inputs listed in Section 5.1.
- The bound of item 7 holds under (H0) (R-006), or under the inputs of (G-gain) (H-111) (R-037).

So at $L = 13$ the weak form of Theorem 2 rests on the inputs of the fixed-level results, of L5 and of item 7, on the Bałaban-level statements of Section 7.7, and on Convention 6.1, the reading of Bałaban's averaging operator that Theorem 6.2 uses.

Theorem B (compactness on a finite torus)**.** *Let $G = SU(2)$, $d = 4$, L odd and large, and $\ell \in 2\mathbb{N}$ fixed. Let $g_0 = g_0(\varepsilon, g)$ be a shooting solution whose effective couplings satisfy $g_k \in]0, \gamma]$ for $0 \leq k \leq K$ and $g_K = g$. Assume the uniform bound (H-U) (H-116) of Definition 4.1. Then the lattice Schwinger functions of the centred plaquette density are relatively compact as $\varepsilon = L^{-K} \to 0$, and every limit family satisfies the torus forms of (OS0)–(OS3).*

This is Theorem 4.2, proved in Section 4. Theorem B does not assert that a limit is nonzero. Nontriviality is the separate Theorem 4.5, which assumes the hypothesis (NT); (NT) is open.

The hypothesis (H-U) is the output of three blocks: the block (B-T2), that is, Theorem A; the sourced versions of Bałaban's bounds, block (B-S) (R-035); and the bridge from the source form to (H-U), block (B-N1) (R-045). If (H-U) is taken from these blocks and not assumed, the chain also uses two further inputs. The first input is the $\kappa \to \kappa/2$ decay recovery of the sourced small-field step (R-046). We reduce it, by $L/2$-rescaling, to the large-field T-step rate loss (R-020), which Bałaban does not print. The second input is the volume hypothesis (Vol), $\ell > 22M$ (Section 2.1). The dependency table (Section 4.6) lists the hypotheses of each block. Section 7 lists what remains open.

The sourced step also uses the unit-scale condition (U) (K-024), $\rho_1 \le c_* r_0/M^a$ at $\lambda_K = 1$. This condition is a choice of the source-gradient constant, made once after r_0 and before K. It is not a hypothesis on test functions. The bridge (B-N1) meets it by rescaling, at a cost C^n. This condition (U) is different from the Route B hypothesis (U).

Inputs from Bałaban's papers. Section 7.7 lists the statements of Bałaban and the external results that the two theorems use. Theorem A uses 4 Bałaban-level statements and one external result. Theorem B uses 21 Bałaban-level statements and four external results. Part (i) of Theorem A, that is, Theorem 2w, uses one of these statements: R-006, Bałaban's inductive assumption (H0), which includes his assertion that the whole-lattice limit of [8, (1.22)] exists. We reduce the use of R-006 in Theorem 2w to named inputs (Section 5.9). The external input of Theorem 2w is the constant-field log coefficient of Savvidy [23], used in (F). The other inputs of Theorem 2w are Bałaban's printed material, results proved here relative to listed inputs, conventions, readings, scope restrictions, and conditions on constants. The numerics of Section 6.9 leave one open question about the single-step β_1; it does not enter (F).

Of the 23 Bałaban-level statements of Section 7.7, Bałaban states 16 without proof: 3 of the 4 behind Theorem A, and 14 of the 21 behind Theorem B. One more statement, R-021, is a set of printed statements of CMP 99, partly with proof sketches; it contains the printed inputs of the condition (C) (H-057). Each other statement that Bałaban states without proof, and each statement that he uses without printing it, has a proof here relative to listed inputs or a reduction to named inputs (Sections 5.9, 5.10, 5.11, and 7.7). This includes the rows R-022 and R-023 (CMP 109 pp. 253, 275), whose reductions leave open lemmas, and the row R-024 (the multiscale geometry of CMP 119 pp. 275–276), whose sourced extension we prove relative to listed inputs. The condition (C) itself is reduced to named inputs (Proposition 5.126), and the volume hypothesis (Vol) discharges its torus-side hypothesis. The lemma (L-LL) (H-137) is Theorem 5.133, and the bound (S_σ) (H-168) is Proposition 5.135.

The propagator and covariance bounds R-001–R-005 reduce to published theorems of CMP 99 and CMP 102. Their joint residue has five parts:

- Bałaban's walk expansions of the final propagators, through R-002 (CMP 99 Theorems 3.7 and 3.9 are written out as Theorems 5.145 and 5.143);
- the complex walk bound (W^c) in the exponential gauge, and the decay of $C^* \Delta_k C$ at real U (R-008) under R-003;
- the regular-class identification R-018;
- the printed inputs R-021–R-024;
- the conventions K-044 and K-046.

What the paper does not give: literal (0.31) at every K (Section 3.3), the limit of $\beta_j(0)$ and a rate (Section 3.9), uniqueness of the limit, infinite volume, rotation invariance, nontriviality, and a mass gap (Section 4.7). For infinite volume the paper records one conditional statement only, Theorem 4.14, which is outside both targets. It gives compactness and OS0–OS3 for a joint limit $(\varepsilon, \ell) \to (0, \infty)$, under the hypotheses of Theorem B at every ℓ and two named ℓ-uniformity lemmas. Theorem B is a compactness result. Under footnote 2 of Jaffe and Witten [15] it is not an admissible solution of the Millennium problem (Section 4.1).

1.4 Conventions of this paper

- *Citations of Bałaban.* We cite Bałaban's papers by journal volume and page, for example CMP 109 p. 259. When Bałaban states a result without proof, we say so and give the page.
- *Readings.* Where a printed passage admits more than one reading, we fix one reading as a numbered Convention and state the competing reading.
- *Relative proofs.* "Proved relative to listed inputs" means that the proof is complete given the named inputs, and that these inputs are hypotheses of the statement. A "reduction" of a statement X to inputs $Y_1, \ldots, Y_n$ is a proof of $Y_1 \wedge \cdots \wedge Y_n \Rightarrow X$; the Y_i stay open unless a reference is given.
- *Open items.* A statement for which we have no proof appears as a hypothesis, an Open Problem, or a Conjecture. A plausibility argument is labelled as such.
- *Computations.* A computer-assisted proof uses interval or ball arithmetic with outward rounding over a continuum of parameters. A point enclosure is a rigorous enclosure at one parameter value. All other numerical values are floating-point values without rounding control; they enter no proof.
- *Lean.* "Formalized in Lean" means formalized in Lean 4 with Mathlib, with no `sorry`, and with the axioms propext, Classical.choice and Quot.sound only (Appendix A).
- *Project notes.* Some results cite a note of the repository as `NAME.md`. A section sign after a note name (for example `KAPPA_BOUNDED` §4) refers to a section of that note, not of this paper. The appendix on provenance lists, for each result, the note that contains the full derivation.

1.5 Relation to other work

Two recent works overlap with this paper.

Related work 1. E. Sussman and L. Alpöge (with Claude), "The Yang–Mills continuum limit along Bałaban's renormalization group: a complete account of Theorem 0" [26], and "SU(2) Yang–Mills theory in four dimensions via Bałaban's renormalisation group: construction, axioms and nontriviality" (draft dated 6 October 2026) [25]. Repository: `https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`, commit `1b92bd0347e3b8763cd8ecc1f96dc3c2a92dd310` (2026-10-27), accessed 2026-10-27. The README states: "This repository contains work-in-progress, written largely by LLM. Its content has not been satisfactorily vetted. All claims should be understood as tentative, pending verification."

Related work 2. OpenAI, "The canonical massive continuum limit of the two-dimensional O(3) model" (October 4, 2026), `github.com/openai/math`, family 215 [18].

The overlaps are as follows.

1. Sussman and Alpöge prove a weak two-sided flow with an additive constant. They do not need per-step positivity, they use a first-passage cutoff, and they use only the containment $g_k \le \gamma$. This is the structure of Theorem 2w.

2. The observation that the weak form of Theorem 2 needs only bounded κ_k, not their convergence, is theirs (the structure of their Lemma 4.19). It entered this work through a summary of their repository, and Lemma DT and Theorem K use it.

3. The result form of Theorem K (L-009), $|S_k - ck \log L| \le \kappa_*$, matches their Lemma 4.2(b), $|\sum_l \beta_l^0 - kc_0| \le c_2$. Our proof of Theorem K (Section 3.6.1) is derived from Bałaban's papers.

4. Lemma DS (L-004), the shooting argument without early positivity, and the cross-scale sweep of Section 3.4.3 were derived without use of their work.

5. Route B's doubling comparison follows the template of [18, Lemma 6.1]. The idea of a coupling band in place of per-step positivity also appears there.

Their public, dated accounts predate this one.

1.6 Organization

Section 2 fixes the notation for both parts of the paper. Section 3 proves the reduction of Theorem 2 (Theorem A). Section 4 proves the compactness theorem under the hypotheses of the named blocks (Theorem B) and contains the dependency table. Section 5 treats the inputs of the sources block. Section 6 collects the numerical results and the computer-assisted proof of (H2). Section 7 lists what remains. Infinite volume, uniqueness, clustering, and the mass gap are outside the scope of this paper. Appendix A lists the Lean formalization, Appendix B indexes the results, and Appendix C maps the numbering of this paper to the names used in the project notes.

2 Setup and notation

This section fixes the notation for the whole paper. Sections 2.1–2.4 serve the coupling flow of Section 3. Section 2.5 serves the compactness theorem of Section 4. Section 2.6 lists the names that would otherwise collide.

2.1 Lattice, measure, and scales

Notation. L is odd, $L > 11$, fixed (Bałaban [8, p. 256]). $K \in \mathbb{N}$, $\varepsilon = L^{-K}$. The physical side $\ell \in \mathbb{N}$ is in units of the unit lattice. $T_\ell^4 = (\mathbb{R}/\ell\mathbb{Z})^4$ and $T_\varepsilon = (\varepsilon\mathbb{Z}/\ell\mathbb{Z})^4$, so $N_\varepsilon = \ell L^K$ sites per direction. The Wilson action is $A(U) = \sum_p P_p(U)$, $P_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U(\partial p) \in [0, 2]$, over unoriented plaquettes p. The measure is $\mu_\varepsilon = Z_\varepsilon^{-1} \exp(-g_0^{-2} A(U)) \prod_b dU_b$ (Haar). x_p is the centre of p.

The bare coupling $g_0 = g_0(\varepsilon, g)$ is any solution of the shooting problem of Lemma 3.6 (Lemma DT) with terminal coupling $g_K = g$. Lemma DT gives existence, not uniqueness. Any choice works below.

Assumption 2.1 (volume hypothesis (Vol), K-032)**.** The torus side obeys $\ell > 22M$, where $M = L^{m_M}$ is the localization cube of [8, p. 257]. This M is the block size of K-025; it is not the M of Δ_M in Section 2.4.

At every step $k \le K$ the local set $\tilde{\square}^5$, of side $11M$ level-k cells, must lift to $\mathbb{Z}^4$. Then the whole-lattice β subtraction leaves only Bałaban's irrelevant class. This needs $N_k > 22M$ level-k cells per side. The number $N_k = \ell L^{K-k}$ is smallest at $k = K$ (Proposition 5.95). (Vol) does not depend on K. The proof of Theorem 2w (L-001) does not use (Vol). The step from Theorem 2w to Theorem 1 (through R-032) and Theorem B use it.

Even form. For Theorem B we take $\ell = 2L^{m_T}$ with $m_T \ge m_M + 1$. This keeps ℓ even and gives $\ell \ge 2LM > 22M$, because $L > 11$. The power form $\ell = L^{N-K}$ with $N \ge K + m_M + 2$ also suffices, but it makes ℓ odd. Bałaban writes both exponents m_T and m_M as m [8]; here we fix them independently.

2.2 Range of L

Bałaban assumes L odd, $L > 11$, for the whole of [8, pp. 251–252]. So his construction of $\beta_j(g)$ for $g > 0$ and his inductive assumption (H0) (H-202) are available only at odd $L > 11$. Every statement below that uses (H0) or $\beta_j(g)$ at $g > 0$ is stated for odd $L > 11$. Otherwise it carries "(H0) and Bałaban's construction at this L" as an explicit hypothesis. Statements about the one-loop quantities $\beta_j(0)$, S_k, and the symbols σ_j hold for odd $L \ge 3$.

2.3 The coupling sequence

$\varepsilon = L^{-K}$, so $\log(L^k\varepsilon)^{-1} = (K-k)\log L$. Let $u_k := 1/g_k^2$. The recursion is $u_{k+1} = u_k - \beta_{k+1}(g_k)$ [8, (0.18), (0.20)]. The increment β_{k+1} depends on the whole history $g_0, \dots, g_k$ [8, p. 298]. $\beta_j(0)$ is the one-loop (Gaussian) increment. By definition it does not depend on the history.

Shooting notation: $u_k = 1/g_k^2$, $t = 1/g^2$, $T(k,K) := \sum_{i=k+1}^{K} \beta_i(g_{i-1})$. The clamp is $\varphi(u) := \max(u, 1/\gamma^2)^{-1/2}$.

Here γ is the upper bound for the couplings in Bałaban's Theorem 1. Bałaban's small constant β_0 (1/7 in [12, p. 389]) is not a coupling (Section 3.1). In Lemma 3.11 (Lemma E) the symbols β_0, β_0' instead denote lower and upper bounds on $\beta_j(0)$.

2.4 One-loop notation

The letter K has two uses. As an integer it is the number of RG steps, $\varepsilon = L^{-K}$. As a vector $K \in \mathrm{BZ} = [-\pi, \pi]^4$ it is a momentum in the Brillouin zone of the coarse unit lattice. Then $k_l = (K + 2\pi l)/L$ are the fine momenta over K. The meaning is always clear from context. The other one-loop objects are defined where they are first used:

- S_k, the $p \to 0$ F^2-coefficient of the composed k-step one-loop integral, and its telescoping into $\beta_j(0)$: Section 3.5.1;
- $c = 11N^2/(12\pi^2)$ and $b := c\log L$: Section 3.5.3;
- the f^m Taylor symbols $\sigma_j^{(m)}(K)$ of the coarse kernel at level j, and d_j, κ_j, κ_W^F: Section 3.5.4;
- Δ_M (Gaussian RG kernel at block size $M = L^j$), Δ_∞, and the bordered matrix $\mathfrak{B}_j(K)$: Sections 3.5.5 and 3.9;
- the one-step minimizer map $H_j(K)$, ρ_1, ρ_2, and the inequality (H2) (R-025): Section 3.7;
- the plane-wave quantities $\beta_{j+1}(p)$, Γ^{ex}, θ_* of Route B: Section 3.6.2.

2.5 Observables and Schwinger functions

Definition 2.2 (observable, Schwinger functions)**.** The observable is $\varphi_p := P_p - c_\varepsilon$, with $c_\varepsilon := \langle P_p \rangle_{\mu_\varepsilon}$. By hypercubic invariance (Section 4.5.6) c_ε does not depend on p. The test space is

$$\mathcal{D}_n := \{F \in C^\infty(T^{4n}) : \text{ every derivative of } F \text{ vanishes on every diagonal } \{x_i = x_j\},\ i \neq j\},$$

with the norm $\|F\|_m := \max_{|\alpha| \le m} \sup |\partial^\alpha F|$. On the compact torus this is the Osterwalder–Schrader (OS) Schwartz seminorm without weights. The lattice Schwinger functions are

$$S_{n,\varepsilon}(F) := \sum_{p_1,\dots,p_n} F(x_{p_1}, \dots, x_{p_n}) \langle \varphi_{p_1} \cdots \varphi_{p_n} \rangle_{\mu_\varepsilon}, \qquad F \in \mathcal{D}_n.$$

These are moments, not cumulants. We insert no field renormalization factor. The product $B_K = \prod_j c_j$ with $c_j = 1 + O(g_{j-1}^3)$ is bounded above and below, because $\sum_j g_j^3 < \infty$. So a factor B_K^n changes only the normalization of the limit.

Remark 2.3 (scaling). At tree level $\langle \varphi_p \varphi_{p'} \rangle \sim \varepsilon^8 g^4 |x_p - x_{p'}|^{-8}$. So $S_{2,\varepsilon}(F) = O(1)$ for $F \in \mathcal{D}_2$ with no rescaling, while $\sum_p f(x_p)\varphi_p$ has variance $\sim \varepsilon^{-4}$. The limit object is a family of distributions off the diagonals, not a random field.

Parity of ℓ. We assume that ℓ is even. For a reflection of the periodic lattice to exchange two halves with both seams on link planes, N_ε must be even. With L odd, $N_\varepsilon = \ell L^K$ is even if and only if ℓ is even. The condition is sufficient for the reflection argument of Section 4; it is not known to be necessary. For odd N_ε every time reflection has one site seam and one link seam. Osterwalder and Seiler treat link-plane reflections [21], and Menotti and Pelissetto treat site-plane reflections [17]. No reference states the mixed case.

2.6 Names

Hypotheses are named by short labels in parentheses, such as (H2) (R-025), (PI) (R-026), or (FX) (R-043). The dependency table (Section 4.6) lists each one. Three names of the project notes collide in one document, and we change them here. Appendix C records the old names.

- The plane-wave theorem (Theorem 3.2 of `BETA_FP_PW`, called "Theorem B" there) is called Theorem $\mathrm{B_{PW}}$ (Theorem 3.15). So "Theorem B" means the main Theorem B.
- The three global conditions of the dependency table are called (GC1) (H-200)–(GC3) (H-201). The labels (G1) (constraint conditions of Proposition 3.12) and (G2) (R-030) (per-cell infinite-volume densities) keep their meaning from Section 3.
- The final sourced bound of the sources block is called ($\mathrm{S_{src}}$). The label K-018 (S) means the smallness condition of Section 3.1.

3 The coupling flow and Theorem 2

This section proves Theorem A. Section 3.1 lists what Bałaban's proof of Theorem 1 uses of the coupling sequence; this is weaker than his (0.31). Section 3.2 states the weak form, Theorem 2w (L-001), that supplies exactly this. Sections 3.4–3.7 prove Theorem 2w from bounded window sums of the one-loop increments, give two routes to these window sums, and reduce the uniform part of both routes to the single inequality (H2) (R-025). Section 3.8 assembles the result. Its output, the trajectory conditions (b), (c-pair), K-018 (S), is the block (B-T2) (H-084) of Section 4.

3.1 What Bałaban's Theorem 1 uses

The statements of Theorems 1 and 2 are quoted in Section 1.2. The stated hypothesis of Theorem 1 is only $g_k \in]0, \gamma]$. The proof uses more. Its working form is [9, pp. 255–256, (2.6)–(2.9)], for scales $m < n$:

- (2.6) $g_n \le (1 + g_n^2 \beta'(n-m))^{1/2} g_m$ and $g_m \le (1+\beta_0) g_n$;
- (2.7) $(\log g_n^{-2})^p \le (1+\beta_0)(\log g_m^{-2})^p$ and $(\log g_m^{-2})^p \le (1 + g_n^2\beta'(n-m))^{\beta_0}(\log g_n^{-2})^p$;
- (2.8), (2.9) the same for $\varepsilon_j = g_j p_0(g_j)$ and R_j.

Here β_0 is Bałaban's small constant (1/7 in [12, p. 389]), not a coupling.

Remark 3.1 (a misnumbered reference). Theorem 1 of [9, p. 262] assumes "(I.0.33)". This is a misnumbered reference to (0.31) of [8].

In u-variables, (2.6)–(2.9) follow from two pairwise bounds:

- (P-up) (H-156) $u_m \le u_n + B'(n-m)$, with no additive constant;
- (P-low) (H-155) $u_m \ge u_n + \beta(n-m) - D'$, with $\beta > 0$, $D' \ge 0$ fixed,

together with the smallness condition

- K-018 (S) $\gamma^{-2} \ge c_{sw} D'$, $c_{sw} = \max(2, 1/(1-(1+\beta_0)^{-2})) = 64/15$ at $\beta_0 = 1/7$, and $\gamma \le \gamma_*(\beta_0, \beta, B', p_0, p_1, r, \nu, \kappa_0)$ (the conditions that literal (0.31) also needs). γ_* is free of the D'-ratio conditions, but it depends on R through B': it contains $\gamma^2 B' \le (1+\beta_0)^2 - 1$ with $B' = 5b/4 + 2R$. The order of constants in Section 3.8 (R before γ) handles this.

β_0 must be fixed before γ.

3.2 The weak form

Theorem 3.2 (Theorem 2w, weak form)**.** *Let $G = SU(2)$, $d = 4$, L odd, $L > 11$ (or L odd ≥ 3, with (H0) (H-202) and Bałaban's construction of $\beta_j(g)$ at this L as hypotheses). Assume (B2) (Section 3.7), the fixed-level inputs (PI), (F), (M_j), V1, (G2) of Section 3.6.1, the history condition L5 (Section 3.4.5), and the bound $|\beta_{k+1}(h,g) - \beta_{k+1}(0)| \le C_\beta g$ on every past h (item 7 of Section 7). Then there are $\gamma > 0$, $g_1 \in]0,\gamma]$, $\beta > 0$, $B' \ge \beta$, $D' \ge 0$ such that for every $K \ge 1$ and every $g \in]0, g_1]$ there is a bare coupling $g_0 = g_0(K, g)$ with*

- *(b) $g_k \in]0,\gamma]$ for $0 \le k \le K$, and $g_K = g$;*
- *(c-pair) for all $0 \le m \le n \le K$: $u_n + \beta(n-m) - D' \le u_m \le u_n + B'(n-m)$;*
- *K-018 (S) holds for γ, D'.*

In particular Bałaban's (0.31) holds up to the additive constant D' on the lower side, with constants $\beta/\log L$ and $B'/\log L$. The two-sided bound fixes $1/g_k^2$ up to constants.

Proof. Theorem K (Theorem 3.14) gives the window bound (5.1) with $R = \kappa_*$. Lemma DT (Lemma 3.6), applied in its history form (Theorem 5.104), gives (b) and (c-pair). The choice of γ in Section 3.8 gives K-018 (S). □

By Theorem 3.19, (H2) (R-025) implies (B2). A second route, Route B, gives the same conclusions from Theorem B_{PW} and Lemma DT, under the hypotheses (U), $(B\theta)$ and the plane-wave defect bounds of Section 3.6.2. Both routes need (H0) at this L and L5. See Section 3.6.

Setting. The torus obeys the volume hypothesis (Vol) (K-032) of Section 2.1: side $\ell > 22M$ final-scale cells, or the even form $\ell = 2L^{m_T}$, $m_T \geq m_M + 1$. The proof of Theorem 2w does not use (Vol), because Lemmas DS and DT use the whole-lattice β_j. (Vol) enters where Theorem 2w is applied on the torus: in Theorem 1 through R-032, and in Theorem B.

Inputs. By Route K, Theorem 2w uses one Bałaban-level statement, R-006 (the inductive assumption (H0)), the inputs R-025–R-031, and the external result X-005 [23]. Section 7.7 lists these inputs.

3.3 Literal (0.31)

- (L1) Literal (0.31) holds for every $K \geq K_0$ and every $g \in]0, \gamma]$ under eventual uniform positivity: there are J and $\beta_* > 0$ with $\beta_j(0) \geq \beta_*$ for all $j > J$, and $\sup_j |\beta_j(0)| < \infty$. This implication is Lemma 3.4 (d); it is formalized in Lean. Theorem K (L-009) and Theorem B_{PW} do not give eventual positivity: they bound window sums, not single increments. Eventual positivity would follow from the convergence route (Section 3.9), which is open.

- (L2) Literal (0.31) holds for every $K \geq 1$ if $\inf_j \beta_j(0) > 0$ and $\sup_j \beta_j(0) < \infty$ (Lemmas D and E). If $\beta_j(0) \to \beta_\infty > 0$, then "every $\beta_j(0) > 0$" is equivalent to $\inf_j \beta_j(0) > 0$ (Lemma DS (L-004), Step 8). This implication is formalized in Lean. Its inputs are open.

- Literal (0.31) is not needed for Theorem 1. Section 3.1 shows that the proof uses (c-pair) and K-018 (S). Literal (0.31) is used directly in one place only, the alternative choice of ε_0 in [9, p. 248], which Bałaban does not adopt.

Remark 3.3 (necessity of positive increments). Literal (0.31) at every $K \geq 1$ forces $\beta_K(0) \geq \beta \log L > 0$ for every K. To see this, take the window $k = K - 1$ and let $g \to 0$, using (H0) (H-202); so L is odd and $L > 11$. So certified early steps are needed only for (L2), not for Theorem 1.

3.4 Shooting lemmas

The shooting lemmas turn bounds on the increments β_j into a trajectory with prescribed terminal coupling. Lemma 3.4 needs eventual positivity of $\beta_j(0)$. Lemma 3.6 needs only bounded window sums, and Theorem 2w (L-001) uses it. The notation is that of Section 2.3. Lemmas DS, DT, D, E are abstract statements about a sequence of functions β_j. Applied to Bałaban's $\beta_j(g)$, they need L odd > 11 (or (H0) (H-202) at this L) and L5.

3.4.1 Lemma DS (shooting without early positivity)

Lemma 3.4 (Lemma DS)**.** Hypotheses. *(i) β_{k+1} is continuous on $[0, \gamma_0]$ and $|\beta_{k+1}(g) - \beta_{k+1}(0)| \leq C_1 g$, with C_1 independent of k, K. (ii) $\sup_j |\beta_j(0)| \leq B$. (iii) There is J with $\beta_j(0) \geq \beta_\infty/2$ for all $j > J$. (This is all the proof uses of "$\beta_j(0) \to \beta_\infty > 0$".)*

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, \beta_\infty/(4C_1))$; $B' := B + \beta_\infty/4$; $D := J \cdot B'$; $D' := J(B + \beta_\infty/2)$; $g_1 := (1/\gamma^2 + D)^{-1/2}$; $K_0 := J + \lceil 8D'/\beta_\infty \rceil$.

Conclusions.

- *(a) $T(k, K) \geq -D$ for all $k < K$ along any couplings in $]0, \gamma]$.*

- *(b) For every $K \geq 1$ and $g \in]0, g_1]$ there is g_0 with all $g_k \in]0, \gamma]$ and $g_K = g$.*
- *(c) $u_K + (\beta_\infty/4)(K-k) - D' \leq u_k \leq u_K + B'(K-k)$. The upper side has no constant.*
- *(d) For $K \geq K_0$: literal (0.31) with lower constant $\beta_\infty/8$, and shooting for every $g \in]0, \gamma]$. If every $\beta_j(0) > 0$: literal (0.31) for all K (Step 8, that is, Lemmas D and E).*

Proof (sketch). The clamped flow $\tilde{u}_{k+1} = \tilde{u}_k - \beta_{k+1}(\varphi(\tilde{u}_k))$ is continuous in u_0. Its tail sums lie in $[-D, B'(K-k)]$ for every u_0, because clamped couplings lie in $]0, \gamma]$. The intermediate value theorem on $[t - D, t + B'K]$ gives $\tilde{u}_K = t$. Then $\tilde{u}_k \geq t - D \geq 1/\gamma^2$ for $g \leq g_1$, so the clamp is never active. The bound is derived for the clamped flow first, so the argument is not circular. For history-dependent β this proof needs β_{k+1} defined and continuous on the whole cube $]0, \gamma]^{k+1}$; Section 3.4.5 removes this need. □

Lean. Lemma DS is formalized in Lean as `YmLean.lemma_DPrime` (`lean/YmLean/LemmaDPrime.lean`). The Lean hypotheses are weaker: continuity on the interval instead of C^1, and (iii) instead of convergence. Step 8 is `lemma_DPrime_step8` (`lean/YmLean/LemmaDE.lean`). These files treat one-argument $\beta_j(g)$; the history form is in Section 3.4.5.

Remark 3.5 (the condition $g \leq g_1$). Without $g \leq g_1$, (b) can fail at small K: take $\beta_1 = \beta_2 = \beta_3 = -0.8$, $K = 3$, $g = \gamma$.

3.4.2 Lemma DT (tail-sum form; no index J)

Lemma 3.6 (Lemma DT)**.** Hypotheses. *(i) as in Lemma 3.4, and the window bound*

$$\textit{(5.1)} \quad T_0(k, K) := \sum_{i=k+1}^{K} \beta_i(0) \in [\, b(K-k) - 2R,\ b(K-k) + 2R\,] \quad \textit{for all } 0 \leq k < K,$$

with $b := c \log L > 0$ and some $R \geq 0$. (Route K gives $R = \kappa_*$; Route B gives $R = C_B$.)

Constants. $\gamma := \min(\gamma_0, \gamma_{Bal}, b/(4C_1))$; $\beta := 3b/4$; $B' := 5b/4 + 2R$; $D = D' := 2R$; $g_1 := (1/\gamma^2 + D)^{-1/2}$.

Conclusions. *(b) of Lemma 3.4, and the pairwise bound*

$$\textit{(c-pair)} \quad u_n + \beta(n-m) - D' \leq u_m \leq u_n + B'(n-m), \quad 0 \leq m \leq n \leq K,$$

with no constant on the upper side.

Proof. $|T - T_0| \leq C_1\gamma(K-k) \leq (b/4)(K-k)$. Add (5.1). Steps 2–5 of the proof of Lemma DS (L-004) use only continuity and the bracket $\tilde{T} \in [-D, B'(K-k)]$; this bracket now comes from (5.1). The pairwise form is the same window argument with $K \to n$. For history-dependent β, Section 3.4.5 replaces the clamped flow by nested intervals. □

Lean. Lemma DT is formalized in Lean as `YmLean.lemma_DDoublePrime` (`lean/YmLean/LemmaDDoublePrime.lean`). The formal statement proves (a-tail), (b) with the clamp inactive on the solution, (c-pair) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, and (d) below. Its hypotheses are weaker than those of Lemma 3.6: continuity instead of C^1; any γ with $0 < \gamma \leq \gamma_0$ and $C_1\gamma \leq b/4$; $R \geq 0$ derived; $K = 0$ allowed. (c-pair) and (d) hold for every trajectory with couplings in $]0, \gamma]$.

In addition the following holds, and it is formalized in Lean. (d) For $n - m \geq 8R/b$, $u_n + (b/2)(n-m) \leq u_m \leq u_n + B'(n-m)$. So (5.1) alone gives literal (0.31), with lower constant $b/2$, only for windows with $n - m \geq 8R/b$. Short windows near K are not controlled (see Section 3.3).

3.4.3 Lemma DT and Bałaban's proof: the cross-scale sweep

We searched all twelve papers of Bałaban's series for uses of the coupling sequence at two different scales, and we read every hit that is not single-scale on the printed page. The comparisons A–E on the pages [11, p. 192, p. 198 (1.94)], [12, p. 381 (twice), p. 386] are of this kind. Comparison A has the factor $(1+\beta_0)^2(k-j)^{\beta_0}$, and there $\delta'_j = g_j A_1 p_1(g_j)$. The five comparisons give four rows of the list below: A is item 11, E is item 12, C and D are item 16, and B is item 18.

Proposition 3.7 (cross-scale sweep, R-034)**.** *There are 19 cross-scale uses of the coupling sequence, all in CMP 119 and CMP 122 I–II; there are none in CMP 95, 96, 98, 99, 102, 109, 116. Each is covered by (b), (c-pair) and K-018 (S) of Section 3.1:*

- *items 1, 7, 10, 12, 17 need (P-up) (H-156) with no constant (for example, (2.6) line 1 is exactly (P-up));*
- *items 2, 3, 11, 13, 15, 16, 18 need (P-low) (H-155) with fixed multiplicative slack $(1+\beta_0)$ or $(1+\beta_0)^2$; item 2 ((2.6) line 2) gives the constant $64/15$; item 11 needs only $2.42D'$;*
- *items 4 ((2.8)–(2.9), ε and R ratios), 6 (CMP 119 (3.7), (3.8), (3.12), (3.13)), and 9 (CMP 122 I (1.52)), the last two of the form $\varepsilon_{k+1} \le (1+\beta_0)\varepsilon_k$, use both sides; they follow as items 1 and 3 do (item 4 from items 1–3);*
- *item 14 (CMP 122 I p. 200, R_{k-N_0+1} against R_k) uses the lower side; it follows as item 3 does;*
- *item 5 (CMP 119 (2.31), (2.46): $\sum_j g_j^{\kappa_0} \le g_n^{\kappa_0-6}$) needs linear growth (P-low) and $\gamma^{-2} \ge 2D'$;*
- *item 0 is CMP 119 Theorem 1 assuming (0.31); its proof works with (2.6)–(2.9);*
- *item 8 (CMP 119 p. 284) is in a sketch and carries no load.*

Remark 3.8 (scope of the sweep). (1) The sweep is a search of the text together with a reading of each hit on the page. It is not a line-by-line reading of every proof. (2) Bałaban's sentence "Similar inequalities hold for other constants, which will be introduced later" [9, p. 256] refers to constants that the sweep did not locate one by one; Proposition 5.85 gives the list. (3) Bałaban says that β_0 "can be chosen arbitrarily small, if g is sufficiently small". Under (P-low) the floor is $\beta_0 \gtrsim D\gamma^2/2$, so β_0 must be fixed first. We found no step that needs $\beta_0 \to 0$.

Convention 3.9 (text readings of the sweep)**.** For the passages Π in (4.37)/(5.1), CMP 119 (2.42), the references (2.12)/(2.23), the estimates of Sect. 3, and the bound lists of the 1989 papers, the sweep uses the scanned text of Bałaban's papers. A different reading of one of these passages on the printed page would require a new check of the items that cite it.

3.4.4 Lemmas D and E

Lemma 3.10 (Lemma D)**.** *If $b \le \beta_j(g) \le b'$ on $]0,\gamma]$ with $b > 0$, then for every K and $g \in]0,\gamma]$ shooting succeeds and $t + b(K-k) \le u_k \le t + b'(K-k)$.*

This is literal (0.31). The proof is an intermediate value argument on the interval $[t+bK,\ t+b'K]$ of initial values. A nested-interval variant holds for β defined only on admissible histories; it is not formalized in Lean. Lemma D is formalized in Lean as `lemma_D` (`lean/YmLean/LemmaDE.lean`).

Lemma 3.11 (Lemma E)**.** *If $\beta_0 \le \beta_j(0) \le \beta'_0$ and $|\beta_j(g) - \beta_j(0)| \le Cg^2$, then for $\gamma \le \sqrt{\beta_0/(2C)}$: $\beta_0/2 \le \beta_j(g) \le \beta'_0 + \beta_0/2$.*

Lemma E is formalized in Lean as `lemma_E`. The same two-line proof gives the C^1 form: if $|\beta_j(g) - \beta_j(0)| \le C_1 g$ and $\gamma \le \beta_0/(2C_1)$, then $|\beta_j(g) - \beta_j(0)| \le \beta_0/2$, and the same conclusion follows.

3.4.5 History dependence (L5)

Bałaban's β_{k+1} depends on $(g_0, \dots, g_k)$. Two facts make this dependence matter.

- *Domain.* Bałaban defines β_{k+1} only on histories for which his Theorem 1 induction has run to step k. By the sweep (Section 3.4.3), that induction uses (2.6)–(2.9), that is (c-pair) and K-018 (S), on the history, not only $g_i \in]0, \gamma]$.
- *The clamp.* The clamped shooting of Lemmas 3.4 and 3.6 evaluates β on clamped histories $\tilde{g}_i = \varphi(\tilde{u}_i)$. A clamped history can violate the growth side of (c-pair): from $\max(a, c) - \max(b, c) \ge \min(a - b, 0)$ one gets only $u_m \ge u_n - D'$, not $u_m \ge u_n + \beta(n - m) - D'$. So the clamped proof needs β on the whole cube of histories, which Bałaban does not supply.

Theorem 5.104 (Lemmas DT-nest and DS-nest) removes the clamp. It uses nested intervals on true trajectories, with domain $D_k = \{u_0, \dots, u_k \ge \gamma^{-2}\}$ and the constants of Lemma 3.6. On such trajectories, (c-pair) up to step k follows from the tail-sum bound (a-tail) of Lemma DT, and K-018 (S) holds by the choice of γ. So every evaluation of β stays inside Bałaban's domain. The continuity input (C-tr) (H-089) is the continuity of β along true trajectories; it is proved relative to Bałaban's small-field step in Section 5.1. The history forms are formalized in Lean as `lemma_D_hist`, `lemma_DPrime_hist`, `nested_shoot`, `lemma_DPrime_hist_adm` and `lemma_DDoublePrime_hist_adm`. These need no continuity on the cube. A Lean counterexample, `βce_no_shot`, shows that separate continuity in each argument is not enough.

3.5 The exact structure of the one-loop sums

The window sums of the shooting lemma are sums of one-loop increments $\beta_j(0)$. This section expresses these sums as level-k quantities of a coarse Gaussian theory. It isolates the remainder κ_k whose boundedness is the content of Theorem K (L-009). It also lists the symbol facts, uniform in the level, that Section 3.6 uses.

3.5.1 Telescoping

Let S_k be the $p \to 0$ F^2-coefficient of the composed k-step gauge-fixed one-loop integral (U1 (1.2)). Then $S_k = \sum_{j=1}^{k} \beta_j(0)$ and $S_0 = 0$ (`BETA_FP` Prop. 1.1). The identity is exact for the $p \to 0$ coefficient at fixed k. It does not hold for $\log Z_k$ at fixed p, because the plane wave is off-shell.

3.5.2 Anchor-free identity

Proposition 3.12 (anchor-free identity; Prop. 2.1 of `BETA_FP_LEMMA43`)**.** *Under (G1) (H-194)* $= \{\Delta D = 0,\ \tilde{Q} D = D_c \iota_a^T\}$*:*

$$\log Z_\Lambda(V) = \big[\log\det X - \tfrac{1}{2}\log\det K\big] - \big[\log\det(D_c^T D_c) - \tfrac{1}{2}\log\det(\Delta_\Lambda(V) + D_c D_c^T)\big].$$

In words: the constrained one-loop term equals the fine Wilson one-loop term in background Feynman gauge minus the one-loop term of the coarse Gaussian theory with kernel $\Delta_\Lambda(V)$. All point-anchor objects cancel, and the constant is exactly 0. A lattice computation with the CMP 109 averaging at random SU(2) backgrounds agrees with the identity to relative error 10^{-14}.

- $\tilde{Q}D = D_c\iota_a^T$ holds exactly at any background (lattice check: 10^{-15}). The condition $\Delta D = 0$ carries load: the raw Hessian gives $O(1)$ errors.

- At the constant abelian field V_f the background is critical. So $\Delta D = 0$ holds exactly, and Proposition 3.12 holds exactly. At a plane wave it holds up to the defect $\mathfrak{d}_Y$ (`BETA_FP_PW` Lemma 2.3).

- The identity concerns per-cell infinite-volume densities. On a torus the constant zero modes must be removed. This is the input (G2) (R-030), which holds at each fixed j relative to the inputs listed in Section 5.1.

3.5.3 The constant c

$c = 11N^2/(12\pi^2) = 2N \cdot b_0/(8\pi^2)$, $b_0 = 11N/3$; for SU(2), $c = 0.371511$. The normalization is $1/g_B^2 = 2N/g^2$. The log coefficient of the fine piece at constant field is $c_2 = c/2$ per $\ln(1/\varphi)$; this agrees with Savvidy [23, eq. (1.4)]. This is the input (F) (R-027). Theorem 5.134 proves it on the lattice by a proper-time argument.

$b := c\log L$: 0.25751 ($L = 2$), 0.40815 (3), 0.51502 (4), 0.59793 (5), 0.72293 (7).

3.5.4 The decomposition $S_k = ck\log L + \kappa_W^F - \kappa_k$

As bookkeeping the split is empty: any c' gives an exact identity if κ_k is defined as the remainder. Boundedness of a remainder-defined κ_k is a restatement of the problem. The content comes from a definition of κ_k by the coarse theory alone.

Proposition 3.13 (`KAPPA_BOUNDED` Prop. 1.1)**.** *Assume (PI) (R-026), (F), (M_j), V1, (G2). With the Wilson level $j = 0$ as reference,*

$$S_j = cj\log L - d_j, \qquad \kappa_j = \kappa_W^F + d_j,$$

$$d_j := -\frac{1}{2n_2}\int_0^1 dt \int_{\mathrm{BZ}} \frac{dK}{(2\pi)^4}\ \mathrm{tr}\big[b_{t,2}\,\delta\sigma_j^{(0)} + b_{t,1}\,\delta\sigma_j^{(1)} + b_{t,0}\,\delta\sigma_j^{(2)}\big],$$

where $\delta\sigma_j^{(m)} = \sigma_j^{(m)} - \sigma_0^{(m)}$ are differences of the f^m Taylor symbols of the coarse kernel at level j and level 0, and $b_{t,m}$ are the Moyal inverse coefficients of the interpolated coarse operator. No limit symbols σ_∞ are used.

- Numerical checks: $\rho(1) = \kappa_W^F$ (Lüscher–Weisz (5.7) [16]); at $L = 2$, $d_1 = -0.372$ and $\kappa_1 = 1.114$; at $L = 3$, $d_1 = c\log 3 - \beta_1 = -0.256 \pm 0.01$ (fits give $-0.254\cdots - 0.258$). These check the definition of d_j through S_j. They do not evaluate (1.1) from the symbols independently.

- Cor. 3.1 of `BETA_FP_LEMMA43`, $\mathcal{C}(\Lambda) = c\log\Lambda + \kappa_W^F - \rho(\Lambda)$, is an exact identity. The statement "the whole $\log\Lambda$ sits in the fine piece" is equivalent to $\rho(\Lambda) = o(\log\Lambda)$. It is not a separate fact.

- Given convergence $\beta_j(0) \to \beta_\infty$, Cesàro gives $\beta_\infty = c\log L - \lim\kappa_k/k$. So the hypothesis ($\mathrm{H}_o$) "$\kappa_k = o(k)$" is equivalent to $\beta_\infty = c\log L$, not weaker. Theorem K (Theorem 3.14) gives κ_k bounded, hence $\kappa_k = o(k)$.

3.5.5 Level-uniform symbol facts

We use the following facts about the coarse symbols (`U3_AW.md`).

- O5/A0: Δ_M is real-analytic on a fixed strip, and $|\partial^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, uniformly down to $K = 0$ (odd L, centred blocks). Prop. B: $c_0|\hat{K}|^{-2}P_T' \le P_T' S_M P_T' \le C_1|\hat{K}|^{-2}P_T'$, with $c_0 = (2/\pi)^{12}$ and $C_1 \le 1363$. The constants are existential.
- Prop. E (evenness): $\sigma_j(f;-K) = \sigma_j(f;K)$ in the midpoint trivialization, for all f and j. A numerical test in $d = 2$ agrees.
- Theorem W (L-308): $\sigma_j^{(1)}(0) = \tau_c = (3ie/2)(E_{10} - E_{01})$ for every $j \ge 0$. This follows from the exact Ward identity $\Delta_j(f)D_f = 0$. The numerical values are 0.750000, 0.75015, 0.75051 at $j = 0, 1, 2$, against 0.75 ($d = 2$ proxy).
- L-071: $|\sigma_j^{(1)}(K) - \tau_c| \le C_A|K|^2$ for real K, uniformly in j (C^0 only).
- Prop. R-003: $\sigma_j^{(m)}$, $m \le 2$, is analytic on the same strip at every fixed j. This is not uniform in j for the derivatives of $m = 1$, nor for $m = 2$.
- Lemma M (L-219): the Maxwell coefficient is exactly 1 at every level. For every $M \in \{1, L, \dots, \infty\}$ the degree-≤ 3 Taylor polynomial of Δ_M at $K = 0$ in midpoint coordinates is $|K|^2\mathbf{1} - KK^T$.

3.6 Bounded window sums: two conditional routes

This section gives two routes to the window bound (5.1) of Lemma 3.6, $|S_K - S_k - b(K-k)| \le 2R$, with $b = c\log L$. Lemma DT (L-005) then gives Theorem 2w (L-001), at odd $L > 11$ (or with (H0) (H-202) at this L as a hypothesis) and with L5. Neither route uses convergence of $\beta_j(0)$, a rate, (N), or certified early steps. The two routes have formally distinct named hypotheses. Their mathematical content is not independent: both reduce their uniform-in-level content to the same inequality (H2) (R-025) on the two-step minimizer map (Section 3.7), together with facts of the type of Theorem A3 (L-071).

3.6.1 Route K: Theorem K

Theorem 3.14 (Theorem K)**.** *Let L be odd, $L \ge 3$. Assume (B2) (Section 3.7) and the fixed-level inputs (PI), (F), (M_j), V1, (G2) listed below. Then $\sup_k |d_k| \le \kappa_* < \infty$. So $\sup_k |\kappa_k| \le |\kappa_W^F| + \kappa_*$ and $|S_k - ck\log L| \le \kappa_*$ for all k. Window sums satisfy (5.1) with $R = \kappa_*$.*

Theorem K is a statement about one-loop quantities. Its use in Lemma DT for Bałaban's $\beta_j(g)$ needs L odd > 11, or (H0) at this L. By Theorem 3.19, (H2) implies (B2).

Proof. 1. By `KAPPA_BOUNDED` Prop. 1.1 (Section 3.5.4), $S_j = cj\log L - d_j$, under (PI) (R-026), (F) (R-027), (M_j), V1, (G2) (R-030).

2. Lemma 2.1: the only K-derivative of $\sigma^{(1)}$ in d_j enters linearly with f^0 coefficients. Integration by parts moves it onto f^0 data. The boundary terms vanish: the integrand is periodic up to a common constant conjugation, and the inner sphere contributes $O(r^2)$. So $\sigma^{(1)}$ is needed only in C^0, which L-071 gives.

3. Prop. 2.2: with (E0) the f^0 bounds (Prop. B, O5/A0, Lemma M (L-219)), (E1) L-071 and Theorem W (L-308), and (B2), every term of the d_j integrand is $\le C|K|^{-2}$, uniformly in j and t. In $d = 4$ this leaves two powers of infrared margin. So $|d_j| \le \kappa_* := (C(L)/n_2)(1 + Y_1^2 + C_A(1 + Y_1) + Y_2) \cdot I_2$.

4. Infrared: the degree -4 part of the coarse f^2 integrand is the same at every level (Lemma M and Theorem W). So the $\tau_c \log$ term cancels exactly in level differences. □

The constants are existential; κ_* is not evaluated.
The fixed-level inputs of Route K are needed at each j separately; no uniformity in j is required.

- (PI) probe independence: the plane-wave p^2 coefficient equals the constant-field f^2 coefficient. It holds at each fixed level relative to the inputs listed in Section 5.1 (R-026).
- (F) the fine log coefficient c_2 at constant field (Section 3.5.3). This is Theorem 5.134.
- (M_j) the formal f^2 coefficient d_j equals the true one at fixed j. This is Theorem 5.102, proved relative to the inputs listed there. A resolvent bound cannot give it directly, because of the Nielsen–Olesen zero at $k_\perp^2 = |ef|$. Theorem 5.102 uses a direct bound on $\log|\det|$ instead.
- V1: the Maxwell coefficient of the per-step determinant equals $\beta_{j+1}(0)$; that is, CMP 109 (1.3)–(1.5) is complete at order g^0. It holds at each fixed level, given (G2), Bałaban's inductive bounds, and the averaging (0.4) (Section 5.1).
- (G2) per-cell infinite-volume densities. It holds at each fixed j relative to the inputs listed in Section 5.1. We do not prove a j-uniform version, and Route K does not need one.
- (H0) Bałaban's inductive assumption: β_{j+1} is smooth on $[0, \gamma]$, "uniformly bounded on this interval together with all derivatives" [8, pp. 263–264, (1.22)]. It is available at odd $L > 11$ only. CMP 116 has no theorem statement and no C^m-in-g bound in words. The smoothness in g is the inductive assumption of CMP 109; CMP 109 Theorem 3 asserts it, and CMP 116 completes the proof of Theorem 3.
- L5: the history-dependent β, its domain, and its continuity (Section 3.4.5; Theorem 5.104).

3.6.2 Route B: Theorem $\mathrm{B_{PW}}$

Theorem 3.15 (Theorem $\mathrm{B_{PW}}$; Theorem 3.2 of `BETA_FP_PW`)**.** *Let L be odd, $L \ge 3$. Assume*

- *(U) there are $p_* > 0$ and C_U with $|\beta_{j+1}(p) - \beta_{j+1}(0)| \le C_U p^2$ for all $j \ge 0$, $0 < p \le p_*$;*
- *(Bθ) there is $\theta_* \in (0, \min(p_*, 1))$ with $B_* := \sup_k |\Gamma^{ex}(\theta_*; L^k)| < \infty$, where Γ^{ex} is the coarse remainder at fixed coarse momentum θ_*;*
- *Lüscher–Weisz (5.7) [16].*

Then $|S_k - ck \log L| \le C_B := B_* + c_2 \ln(1/\theta_*^2) + |\kappa_W^F| + \sup_{p \le \theta_*} |\eta(p)| + C_U \theta_*^2/(L^2 - 1)$, and window sums satisfy (5.1) with $R = C_B$.

The condition $\theta_* < 1$ makes $c_2 \ln(1/\theta_*^2) \ge 0$; without it the term must be replaced by its absolute value. The identity 3.0 of `BETA_FP_PW` is a tautology ($\Gamma^{ex} := \hat{\Pi} - S$), so all content sits in (Bθ) and (Cθ). A lattice computation on a 4^4 lattice confirms the determinant identity of Prop. 2.2 there to $4 \cdot 10^{-13}$.

Key step. Lemma 1.3: $|S(\theta; L^k) - S_k| \le C_U \theta^2/(L^2-1)$ from (U). The massless coarse side is then needed only at a fixed $\theta_* > 0$, where there is no infrared problem. Route B uses no (W), (PI), (M), infrared weights, or spectral theory.

Remark 3.16 (a possible approach to $(B\theta)$). We do not prove $(B\theta)$. The following plausibility argument (`BETA_FP_PW` §4) reduces it to three items. The first is Lemma 4.1(i), the convergence at trivial background, which follows from O5 at odd L. The second is a vertex bound at fixed θ_*; the fixed-θ data hypothesis $(D\theta)$ is needed only in its "bounded" form. The third is boundedness in k of the defect $\mathfrak{d} = \mathfrak{d}_Y + \mathfrak{d}_{gf}$.

- $\mathfrak{d}_Y$ is the defect of Prop. 2.1 at the non-stationary plane wave. A plausibility argument gives $O(a\Lambda^{-2})$ relative size at fixed θ, from the zero capacity of point anchors in $d = 4$; the region $|K| \lesssim \Lambda^{-1}$ is not treated.
- $\mathfrak{d}_{gf}$ compares the composite gauge with the D_k^T gauge at the plane wave. A plausibility argument gives $O(\theta^2)$ relative size; uniformity in k is not shown. Data: 0.953 (direct) against 0.9595 (recursion) at $L = 2$, $p_0 = 0.2$. The difference is below the tabulation error.

Identification. Route B's $\beta_j(0)$ is the per-level plane-wave increment, the quantity that the code computes. It matches Bałaban's second-moment β up to the expansion point (fine wave against constrained minimizer). A plausibility argument shows that this changes S by $O(\theta^2)$ relative. This identification and the bound on $\mathfrak{d}_{gf}$ need expansion-point bounds uniform in k. So Route B does not remove the (PI)-type content of Route K; it moves it into a uniform-in-k input.

Three structural points follow.

1. $\theta_* < 1$ is required (above).
2. The identification of β is a uniform-in-k input, not a fixed-level input.
3. (U) is a uniform-in-j analyticity of the level data. It plausibly needs the plane-wave analogue of a uniform C^1 bound on $\sigma^{(1)}$, that is, uniform p-smoothness of the a^1 vertex. This is the same uniform first-order vertex smoothness as the condition $(C1\sigma)$ of Route K, which (H2) implies (Theorem 3.18).

3.6.3 The hypotheses of Route B

Remark 3.17 (a possible approach to (U) and $(B\theta)$). `BETA_FP_U.md` §2–§4 gives a possible approach to the two uniform hypotheses of Theorem $\mathrm{B_{PW}}$.

- (U) from (T1) and (H2) (its Theorem 2.4, odd $L \ge 3$, with A0, R-001, Lemma 2.2). Here (T1) is $\sup_j \sup_q |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of L-071, for which there is a plausibility argument. (H2) is the inequality of Section 3.7; `BETA_FP_U` writes it as $\tilde{h}_{(2)} < L$.
- $(B\theta)$: $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ exactly (its Prop. 3.1). The term γ_{rec} is bounded in k given Theorem 2.4 (its Prop. 3.2). The term $\mathfrak{d}_F$ is bounded under the hypothesis (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$. At fixed θ, $\mathfrak{d}_F$ is $O(1)$ relative, not $O(\theta^2)$; only boundedness is used.
- Its Corollary 4.1: given (T1), (H2), (DF), and Lüscher–Weisz (5.7), $|S_k - ck \log L| \le C_B$, so (5.1) holds with $R = C_B$.
- Theorem 2w by Route B then needs in addition the identification of the plane-wave β with Bałaban's β up to the expansion point, uniform in k, and (H0) (Lemma DT (i)).

In this approach Route B uses (T1), (H2), (DF), Lüscher–Weisz (5.7), the identification of β, and (H0), together with L5, (G2) and odd L, as Route K does. The fixed-level inputs (PI), (M_j), V1 of Route K do not appear; (DF) and the identification of β replace them.

3.6.4 Comparison

	Route K	Route B
uniform-in-level input	(B2): $\sup_j \sup_K \lvert\sigma_j^{(2)}\rvert < \infty$; (H2) $\Rightarrow$ (B2) (Theorem 3.19)	(U), $(B\theta)$, and the β identification and $\mathfrak{d}_{gf}$ bound, both with k-uniform expansion-point bounds; a possible approach derives (U) from (T1) and (H2), and $(B\theta)$ from Theorem 2.4 and (DF)
single uniform inequality	(H2)	(H2)
fixed-level inputs	(PI), (F), (M_j), V1, (G2)	Lüscher–Weisz (5.7), $\mathfrak{d}_Y$, $\theta_* < 1$
uses (W), L-071	yes	no; the possible approach uses (T1), the plane-wave analogue of L-071
logic	Theorem K, conditional on (B2) and the fixed-level inputs	Theorem $\mathrm{B_{PW}}$, conditional on (U) and $(B\theta)$
common	(H0) (H-202) at odd $L > 11$, L5, (G2), V1-type identification, Bałaban's proofs, SU(2)	

Both routes share (H0), L5, (G2), odd L (> 11 for (H0)), and the identification of the computed $\beta_j(0)$ with Bałaban's $\beta_j(0)$ (V1; in Route B also the expansion point). The uniform-in-level hypotheses, (B2) against (U) and $(B\theta)$, are formally distinct. Their content overlaps: both rest on (H2) (Section 3.7). In Route K, (H2) gives $(C1\sigma)$ (L-011) and then (B2) (L-012). In Route B, (H2) and (T1) would give (U), and (H2) would give the vertex bound behind $(B\theta)$. So a proof of (H2) serves both routes. One closed route suffices for Theorem 2w.

3.7 The hypothesis (B2) of Route K, and the single uniform inequality (H2)

Both routes need a bound on the coarse symbols that is uniform in the level. This section states that bound for Route K and reduces it to the inequality (H2) (R-025). The numerical values of (H2) are in Sections 6.4–6.11. At $L = 13$, Theorem 6.2 (Section 6.7) proves (H2).

(B2): $\sup_{j\geq 0} \sup_{K\in \mathrm{BZ}} \lvert\sigma_j^{(2)}(K)\rvert < \infty$. A weaker form suffices, (B2w) (H-184): $\sup_j \int_{\mathrm{BZ}} |K|^{-2} \lvert\sigma_j^{(2)}(K)\rvert \, d^4K < \infty$. The reason is that $\sigma^{(2)}$ enters d_j only undifferentiated (`KAPPA_BOUNDED` Remark 2.3).

The symbol $\sigma^{(2)}$ obeys the one-step recursion (`U3_AW` (1.2)) $\sigma_j^{(2)} = L^{-2}\mathcal{A}_j \sigma_{j-1}^{(2)} + Q_j[\sigma_{j-1}^{(1)}] + g_j^{(2)}$, with $\mathcal{A}_j s = L^{-2} H_j^\dagger (\oplus_l s(k_l)) H_j$ and $H_j(K)$ the one-step gauge-fixed minimizer map. Notation: $\rho_1 := \sup_{j,K} \|H_j(K)\|/L$ (one step); $H_{(2),j}(K) := (\oplus_l H_{j-1}(k_l)) H_j(K)$ (two steps), $\rho_2 := \sup_{j\geq 2} \sup_{K \text{ real}} \|H_{(2),j}(K)\|/L^2$.

3.7.1 Known bounds

- The $K = 0$ bound: $\sup_j \lvert\sigma_j^{(2)}(0)\rvert \leq C(|e|C_A + e^2)$. L-071 and scaling give $\lvert\partial^2 \sigma_j^{(1)}(0)\rvert \leq 2C_A$ by polarization, and the slaving identity (W-2) (H-178) expresses $\sigma_j^{(2)}(0)$ through these second

derivatives.

- Each fixed j: $\sigma_j^{(2)}$ is analytic on a fixed strip (Prop. R-003), hence bounded. The content of (B2) is uniformity in j.

- Prop. 4.1: if (h) $h := \sup_{j,K} \|H_j(K)\| < L^2$ and ($C1\sigma$) $\sup_j \sup_K |\partial_K \sigma_j^{(1)}(K)| < \infty$, then (B2) holds. Proof: in the form norm $\pm s \le N(\Delta + 1)$, $N_j \le (h^2/L^4)N_{j-1} + Z_2$, and Z_2 is finite by L-071, ($C1\sigma$), and the fixed f^0 vertices.

- The mechanism of L-072 alone does not give (B2). The obstruction is the missing uniform C^1 bound on $\sigma^{(1)}$. It enters twice: in Q_j (the term $L^{-3}\partial_K \sigma_{j-1}^{(1)}$) and in the longitudinal part of $\sigma^{(2)}$ given by (W-2). L-071 is C^0 only.

3.7.2 The one-step bounds at $L = 3$

- The condition (HS) $h < L\sqrt{L-1}$ fails at $L = 3$. The measured values $\max_K \|H_j(K)\|/L$ are 1.652, 1.736, 1.757, 1.760 for $j = 1, 2, 3, \infty$, and the threshold is $\sqrt{2} = 1.414$. A gauge choice cannot help: H_j is already minimal-norm in its gauge orbit, for gauge fixings that select an orbit.

- The one-step sup-norm C^1 recursion at $L = 3$ has the factor $\rho_1^2/L = 0.909, 1.0046, 1.0289, 1.0325$ for $j = 1, 2, 3, \infty$ (numerical). So it exceeds 1 for $j \ge 2$ ($\rho_1 > \sqrt{3} = 1.7321$). At $K = 0$ the factor is L^{-1}. The bad region is the face $K_3 = K_4 = \pi$, $K_1 = K_2 \approx 1.86$–1.92. This shows that the one-step sup-norm bound fails at $L = 3$. It does not show that a C^1 bound fails; a weighted norm is not excluded.

3.7.3 (H2) and the two-step theorems

$$\text{(H2)} \quad \rho_2 = \sup_{j \ge 2} \sup_{K \in \mathrm{BZ\ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L.$$

(H2) is a plain operator-norm inequality on real K. It uses no strip, no Δ-relative weight, and only $n = 2$.

Theorem 3.18 (L-011)**.** *(H2) implies ($C1\sigma$), with contraction factor $q_1 = \rho_2^2/L^2$ on the derivative.*

Theorem 3.19 (L-012)**.** *(H2) implies (B2), hence (B2w), in plain sup norm, with factor $q_2 = \rho_2^2/L^4$.*

Here ($C1\sigma$) is $\sup_j \sup_K |\partial_K \sigma_j^{(1)}(K)| < \infty$ (Section 3.7.1). The proofs (`B2_TWOSTEP.md` §1–§3) rest on the following points.

- The two-step map. $\mathcal{A}_j \mathcal{A}_{j-1} s = L^{-4} H_{(2)}^{\dagger}(\oplus s) H_{(2)}$, with $dk_{l,l'}/dK = L^{-2}$.

- Theorem 3.18. The chain-rule gain is exactly L^{-2} per two steps on real K. So the threshold is $\rho_2 < L$, not $\sqrt{L^2 - 1}$: no strip-Cauchy loss enters. The large constants ($b(L)$ of O6, η_1, η_2, G_1) enter only the source term, not the contraction factor.

- Theorem 3.19. This contraction needs only $\rho_2 < L^2$, which (H2) implies. In one step it is the condition (h) of Prop. 4.1.

- (H2) and (H_A). (H2) is the plain operator norm of the same matrix as the condition (H_A) of `U3_AW` at $n = 2$. It is weaker: the threshold is L (3 at $L = 3$) against $\sqrt{L^2 - 1}$ (2.828), on real K only, and for $n = 2$ only. We do not use (H_A).

- The form of Q_j (G3). The proofs use the form of Q_j in `U3_AW` (1.2)/(3.3), in which $\partial\sigma^{(1)}$ enters linearly. We take the formula (3.3) of `U3_LEMMA51` as given there; we do not rederive it from the Moyal rule here.

- We do not need C^2 and C^3 bounds on $\sigma^{(1)}$ for (B2).

- Submultiplicativity: $\rho_2 \le \rho_1^2$ always. At $L = 3$ the gain of the two-step map is that this inequality is strict by a factor 1.50 (2.064 against 3.098).

3.7.4 One-step sufficiency

$\|H_{(2),j}(K)\| \le \max_l \|H_{j-1}(k_l)\| \cdot \|H_j(K)\| \le L^2\rho_1^2$. So $\rho_1^2 < L$ implies (H2). Then the one-step C^1 recursion of `B2_TWOSTEP` (2.2) contracts with factor ρ_1^2/L. The factorization is formalized in Lean as `BlockNorm.rho2_le`. At $L = 3$ the measured ρ_1^2 exceeds L, so the one-step route fails and the two-step map is needed. At $L = 5$ and $L = 7$ the measured ρ_1^2/L is below 1. The values are in Section 6.5. At $L = 13$ Theorem 6.2 gives $\sup_{j,K} \rho_1^2 \le 10 < 13$, so (H2) holds there with $\rho_2 \le 10$.

3.7.5 The remaining inequality

At a fixed odd $L \ge 3$ the single uniform hypothesis is (H2).

- Route K: (H2) $\Rightarrow$ ($C1\sigma$) (L-011) $\Rightarrow$ (B2) (L-012).

- Route B: in the possible approach of Section 3.6.3, (H2) and (T1) give (U), and (H2) gives the vertex bound for γ_{rec} in ($B\theta$) (`BETA_FP_U` Theorem 2.4, Prop. 3.2).

- (H2) at $L = 13$, inside Bałaban's range $L > 11$: Theorem 6.2 gives $\rho_1(j, K)^2 \le 10 < 13$ for all $j \ge 1$ (including ∞) and all K, hence $\rho_2 \le 10 < 13$, under Convention 6.1. The same holds at $L = 15, 17, 21$ (Proposition 6.4). At $L = 3, 5, 9$ only floating-point values exist (at $L = 9$ at one point).

- Alternatives: ($C1\sigma$) and (h) imply (B2) (Prop. 4.1). For (H_A) $\Rightarrow$ (B2) there is only a plausibility argument. One can also aim at (B2w) directly. No per-momentum $d = 4$ constant-field code exists, so (B2) itself has not been scanned numerically.

3.8 Conclusion of the reduction

Theorem 2w (L-001) follows from the inputs of Route K, or from those of Route B. Precisely:

Route K. (B2), which follows from (H2) (R-025) (Theorem 3.19); the fixed-level inputs (PI) (R-026), (F) (R-027), (M_j), V1, (G2) (R-030); (H0) (H-202), Bałaban's inductive assumption at odd $L > 11$, whose use here reduces to named inputs (Section 5.9); and L5 (Theorem 5.104). These give Theorem K (L-009) with $R = \kappa_*$, hence (5.1), hence Lemma DT (L-005), hence Theorem 2w. At $L = 13$, Theorem 6.2 gives (H2).

Route B. (U) and ($B\theta$) with $\theta_* < 1$, Lüscher–Weisz (5.7), bounded $\mathfrak{d}_Y$ and $\mathfrak{d}_{gf}$, the identification of β (uniform in k), (G2), (H0) at odd $L > 11$, and L5 give Theorem $\mathrm{B_{PW}}$ with $R = C_B$, hence (5.1), hence Lemma DT, hence Theorem 2w. The possible approach of Section 3.6.3 replaces (U) and ($B\theta$) by (T1), (H2), (DF) and Lüscher–Weisz (5.7).

Single uniform hypothesis. Both routes need (H2). Theorem 6.2 proves (H2) at $L = 13$, an odd $L > 11$. This gives (B2), the only uniform-in-level hypothesis of Route K (Theorems 3.18, 3.19). In the possible approach to Route B it reduces (U) to (T1); (DF) and the identification of β remain.

Then Bałaban's Theorem 1. Theorem 2w gives (b) and (c-pair). With γ shrunk so that K-018 (S) holds, every one of the 19 cross-scale uses in CMP 119 and CMP 122 I–II is covered (Section 3.4.3), and the single-scale uses need only $g_k \le \gamma$. So Theorem 1 holds for the bare coupling $g_0(K, g)$, for every K and every $g \in]0, g_1]$, without Bałaban's coupling assumption. This needs L5 in the form of Section 3.4.5: β must exist on every history that the shooting visits. It also rests on Bałaban's published proof of Theorem 1 (R-013).

The sweep uses Bałaban's sentence that similar inequalities hold for the later constants (R-012). Proposition 5.85 lists these constants, with the order entries K-029–K-031. The step to Theorem 1 also needs the finite-volume identification R-032, which rests on R-015. Proposition 5.87 gives the torus coefficient within $C_j N_j^6 e^{-m_j N_j/2}$ of the whole-lattice one, given (Loc) (H-141); exact equality cannot be derived. Proposition 5.95 gives the form of R-032 that we use, proved relative to K-032–K-038. So the step from Theorem 2w to Theorem 1 rests on 4 Bałaban-level statements (R-006, R-012, R-013, R-015), on the inputs R-025–R-032 and R-034, and on X-005.

Order of constants. Fix $\beta_0, p_0, p_1, r, \nu, \kappa_0$ (Bałaban). Fix L (odd, > 11), then R from Route K or B, then $b = c \log L$, then $\gamma \le \min(\gamma_0, \gamma_{Bal}, b/(4C_1), \gamma_*)$ with $\gamma^{-2} \ge c_{sw} \cdot 2R$. Then $g_1 = (\gamma^{-2} + 2R)^{-1/2}$. Bałaban's phrase "for a sufficiently small positive g" fits $g \le g_1$.

What this does not give.

- Literal (0.31): only for windows of length $\ge 8R/b$. Literal (0.31) at every K needs (L1) or (L2) of Section 3.3, which need individual increments, not window sums. Theorem 1 does not need it.
- The value $\lim \beta_j(0)$ and any rate: Section 3.9.
- Even L, or odd $L \le 11$ without (H0) as an extra hypothesis (Bałaban assumes L odd > 11 throughout CMP 109); SU(N) with $N > 2$.
- Monotone couplings: g_m can exceed g_n for $m < n$ by the bounded amount $u_m \ge u_n - D'$. Section 3.4.3 shows that the cross-scale uses in Bałaban's proof tolerate this under K-018 (S), within the scope stated there.
- A continuum limit. Section 4 treats it separately.

3.9 The convergence route

Theorem 1 does not need this route. It would add: (i) existence of $\beta_\infty = \lim \beta_j(0)$; (ii) its value $c \log L$; (iii) a rate; (iv) eventual positivity, hence literal (0.31) for $K \ge K_0$ (Lemma DS (L-004) (d)); (v) with certified early steps, literal (0.31) at every K (Section 3.3 (L2)). It also implies (B2), since convergence of $\sigma_j^{(2)}$ implies boundedness. So it needs strictly more than Route K.

3.9.1 Components

- U3 Theorem 4.1 with O5: $|\partial_K^\alpha(\Delta_M - \Delta_\infty)| \le CM^{-2}$, $|\alpha| \le 2$, uniformly down to $K = 0$, for odd L. Numerically $|\Delta_M - \Delta_\infty| = 1.007,\ 0.129,\ 0.0147$ at $M = 3, 9, 27$.
- O6: $\|\mathfrak{B}_j(K)^{-1}\| \le b(L)$ for all j, K, for the one-step constraint at trivial background. The constant is crude ($b(3) \approx 7.5 \cdot 10^{33}$). The measured corner values 103.5, 176.7, 193.0, 195.3 ($j = 0, 1, 2, \infty$, $L = 3$) use a surrogate Λ.

- Theorem 6.1: assume V1. Then $\beta_{j+1}(0) = \Phi(\mathcal{D}_j)$ is Lipschitz in the level data, in the Taylor-data norm.
- Lemma 5.1 (convergence of $\sigma_j^{(1)}$, $\sigma_j^{(2)}$ in C^2, $|\alpha| \le 2$):
  - in the neutral sector, the value bound holds with rate L^{-4j}; for the K-derivatives, `U3_LEMMA51` gives a possible approach;
  - (W) holds (`U3_AW` Theorem W (L-308)); it removes the marginal direction of the f^1 recursion;
  - (A) holds in qualitative form, and L-071 holds. (A) in full (strip/C^3 for $m = 1$, all of $m = 2$) would follow from (H_A), which is open. The route through (HS) fails at $L = 3$ (Section 3.7.2);
  - for convergence of $\sigma^{(1)}$ at rate jL^{-2j}, `U3_AW` §4.7 sketches a possible approach; convergence of $\sigma^{(2)}$ is open and needs (A) in full;
  - the single-block-constraint version of Lemma 5.1 is false for $m = 2$ in $d = 4$. Bałaban's composed constraint is not affected.
- Prop. 3.6: assume V1 and (A) in full. Then $|\beta_{j+1}(0) - \beta_\infty| \le C(1 + j)^2 L^{-2j}$. The proof identifies the σ-Taylor data with the data of (6.1). The clean rate L^{-2j} of (N) is not needed, and we do not expect it to hold.
- Value of the limit. Given convergence, Theorem K (L-009) or Theorem B_{PW} gives $\beta_\infty = c \log L$ at once: $S_k/k \to \beta_\infty$ by Cesàro, and $S_k/k = c \log L + O(1/k)$. Two other approaches to the value exist. The first is the variant of Theorem 3.1 of `BETA_FP_H`, with (H) restated at the constant field; given (W) it reduces to Lemma 5.1 (1.2) alone, plus (PI), (F), and (M), where for (M) there is only a plausibility argument. The second is `BETA_FP_PW` Theorem 3.1 (limit form), given (U), ($C\theta$), and Lüscher–Weisz (5.7), which we offer as a possible approach.
- Rate of $\mathcal{C}(\Lambda)$: a plausibility argument through the slaving of `U3_AW` §3.4(b) gives $O(\Lambda^{-2} \log \Lambda)$, given Lemma 5.1 with rates.

3.9.2 Early steps and j_*

Under Prop. 3.6 and $\beta_\infty = c \log L$, $\beta_{j+1}(0) \ge (c \log L)/2$ for $j \ge j_* := \min\{J : C(1 + j)^2 L^{-2j} \le (c \log L)/2 \text{ for } j \ge J\}$. Heuristic sizes: $C \sim b^3$ gives $j_* \approx 110$ ($L = 3$) and ≈ 70 ($L = 13$); $C \sim b^6$ (one-step inverses inside C_{data}) gives ≈ 218 and ≈ 138. With the measured $\|\mathfrak{B}^{-1}\| \approx 195$: $j_* \approx 10$ ($L = 3$), ≈ 4 ($L = 13$). A proof of $\beta_1(0), \dots, \beta_{j_*}(0) > 0$ needs interval arithmetic at odd $L \ge 13$. Dense methods need blocks of $\approx 3.4 \cdot 10^5$ rows per momentum; the umklapp code reduces this to 12×12 blocks at L^4 momenta. This is needed only for (L2), not for Theorem 1.

4 The finite-torus existence theorem

This section proves Theorem B. The trajectory of Section 3 enters as the block (B-T2) (H-084). Section 4.1 fixes what "exists" means here. Section 4.2 states the uniform bound (H-U) (H-116) that the blocks must give. Sections 4.3 and 4.4 state the theorems. Section 4.5 proves them modulo the named blocks. Section 4.6 gives the dependency table of the paper. Section 6 collects the numerical evidence for the hypotheses.

The theorem is a reduction to named blocks. It is not unconditional. The notation is that of Sections 2.1 and 2.5.

4.1 What "exists" means here

Jaffe and Witten write in footnote 2 of [15]: "We specifically exclude weak-existence (compactness) as the solution to the existence part of the Millennium problem, unless one also uses other techniques to establish properties of the limit (such as the existence of a mass gap and the axioms)."

We use the following reading. "Exists" means: (i) the family $\{S_{n,\varepsilon}\}$ is relatively compact (uniform bounds); (ii) at least one limit point exists along a sequence $\varepsilon_k \to 0$; (iii) every limit point has the properties (OS0)–(OS3) below, proved on the limit and not assumed. We claim no uniqueness. Convergence of the full sequence needs E13 (comparison of two trajectories; (E14-d) (H-099), D6), which this paper does not give.

This reading is weaker than the parenthesis of the footnote. The theorem below gives no gap and only the torus forms of four axioms. It is the finite-torus step of Jaffe–Witten §6.5: "One must then verify the existence of limits of appropriate expectations of gauge-invariant observables as the lattice spacing tends to zero and as the volume tends to infinity". Only the first limit is taken here.

The problem description [15] does not state the OS axioms. It cites them as OS (CMP 31 (1973), CMP 42 (1975)) [19, 20] and requires "axiomatic properties at least as strong as those cited". We use OS II in the form stated in [18]: "The linear growth condition requires a bound $|S_n(F)| \le \sigma_n |F|_{s_n}$ with fixed s and $\sigma_n \le A(n!)^D$ for fixed A, D". Restriction to Schwartz tests that are flat on the coincidence diagonals gives the test class of that reference.

4.2 Hypothesis delivered by the blocks

Definition 4.1 (Uniform bound (H-U), H-116)**.** There are $s \in \mathbb{N}$ and $A, D < \infty$ such that for all $K \ge K_0$, all $n \ge 1$, and all $F \in \mathcal{D}_n$:

$$|S_{n,\varepsilon}(F)| \le \sigma_n \|F\|_{sn}, \qquad \sigma_n := A(n!)^D, \quad \varepsilon = L^{-K}.$$

The derivative order sn grows linearly in n. The OS linear growth condition admits this. The blocks (B-T2) (H-084), (B-S) (R-035), and the bridge (B-N1) (R-045) of Section 4.5 give (H-U).

4.3 Theorem (finite-torus existence, compactness form)

Theorem 4.2 (finite-torus existence, compactness form)**.** *Let $G = SU(2)$, $d = 4$, L odd and large, $\ell \in 2\mathbb{N}$ fixed. Let $\gamma > 0$ be Bałaban's small constant and $g \in]0, \gamma]$ small. For $K \ge K_0$ let $\varepsilon = L^{-K}$, and let $g_0 = g_0(\varepsilon, g)$ be a shooting solution (Lemma 3.6) whose effective couplings satisfy $g_k \in]0, \gamma]$ for $0 \le k \le K$ and $g_K = g$. Assume (H-U) (H-116). Then:*

(a) Compactness. Every sequence $K_m \to \infty$ has a subsequence K_{m_i} such that, with $\varepsilon_i = L^{-K_{m_i}}$, the limit $S_n(F) := \lim_i S_{n,\varepsilon_i}(F)$ exists for every $n \ge 1$ and every $F \in \mathcal{D}_n$.

(b) Every limit family $(S_n)_{n\ge1}$ obtained in (a) has the following properties.

- *(OS0) Regularity. S_n is linear on $\mathcal{D}_n$, and $|S_n(F)| \le A(n!)^D \|F\|_{sn}$. In particular S_n restricted to $C_c^\infty(T^{4n} \setminus \bigcup_{i<j}\{x_i = x_j\})$ is a distribution of order $\le sn$.*
- *(OS1) Invariance. $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for every $a \in T_\ell^4$ (all continuous translations), and $S_n(F \circ R^{\times n}) = S_n(F)$ for every R in the hypercubic group B_4 (signed permutations of the axes, acting about a lattice site).*
- *(OS2) Reflection positivity. Let $\theta(x_0, \vec{x}) = (-x_0, \vec{x})$ on T_ℓ^4 and $T_+ := \{x : x_0 \in]0, \ell/2[\}$. For every finite family $F_0 \in \mathbb{C}$, $F_n \in \mathcal{D}_n$ with $\operatorname{supp} F_n \subset T_+^n$ $(1 \le n \le N)$:*

$$\sum_{n,m=0}^{N} S_{n+m}(\Theta F_n \otimes F_m) \ge 0, \quad (\Theta F_n)(x_1, \dots, x_n) := \overline{F_n(\theta x_n, \dots, \theta x_1)}, \quad S_0 := 1.$$

- *(OS3) Symmetry and reality.* $S_n(F \circ \pi) = S_n(F)$ *for every permutation* π *of the* n *arguments, and* $S_n(F) \in \mathbb{R}$ *for real* F*.*

(c) Nontriviality is a separate statement, Theorem 4.5.

Remark 4.3 (inputs of (H-U)). If (H-U) is derived from the blocks of Section 4.5 and not assumed, the derivation also uses the recovery $\kappa \to \kappa/2$ of the decay rate in the sourced small-field step D2 (R-046). Bałaban does not state this recovery, and it does not follow from the other hypotheses. We reduce it to the large-field T-step rate loss R-020 (Section 7.3). The sourced blocks also use the unit-scale condition (U) (K-024), $\rho_1 \le c_* r_0/M^a$ at $\lambda_K = 1$, where r_0 is the Cauchy margin of the constant-source family. This condition is a choice of constants, made once after r_0 and before K. The bridge (B-N1) meets it by the rescaling $\lambda_i = 2N_U(f_i)$ at a cost C^n; s and D do not change. This (U) is not the Route B hypothesis (U).

Remark 4.4 (what is claimed about measures). We claim no probability measure. The limit is a family of Schwinger functions on $\mathcal{D}_n$. A measure on distributions would need finite moments of smeared fields, and $\mathrm{Var} \sum_p f(x_p)\varphi_p \sim \varepsilon^{-4}$ (Section 2.5). The properties (OS0)–(OS3) on $\mathcal{D}_n$ are the inputs of OS reconstruction.

4.4 Theorem (nontriviality), conditional on (NT)

Theorem 4.5 (nontriviality, conditional on (NT) of Section 4.5.9)**.** *Assume the hypotheses of Theorem 4.2 and Hypothesis (NT). Then every limit family has the following properties. (i)* $S_2 \neq 0$*: there are* $f_1, f_2 \in C^\infty(T^4)$ *with disjoint supports and* $S_2(f_1 \otimes f_2) \neq 0$*. (ii)* S_2 *is not the two-point function of a scale-invariant (in particular free Maxwell) composite:* $d^8 S_2(x, y)$*,* $|x - y| = d$*, is not asymptotically constant as* $d \to 0$*.*

Remark 4.6 (form of (ii)). The proof of (ii) uses only the two-sided bound (0.31). It gives the envelope $d^8 S_2 \to 0$ and no asymptotic form. The stronger form $d^8 S_2 \sim c\, g(d)^4$ with $g(d)^2 \sim (\beta \log(1/d))^{-1}$ would need the convergence $\beta_j \to \beta_\infty$, which we do not prove.

4.5 Proof modulo blocks

Structure. (B-T2) (H-084) gives the trajectory. (B-S) (R-035) gives sourced bounds uniform in ε. (B-N1) (R-045) turns them into (H-U) (H-116). (B-C) (H-074) gives (a). (B-REG) (H-082), (B-TI) (H-085), (B-RP) (H-083), and (OS3) give (b). (B-NT) (H-081) gives Theorem 4.5. This section proves (B-C), (B-REG), (B-TI), (B-RP), and (OS3). Their proofs use only (H-U) and exact lattice symmetries.

4.5.1 (B-T2) Theorem 2: the running coupling

Needed: the hypothesis $g_k \in]0, \gamma]$ for all k of Bałaban's Theorem 1 (CMP 122 II p. 355), quoted in Section 1.2. Bałaban's Theorem 2 (CMP 109 p. 259) [8] states: for small g there is $g_0(\varepsilon, g)$ with all $g_k \in]0, \gamma]$, $g_K = g$, and (0.31): $1/g^2 + \beta \log(L^k \varepsilon)^{-1} \le 1/g_k^2 \le 1/g^2 + \beta' \log(L^k \varepsilon)^{-1}$. Bałaban writes that its proof "has not been published yet".

Section 3 proves the weak form, Theorem 2w (L-001), by Route K (Theorem K (L-009) and Lemma DT (L-005)). The proof is conditional on (H2) (R-025) and on the fixed-level inputs (PI) (R-026), (F) (R-027), (M_j) (R-028), V1, (G2) (R-030), on L5, and on item 7. Item 7 follows from (H0) (H-202) at odd $L > 11$, through a reduction with residue N1–N9 (Section 5.9), or from (G-gain) (H-111) with the running-maximum lift. Item 9 (K-independence) holds (Section 7.2). Theorem 2w suffices for Theorem 1 by the cross-scale sweep (Section 3.4.3). (M_j) and L5 are proved relative to

listed inputs (Theorems 5.102 and 5.104), and (F) is proved (Theorem 5.134). Literal (0.31) is not obtained; it needs eventual positivity of $\beta_j(0)$. Route B (Theorem B_{PW}, L-088) is a formally distinct second route. Its logic is proved, conditional on the hypotheses (U), (DF), (T1) of Section 3.6.3, which rest on (H2). The reduction uses Lemma DS (L-004), Lemma DT, and Lemmas D and E; their one-argument forms are formalized in Lean.

At $L = 13$, (H2) is a theorem (Theorem 6.2). So (B-T2) is conditional on the fixed-level inputs, the inputs of L5, and item 7; see the dependency table (Section 4.6).

4.5.2 (B-S) Sourced bounds E14-a..g

Needed: a version of Bałaban's Theorem 1 with the (C1) source, uniformly in ε. The source is the local bare coupling $w_p = 1/g_0^2 - f(x_p)$, with f real and $|\partial^m f| \le r$ for $m \le 5$ (class F_r). Output form ((N1) (H-224)): $\log Z_\varepsilon[f] - W_\varepsilon[f]$ is C^∞ in the source amplitudes z, and $|\partial_z^\alpha(\log Z_\varepsilon[f] - W_\varepsilon[f])| \le C^{|\alpha|}\alpha!\, r^{-|\alpha|}$ for $f = \sum_i z_i f_i$ with disjoint supports. Here $W_\varepsilon[f]$ is a local field-independent contact functional. It may diverge. For disjoint supports it contributes zero to every mixed derivative of order ≥ 2.

Estimates needed: (E14-a) (H-096), the sourced small-field step; (E14-b) (H-097), the sourced large-field step and R operation; (E14-c) (H-098), derivative bounds in f. Not needed: (E14-d) (H-099), fixed-torus convergence (= E13); (E14-g) (H-102), tensor sources. (E14-e) (H-100), the transfer of RP and symmetry, is proved in this section (Sections 4.5.6, 4.5.7). (E14-f) (H-101), nontriviality at a fixed layer, is needed only for Theorem 4.5. The dependency table (Section 4.6) gives the form of each sub-block. (B-S) is a hypothesis of Theorem B; Section 7.3 records its reduction.

4.5.3 (B-N1) Bridge: source-form bounds ⇒ (H-U)

The source form alone does not give $|S_n(F)| \le C^n n! \|F\|_{s,n}$ with a fixed s for F flat on the diagonals. The bridge uses the growth sn of the OS index instead.

Proposition 4.7 (bridge (B-N1))**.** *Assume the sourced Gevrey bound (S) (R-045) in the following form: for $f = \sum_i z_i f_i$ with disjoint supports, the mixed z-derivatives of $\log Z_\varepsilon[f] - W_\varepsilon[f]$ of order n are bounded by $C^n(n!)^{1+\eta}$ in units of the C^1 norm $\max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$, uniformly in the position and size of the supports and in ε. Then (H-U) holds with $s = 6$ and some $D > 1$. Under the fallback order $2+\eta$ in place of $1+\eta$, (H-U) holds with $s = 6$ and some $D > 2$.*

Proof (sketch). We give the structure of the argument. The sketch uses the class F_r with five derivatives; it gives the cruder exponents δ^{-5n} and $s \ge 20$. The C^1 norm of the hypothesis gives $s = 6$ by the same steps.

1. Moments from cumulants. $\langle \prod_i \phi(f_i) \rangle = \sum_{\text{partitions}} \prod_{\text{blocks}} \kappa(\cdot)$. Singleton cumulants vanish (φ is centred). For disjoint supports the source form bounds every cumulant by $C^{|B|}|B|!\, r^{-|B|}$, so the moment by $C^n n!\, r^{-n}$.

2. Scale-δ tests. A bump of side $\delta \le 1$ and amplitude a has $|\partial^5 f| \sim a\delta^{-5}$; it lies in F_r if $a \le cr\delta^5$. So n disjoint unit-amplitude scale-δ bumps have $|\text{moment}| \le C^n n!\, \delta^{-5n}$. This is polynomial in δ^{-1}.

3. Whitney decomposition. Split T^{4n} by a smooth dyadic partition in $\rho(x) := \min_{i<j} |x_i - x_j| \in [\delta, 2\delta]$, $\delta = 2^{-m} \ge \varepsilon$. On each piece cover by cubes of side $\delta/100$; cube tuples are pairwise disjoint. Expand $F\cdot$(cutoffs) in a Fourier series on each cube tuple; the coefficients decay

faster than any power; each term is a product of disjointly supported scale-δ bumps. Flatness gives $|\partial^\beta F(x)| \le C_k \|F\|_{k+|\beta|} \rho(x)^k / k!$. There are $\le C^n \delta^{-4n}$ cube tuples. Derivatives of the cutoffs cost at most $(Cn^2/\delta)^M$; with $M = O(n)$ this is $\le C^n (n!)^c \delta^{-M}$. Complex Fourier modes against real sources cost a factor 2^n, and the derivatives of the Fourier summation are absorbed in s.

4. Bound per dyadic scale: $\le C^n (n!)^{D'} \delta^{k-9n-M} \|F\|_{k+M}$. Choose $k = sn - M$ with s large enough so that the δ-exponent is $\ge n$; the sum over dyadic $\delta \in [\varepsilon, 1]$ converges uniformly in ε.

5. Sub-lattice region $\rho(x) \lesssim \varepsilon$: there are $\le C^n \varepsilon^{-4n}$ plaquette tuples, each moment is $\le 2^n$, and $|F| \le \|F\|_k \varepsilon^k$; this is $\le C^n \|F\|_{sn}$ for $k = sn \ge 4n$.

The full argument, with the C^1 norm, the order $(n!)^{1+\eta}$, and the smooth min-distance partition, gives $s = 6$ and $D > 1$. □

Section 7.3 lists the inputs of (S): the uniformity in the support size and in ε, and the (E14-c) (H-098) clause, which reduces to (S). Given (FX) (R-043), the step (S_{src}) ⇒ (H-U) with $s = 6$, $D > 1$ holds (Section 4.6, row B-S.D4).

4.5.4 (B-C) Compactness

Proposition 4.8. *Assume (H-U). Then every sequence $\varepsilon_m \to 0$ has a subsequence along which $S_{n,\varepsilon}(F)$ converges for every n and every $F \in \mathcal{D}_n$. The limit S_n is linear and satisfies $|S_n(F)| \le \sigma_n \|F\|_{sn}$.*

Proof. 1. Separability. $C^{sn}(T^{4n})$ with $\|\cdot\|_{sn}$ is separable (trigonometric polynomials with rational coefficients are dense; Fejér means converge in C^k). A subset of a separable metric space is separable. So $\mathcal{D}_n$ has a countable subset Q_n dense for $\|\cdot\|_{sn}$. 2. Diagonal extraction. $Q := \bigcup_n Q_n \times \{n\}$ is countable; enumerate it as $(F^{(1)}, n_1), (F^{(2)}, n_2), \ldots$. Each sequence $S_{n_t,\varepsilon_m}(F^{(t)})$ is bounded by $\sigma_{n_t} \|F^{(t)}\|_{sn_t}$ (H-U), so it has a convergent subsequence. Extract successively and take the diagonal subsequence ε_i. 3. Extension. Let $F \in \mathcal{D}_n$, $\eta > 0$, and $G \in Q_n$ with $\|F - G\|_{sn} \le \eta$. By (H-U) and linearity, $|S_{n,\varepsilon_i}(F) - S_{n,\varepsilon_j}(F)| \le 2\sigma_n \eta + |S_{n,\varepsilon_i}(G) - S_{n,\varepsilon_j}(G)|$. The last term $\to 0$. So the sequence is Cauchy; call the limit $S_n(F)$. 4. Linearity passes to pointwise limits, and $|S_n(F)| = \lim |S_{n,\varepsilon_i}(F)| \le \sigma_n \|F\|_{sn}$. □

Remark 4.9. (i) The same subsequence serves all n; (OS2) needs this. (ii) The proof does not use E13. (iii) The bounded normalizations B_K and c_ε can be included in the extraction.

4.5.5 (B-REG) Regularity (OS0)

By step 4 of the proof of Proposition 4.8 the limit satisfies $|S_n(F)| \le \sigma_n \|F\|_{sn}$, $\sigma_n = A(n!)^D$. On the compact torus the norm $\|F\|_p$ replaces the Schwartz seminorm: the weights of the seminorm control decay at infinity on $\mathbb{R}^{dn}$, and the torus has no infinity. This is the quoted OS II form. Distribution property: for $F \in C_c^\infty$ of the off-diagonal open set, $F \in \mathcal{D}_n$, and the bound is continuity in the C^{sn} topology. So S_n is a distribution of order $\le sn$ there.

OS II is formulated on $\mathbb{R}^{dn}$. On T^{4n} we use the axioms in their torus form (periodic test functions). This is a change of setting, not a consequence of OS II.

4.5.6 (B-TI) Translation and hypercubic invariance (OS1)

Lattice, exact. Put the origin at a lattice site.

- Translations $a \in \varepsilon\mathbb{Z}^4$. The map $U \mapsto U^a$, $U^a(b) := U(b-a)$, permutes the links, preserves the product Haar measure and the Wilson action, and $P_p(U^a) = P_{p-a}(U)$. So $\langle \prod_i \varphi_{p_i+a} \rangle_\varepsilon = \langle \prod_i \varphi_{p_i} \rangle_\varepsilon$ and $S_{n,\varepsilon}(F \circ \tau_a^{\times n}) = S_{n,\varepsilon}(F)$. The coupling $g_0(\varepsilon)$ is one number, so nothing in μ_ε depends on position.

- Hypercubic $R \in B_4$ (signed permutation matrices, $|B_4| = 384$). R maps sites to sites, links to links (possibly reversed), plaquettes to plaquettes, and $x_{Rp} = Rx_p$. Define $(RU)(b) := U(R^{-1}b)$ with $U(-b) = U(b)^{-1}$. Haar measure is invariant under $U \mapsto U^{-1}$ and $\operatorname{Re}\operatorname{tr} W^{-1} = \operatorname{Re}\operatorname{tr} W$ for $W \in SU(2)$, so A and the measure are invariant and $P_{Rp}(RU) = P_p(U)$. Hence $S_{n,\varepsilon}(F \circ R^{\times n}) = S_{n,\varepsilon}(F)$.

- These identities hold at every ε. $F \circ \tau_a^{\times n}$ and $F \circ R^{\times n}$ are again in $\mathcal{D}_n$. They pass to the limit along the subsequence of Proposition 4.8: $S_n(F \circ R^{\times n}) = S_n(F)$, and $S_n(F \circ \tau_a^{\times n}) = S_n(F)$ for $a \in \varepsilon_i\mathbb{Z}^4$ for all large i.

Continuous translations. Let $a \in T_\ell^4$. Pick $a_i \in \varepsilon_i\mathbb{Z}^4$ with $|a - a_i| \le \varepsilon_i$ (componentwise rounding gives $\le \varepsilon_i\sqrt{4}/2 = \varepsilon_i$). Write $F_a := F \circ \tau_a^{\times n}$. The mean value theorem along the diagonal direction gives $\|F_a - F_{a_i}\|_{sn} \le 4n|a - a_i|\,\|F\|_{sn+1}$. Then

$$|S_n(F_a) - S_n(F)| \le |S_n(F_a) - S_{n,\varepsilon_i}(F_a)| + |S_{n,\varepsilon_i}(F_a) - S_{n,\varepsilon_i}(F_{a_i})| + |S_{n,\varepsilon_i}(F_{a_i}) - S_{n,\varepsilon_i}(F)| + |S_{n,\varepsilon_i}(F) - S_n(F)|.$$

The first and last terms $\to 0$ by Proposition 4.8. The third is 0 by lattice invariance. The second is $\le \sigma_n \cdot 4n\,\varepsilon_i \|F\|_{sn+1}$ by (H-U) and linearity. Let $i \to \infty$. F is smooth, so $\|F\|_{sn+1} < \infty$. Rotations outside B_4 are not available (Section 4.7).

4.5.7 (B-RP) Reflection positivity (OS2)

Lattice theorem. We use the theorem of Osterwalder and Seiler [21] in its usual form.[2] For the Wilson action with compact gauge group and any $\beta \ge 0$, and θ the reflection in a lattice hyperplane that bisects the links orthogonal to it, $\langle \Theta(\Phi)\Phi \rangle \ge 0$ for every function Φ of the link variables in the positive half-lattice, where $(\Theta\Phi)(U) := \overline{\Phi(\theta U)}$. Jaffe–Witten [15] write: "Reflection positivity holds for the Wilson approximation [36], a major advantage". The template [18] uses the same input: "At every cutoff the nearest-neighbour measure is reflection positive across microscopic site and link seams." The standard proof fixes the crossing links to 1 by a gauge transformation. The action then splits as $A_+ + \theta A_+ + \sum_{\text{crossing}} \operatorname{Re}\operatorname{tr}(X_p \theta(X_p)^*)$, and $\exp(\beta \operatorname{Re}\operatorname{tr}(XY^*))$ is a kernel of positive type (a positive combination of characters), which gives a sum of squares. On the periodic lattice there are two fixed planes, $x_0 = \varepsilon/2$ and $x_0 = \varepsilon/2 + \ell/2$. Both are link planes, because N_ε is even (ℓ even, Section 2.5). We attribute the periodic two-seam version to [21], as is standard; we do not quote it.

Remark 4.10 (parity of ℓ). The hypothesis that ℓ is even is sufficient, not necessary. With L odd, $N_\varepsilon = \ell L^K$ is even if and only if ℓ is even. For odd N_ε every time reflection has one site seam and one link seam. Osterwalder and Seiler [21] treat link-plane (between-site) reflections. Menotti and Pelissetto [17] treat site-plane reflections. Their abstract states: "We extend the proof by Osterwalder and Seiler of reflection positivity of lattice gauge theories with fermions to the case in

[2] The reference list of Jaffe–Witten gives the title "Gauge theories on the lattice"; the bibliography entry follows the title in Annals of Physics.

which the reflection is performed with respect to planes containing sites." and "Our proof applies to all observables and relies on the particular structure of the Wilson action." The mixed case (one site seam and one link seam on a periodic lattice) is very likely routine, because each mechanism is local to its seam, but we know no reference that states it. So odd ℓ is not excluded; the hypothesis ℓ even only avoids the mixed case.

Passage to the limit. We assume (H-U) and the lattice theorem. Setup. $\theta_\varepsilon(x) := \theta(x) + \varepsilon e_0 = \theta(x - \varepsilon e_0)$, with fixed planes $x_0 = \varepsilon/2,\ \varepsilon/2 + \ell/2$. The positive half Λ_+ consists of the links with both ends in $\{x_0 \in \{\varepsilon, 2\varepsilon, \dots, \ell/2\}\}$. Fix $F_0 \in \mathbb{C}$ and $F_n \in \mathcal{D}_n$, $\operatorname{supp} F_n \subset T^n_+$, $1 \le n \le N$. The supports are compact in the open set, so there is $\delta > 0$ with $x_0 \in [\delta, \ell/2 - \delta]$ on all supports.

Step 1 (lattice form). For $\varepsilon < \delta/2$ every plaquette with centre in a support has all four links in Λ_+. Define $\Phi := F_0 + \sum_{m=1}^{N} \sum_{p_1 \dots p_m} F_m(x_{p_1}, \dots, x_{p_m}) \prod_i \varphi_{p_i}$. Φ depends only on the links of Λ_+. Since φ_p is real and $P_p(\theta_\varepsilon U) = P_{\theta_\varepsilon p}(U)$, re-indexing gives $\Theta\Phi = \bar{F}_0 + \sum_n \sum_q \overline{F_n}(\theta_\varepsilon x_{q_1}, \dots, \theta_\varepsilon x_{q_n}) \prod \varphi_{q_i}$. Hence

$$0 \le \langle \Theta\Phi\, \Phi \rangle_\varepsilon = \sum_{n,m=0}^{N} S_{n+m,\varepsilon}(\Theta_\varepsilon F_n \otimes F_m), \quad \Theta_\varepsilon F_n := (\Theta F_n) \circ \tau^{\times n}_{-\varepsilon e_0},\ S_{0,\varepsilon} := 1.$$

(The argument order inside ΘF_n does not matter, by (OS3).) Each $\Theta_\varepsilon F_n \otimes F_m$ lies in $\mathcal{D}_{n+m}$: its two factors have supports at distance $\ge 2\delta - \varepsilon > 0$, and each factor is flat on its own diagonals.

Step 2 (seam shift). $\|\Theta_\varepsilon F_n \otimes F_m - \Theta F_n \otimes F_m\|_{s(n+m)} \le 2^{s(n+m)} \cdot 4n\,\varepsilon\, \|F_n\|_{s(n+m)+1} \|F_m\|_{s(n+m)}$. By (H-U) the difference of the $S_{n+m,\varepsilon}$ values is $\le \sigma_{n+m}$ times this, which $\to 0$.

Step 3 (limit). Along the subsequence of Proposition 4.8, $S_{n+m,\varepsilon_i}(\Theta F_n \otimes F_m) \to S_{n+m}(\Theta F_n \otimes F_m)$ for each of the finitely many (n, m). So $\sum_{n,m} S_{n+m}(\Theta F_n \otimes F_m) = \lim_i \sum_{n,m} S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m) \ge 0$.

Lemma 4.11 (positivity of a finite form survives pointwise limits)**.** *Let $a^{(i)} \in \mathbb{C}^{(N+1)\times(N+1)}$ with $\sum_{n,m} \bar{c}_n c_m a^{(i)}_{nm} \ge 0$ for all c and all i, and $a^{(i)}_{nm} \to a_{nm}$ for every (n, m). Then $\sum \bar{c}_n c_m a_{nm} \ge 0$ for all c.*

Proof. For fixed c the left side is a finite linear combination of the entries, so it converges to $\sum \bar{c}_n c_m a_{nm}$; $[0, \infty)$ is closed. □

Step 3 is this lemma with $a^{(i)}_{nm} := S_{n+m,\varepsilon_i}(\Theta_{\varepsilon_i} F_n \otimes F_m)$ (replace F_n by $c_n F_n$), together with Step 2.

4.5.8 (OS3) Symmetry and reality

$S_{n,\varepsilon}(F \circ \pi) = S_{n,\varepsilon}(F)$: the φ_p are commuting real random variables; re-index the sum. For real F, $S_{n,\varepsilon}(F) \in \mathbb{R}$. Both identities are linear and pass to the limit. $S_0 = 1$ by definition.

4.5.9 (B-NT) Nontriviality: what is needed

- A two-point function alone never excludes a Gaussian: every positive-definite S_2 is the covariance of some Gaussian generalized field.
- The plaquette density is a composite ($\approx \frac{1}{4} \operatorname{tr} F^2 \times \varepsilon^4$). Even in the free theory its connected $n \ge 3$ functions are nonzero (Wick squares). So $S^c_n \ne 0$ for some $n \ge 3$ does not prove interaction.

- The meaningful test excludes (a) the zero theory and (b) the free Maxwell composite. The free composite has, at short distance, the scale-invariant form $c_0 d^{-8}$. Asymptotic freedom predicts $c\, g(d)^4 d^{-8}$ with a running coupling.

Free-theory prediction along this sequence. At tree level with the bare coupling, $\langle \varphi_p \varphi_{p'} \rangle \approx c\, \varepsilon^8 g_0^4 |x-y|^{-8}$. Since $g_0(\varepsilon)^2 \le (1/g^2 + \beta \log \varepsilon^{-1})^{-1} \to 0$ by (0.31), the bare-coupling prediction for S_2 at fixed separation tends to 0. The claim is that the true S_2 does not.

Hypothesis 4.12 ((NT), the fixed-layer expansion (E14-f))**.** There are $c > 0$ and C such that for all K, all $f_1, f_2 \in C^\infty(T^4)$ with disjoint supports at distance $\ge d$, $d \ge L\varepsilon$, and $k(d)$ the layer with $L^k \varepsilon \le d < L^{k+1}\varepsilon$:

$$S_{2,\varepsilon}(f_1 \otimes f_2) = c \iint f_1(x) f_2(y)\, g^4_{k(|x-y|)} |x-y|^{-8}\, dx\, dy\, (1+R), \quad |R| \le C\big(g^2_{k(d)} + (d/\ell)^2 + (\varepsilon/d)^2\big).$$

(NT) is a corollary of (E14-a) at a fixed layer, which (E14-f) would supply. The term $(\varepsilon/d)^2$ accounts for the lattice artifacts: the lattice ring differs from the continuum ring by a term of this size. Equivalently one may drop this term and require $K \ge K_0(d)$. The coefficient c is the tree-level coefficient in the composite, $c \propto \sum$(free field-strength covariance)^2, a sum of squares, so $c > 0$. In Bałaban's normalization, with unoriented plaquettes, $c = 9/\pi^4$.

Proof of Theorem 4.5 from (NT) and (B-T2). (i) Take $f_1, f_2 \ge 0$, not identically 0, with supports at distance $\ge d_0$ and with support diameters small compared with ℓ. Then $g^2 + (|x-y|/\ell)^2$ is small on the whole support pair, and $|R| \le \frac{1}{2}$ for ε small. By (0.31) at layer $k(d)$ with $d \in [d_0, \ell]$: $1/g_k^2 \le 1/g^2 + \beta' \log(L^k\varepsilon)^{-1} \le 1/g^2 + \beta' \log(L/d_0)$, uniformly in ε. So $S_{2,\varepsilon}(f_1 \otimes f_2) \ge \frac{1}{2} c (1/g^2 + \beta' \log(L/d_0))^{-2} \iint f_1 f_2 |x-y|^{-8} > 0$ uniformly in small ε, and the limit has the same bound. (ii) For bumps concentrated at separation $d \to 0$ (with d a fixed multiple of their width), $d^8 S_2 \le (1 + C g^2_{k(d)})\, c\, (1/g^2 + \beta \log(1/d))^{-2} \to 0$ as $d \to 0$ by the lower bound in (0.31), while $d^8 S_2 > 0$ by (i). A scale-invariant composite would give a nonzero constant limit. So the limit is neither zero nor the free composite. The proof uses only the two-sided bound (0.31). □

Remark 4.13. Part (ii) needs (NT) uniformly down to $d \ge L\varepsilon$, with the remainder controlled by $g^2_{k(d)}$; that is, the fixed-layer expansion at every layer. If (E14-f) holds only at a fixed layer j, then (i) holds, and (ii) holds only in the weaker form "$d^8 S_2(d)$ lies between the two (0.31) envelopes for $d \in [L^{-j}, 1]$".

4.6 Dependency table

The table gives every block of Theorem B and every named hypothesis of Theorem A, with the form in which this paper establishes it. Three global conditions apply to every row. (GC1) (H-200): Bałaban's published proofs CMP 95–122 (1984–89) are correct as published; this paper does not re-check them line by line. (GC2): Bałaban's inductive assumptions (CMP 109 pp. 263–264) hold with the stated uniformity. (GC3) (H-201): L odd, $L > 11$, M large, γ small as Bałaban requires; ℓ even (sufficient, Remark 4.10). Rows "item n" are the named hypotheses of Section 3; n is their item in Section 7. The words in the third column mean the following. "Proved" means proved in this paper or in the cited section. "Proved relative to listed inputs" means proved from the inputs in the last column. "Reduced to X" means that the statement follows from X, and X is not proved here. "Hypothesis" and "open" mean that the statement is neither proved nor reduced. "Numerical" means numerical evidence only. Appendix B gives the project note of each row.

Block	Content	Form of the result	Hypotheses carried
B-T2	Theorem 2w (L-001) (Theorem 3.2): trajectory $g_0(\varepsilon, g)$, (b), (c-pair), K-018 (S)	proved by Route K (Theorem K (L-009) + Lemma DT (L-005)), conditional on the inputs at right. Literal (0.31) is not obtained	R-025 (H2), R-026 (PI), R-027 (F), R-028 (M_j), V1, R-030 (G2), L5, item 7 (from H-202 (H0) or from H-111 (G-gain) with the running-maximum lift)
B-T2.a	Lemmas D, DS, DT, E (shooting)	proved; formalized in Lean (DS, D, E, DT in one-argument form; history form in LemmaDHist.lean, LemmaDHistNested.lean)	continuity of β_{k+1} in the history
B-T2.b	Sweep: (c-pair) + K-018 (S) $\Rightarrow$ the inputs of Bałaban's Theorem 1 (Section 3.4.3; item 10)	proved for all 19 cross-scale uses, including the CMP 122 additions H-266–H-280	the readings of Bałaban's papers in Section 7.4
B-T2.1	U3 Thm 4.1 (Gaussian RG, rate L^{-2j}) + O5	proved	trivial background
B-T2.2	U3 Lemma 5.1; (A), (W)	neutral sector proved. (W): numerically exact; (A): numerical, qualitative and quantitative; (HS) false at $L = 3$. Not used by Theorem 2w (L-001) once (H2) holds; used only by the convergence route	(A) in full, (N) (plausibly false)
B-T2.3	O6 bordered-matrix invertibility	one-step form proved; the composed constraint is reduced to one-step O6 + (A); Route K does not use the composed constraint	—
B-T2.4	$c = 11N^2/(12\pi^2)$, anchor-free identity Prop. 2.1, telescoping	proved. $\beta_\infty = c \log L$ itself is replaced by window sums (Theorem K (L-009), Route B)	R-027 (F) (continuum side cited, [22])
B-T2.5	κ_k bounded (Theorem K (L-009))	proved, conditional on (B2); (B2) proved, conditional on R-025 (H2)	R-025 (H2), R-026 (PI), R-027 (F), R-028 (M_j), V1, R-030 (G2)
B-T2.H2	R-025 (H2): uniform two-step inequality $\rho_2 < L$ (Section 3.7.3; item H)	at $L = 13$: Theorem 6.2, a computer-assisted proof by interval arithmetic with outward rounding, $\rho_2 \le \sup \rho_1^2 \le 10 < 13$. Premises: the reading of (0.4), the gauge choice, Prop. A at $M \ge 27$, L-169 (Lemma G), and the correctness of the interval code `bb_arb.py`. At $L = 3, 5, 9$: numerical; at $L = 3$ one point is enclosed, $\rho_2 \in [1.9993, 2.1273] < 3$	— (cell-certificate cross-check at $L = 13$: mode $M = 1$ partial, Section 6.9)
B-T2.H1	one-step route at $L = 13$: $\sup \rho_1^2 < L \Rightarrow$ R-025 (H2)	$\sup_{j,K} \rho_1^2 \le 9.9995 < 13$ by the majorant B_{13} and interval arithmetic; the factorization is proved and formalized in Lean (`rho2_le`); the majorant core is formalized in Lean (`Tower.lean`)	—
B-T2.B	Route B (Theorem B_{PW})	logic proved, conditional on the hypotheses (U), (DF), (T1) (Section 3.6.3)	(U), (DF), (T1), R-025 (H2), ($B\theta$), R-007, LW (5.7), H-202 (H0)

Block	Content	Form of the result	Hypotheses carried
item 1	(B2) / H-184 (B2w) (Section 3.7)	reduced to R-025 (H2); holds at $L = 13$ (Theorem 6.2); open at other L	R-025 (H2)
item 1	R-025 (H2) $\Rightarrow$ $(C1\sigma)$ $\Rightarrow$ (B2) (Theorems 3.18, 3.19)	proved, conditional on R-025 (H2)	R-025 (H2)
item 1	$(C1\sigma)$ + (h) $\Rightarrow$ (B2)	implication proved; $(C1\sigma)$ $\Leftarrow$ R-025 (H2); (h) numerical at $L = 3, 5$	—
item 1	$(\mathrm{H}_A) \Rightarrow$ (A) $\Rightarrow$ (B2)	a possible approach (U3_AW §4.7); not needed once (H2) holds	—
—	(HS) (Section 3.7.2)	false at $L = 3$	—
item H	$\rho_1^2 < L \Rightarrow$ R-025 (H2) (one step, $L \geq 5$)	arithmetic; $\rho_1^2/L = 0.771$ at $L = 5$ (numerical)	—
item H	certification tail j_0	derived; $j_0 \approx 8$ ($L = 3$), $j_0 \approx 6$ ($L = 5$)	—
item 2	R-026 (PI)	proved at each fixed level relative to listed inputs: (I1) for the composed fine matrix at every j, (I2), and (I3) for the full hypercubic group (the lattice-symmetry input)	definitions (Avg), (Chart), (Bkg)
item 3	R-027 (F)	proved (Theorem 5.134), by proper time, relative to textbook facts and the proved lemmas H-137 (L-LL) (Theorem 5.133) and H-148 (MPT); a separate numerical tension, β_1 against $c \log L$, concerns another family (Section 6.9)	—
item 4	(M_j): H-244 (S1)–H-247 (S3), Prop. 4.4	proved at fixed j relative to listed inputs (Theorem 5.102). H-244 (S1), H-247 (S3), H-245 (S2)$\vert_{t=0}$, H-246 (S2c) hold at fixed j given (I3); (I3) is proved, given the identification of the one-step objects; $(\mathrm{G}_{2\mathrm{D}})$ is proved for $0 < \vert f\vert \leq ((1-c)/100)^2/2$ (Theorem 5.98), and this f-range suffices	the KAPPA (1.1) coefficients, which are the Moyal coefficients at $\sigma = 0$
item 5	V1	proved at fixed level, history-free, conditional on the inputs at right	R-030 (G2), Bałaban's inductive bounds, averaging (0.4)
item 6	R-030 (G2)	proved at fixed j relative to listed inputs, including (Gap_f) and $\tilde{Q}\tilde{Q}^T > 0$; no uniformity in j (not needed)	the listed inputs of R-030 (Section 7.7)
item 7	$\vert\beta_{k+1}(h, g) - \beta_{k+1}(0)\vert \leq C_\beta g$ on every past	follows from H-202 (H0) (Bałaban, odd $L > 11$) through the reduction H0-red with residue N1–N9 (N2 = G3, proved relative to listed inputs, Section 5.9), or from H-111 (G-gain) with the running-maximum lift (proved)	N1, N3–N9 of H0-red; the inputs of H-111 (G-gain) if item 7 is taken from it
item 8	L5: history form and domain (Section 3.4.5)	proved relative to listed inputs by Route 2 (Theorem 5.104)	R-013, R-018, CMP 102 Sect. G, the causality reading H-047

Block	Content	Form of the result	Hypotheses carried
item 9	$\beta_j(g)$ independent of K	holds: β_j is the whole-lattice coefficient by definition, and DS/DT run at fixed K. The residual volume question is R-032, an input of Theorem 1. Torus and whole-lattice coefficients differ by at most $C_j N_j^6 e^{-m_j N_j/2}$ (Proposition 5.87; H-141 (Loc) proved, Theorem 5.93); R-032 is replaced and proved under K-032 (Vol) (Proposition 5.95)	(K1), (K2) = item 7, (K3)
item 10	readings of Bałaban's text (Section 7.4)	page readings, with the misprints corrected (former R-048, R-049, R-051); R-012 holds as a list (Proposition 5.85; K-029–K-031)	R-020
item 11	odd L; $L > 11$ in Bałaban	input	CMP 109 pp. 251–252
item 12	SU(2)	scope	—
item 13	Lean history form	formalized: LemmaDHist.lean, LemmaDHistNested.lean (`lemma_DPrime_hist_adm`, `lemma_DDoublePrime_hist_adm`); that Bałaban's β_j meets the continuity hypotheses is an input	—
B1, BH	Route B: R-025 (H2) + (T1) $\Rightarrow$ (U)	a possible approach (BETA_FP_U Theorem 2.4); (U) is a hypothesis of Route B	R-025 (H2), (T1)
B1	Route B: (T1)	hypothesis; a possible approach is sketched	—
B3	Route B: $(B\theta) = \gamma_{rec} + \mathfrak{d}_F$	γ_{rec} bounded given (U) (a possible approach); (DF) hypothesis	(DF), (U)
B4	Route B: LW (5.7)	cited [16]	—
B5	Route B: R-007 (β identification, uniform in k)	hypothesis; a possible approach is sketched	—
B7	Route B: $\theta_* < 1$	trivial condition	—
B-T2.6	early steps $\beta_j(0) > 0$	needed only for literal (0.31) at every K, not for Theorem 1; numerical, no certificate; $\beta_1(0)$ computed up to $L = 13$ ($N_g = 8$ at $L \geq 9$)	$L > 11$ is Bałaban's range
B-S	sourced Theorem 1 (E14-a,b,c)	open as a whole. Reduced to its parts (rows below) and the reading (W6*) (H-361) of Bałaban's pages; joint holomorphy holds given the bounds	B-T2 for the sourced trajectory
B-S.D1	one sourced step, Gaussian level	proved	—
B-S.D2	sourced small-field step H-096 (E14-a)	parts (a), (b), (d), (f) proved; part (c) is replaced by row B-S.D2C; part (e) reduced	analyticity in the input activities; $\kappa \to \kappa/2$ (R-046, uses R-020); K-024 (U), a choice of constants

Block	Content	Form of the result	Hypotheses carried
B-S.D2C	part (c) with frozen source per domain, L-054 (Lemma A), $h \equiv 0$	L-054 (Lemma A) and the recursion proved; R-036 (I) holds exactly with the adapted gauge-fixing weight K-011	domain hypothesis (D); (Hβ)
B-S.Hβ	two-loop bound $\sup \sum_j \lvert B'_j \rvert \le \frac{1}{2}$	proved, conditional on R-036 (I) and R-037 (H-hist); R-036 (I) is proved relative to listed inputs (Lemma I36)	R-036 (I), R-037 (H-hist)
B-S.hist	R-037 (H-hist), history sensitivity	reduced, in the small-field and the large-field half, to R-044 (W4), Facts A/B, R-036 (I)/L3, and H-111 (G-gain) in both slots; L-189 (Lemma HH)	H-111 (G-gain), small and large field
B-S.Ggain	H-111 (G-gain): gain $g_k \lvert B \rvert$ in the small-field step	small field: pointwise Cauchy bound and uniform C_β on $]0,\gamma]$ proved, including the vacuum subtraction. The χ_k surface term is L-174 (Lemma G3), proved relative to listed inputs with (Q) assembled from R-008, R-009 (Theorem 5.91). Large field: reduced, by one Cauchy step on the large-field disc. Non-monotone pasts: the running-maximum lift (proved). L-178 (Lemma G4) is not used	(M_r), K-026, K-027, CMP 102 (189), the H-145 (M-B) bookkeeping; not at $g = 0$
B-S.G3	L-174 (Lemma G3): χ_k surface term, uniform in g	proved relative to listed inputs (Theorem 5.91); G_Γ is bounded uniformly in M	R-008, R-009, the (2.15)–(2.25) chain, K-026, K-027, N3, N6, N9, (M_r); the s-rule convention K-046
B-S.Q	(Q): covariance lower bound $c_- > 0$	holds in the joint (X, B) form from R-008 + R-009 (Theorem 5.91), with G_Γ in Λ. Alternative route: reduced to CMP 99 (3.132) (R-001, whose kernel-decay form reduces via L-125 (Lemma CT) to CMP 99 Thms 3.1, 3.3, 3.12), (E_U), and LOC; the Woodbury identity is proved; the free-case constants are numerical	residue of R-001: H-057 (C) reduced (Proposition 5.126), torus-side hypothesis discharged by K-032 (Vol), printed inputs R-021, conventions K-044, reading $a = 1$
B-S.EU	(E_U): uniform block extension	$K_0(4) = 195.19$ ($N = \infty$), numerical; $K \le K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$ (proof sketch); (E_U) reduced to kernel-moment bounds that Bałaban does not print	k-uniform kernel moments; the unprinted constant C'_1
B-S.LOC	LOC: localized covariance $C^{(k)}(Z_0, 0)$	proved, conditional on the inputs at right (the symmetry (S) is not needed)	R_2 bound (2.16) (R-003); γ_0 for general U (R-004, $\gamma_0 \ge m_L/(48d^2L^d)$); α_0, α_1 small
B-S.thr	R-039 (H-thr): threshold shell sum	L-285 (Lemma T) and the trajectory sum proved, conditional on R-038 (H-hist2); R-038 reduced to the real uniform R-037 (H-hist) with no new input	R-038 (H-hist2), M3 weights, frozen integers (L3)

Block	Content	Form of the result	Hypotheses carried
B-S.hist2	R-038 (H-hist2): mixed second history derivatives	reduced to the real uniform R-037 (H-hist), no new input (Route 2(a); decay $\lambda_C^{m/2}$; the product form λ^m is false). The β_j part (threshold-history mixed derivative) is open	real uniform R-037 (H-hist)
B-S.D3	sourced large-field step and R operation H-097 (E14-b)	proved relative to the inputs at right (route (a); L-164 given R-044 (W4), (Eps), F1, (D)). R-043 (FX) is a separate hypothesis, row B-S.FX	R-043 (FX), H-191 (FX2); K-024 (U), a choice of constants
B-S.FX	R-043 (FX): Q-059(b), Q-060, Q-061, Q-062, R2	reduced to R-042 (N3) and R-044 (W4). Q-059(b) proved (Lemma [P], no additive constant). Q-060 proved (the R-denominator is the bare Wilson integral (1.100)). Q-061 proved relative to R-042. R2 ($\kappa_{fin} \geq \kappa_{final}$) reduced, in volume form, to $R_{K+1} \geq L$ and C_c independent of g, both page readings. Q-062 (unbounded components in (1.103)) is avoided by Bałaban's main form (1.98) (K-014). The coarsest-scale H-191 (FX2) radius is not a separate input (Lemma G-1)	R-042 (N3), R-044 (W4); CMP 102 (189) (R-005)
B-S.N3	N3 (R-042): HBETA strip bound $\lvert\delta w\rvert \leq \frac{3}{2}\tau$	reduced to R-040/R-041 (H-hol) + real R-037 (H-hist) + H-120 (H2-hist); the inputs R-037, R-040 route through Proposition A	R-040/R-041 (H-hol), R-037 (H-hist), H-120 (H2-hist)
B-S.N3A	condition (a) of L-232 (Lemma N3)	proved (slack $> 5\times$)	order of constants $L \to$ Bałaban's constants $\to$ $\gamma \to g \leq \gamma/2 \to r_0 \to K$
B-S.hol	R-040 (H-hol), small field: β_j holomorphic on $\lvert g - x\rvert < \rho_0 x$	reduced to Bałaban's conditions with modified constants; $\lvert\beta_j\rvert \leq M_\beta$ reduced to Proposition A, K_U (R-005, R-019), and K_2	(C1-m)–(C4-m), H-185 (C-rad), (2.24-m)
B-S.holLF	R-041 (H-hol), large field (frozen thresholds at x^*)	reduced to H-143 (M-A), H-145 (M-B), H-146 (M-C), R-038 (H-hist2); H-146 (M-C) is itself reduced (row M-C)	H-143 (M-A)–H-147 (M-D)
M-A	equipartition $\langle A - A_{\min}\rangle_u \leq (n_i/u)(1+\eta_x)$	proved given (H−); H-144 (M-A2) proved; (H−) reduced to CMP 102 (189) (R-005)	CMP 102 (189); (C1), (C2)
M-B	(1.80)–(1.98) bookkeeping, changed constants	proved with $p_0 \geq 17/(\gamma_0 A_1^2)$, $\theta_W + \tau_C \leq 0.2$, $\beta_0 < 1/3$	H-143 (M-A), H-146 (M-C)
M-C	complex small-field step	reduced to the printed conditions of CMP 116 §2 (B1–B23 and four more), Lemma 2, (1.36), p. 12 (2.5)–(2.6), with the shrinks $\varepsilon_1/(4\lambda_V)$, $\gamma/(8\lambda_V)$; H-134 (I-X), H-133 (I-V0), H-131 (I-ord), and (I-ε) in both layers proved; K-025 (M1) is a convention	two readings: the unprinted scaling of α_2–α_4, and q_0 versus p_0 (each branch closes)

Block	Content	Form of the result	Hypotheses carried
M-D	frozen map against Bałaban's map	not a convention: it needs the derivative of the frozen map at x^*, which is reduced to R-037 (H-hist) + R-039 (H-thr)	R-037 (H-hist), R-038 (H-hist2)
B-S.D4	real-source derivative bounds H-098 (E14-c)	D4.1, D4.3, D4.4 and L-111 (Lemma C) (order $1+\eta$, given R-043 (FX)) proved; $(\mathrm{S}_{\mathrm{src}}) \Rightarrow$ H-116 (H-U) with $s = 6$, $D > 1$, given R-043 (FX)	R-043 (FX), S1 at all scales
B-N1	source form $\Rightarrow$ H-116 (H-U) on $\mathcal{D}_n$	proved given the sourced Gevrey bound R-045 (S) (Proposition 4.7); the uniformity of R-045 (S) in support size and ε is reduced (h-independent) to the inputs at right; the H-098 (E14-c) clause is reduced to R-045 (S)	R-043 (FX) main form, S1 at all scales, L-127 (Lemma CS)
B-C	compactness (diagonal extraction)	proved (Proposition 4.8)	H-116 (H-U)
B-REG	(OS0)	proved (Section 4.5.5)	H-116 (H-U); torus form of OS II
B-TI	(OS1) all of T^4 and B_4	proved (Section 4.5.6)	H-116 (H-U); origin at a site
B-RP	(OS2)	lattice: cited [21]; passage to the limit: proved (Section 4.5.7)	ℓ even (sufficient); H-116 (H-U)
OS3	permutation symmetry, reality	proved (Section 4.5.8)	—
B-NT	nontriviality (Theorem 4.5)	open: needs (NT) = H-101 (E14-f); the derivation from (NT) is proved	(NT) at every layer for (ii)
B-S.A	CMP 109 and CMP 119 transfer assertions R-022 (p. 253), R-023 (p. 275), R-024 (Q-049, pp. 275–276, multiscale Ω-geometry); Q-048 (NEW-2) inside R-016	R-022 and R-023 reduced to page readings (their lemmas follow from the Dressing Lemma); R-024 reduced, with the sourced extension proved relative to its inputs; Q-048 proved relative to $\mathrm{Q1}^{\Omega}$ and a Schur step	R-024: (Sep) on the R-operation-modified $T^{(j)}$ (a reading), $(\mathrm{S}^{\Omega}_{H})$ (proof sketch); readings (R-022, R-023)
E13	convergence of the full sequence	not needed here; open	H-099 (E14-d), D6
IV	infinite volume, joint limit $(\varepsilon, \ell) \to (0, \infty)$ (Theorem 4.14); outside Theorem B	a conditional statement (Theorem 4.14); L-320 proved relative to listed inputs; L-331 proved relative to construction readings; L-323 proved relative to listed inputs, with the readings I8 and (D) on $\tilde{U}^c$; L-332–L-334 not asserted	the hypotheses of Theorem 4.14

The following inputs separate Theorem 4.2 from an unconditional statement, modulo GC1–GC3.

(a) Theorem 2w. (H2) (R-025) holds at $L = 13$ (Theorem 6.2). B-T2 rests on R-006, through the reduction H0-red (its item N6 relative to the small-field step of R-013, H-040, and N9), and on the residue of L5 (R-031): R-013, R-018, CMP 102 Sect. G, and the causality reading H-047. The inputs (PI), (M_j), V1, (G2), (F), and L5 are proved relative to listed inputs (Sections 5.1 and 5.11; Theorems 5.102, 5.104, 5.134).

(b) The step to Theorem 1 needs the sweep (proved; R-012 holds as a list) and the finite-volume

identification R-032 (replaced form, proved relative to K-032–K-038; R-015 reduced to conventions).

(c) The off-shell item H-358 of R-010 and R-014: it is reduced to seven items of R-010; of the literal readings of (189), reading (i) holds and readings (ii), (iii) are open. R-014 is asserted by Bałaban.

(d) R-019: the reading of the complex form of CMP 98 Prop. 5 (p. 43). R-018, including the residue (ii) of the R-operation geometry.

(e) R-024: the separation (Sep) on the R-operation-modified $T^{(j)}$ (a strong reading) and $(\mathrm{S}^{\Omega}_{H})$ (proof sketch). R-022 and R-023: page readings.

(f) B-S: the reading (W6*) (H-361); the sub-items of R-039 (the item (ii) of $(\mathrm{R1_f})$ (H-241), (R2)–(R4), the large-field coarea Jacobian); the β_j part of R-038; the inputs of R-042 (N3); the residue of R-020 ((H′-def), (O1)).

(g) The residue of R-001–R-005: the printed inputs of (C) (H-057) (R-021); Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10 (written out as Theorems 5.145 and 5.143); WG-1; the readings and conventions of the write-outs (Theorems 5.76, 5.77, 5.78); the inputs of Theorems 5.79, 5.80, 5.82; the residue of R-018 (Proposition 5.108).

The blocks proved in this section are short and are not among these inputs.

4.7 What the theorem does not give

Each item needs a separate argument that this paper does not give.

- Rotation invariance. Only B_4 and the torus translations are proved. Full $O(4)$ is not a symmetry of T^4_ℓ. On the torus the inclined comparison gives only the tilted-axis reflection symmetry $R_{2\theta}s$, never a rotation. The template [18] reaches rotations only after the limit torus $\to$ plane, which it controls with its mass gap.

- Infinite volume. ℓ is fixed. Section 4.8 records one conditional statement for a joint limit, Theorem IV (L-196), outside this theorem. Jaffe and Witten tie the limit $T^4 \to \mathbb{R}^4$ to a volume-uniform gap: "New ideas are needed to prove the existence of a mass gap that is uniform in the volume of space-time. Such a result presumably would enable the study of the limit as $T^4 \to \mathbb{R}^4$." We know no public mechanism without a gap. With the constants of this paper, a unit-scale cluster expansion uniform in ℓ would give exponential clustering at a rate of order one in unit-scale units, against the correlation length $\sim e^{1/(2b_0 g^2)}$ that asymptotic freedom predicts. So such an expansion is expected to be impossible.

- Mass gap. Not addressed. On a finite torus a "gap" of the transfer matrix is not the Clay mass gap.

- Uniqueness. Different subsequences may give different Schwinger functions. Uniqueness needs E13, which Bałaban does not have. Uniqueness of the vacuum after reconstruction is also not given (no clustering on a torus).

- Vacuum state. Time is periodic. So OS reconstruction from torus reflection positivity gives a transfer matrix on T^3 and a thermal state at inverse temperature ℓ, not a vacuum state.

- Also not given: clustering (OS4); a probability measure for the density field; Wilson-loop observables (class C3); the stress tensor and the operator product expansion (class C2 and beyond); independence of the limit from the block-averaging choices of Bałaban's RG (only the lattice measure enters $S_{n,\varepsilon}$, but $g_0(\varepsilon)$ is chosen through Bałaban's flow).

- Status of the whole statement. Even with every block closed, the theorem is conditional on Bałaban's published 1984–89 proofs (GC1) (H-200). Under Jaffe–Witten footnote 2 it is a compactness ("weak-existence") result on a torus with the torus forms of OS0–OS3. It is not a solution of the Millennium problem.

4.8 Infinite volume: one conditional statement

This subsection is outside both targets of the paper. It records one conditional statement about the limit $\ell \to \infty$. The statement adds nothing to Theorems 4.2 and 4.5, and no binding set of Section 7.7 contains it.

Theorem 4.14 (Theorem IV, infinite-volume existence, compactness form; conditional)**.** *Let $G = SU(2)$, $d = 4$, L odd, $L > 11$, fixed. Let $g \in]0, \gamma]$ with γ as in Theorem 2w (L-001), and let $g_0(K, g)$ be the bare coupling of Lemma DT (L-005). Let the torus sides range over $\ell \in \mathcal{L}(g) = \{2L^m : 2L^m \geq \ell_0(g)\}$. Assume:*

(i) the hypotheses of Theorem 4.2 (Theorem 2w, Bałaban's Theorem 1, the sourced chain S1, and (FX) (R-043) Steps A, C, D) on every torus of side $\ell \in \mathcal{L}(g)$;

(ii) L-320 (FX-V) (H-106): the polymer representation of $\Delta(f, V) = \log(\hat{\rho}^f_K/\hat{\rho}^0_K)(V)$ holds with constants uniform in the unit-scale field V, in K, and in ℓ;

(iii) L-323 (S1-pt) (H-165): the oscillation bound $\operatorname{osc}_V \partial_t \Delta(0, V) \leq C\, N(f)(1 + |\operatorname{supp} f|_1)$ holds with C independent of ℓ;

(iv) L-319 (U_ℓ): the constants of the scales $k < K$ are independent of ℓ.

Named among these hypotheses: (N3) (R-042); (D), the domain hypothesis of L-054 (Lemma A) (row R-044); (Reg), regularity at every scale m of the background $U_m(V_K)$ of a χ_K-small V_K (no ledger row); (FX) in Bałaban's main R-step form K-014; and two readings used by L-323: (D) on the analyticity spaces $\tilde{U}^c_j(X, \tilde{\alpha}_0, \tilde{\alpha}_1)$ of the R-terms (CMP 122 II p. 390), and I8 (no boundary term B' survives to the unit scale under K-014). Neither (1.103)–(1.104) nor Q-062 is used. Then:

(a) There is a sequence $(K_i, \ell_i) \to (\infty, \infty)$ along which the Schwinger functions $S_{n,\varepsilon_i,\ell_i}(F)$ of the centred plaquette density converge for every $n \geq 1$ and every compactly supported $F \in {}^0\mathcal{S}_n(\mathbb{R}^{4n})$ (Schwartz functions that vanish with all derivatives on all diagonals). The limit S_n extends to ${}^0\mathcal{S}_n$.

(b) Every limit family satisfies OS0 in the form $|S_n(F)| \leq A^n (n!)^{1+2\eta} |F|_{sn+1}$ with $s = 20$; OS1-T (all translations of $\mathbb{R}^4$); OS1-H (the hypercubic group B_4 about every point); OS2 (reflection positivity for the half space $x_0 > 0$); and OS3.

(c) Assume also L-331 (Loc-f) (H-142). Then every limit is determined by Bałaban's $\mathbb{Z}^4$ activities and a limit point ν_∞ of the unit-scale measures (the identification (3.2) of `INFINITE_VOLUME_FABLE.md`).

The lemmas L-321, L-322, and L-324–L-330 used in the proof are proved in `INFINITE_VOLUME_FABLE.md`. Hypothesis (ii), L-320, holds relative to the hypotheses of Theorem 4.2 on T_ℓ for every $\ell \in \mathcal{L}(g)$ (including R-042 through Q-061), the narrowed form (U_ℓ) of (H-U), and the Kotecky–Preiss theorem; Step A pointwise in V is printed in CMP 122 II (1.73)–(1.75), p. 380, and in the U-free right side of (1.79), p. 383. Hypothesis (iii), L-323, holds relative to the sourced chain S1 at all scales (R-035), (5.1) of the constant-source family, L-054 (Lemma A) with (D) including $\tilde{U}^c$, (Reg), the boundedness of B_K with (c-pair), I8, and R-042 (both forms). The additional hypothesis L-331 of (c) holds relative to readings of the construction. For hypothesis (iv), L-319, a possible approach rests on named results of Bałaban (the CMP 99 and CMP 102 constants for domains of any shape, R-004, CMP 95 Prop. 1.2 and (1.67), CMP 89 Sect. 5) and on Lemma G, which replaces Bałaban's Case 2 step on CMP 122 II p. 387 (row R-013 of Section 7.7); Bałaban prints the torus independence of β, γ_0, q_0, q_1, B_0–B_3, B_5, δ_0, the R-formula, p_0 given A_0, and the outputs E_1, R_1, $E_\pm$. In (b), $s = 20$ comes from the singleton volume factor of the route through ν_∞ ($q = 10$).

Not asserted: L-332 (convergence along the full sequence in ℓ; infrared); L-333 (O(4) covariance); L-334 (OS4 and a mass gap); any observable other than the centred plaquette density; nontriviality (it needs (NT) uniform in ℓ). Theorem IV is separated from an unconditional weak-existence statement by every open input of Theorem 4.2 (S1, the sub-items of R-039, R-042, (W4) (R-044), (Eps), (D)), by the hypotheses (ii)–(iv), by the readings I8 and (D) on $\tilde{U}^c$ of L-323, and by the (S_∞) (H-169) bridge. The result is weak existence only. Under Jaffe–Witten footnote 2 it is not a solution of the Millennium problem: it gives no uniqueness, no O(4) covariance, no clustering, and no gap.

Remark 4.15 (OS4 for joint limits). Theorem IV does not assert OS4. Its limits are joint limits $(K_i, \ell_i) \to (\infty, \infty)$. OS4 for such limits would need a hypothesis that is uniform in K, (Gap_U): for every pair of disjoint bumps f_1, f_2,

$$\lim_{R\to\infty} \limsup_{K,\ell\to\infty} \sup_{R\le|a|\le\ell/4} \left|\mathrm{Cov}_{\nu_{K,\ell}}(w_1, w_{2,a})\right| = 0,$$

where $\nu_{K,\ell}$ is the unit-scale probability measure on the torus of side ℓ at level K, and w_1, $w_{2,a}$ are the first-order responses to f_1 and to the translate $\tau_a f_2$. The fixed-K decay condition (Gap_K), even for every K, does not suffice for an arbitrary joint sequence. K-uniform (Gap_K) suffices for iterated joint limits ($\ell \to \infty$ at fixed K first, then $K \to \infty$), and Theorem IV(a) can always be realized along such a sequence (proof sketch). (Gap_U) is the condition for every joint limit. This paper uses none of these statements.

5 The sources block

This section states the inputs of the two targets. Each result is stated with its hypotheses and followed by a sketch of its proof. A hypothesis that this paper does not prove is named by its row in the dependency table (Section 4.6). Labels such as R-036, H-111, K-024, L-182, Q-059, and X-004 are the row labels of that table. A reading of an unclear printed passage is stated as a numbered Convention. The order follows the dependency chain. First come the inputs of Theorem 2w (L-001) (Target 1). Then come the inputs of the finite-torus theorem (Target 2), in the order of steps 2–9 of its dependency chain. The provenance of each result is in the appendix table.

5.1 Target 1: inputs of Theorem 2w

Item 7: the running-maximum lift. Item 7 asks for the own-slot bound of (G-gain) (H-111) on every past, also on the clamped, non-monotone pasts that the shooting evaluates.

Convention 5.1 (reading of β in item 7)**.** We read β_{k+1} as the coefficient defined from the pure part of the step, as in Proposition 5.16. On the competing literal reading, β_{k+1} is defined from the whole $E^{(k+1)}$, and the possible g-dependence of the newest R-input function $R^{(k)}$ is open.

Proposition 5.2 (item 7 on every past)**.** *Assume either (H0) (H-202) (ledger R-006), or the inputs of (G-gain) listed in ledger R-037. Assume Convention 5.1 and the side condition $4\gamma^{\kappa_0} \le E_0$. Then for every step k, every past $h \in]0,\gamma]^k$, monotone or not, and every $g \in [0,\gamma]$,*

$$|\beta_{k+1}(h,g) - \beta_{k+1}(0)| \le C_\beta g, \qquad C_\beta = C_\Pi S_0,$$

where $C_\Pi S_0$ is the constant of (G-gain). The bound is uniform in k, K, and the past.

Proof (sketch). The proof of (G-gain) uses monotonicity at one place only: Step 2 of L-164, where $\max_{j\le k} 4g_j^{\kappa_0} = 4g_k^{\kappa_0}$ compares bound sizes. Put $G_k = \max_{i\le k} g_i$, the running maximum. The induction $a_j \le 2G_j^{\kappa_0}$ then holds for every past under $4\gamma^{\kappa_0} \le E_0$, a condition of Bałaban's type. The constant does not change, because $E_0 + 4\gamma^{\kappa_0} \le 2E_0$ is an input radius that (G-gain) already covers. The value at zero own coupling, (K1), is free of the history for any input, because the exponent vanishes at $g_k = 0$ (CMP 109 p. 268). The endpoint $g \to 0$ is a dominated-convergence step, which we do not write out line by line. □

Hypotheses. Under (H0) (ledger R-006), the use here reduces to the residue N1–N9 of Section 5.9. Under (G-gain), the inputs of ledger R-037 are (Q)/c_-, (M_r), CMP 102 (189), and the (M-B) (H-145) bookkeeping. The result also uses L5, the evaluation of β off Bałaban's trajectory (item 8, ledger R-031). We assume, in addition, that Bałaban's R-operation bounds CMP 119 (2.31) and the off-trajectory small-field induction of `SOURCES_SM.md` §5.1 contain no further use of the order of the couplings.

Item 9: K-independence of $\beta_j(g)$. Item 9 is a statement about the volume, not an identity about histories.

Proposition 5.3 (the window on every admissible history)**.** *By CMP 119 (2.24), p. 253, and p. 279, β_j is the whole-lattice coefficient of CMP 109, whose Π in (4.37) is a sum over $\mathbb{Z}^4$. So the β_j of the recursion does not depend on K. Assume: (K1) $\beta_j(h,0) = \beta_j(0)$ for every admissible past; (K2) $|\partial\beta_j/\partial g_{j-1}| \le C_\beta$; (K3) the window bound $\sum_{i=k+1}^{n} \beta_i(0) \in [b(n-k) - 2R,\, b(n-k) + 2R]$ of Theorem K (L-009). If $\gamma \le b/(4C_\beta)$, then $T(m,n) \in [(3b/4)(n-m) - 2R,\, (5b/4)(n-m) + 2R]$ on every admissible history.*

Proof (sketch). Lemmas DS and DT run at one fixed K. They never compare two trajectories or two values of K. So only their constants must be free of K. The bound $|\beta_{k+1}(h_k) - \beta_{k+1}(0)| \le C_\beta g_k$ follows from (K1) and the mean-value theorem in the own slot. It gives the window bracket pointwise on each clamped history of the intermediate-value argument. □

Hypotheses. (G-gain) alone gives C_β only on monotone slot histories. So (K2) needs item 7, which Proposition 5.2 gives from (H0) (ledger R-006) or from (G-gain) (ledger R-037). (K3) rests on the inputs of Theorem K. The result is used together with L5, the domain and continuity of β off Bałaban's trajectory (item 8, ledger R-031). Theorem 5.104 proves L5 relative to listed inputs, by nested intervals on true trajectories. The finite-volume identification of the step (ledger R-032, with R-015, narrowed by Proposition 5.87) is an input of Theorem 1, not of the shooting.

Item 5: V1 (ledger R-029).

Convention 5.4 (the averaging operator)**.** The averaging of the step is CMP 109 (0.4), the averaging that the numerics use. The competing choice is (0.12), which Bałaban's step uses as printed.

Proposition 5.5 (V1 at fixed level)**.** *Assume Bałaban's inductive bounds, (G2) (ledger R-030), and Convention 5.4. Work inside Bałaban's construction with the cut CMP 109 (2.9). Then the term $E^{(k+1)}$ of (2.13) vanishes identically at $g_k = 0$. Hence $\Pi^{(k+1)}(0)$ and $\beta_{k+1}(0)$ come from $\log Z^{(k)}$ alone. The coefficient $\beta_{k+1}(0)$ is the Maxwell coefficient of the one-step Gaussian determinant. It is free of the history, and it equals the determinant of the numerics up to a constant.*

Proof (sketch). The proof reads the definitions CMP 109 (1.3)–(1.22) and (2.10)–(2.15) on the page. It then passes from Bałaban's exponential gauge fixing to the bordered Hessian of the numerics. The steps are gauge-fixing independence, the constraint Jacobian, the expansion point, and the normalization. □

Item 6: (G2) (ledger R-030).

Proposition 5.6 ((G2) at fixed level)**.** *Fix the level j. Assume that $\tilde{Q}$ is surjective and that the averaging is analytic in the background. Then: (i) the constrained one-loop log-determinant on a torus of side N has no zero modes; (ii) its density per coarse cell converges to the infinite-lattice density with error $\leq C_j e^{-\delta_j N}$, also for every derivative in the plane-wave amplitude; (iii) the p^2-coefficient commutes with $N \to \infty$; (iv) if in addition the gap input (Gap_f) holds and the density is C^4 in f, then the f^2-coefficient at the constant field commutes with $N \to \infty$. The massless pieces of Prop. 2.1 are treated fibre by fibre, with a zero-mode constant independent of N.*

Proof (sketch). The proof uses a Combes–Thomas decay bound and the method of images. The constants are fixed-j constants, with $\delta_j \gtrsim L^{-2j}$. □

We do not prove a version that is uniform in j.

(M_j) (ledger R-028): the lattice spectral inputs.

Proposition 5.7 ((S1)–(S3) at fixed level)**.** *Fix $j \geq 1$ and odd $L \geq 3$. Assume the Peierls input (I3) (Proposition 5.9). Then for small flux f the following hold: (S1) (H-244) $\mathcal{K}_t(f) \geq -C_1|f|$ for $t \in [0,1]$; (S3) (H-247) $\pm V \leq C_3[(\mathcal{K}_t+\mu)^2+f^2]$, in the subdivided form of `KAPPA_BOUNDED.md` §1.3; (S2) (H-245) at $t = 0$; (S2c) (H-246) the coarse comparison $N_0(\lambda-\eta_1) \leq N_t(\lambda) \leq N_0(\lambda+\eta_1)$ with $\eta_1 = O(f^2)$; and region 2 of Prop. 4.4 of `BETA_FP_M.md`. Corollary 7.1 then reduces (M_j), the limit of $[\Gamma_c(\Lambda_j; f) - \Gamma_c(1; f) - (f = 0 \text{ value})]/(n_2 f^2)$ to d_j, to three inputs: the near-zero input (Z_j), itself reduced to the 2D gap (G_{2D}); region 1 of Prop. 4.4 of `BETA_FP_M.md`; and the differentiation under the integral in the neutral sector.*

Proof (sketch). The proofs work in the twisted algebra of operators that commute with the magnetic translations. The tools are Peierls products, Moyal expansions with kernel-norm remainders, a Nenciu-type inverse, and operator bounds from the vanishing order of symbols. In Lemma 1.4, ∇^β is one twisted convolution. At $t = 0$ a fibre decomposition over $k_\perp$ separates the operator. A gauge-copy lemma (Lemma 5.8) shows that the near-zero density of states needs only (G_{2D}): the transverse charged 2D SU(2) Wilson Hessian at constant flux has no spectrum in $(-c|f|, c|f|)$. We do not use the explicit power of L in C_j of Prop. 3.2, because it depends on conventions. □

Hypotheses. Proposition 5.9 proves (I3). Numerically, at six rational fluxes, the gap of (G_{2D}) is about $|ef|$. Theorem 5.98 proves (G_{2D}) in a narrow range (Remark 5.99), and Theorem 5.102 proves (M_j) relative to listed inputs.

The Peierls input (I3) (ledger R-028).

Convention 5.8 (identification of the one-step objects)**.** The one-step objects of this paper are the linearization of the averaging (0.4), or of (0.12) with Federbush's average, at the constant-flux background.

Proposition 5.9 ((I3))**.** *Let L be odd, $L \geq 3$, and let the averaging be Bałaban's (0.4), or (0.12) with Federbush's average. Assume Convention 5.8. Then the linearized averaging $\tilde{Q}_f$ at the constant-flux background is entrywise analytic in complex f on $|f| < f_0$, with $f_0 = \pi/8$ for (0.4) and $f_0 = \min(\pi/8, 1/8)$ for (0.12). The input (I3) holds at every step $i \leq j$, with radius and constants independent of i and j (Theorem 7.1 of `SOURCES_T2_I3.md`). It holds in the stated two-block range and does not use (I1) (H-207) for the fixed vertices.*

Proof (sketch). The proof linearizes the averaging explicitly at the flux background (Prop. 5.3). It bounds each charged entry of $\tilde{Q}_f$ and of D_f by its reduced kernel (Props. 5.4, 6.1). Then it assembles the bordered matrix of one step. □

For (M_j), the 2D gap (G_{2D}) and region 1 of Prop. 4.4 of `BETA_FP_M.md` are treated in Theorems 5.98 and 5.102.

5.2 Target 2, step 2: the sourced Theorem 1

The E14 reduction of (B-S) (ledger R-035).

Proposition 5.10 (structure of the sourced construction)**.** *Let $\rho_0^f = \exp(-\sum_p (g_0^{-2} - f_p) P_p(U))$ be the plaquette-density source.*

(a) The source is a local bare coupling. The derivatives of $\log Z$ in f are the cumulants of the plaquette density. So the Schwinger functions on test functions flat on the diagonals reduce to a sourced version of Bałaban's Theorem 1.

(b) The sourced Theorem 1 consists of a pointwise coupling flow (S-A) (H-159), sourced regular terms (S-B) (H-160), a local contact functional $W_k[f]$ (S-C) (H-161), a local β-function (S-D) (H-162), and the normalization (S-E) (H-163). The printed CMP 119 (2.24) is $1/g_{j-1}^2(x) = 1/g_j^2(x) + \beta_j(g_{j-1})\varphi_j(x)$, and by (2.25) Bałaban evaluates β_j at the constant coupling. So (S-A) and (S-D) have no counterpart in Bałaban's construction.

(c) The estimates needed beyond Bałaban's are (E14-a) (H-096), the sourced small-field step; (E14-b) (H-097), the sourced large-field step and R operation; and (E14-c) (H-098), the derivative bounds in f. (E14-d) (H-099) = E13 is needed only for convergence along the full sequence, and (E14-f) (H-101) only for nontriviality.

(d) The source of the dimension-4 density is marginal at every degree. Its field-independent part is relevant and is removed as local contact terms. The first field-dependent gradient term has dimension 6.

(e) At the Gaussian level, put $K_f = (\Delta^{(1)})^{-1/2} \Delta^{(f)} (\Delta^{(1)})^{-1/2}$ with $\|K_f\| \leq r$. The source enters one step through $-\frac{1}{2}\log\det(1 - g_0^2 K_f)$, and the linear response per step is $c_j = 1 + \frac{1}{2} g_{j-1}^3 \beta_j'(g_{j-1})$. If the source renormalization obeys a one-step recursion, then $\prod_j c_j$ is uniformly bounded.

Proof (sketch). Parts (a)–(c) compare the sourced weight with CMP 119 (2.24)–(2.25). Part (d) is a dimension count. Part (e) writes out one step at the Gaussian level. □

Hypotheses and limits. The product $\prod_j c_j$ is not shown to converge. The uniform bound for a general F flat on the diagonals is not derived from the source form. (B-S) as a whole, with (E14-a, b, c), is open (ledger R-035, with its parts R-036–R-044 and R-046). The (E14-c) clause of the bridge (B-N1) is ledger row R-045. (NT) = (E14-f) is needed for Theorem 4.5. E13 = (E14-d) is not needed for Theorem B.

One sourced step at one loop (D1).

Proposition 5.11 (the sourced step at one loop)**.** *Let the plaquette-density source enter as the local bare coupling $w_p = 1/g_0^2 - f_p$. Then at one loop, for every background:*

(a) the multiplicative source renormalization is $c_1 = 1$ exactly;

(b) the first correction is $c_1 - 1 = \frac{1}{2}g^3\beta_1'(g)$; it is $\beta_1^{(2)}g_0^4 + O(g_0^6)$ if β_1 is even, and $O(g_0^3)$ under a C^1 bound only;

(c) the field-independent terms are the tadpole $\frac{1}{2}(9L^4 - 9)g_0^2$ per coarse cell ($\frac{3}{4}(1 - L^{-4})g_0^2$ per plaquette) and a contact term $C(q)$ that is analytic at $q = 0^+$;

(d) the first field-dependent gradient terms have dimension 6, by index parity.

Proof (sketch). The source enters the step only through the Hessian $\Delta_f = \sum_p w_p H_p$ and a current term $\langle J_f, B\rangle$ at Bałaban's unweighted minimizer. Gauge fixing, ghosts, and the constraint carry no f. For constant f the quadratic form scales by $1 - g_0^2 f_0$. So the determinant changes by a field-independent constant over $9L^4 - 9$ modes per cell. The error of the expansion point has the small parameter Lp. □

Per-momentum formulas for the tadpole, $C(q)$, and the mixed derivative are evaluated by `numerics/d1_check.py`. They agree with real-space finite differences on the periodic 6^4 lattice. Numerically, $|c_1 - 1| \le 5 \times 10^{-4}$. The data do not show geometric convergence in the grid, and we do not use the numerical values of γ_2. The non-Gaussian sourced step and the pointwise flow are the subject of Propositions 5.12–5.16 and of (B-S) (ledger R-035).

The sourced small-field step (E14-a).

Proposition 5.12 (sourced small-field step)**.** *Let the source be block-constant, $\bar{f}$, at the scale of the step. Assume that the step is analytic in its input activities, (W4) (ledger R-044), and the hypothesis on the field-dependent domains Λ_j in W (ledger R-044). Then the sourced small-field step is Bałaban's step with the following changes:*

(i) extra source vertices of order g_k, g_k^2, g_k^3, all in his $P^{(k)}$ class, each $\le O(g_k)(1 + |B|^2)$ on $|B| < \varepsilon_1/g_k$;

(ii) a propagator for the sourced Hessian with Bałaban's decay and regularity bounds, with constant $3c_1$, $c_1 = O(\delta_0^{-5})c_0$, and rate $\delta_0/2$;

(iii) a field-independent vacuum term W that factors out in the small-field region.

The source polydisc has radius $r/(2n)$. The remainder bounds reduce to Bałaban's estimates, with a freezing property and a constant-source family.

Proof (sketch). The source term is evaluated at Bałaban's unweighted minimizer. So his measure and propagators are used without change. The sourced propagator follows from a weighted Neumann series in his cube-to-cube operator norms (CMP 116 (1.7)). The interpolation Lemma 7.1 ties the real-source derivative bounds to polystrip holomorphy. □

The pointwise flow $1/g_{j+1}^2(x) = 1/g_j^2(x) - B_{j+1}(f(x))$ with $h \equiv 0$ (step (c)) does not follow from CMP 109 Sects. 3–4 verbatim: (4.34)–(4.37) use Euclidean symmetry, and their cubes are at scale k, not j. Proposition 5.13 proves step (c) by another route.

Hypotheses. The recovery $\kappa \to \kappa/2$ is Proposition 5.19 (ledger R-046, a reduction that uses R-020). (Hβ) is reduced in Proposition 5.18. The scale condition (U) is the constant choice K-024, not an open input (Proposition 5.20).

Step (c): the frozen source per domain.

Proposition 5.13 (Lemma A (L-054) and the pointwise recursion)**.** *Assume the domain hypothesis (D) (ledger R-044) and (Hβ). For $X \in D_j$ let b'_X be the point of X nearest to b_X, and freeze each step term $E^{(j)}(X, \cdot\,; f)$ at $c_X = f(b'_X)$. Fix $\beta > 0$ in (4.18). Then:*

(i) (Lemma A, L-054.) The freezing error obeys $|D_X| \le E_0 C M^3 (\rho_1/r)(L^j\eta)^5 \lambda_k e^{-\kappa d/2}$. It is irrelevant with exponent 1 and does not grow in k.

(ii) Assume the identification (I) (ledger R-036) of the constant-source increment $B_j(c)$ with Bałaban's β_j on the trajectory from $1/g_0^2 - c$, with the adapted gauge-fixing weight $1/u_k(c)$ (K-011). Then $1/g_{j+1}^2(x) = 1/g_j^2(x) - B_{j+1}(f(x))$ with $h \equiv 0$, and the gradient corrections are irrelevant with exponent $\min(\alpha, \beta, 1)$.

Without the adapted weight, (I) holds up to $O(e^{-c\varepsilon_1^2/g_k^2})$.

Proof (sketch). The non-constant source is treated at scale j, per localization domain, as a gauge-invariant difference. This difference is bounded in an axial gauge on X, where each field costs two powers of $L^j\eta$. Bałaban's CMP 109 Sects. 4–5 are then used only for constant sources, where they apply at the bare coupling $1/g_0^2 - c$. The proof does not use a cancellation of the cubic first-moment term of (4.34). □

Hypotheses. (D) is suggested by the remark on CMP 109 p. 263, which cites CMP 102 Prop. 9 (ledger R-005); it is ledger R-044. (Hβ) is Lemma H (L-182), Proposition 5.18. (I) with large-field thresholds is ledger R-036. The adapted gauge-fixing weight changes the construction (ledger K-011).

The sourced large-field step (E14-b).

Proposition 5.14 (sourced large-field step)**.** *Assume (H-thr) (ledger R-039), the real uniform (H-hist) (ledger R-037), the strip holomorphy N3 of u_k (ledger R-042), (W4) and the field-dependent domains Λ_j (ledger R-044), and (FX) (ledger R-043) with (FX2) (H-191) for the final step. Then the sourced large-field step is Bałaban's step with these changes: all thresholds are frozen at the reference couplings; the local coupling is $w_j = w_{j-1} - B_j(f)\varphi_j$; an R operation resets it on Z to $u_k(f(x))$ (CMP 122 II p. 365); and the local gauge-fixing weight is adapted. Moreover:*

(a) the reset term is bounded by Bałaban's sup bound (CMP 122 II p. 366);

(b) real sources keep the positivity of the large-field weights under $5\gamma^2 r \le \frac{1}{2}$ (Proposition 5.15);

(c) (Lemma D3.1.) One positive denominator has the local holomorphy radius $c_\tau r/(E_0|Y|)$;

(d) the identification (I) is off by a threshold term, the input (H-thr);

(e) the lower bound on w_k holds relative to Bałaban's local coupling;

(f) the cumulant bound has Gevrey order $(n!)^{2+\epsilon}$.

Proof (sketch). We list the places (P1)–(P8) where Bałaban uses a constant coupling in the large-field step. Here (P8) (H-240) is the t-parameter term of CMP 119 p. 271. At each place we state the change and reduce it to a quoted estimate of Bałaban (CMP 119 (2.23)–(2.31), CMP 122 II (1.73)–(1.75)). We do not repeat CMP 122 I–II line by line. The old R-terms are merged into E_k, so freezing through R needs a separate argument (Proposition 5.15). □

Hypotheses. (U) is the constant choice K-024.

Revision of D3: (P3), (P8), freezing through R, Gevrey order.

Proposition 5.15 ((P3) (H-235), (P8), and the R-induced terms)**.** *Assume $\rho_1 \le r/(4C_hM)$. Then:*

(a) (Lemma P3, L-245.) The sourced reset coefficient differs from Bałaban's by at most $\delta_r = 4r + 2C_hM\rho_1\lambda_k$, additively and pointwise. So his sup bound of CMP 122 II p. 366 holds with one changed constant. The large-field factors survive with $\gamma_0 \to \gamma_0(1-\theta) \ge \gamma_0/2$, $\theta = \gamma^2\delta_r \le 5\gamma^2 r \le \frac{1}{2}$.

(b) The t-parameter term splits as $d_B + d_S$ with $|d_S| \le 2r$. The term h_k is booked as irrelevant. The term d_S changes the covariance $C^{(k)}$ at every t, with relative size $O(\gamma^2 r)$.

(c) Freezing through R splits into a pure part (L-162) and an R-induced part (L-163). The bound L-163 is a new bound; it is not Bałaban's bound for merged R-terms. Proposition 5.16 proves it (L-164).

(d) Assume (B-N1) (ledger R-045). Then the cumulant bound has Gevrey order $(n!)^{2+\epsilon}$, which is admissible with $D = 3$.

Proof (sketch). Positivity is kept by a split against Bałaban's local coupling. The (P8) term is split into a piece with the support of Bałaban's deviation, booked as a boundary term, and the source weight of D2. An auxiliary parameter splits the regular output into the pure and R-induced parts. The Gevrey count uses a Lah-number sum over partitions. □

Hypotheses. The result is used with the strip holomorphy of u_k (N3, ledger R-042), (H-thr) (ledger R-039), the real uniform (H-hist) (ledger R-037), and (W4) (ledger R-044). The covariance change in (P8) is part of ledger R-044; it is treated by analogy with R-016 (Proposition 5.16).

The covariance change along t and the merged R-outputs.

Proposition 5.16 (the t-measure and L-164)**.** *Assume (W4), (Eps), L-162, the domain hypothesis (D), and the conditions $32\varepsilon' \le 1$, $S \ge 1$, $\kappa_0 > 2$ (K-017). Then:*

(a) Along the t-interpolation the measure is Bałaban's CMP 119 (3.27) with covariance $C_f^{(k)}$, independent of s and t. The cross term $g_k^2\langle A, \mathcal{W}_S A_k\rangle$ goes to $B^{(k+1)}$. If $\tau \le 3c_1c_H\gamma^2 r \le \frac{1}{2}$, then $\|C_f - C_0\|_\delta \le 2c_1\tau$, and the large-field factors survive with $\gamma_0 \to \gamma_0/4$.

(b) (L-164.) The R-induced part obeys $a_m \le 32\varepsilon' g_m^{\kappa_0}$, with no compounding. On regular configurations, $|E_R^{(m)}(X,U) - E_R^{(m)}(X,1)| \le CM^2 \cdot 32\varepsilon'(L^m\eta)^4 g_m^{\kappa_0} e^{-\kappa' d_m(X)}$. This is a marginal term of Bałaban's R-type form, summable in m for $\kappa_0 > 2$.

Proof (sketch). Transported inputs keep their scale label and never pass through the generated map G. So a Schwarz-lemma step on $G - G(0)$ gives $a_{k+1} \le 8\varepsilon' b_k$ per generation. Here $\varepsilon' = O(\alpha_2^{-1}\varepsilon_1)$ is Bałaban's factor (CMP 109 p. 280); it contains his large constant of p. 280. The covariance bounds come from the weighted Neumann series of Proposition 5.12. Positivity comes from domination by a rescaled C_0 Gaussian. The domination uses the marginal region (or Anderson's inequality), so that no volume factor appears. □

Hypotheses. (W4) and (D) are ledger R-044. Theorem SM (L-281) reduces (W4) to the printed (C1)–(C5). (Eps) follows from Theorem SM and R-010. CMP 116 p. 3 covers C_f only by analogy; the extension R-016 to the fluctuation covariance supplies it. The conditions $32\varepsilon' \le 1$, $S \ge 1$, $\kappa_0 > 2$ change the construction (ledger K-017). L-164 does not need a monotone history: the running maximum of Proposition 5.2 removes this assumption, with the side condition $4\gamma^{\kappa_0} \le E_0$.

RP and symmetry transfer, and the cumulant bound ($\mathbf{S_{src}}$).

Proposition 5.17 (transfer and (S_{src}))**.** *(a) (Lemma D4.1.) The sourced RG output equals the bare integral $Z_\varepsilon[f]$.*

(b) (Lemmas D4.2–D4.4.) Reflection positivity and the lattice symmetries are properties of the lattice measure. The block geometry is invariant only under translations by $M\mathbb{Z}^4$ or coarser.

(c) (Lemma C, L-111.) Assume (FX) (ledger R-043) in Bałaban's main form, with the coarsest-scale (FX2) radius. Then the cumulants of real sources obey (S_{src}):

$$|\kappa_n(f_1,\dots,f_n)| \le (C_S\sigma^{-a})^n (n!)^{1+\eta} \prod_i N(f_i), \qquad N(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1),$$

for $\sigma \in [16\varepsilon, \ell/8]$, supports pairwise at distance $\ge \sigma$, minimal diameter $\le \sigma$, and C_S independent of K. Without the coarsest-scale (FX2) radius the bound holds with an extra factor $(n!)^{2p_R/p_0}$, which is still admissible for the OS axioms.

(d) (S_{src}) implies (H-U) (H-116) with $s = 6$ and $D > 1$.

Proof (sketch). Each RG operation is an identity of integrals: Haar invariance, δ-function blocking, partitions of unity, convergent expansions, and $\int dV\, R\rho = \int dV\, \rho$ for the R operation of CMP 122 I (1.102). Complex sources follow by the identity theorem. Lemma C takes one Cauchy derivative in each of n separated source amplitudes, on a polydisc whose radius does not depend on n. So the sum over polymer sizes gives a single $n!$. □

Hypotheses. (FX) is a new hypothesis; it is not contained in CMP 116–122. Its remaining input is N3 (ledger R-042). The bound also uses S1 at all scales, part of (B-S) (ledger R-035), and the (E14-c) clause (ledger R-045).

Lemma H: the two-loop bound (Hβ).

Proposition 5.18 (Lemma H (L-182))**.** *Assume (I), (E), and (H-hist); the column-sum form of (H-hist) suffices. Let β_j be the infinite-volume function, independent of K. Let $g = g_K$, $x := C_1 A\gamma/\beta \le \ln(3/2)$, and $3r < g^{-2} - \gamma^{-2}$. Then every real c with $|c| \le 2r$ keeps all $g_k(c) \in]0, \gamma]$, and*

$$\sum_{j \le K} |B_j'(c)| \le e^x - 1 \le \tfrac{1}{2}, \qquad \tfrac{1}{2} \le |v_k(c)| \le \tfrac{3}{2},$$

uniformly in K. With $\beta = \beta_0/2$ the smallness condition reads $\gamma \le \ln(3/2)\,\beta_0/(2C_1A) \approx 0.20\,\beta_0/(C_1A)$.

Proof (sketch). In $u = 1/g^2$ the chain rule gives $\partial\beta_j/\partial u_i = -\frac{1}{2} g_i^3\, \partial\beta_j/\partial g_i$. So a C^1 bound costs $O(g_i^3)$ per step, and $\sum_i g_i^3 \le 2\gamma/\beta$ along the trajectory. A Gronwall product on the variational equation gives $e^x - 1$. A bootstrap on the component of $\{|c| \le 2r : u_K(c) > \gamma^{-2}\}$ that contains 0 keeps every c-trajectory in $g \le \gamma$. The Gronwall algebra is formalized in Lean (`LemmaH`). □

Hypotheses. (I) is ledger R-036. Lemma I36 proves (I) relative to the frozen thresholds K-001, the adapted gauge-fixing weight K-011, Lemma U of Proposition 5.13, and (Hβ) (`SOURCES_SWEEP_F.md` §3). The real uniform (H-hist) is ledger R-037. It reduces to (Q)/c_-, (M_r), CMP 102 (189), and the (M-B) (H-145) bookkeeping. Of these, (M_r) holds exactly (Proposition 5.106) and (M-B) is written out (Proposition 5.37); CMP 102 (189) and the coarea Jacobian are open. (E) is the uniform positivity of the one-loop $\beta_j(0)$. Complex c is the strip bound N3 (ledger R-042, Proposition 5.29).

The decay recovery $\kappa \to \kappa/2$ (ledger R-046).

Proposition 5.19 (recovery of the decay rate)**.** *The decay loss $\kappa \to \kappa/2$ of the sourced step occurs at every step in three places: the interpolated inputs of the freezing proof (`SOURCES_D2.md` §6.3), the extraction remainders (`SOURCES_D2C.md` §3), and the R-induced terms of L-163 (Proposition 5.15). Let $\theta := 1 - m_{\mathrm{tot}}/\kappa$ be the kept fraction per cycle; $\theta \ge \frac{1}{2}$ for $\kappa \ge 2m_{\mathrm{tot}}$. Assume R-020 and CMP 116 (2.36), and assume*

(K1$_\delta$) (H-216) $\delta \le \frac{1}{13}(1 - 2(1+4\beta)/(\theta L))$, which is $\delta \le 5/169 \approx 0.0296$ at $L = 13$, $\beta = \frac{1}{4}$, $\theta = \frac{1}{2}$;

(K2) (H-217) $\theta(1-\beta)L/2 \ge 1 + 3\beta$.

Then the loss is recovered by $L/2$-rescaling, without compounding. The choice $\delta = 0.025$ satisfies (K1$_\delta$) at $L = 13$, and $\delta = 0.03$ does not (Lean `K1_numeric_13`). At the last scale the final expansion runs at $\kappa_{\mathrm{eff}} = \theta\kappa/(1+4\beta) = \kappa/4$, with the M-condition (K3) (H-218) $M \ge 12L(4.5\kappa/C_1)^{1/4}$; (K3) follows from the M-condition of Proposition 5.28. The lower bound on κ is $\kappa \ge 1536$ at $d = 4$, $\beta = \frac{1}{4}$.

Proof (sketch). Bałaban proves only $2d_k(Z) \ge Ld_{k+1}(Z')$ (CMP 116 (2.36)). So the gain per step is $L/2 = 6.5$, and his printed choice $(1 - 10\delta)\frac{1}{2}L = 1$ in (2.41) spends all of it. The recovery therefore takes δ smaller, at the cost of a larger κ, a lower bound about 4.3–4.7 times the unsourced one. This moves only constants fixed after κ. The three losses stack within one step. Each loss is a polynomial absorption with additive cost, so the kept fraction per cycle is θ. The R operation drops $4\beta\kappa$ of rate (CMP 122 II (1.66)–(1.100)). The rescaling pays these drops and the T-step fraction, at most 13δ: 10δ printed for the small-field step and at most 3δ, not printed, for the large-field step (ledger R-020). The volume charge of (1.93) is paid by p_0 through the condition before (1.95). □

Hypotheses. R-020 is the large-field T-step rate loss of CMP 119 Sect. 3, which Bałaban does not print; the homogeneity of the large-field T step in the input rate rests on it. Bałaban states CMP 116 (2.36) without proof.

The scale condition (U) (ledger K-024).

Proposition 5.20 ((U) is a constant choice)**.** *Let $\lambda_K = 1$ (physical units). Assume the (Hβ) gradient bound $|\nabla f_k| \le \frac{3}{2}\rho_1$. Then the r in (U) is the Cauchy margin r_0 of the constant-source family, not a test-function amplitude. So (U) is the constant choice $\rho_1 \le c_* r_0/M^a$, made once with $\rho_1 := \min(c_* r_0/M^a,\, r_0/(4C_h M))$, after r_0 and before K. This choice is compatible with the bridge (B-N1): the bridge evaluates the generating functional only on sources rescaled into the domain of (FX2), at the price of a factor $(4/\rho_1)^n$ per n-point bound, with s and D unchanged.*

Proof (sketch). The bridge uses n-linearity, $\kappa_n(f_1,\dots,f_n) = \prod_i \lambda_i\, \kappa_n(f_1/\lambda_1,\dots,f_n/\lambda_n)$, with $\lambda_i = 2N_U(f_i)$ and $N_U(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$. The factor 2 comes from the gradient radius $\rho_1/2$ of (FX2). Pairwise σ-separated supports make the sup and the gradient of $\sum_i z_i f_i$ a maximum over single terms. So the whole Cauchy polydisc stays in the admissible domain, and the sup radius of the activity polydisc is $\tau_m(\Gamma) \le r$. Overlapping supports never reach the bound (S) of R-045. Small widths $h \ge \varepsilon$ are paid by the norm growth $1/h$ per bump, which the flatness of F absorbs, while (U) costs only an h-independent factor. A scale-matched reading of `SOURCES_D2.md` l. 217 would create a conflict; we do not use it. □

Hypotheses. For real c, the (Hβ) gradient bound follows from Lemma H (Proposition 5.18) and the chain rule, given (I) (ledger R-036) and (H-hist) (ledger R-037). For complex c it follows from Lemma N3 on the strip (ledger R-042, Proposition 5.29). The result is used with the $\kappa \to \kappa/2$ recovery (ledger R-046, Proposition 5.19).

5.3 Step 3: the clauses of (S) and the hypothesis (FX)

The source bound R-045 (S) must hold uniformly in the width and the position of the bumps.

Proposition 5.21 (Uniformity of R-045 (S) in the bump width)**.** *Assume S1 at all scales, including $m = 0$; (FX) (ledger row R-043) in Bałaban's main form (1.98) (choice K-014); the coarsest-scale (FX2) (H-191) radius; the bound of `SOURCES_D2.md` §4.3; and the residue of R-043. Let $N_U(f) = \max(\|f\|_\infty/r, \|\nabla f\|_\infty/\rho_1)$. Then for real sources with pairwise separation $\sigma \ge 16\varepsilon$ and $\min_i \operatorname{diam}\operatorname{supp} f_i \le \sigma$,*

$$|\kappa_n(f_1,\dots,f_n)| \le (C_S \sigma^{-a})^n (n!)^{1+\eta} \prod_i N_U(f_i),$$

uniformly in the bump width $h \in [\varepsilon, \ell/512]$ and in the position of the bumps. Every factor of $C^n(n!)^{1+\eta}$ is independent of h. In the bridge $D/\sigma = 3/8$, so the cube-count factor is $(2.75)^4 \approx 57$.

Proof (sketch). We trace one rescaled bump through L-111 (Lemma C) of Proposition 5.17. There is one h-dependent term: the cube count $Q_m \le (2 + D/(M\lambda_m))^4$ at scales finer than the bump. With $D \le \sigma$ this count is at most $(2 + 2D/\sigma)^4 u^4$, $u = \sigma/(2M\lambda_m)$. The tree decay $e^{-3\kappa u/8}$ of the polymers that meet all supports cancels it. The norm N_U and the flatness of the test function pay the gradient cost. □

Hypotheses. S1 at all scales is part of (B-S) (ledger R-035). (FX) is reduced in Proposition 5.22; its remaining input is N3 (ledger R-042). Without the coarsest-scale (FX2) radius the order is $2+\eta$ and $D > 2$. The bound of `SOURCES_D2.md` §4.3 is reduced, not proved. Section 5.12 reduces the (E14-c) (H-098) clause of R-045 to R-045 (S).

The hypothesis (FX) asks for convergence of the final expansion over large-field geometries.

Proposition 5.22 (Reduction of (FX))**.** *Let (FX) be the convergence of the final Mayer expansion of the sum over large-field geometries plus the unit-scale integral, with polymer radius $c \cdot r$ in the source (FX2), uniform in K and ε. Assume Bałaban's large-field bounds CMP 122 II (1.79)–(1.89) and (1.97), the convergence criterion of CMP 116 (2.38)–(2.41) with Cammarota's theorem (ledger X-004), the tree decay Q-059, the gaps Q-060 and Q-061, and the bound R2 ($\kappa_{\mathrm{fin}} \geq \kappa_{\mathrm{final}}$). Use the main form (1.98) of the R step (choice K-014). Then (FX) holds. The fallback order is at most $1 + \eta + 2/3$.*

Proof (sketch). The construction is a final R-step with $Z'' = \emptyset$, followed by the unit-scale cluster expansion of CMP 116 §2. Dimock uses the same scheme for φ^4_3 (Dimock III, Theorem 2). We read CMP 122 I (0.2)–(0.6) and CMP 122 II (1.71)–(1.104). Bałaban already exponentiates the sum over renormalizable large-field components inside each R step. Only the components that pay for later steps stay as a sum, and at the last scale there are no later steps. The per-step expansion sums only components of diameter $\leq 100 M R_k$, counted by position ((1.96)). The last step sums components of unbounded size, so it needs the tree decay Q-059. The bound (1.79) controls the fine-history sum inside a component, because Dimock's power-decay argument ((209)–(211)) does not transfer to $d = 4$. The source enters the large-field operations only through the upper bound (1.73). The page data used are: (1.80) has R_n^{d+1} and a local K; (1.85) has $A_1^2 p_0^2$; the condition after (1.94) ends in $-\frac{3}{2} p_0(g_k)$. □

Hypotheses. Q-059(b) is Proposition 5.23. Q-060 and Q-061 are Proposition 5.25; Q-061 holds relative to N3. R2 is Proposition 5.28. The main form K-014 avoids Q-062. So the remaining input of R-043 is N3 (ledger R-042), through Q-061.

The gap Q-059(b) is a geometric comparison of tree lengths across one scale.

Proposition 5.23 (Lemma P: tree length under one blocking step)**.** *For a connected union Y of MR_j-cubes, $d'_{j+1}(S(Y)) \geq c_* d'_j(Y)$ with $c_* = [3 \cdot 4^d \sqrt{d} (\rho + 2)^d]^{-1}$, $\rho = L R_{j+1}/R_j$, and no additive constant. With the $n = K + 1$ term of CMP 122 II (1.80) this gives $\kappa_K(Y) \geq \kappa_{\mathrm{fin}} d_K(Y)$ for every non-renormalizable final component Y. This is Bałaban's assertion of the term $-\kappa_1 d_k(X)$ on CMP 122 II p. 387, with $\kappa_1 = \kappa_{\mathrm{fin}}$.*

Proof (sketch). The proof combines three elementary facts. A tree of length ℓ meets at most $4^d(1 + 2\ell)$ closed unit cubes. Next, $d'_j(Y) \leq \sqrt{d}(n_f - 1)$. A coarse cube meets at most $(\rho + 2)^d$ fine cubes. With these facts we count at least 21^d coarse cubes in $S(Y)$. The halo $S(Y) \setminus Y$ carries no large fields (p. 384), so p_0 absorbs nothing. We use only the first term of (1.80); all its terms are non-negative. The comparison R3 is Bałaban's own step (1.86)→(1.87). □

Remark 5.24. A bound of the form "factor $1/L$ plus a constant" fails in general. The counterexample needs $d \geq 2$, a constant independent of L, and L large.

Hypotheses. The comparison R2 ($\kappa_{\mathrm{fin}} \geq \kappa_{\mathrm{final}}$) is Proposition 5.28 (choice K-015). The grid alignment R1 changes constants only.

The gaps Q-060 and Q-061 concern the R-denominator and the sourced weight inside the large-field operations.

Proposition 5.25 (Gaps Q-060 and Q-061 of (FX))**.** *(Q-060.) The R-denominator of CMP 122 I (1.100) is one small-field integral per renormalizable component, with the bare Wilson action $-g_k^{-2} A(\zeta_i, \cdot)$ and no E-, R-, or B-terms. It is holomorphic and non-vanishing on a source polydisc of radius $c \cdot r$, with c independent of $|\Lambda_i|$, k, and ε. (Q-061.) Assume N3 (ledger R-042), that is the strip bound $|\delta w| \leq \frac{3}{2}\tau$. Inside the large-field operations T' the sourced weight obeys $\mathrm{Re}(w + \delta w) \geq (1 - \theta_W)(g''_k)^{-2}$. So the upper bounds (1.73), (1.79), (1.89), (1.92), (1.97), (1.100) hold with*

$\gamma_0 \to \gamma_0(1-\theta_W)(1-\tau_C) \geq \gamma_0/4$ *and* $p_0 \geq 16/(\gamma_0 A_1^2)$. *The cleaner condition* $p_0 \geq 8/(\gamma_0 A_1^2)$ *gives* $\gamma_0' \geq \frac{3}{4}\gamma_0$.

Proof (sketch). For Q-060 the Wilson part of the source perturbation obeys $|\sigma_W| \leq \tau\Phi(g_k)$ with $\Phi(g) = C(M)g^2 p_0(g)^2 R(g)^4 \to 0$. The form (1.102) also allows an f-independent denominator. For Q-061 the plaquette terms $P_p \geq 0$ of the real G-valued configuration inside T' (CMP 122 II p. 379) give domination in modulus by a positive operation with a smaller coupling prefactor. The thresholds stay fixed. □

Remark 5.26. In CMP 122 I (1.103) Bałaban divides by $T_k'(Y_i)1$ and gives no proof that this quantity is positive (gap Q-062). The step (1.103)–(1.104) is an optional final remark. We keep the main form (1.98) (choice K-014), so Q-062 does not enter. If (1.104) were used, the fallback order would be $2+\eta$.

The comparison R2 is needed in volume form.

Convention 5.27 (readings of CMP 122 II for R2)**.** We read CMP 122 II so that $R_{K+1} \geq 1$ and the constant C_c of (1.80) does not depend on g (the page reading R-049, Section 5.12).

Proposition 5.28 (Gap R2 of (FX): $\kappa_{\text{fin}} \geq \kappa_{\text{final}}$)**.** *Assume Convention 5.27, and let* $C_1 := \max(1, C_c)$ *be the enlarged constant of (1.80). Then* $\kappa_K(Y) \geq \kappa_{\text{vol}} \# M(Y)$ *with* $\kappa_{\text{vol}} = \sqrt{d}\,\kappa_{\text{fin}} = C_1 M^d R_{K+1}/(3 \cdot 4^d (L+2r)^d)$, $r = R_K/R_{K+1}$. *Moreover* $\kappa_{\text{vol}} \geq \kappa_{\text{final}} := 2\kappa + \sqrt{d}(1+2\beta)\kappa + c_d$ *once* $M \geq 12L(3\kappa_{\text{final}}/C_1)^{1/d}$, *uniformly in* K, R, *and* g.

Proof (sketch). Bałaban's order of constants is: L first, $\kappa \geq \kappa_0$ before $M \geq M(\kappa)$, p_0 and γ last. So $\kappa_{\text{final}} = \kappa_{\text{final}}(d, \kappa, \beta)$ is fixed before M (route (a)), after C_1 is enlarged (route (b)). The final expansion charges large-field domains per M-cube, and the tree rate of Proposition 5.23 alone loses R_K^{d-1}. So we use the volume rate. The term 2κ is the (1.93) decay-recovery charge $(1+\beta+\beta/(3\cdot 2^d))\kappa$ per M-cube. The term $\sqrt{d}(1+2\beta)\kappa$ is a surplus. □

Hypotheses. The value κ_{final} is the choice K-015; it is not sharp.

5.4 Steps 4–5: N3, the holomorphy bound (H-hol), and the large-field inputs

Take the source additive in $u = g^{-2}$: $w_0 = g_0^{-2} - f$, with τ in absolute u-units. Then N3 is Lemma H (L-182) of Proposition 5.18 for complex c on $Q = \{|\operatorname{Re} c| \leq 2r,\ |\operatorname{Im} c| \leq \tau\}$, pointwise in x.

Proposition 5.29 (Lemma N3 (L-232), the strip bound)**.** *Assume (I) (ledger row R-036); (E); the real (H-hist) (ledger row R-037); (H-hol) (ledger rows R-040, R-041) on relative polydiscs; (B-hol) (H-078); and, in the history form, the complex history bound (H-hist*$_C$*) on the complex polydisc, for example through (H2-hist) (H-120). Assume condition (a),* $3r + \frac{3}{2}\tau < g^{-2} - \gamma^{-2}$, *with* $g = g_K$. *Then* $|\delta w_k| \leq \frac{3}{2}\tau$ *for all* $k \leq K$, *uniformly in* K *and* ε. *The constant* $\frac{3}{2}$ *can be any* $1+\eta$.

Proof (sketch). Step 0 bounds $\partial\beta_j/\partial u$ on the thin strip. It uses Bałaban's real C^1 bound at $\operatorname{Re} u$ plus a Cauchy correction of order $g^4|\operatorname{Im} u|$. Here $d\beta/du = O(g^3)$, or $O(g^4)$ for even β. A complex Gronwall step with $\sum_i (\operatorname{Re} u_i)^{-3/2} \leq 4\gamma/\beta$ closes a bootstrap in $\operatorname{Im} c$. Cauchy estimates on relative discs alone give a factor $1 + Cg^2$ per step and a bound that grows with K. So the real derivative bound is necessary for this argument. □

Hypotheses. Condition (a) limits r by the gap between $g = g_K$ and γ; it does not follow from γ small. Proposition 5.30 proves it. (I) is open (ledger R-036). (E) is the uniform positivity of the one-loop $\beta_j(0)$. The real (H-hist) is reduced in Section 5.6. (H-hol) is reduced in Propositions 5.31

and 5.32. (H2-hist) or its Route 2(a) is ledger R-038 (Proposition 5.48); it is sufficient, not necessary. The booking $h \equiv 0$ is K-012, and the additive normalization is K-013. Step 3 in the DT form is treated in Section 5.11 (ledger R-034).

Proposition 5.30 (Condition (a) of Lemma N3)**.** *Fix the order of constants $L \to$ Bałaban's constants $\to \gamma \to g \to (r_0, \tau_0) \to K$ (choice K-005), and let $g = g_K \le \min(\gamma/2, g_1)$. Then condition (a) holds at the uniform radius*

$$r_0 = \tau_0 = \min\bigl(1/8,\ r_0^{\mathrm{D2}},\ r_0^{\mathrm{D3}},\ r_0^{\mathrm{FX}},\ \gamma^{-2}/10,\ 1/(6\gamma^2 c_1 c_H)\bigr),$$

since $\frac{9}{2} r_0 \le \frac{9}{16} < 3 \le g^{-2} - \gamma^{-2}$ for $\gamma \le 1$, a margin of 5.33*. The radius does not depend on K. In u-units it is the physical sup-norm radius of the source.*

Proof (sketch). The choice $g \le \gamma/2$ is admissible for every K: see Step 8 of `LEMMA_DPRIME.md` in the L-007 (Lemma E) form, and `KAPPA_BOUNDED.md` §5.3(b) in the DT form. Nothing forces g_K near γ. In the L-007 (Lemma E) form u_k decreases and g_K is the largest coupling. In the DT form only $u_k \ge u_K - D'$ holds, and g_0 can exceed g_K; Section 5.11 treats this form. The radius $r = \tau = \gamma^{-2}/2$ satisfies (a) but violates conditions (b)–(d) of Lemma N3 and the radius bounds of Propositions 5.15 and 5.16. □

The small-field half of (H-hol) is a holomorphy statement for β_j on relative polydiscs.

Proposition 5.31 ((H-hol), small field)**.** *Assume Bałaban's conditions with modified constants: the conditions (C1-m)–(C4-m) with $\varepsilon_1 \to 1.235 c\varepsilon_1$, the radius condition (C-rad) (H-185) $16 n \rho_0^2 (LM)^4 \le 1$, and (2.24-m); here (C1)–(C5) are our labels for conditions of CMP 116, not Bałaban's printed labels. Assume (F1) at complex couplings, a γ-independent constant C_1 in the real (H-hist) (ledger R-037), and the large-field half (ledger R-041). Then for the small-field chain β_j extends holomorphically and single-valued to the polydisc $|g_i - x_i| < \rho_0 x_i$, with $\rho_0 = \min(0.1, (LM)^{-2}(8n)^{-1/2})/\sqrt{2}$ (ledger K-006). The bound $|\beta_j| \le M$, uniform in j, k, and volume, holds relative to a majorant of the non-Gaussian part.*

Proof (sketch). In reference units $\tilde{B} = xB$ the complex coupling enters only through $w = (x/g)^2$. It enters as the scale of the Gaussian and as the coefficient of the u-prefactored vertices. The cut $|\tilde{B}| < \varepsilon_1$ does not depend on g. A marginal-domination lemma (Lemma D) bounds the complex Gaussian by the real one at $\operatorname{Re} w$, at a cost $(|w|/\operatorname{Re} w)^{1/2} \le e^{2\rho^2}$ per real mode. The count of real modes includes the X-modes, which gives the factor $\sqrt{2}$ in ρ_0. So the volume-uniform bound is made per polymer inside CMP 116 (2.14)–(2.17). Lemma N3 accepts any fixed ρ_0; a radius $\rho \le 1/3$ is not reached and not needed. □

Hypotheses. The X-mode count is settled in the row M-C of the dependency table. (F1) is reduced to Theorem SM (L-281) at complex couplings; from it (H2-hist) follows with $a_m = \lambda_C^{m/2}$. The large-field half is Proposition 5.32.

The large-field half freezes the thresholds at Bałaban's real trajectory x^*.

Proposition 5.32 ((H-hol), large field)**.** *Freeze the R-ratio density, the thresholds, R_k, and N at Bałaban's real trajectory x^* (ledger K-001). Assume (M-A) (H-143), (M-B) (H-145), (M-C) (H-146), and (M-D) (H-147). Then the large-field half of (H-hol) holds on $P_{\rho_0}(x^*)$. The large-field integrals are entire in $u = g^{-2}$ in finite volume. They are dominated in modulus with $\gamma_0 \to \gamma_0'' \ge 0.82\gamma_0$ at $\rho_0 = 0.1$, and F2 holds on the polydisc with $32\varepsilon' \to 40\varepsilon'$.*

Proof (sketch). The radius $(100MR_k)^{-4}$ of W4 is a radius in the input activities. The real (1.100) denominator does not contain the input activities. We remove the coupling from the denominator by freezing the ratio at x^*. The mismatch costs $O(1)$ per unit cube, inside Bałaban's budget $p_0(g) \gtrsim R(g)^{d+2}$. Here R_k is unbounded, of order $(\log K)^r$, but it is largest at $k = 0$ and non-increasing in k. The constant is $\theta = 0.174$. F2 compares the bound sizes $(x_j^*)^{\kappa_0}$, which are ordered on the real trajectory, so no ordering of complex couplings is needed. A slot-dependent radius $\rho_j \propto x_j^2$ would violate the hypotheses of Lemma N3. □

Hypotheses. (M-A) is reduced in Proposition 5.33 to (H−) and CMP 102 (189) (ledger R-005, with R-018 and R-019); its bookkeeping item (M-A2) is Proposition 5.36. (M-B) is Proposition 5.37, with $p_0 \geq 17/(\gamma_0 A_1^2)$ and $\theta_W + \tau_C \leq 0.2$. (M-C) is Proposition 5.38, with the items (I-ε), (I-V0) (H-133), and K-025 (M1) treated in Propositions 5.40–5.44. (M-D) is not a convention: the frozen map equals Bałaban's map only at x^*, and its derivative there is (H-hist) plus (H-thr) (ledger row R-039). So (M-D) needs the real (H-hist) (R-037), (H-thr) (R-039), and (H-hist2) (ledger row R-038).

Proposition 5.33 ((M-A), the equipartition bound)**.** *Assume the Hessian lower bound (H−) along segments, with c_H uniform in k, volume, and background. Consider the cut-off ratio density of CMP 122 I (1.100) at real weight $0 < u \leq x^{-2}$. Let $n_i = (N^2-1)L_i$ count the real coordinates on the free links, and $\eta_x = 0.68M_0^2x^2p_0(x)^2 + 2c_HM_0^2x\,p_0(x)^3$. Then $\langle A - A_{\min}\rangle_u \leq (n_i/u)(1+\eta_x)$, with constant 1; this constant is sharp when the minimizer lies on the boundary. As a corollary the freeze-mismatch average is at most $e^{0.1906\,n_i(1+\eta_x)}$, that is $e^{0.77(N^2-1)(1+\eta_x)}$ per scale-k unit cube ($e^{2.3}$ for SU(2)), additive in the volume.*

Proof (sketch). A virial identity on the convex cutoff ball, centred at the true minimizer X^* of the exponent, bounds $u\langle (X-X^*)\cdot\nabla G\rangle_u$, $G = A - A_{\min}$, by n_i plus a Haar-Jacobian term. Lemma J (L-201) bounds this term by $0.68N(N-1)r^2$ per link for $r \leq \frac{1}{2}$. A segment bound from (H−) (L-268 (Lemma S)) turns this into the bound on $\langle G\rangle_u$. Integration of $\frac{d}{du}\log D^G = -\langle G\rangle_u$ gives the corollary. □

Remark 5.34. The same bound with the full A in place of $A - A_{\min}$ is false in general, because the background minimum $A_{\min}$ is $O(\varepsilon^2)$ per block.

Hypotheses. Proposition 5.36 reduces (H−) to CMP 102 (189) (ledger R-005), together with R-018, R-019, and its conditions (C1), (C2). The bound feeds ledger R-041.

Convention 5.35 (pairing of $e^{uA_{\min}}$)**.** We read Bałaban's chain as removing the factor $e^{uA_{\min}}$ by algebra. Since $A_{\min}$ may be only piecewise analytic in the background, we keep $e^{uA_{\min}}$ paired with an analytic factor or inside modulus bounds. The competing reading treats $e^{uA_{\min}}$ as an analytic factor by itself.

Proposition 5.36 ((M-A2) and (H−))**.** *(i) Every normalized ratio of CMP 122 I (1.100) sees only $G = F - A_{\min}$. The identity $N_x/D_x = e^{-uF}\,[e^{uA_{\min}}/D^G(x^{-2})]\,e^{(u-x^{-2})G}$ holds exactly and keeps $e^{uA_{\min}}$ at the coupling u. So the mismatch costs at most $e^{1.07(N^2-1)(1+\eta_x)}$ per unit cube. This is (M-A2). (ii) Assume (C1) $M_0\varepsilon_k \leq C_1\varepsilon_1$; (C2) the localized map U_{k,X_i} obeys the CMP 102 bounds; (C3) CMP 102 (189); ledger R-018 (including the k-uniformity of C_2, C_4, B_0) and R-019; and Convention 5.35. Then (H−) holds in the form $D^2F[h,h] \geq -c_H\varepsilon_k\sum_b \|h_b\|_2^2$, with c_H uniform in k, volume, and regular backgrounds; c_H depends on M^5 and M_0.*

Proof (sketch). Since $A_{\min}$ does not depend on V', $D^A(s) = e^{-sA_{\min}}D^G(s)$. The ratio must pair a numerator at u with a normalizer at u, not at x. For (H−), the first Hessian term is ≥ 0. The

chain rule per plaquette is combined with one Cauchy step on $\delta\mathcal{H}/\delta B$ minus its linear part, on the complex ball of CMP 102 (172). The $1/\varepsilon$ of the radius cancels the ε of CMP 102 (187). The kernel decay of (189) closes a Schur test, with $c_N = O(1)(B_0C_4 + C_2)B_5$. □

Hypotheses. Bałaban states CMP 102 (189) without proof (CMP 102 p. 308: "we do not perform these calculations here"). He uses it in his own chain. Ledger R-005 records its reduction to R-001. Section 5.11 treats (C1) and (C2).

Proposition 5.37 ((M-B), the large-field bookkeeping)**.** *Assume (M-A) (the freeze-mismatch cost C_m), (M-C) (the complex-u activities α^c and E-terms), and the link from F2 to the activities α, which we take as a hypothesis. Freeze the thresholds at the real trajectory x^*. Then Bałaban's large-field bookkeeping CMP 122 II (1.73)–(1.99) holds under the complex-coupling perturbation. The per-cube factors* $\exp(-p_0(x_j^*))$ *and the per-component factors* $\exp(-2p_0(x_j^*))$ *keep the same ε_j, δ_j, δ'_j, R_j. With Bałaban's output factor kept (option B), the only new condition is $p_0 \geq 17/(\gamma_0 A_1^2)$, under the margin $\theta_W + \tau_C \leq 0.2$ (ledger K-002). The cost constant becomes $C'_c = C_c + C_m + C_G + C_E$, a fixed $O(1)$ per site inside the $O(1)\log g_j^{-2}$ budget. The condition $p_0 \geq 8\kappa(100R_k)^d + 4$ of p. 389 is unchanged.*

Proof (sketch). We list each line of (1.73)–(1.99) that carries γ_0, p_0, ε_j, δ_j, A_1, or a per-cube factor (L1–L24). Each line is redone with $\gamma_0 \to g^*\gamma_0$. This is legitimate: each factor bounds a positive Wilson action or quadratic form, and its modulus at complex u is dominated with prefactor $\operatorname{Re} u \geq 0.8264\,(x^*)^{-2}$. The perturbation weights are additive, so $g^* = 0.82645 - \theta_W - \tau_C$. The combined factor of p. 381 then needs $p_0 \geq c/(\gamma_0 A_1^2)$ with $c = 2/(g^* - \frac{1}{2})$. The constant 17 needs $\theta_W + \tau_C \leq 0.2$; the strict limit is 0.2088, and at $\frac{1}{4}$ one needs $c = 26.2$. Page 383 prints $\frac{1}{2}\gamma_0$ and N^{β_0}, and any fixed $g^* > 0$ is accepted there. The budget constant $C'_1 = \max(1, C'_c)$ (route (b) of Proposition 5.28) multiplies (1.82), (1.88) by a fixed factor. Large p_0 and small γ absorb it. □

Hypotheses. (M-A) is Proposition 5.33, and (M-C) is Proposition 5.38. The block feeds ledger R-041.

Proposition 5.38 (Lemma MC (L-223), the complex small-field step)**.** *Assume that Bałaban's constants satisfy the conditions B1–B23 of CMP 116 §2 and four further printed conditions, with Lemma 2, (1.36), and (2.5)–(2.6) of p. 12. Use the shrinks $\varepsilon'_1 = \varepsilon_1/(4\lambda_V)$, $\gamma' = \gamma/(8\lambda_V)$, $\alpha' = \alpha/4$, $\lambda_V = \max(c, 1.235)$, the radius $\rho_0 = \min\{0.1, (LM)^{-2}(0.45a/(16n))^{1/2}\}$, and the condition K-025 (M1). Assume (I-ε): the smaller ε_1, α_0, α_1 keep CMP 116 §1 and the Part I induction valid. Then the small-field step is holomorphic in complex g with $|g/x - 1| < \rho_0$, for every real $x \leq \gamma'$ and inputs in the ball cE_0. The total output is $\leq E_0$, and Lemma 3 (2.38) holds with Bałaban's $C_3\varepsilon_1$. Hence β_j is holomorphic on these discs and bounded uniformly in j, k, and volume.*

Proof (sketch). We read the 23 inequalities of CMP 116 §2 on the page images. Each one, at complex $w = (x/g)^2$ and primed constants, follows from Bałaban's own. The vertex is affine in w with real-domain coefficients. So the τ-circle (2.18) absorbs its factor λ_V, with $E_0 \to \lambda_V E_0$ and a smaller ε_1. The cut stays real. The Gaussian cost $16n\rho^2$ per site and the α_5 items fit into the budget $(LM)^4\alpha_i \leq \frac{1}{4}$ under K-025 (M1) and (M2). By p. 12 the X-measure is the unit Gaussian, which settles the item (I-X) (H-134). The vertex part (I-V0) of the bracket is Proposition 5.40. □

Hypotheses. (I-ε) and the order (I-ord) (H-131) are Propositions 5.41 and 5.43. K-025 (M1) is a choice of M (Proposition 5.44). The block feeds ledger R-041.

Convention 5.39 (the measure $dB'\sigma(B')$ of CMP 116 pp. 9–10)**.** We read the integral over $dB'\sigma(B')$ in CMP 116 pp. 9–10 as a Haar integral. The page prints only $dB'\sigma(B')$.

Proposition 5.40 ((I-V0), the vertex part of (M-C))**.** *Assume Convention 5.39. The w-dependence of the bracket $\{\dots\}$ in CMP 116 pp. 9–10 is only the E_k difference. It is linear in w, and the rest of the bracket is free of w. The terms* $\operatorname{Tr}\log$ *and* $\log\sigma$ *sit in $P^{(k)}$ with prefactor 1, and* $\log\sigma$ *and the Jacobian do not depend on w. The five $1/g_k^2$ terms each have the form $u\cdot f(C\tilde{B})$, so they are affine in w. The shift restores (2.19), (2.20), and (2.26) exactly, up to the ε_1^2 cut factor of (2.26), which belongs to (I-ε).*

Proof (sketch). We read CMP 116 pp. 9–11 and 16–17 on 300 dpi page images. We check each w-dependent term against the printed display. □

Hypotheses. The item (I-lin) (H-130) is a mild condition (balls per piece).

Proposition 5.41 ((I-ε) and (I-ord), small-field layer)**.** *Every condition on ε_1, α_0, α_1 in CMP 116 §1–§2 and CMP 109 §1, §3 is an upper bound, a ratio of degree one, or a lower bound relative to a later constant. None is two-sided. So the shrinks of Lemma MC (Proposition 5.38) keep CMP 109 and CMP 116 valid. The ε_1^2 cut of (2.26) is restored exactly by $\gamma' = \gamma/(8\lambda_V)$. Scaling (A_0, A_1) by $1/\theta$ restores CMP 122 II pp. 380–381 exactly. (I-ord): the order*

$$d, L \to \delta \to \kappa \to \kappa_1 \to \textit{walk constants} \to M \to \beta, \alpha' \to \alpha_1 \to \alpha_0 \to \alpha_3 \to \alpha_2 \to E_0 \to \varepsilon_1 \to \rho_0 \to A_1 \to A_0, p_0 \to \gamma$$

has no cycle.

Proof (sketch). We list the conditions page by page and classify each one; five conditions beyond the first list are also of an allowed type. In CMP 119 (2.28), $\alpha_{i,j} = g_j C_i (\log g_j^{-2})^{q_i}$ with C_0, C_1 "sufficiently large". A smaller γ alone does not restore (L1) in the large-field layer. Proposition 5.43 treats that layer. □

Convention 5.42 (unprinted constants of CMP 119/122)**.** (a) The constants α_2–α_4 and γ_2 are not printed in CMP 119/122. We fix their scaling by one of three branches; the proof of Proposition 5.43 holds on each branch. (b) CMP 119 does not print whether $q_0 > p_0$ or $q_0 = p_0$. Both readings give the result.

Proposition 5.43 ((I-ε), large-field layer)**.** *Assume Convention 5.42. The shrink of Lemma MC reaches $\alpha_{i,j}$ (CMP 119 p. 262 takes the restrictions from CMP 109). Every printed condition on C_i, q_i is a lower bound, a ratio ($q_0 \le q_1$), or an upper bound on g; none bounds C_i from above. Lemma MC needs only a factor-4 slack, so C_i and q_i stay fixed and γ is lowered. The order continues $\rho_0 \to$ exponents $\to A_1 \to A_0 \to C_0 \to C_1 \to \gamma$, with no cycle.*

Proof (sketch). The condition C3 is a printed inclusion with an unprinted constant $c(B_3)$. The inequality $q_0 \ge p_0$ is forced. The p. 381 factor is $\exp(-\gamma_0 A_1^2 p_1^2(g_j))$, with no extra weight. One inequality on ν reads $\frac{1}{2}\nu \le p_0 - p_1 - 1$. The condition $\gamma' < \min(e^{-q_0}, e^{-q_1})$ is a further upper bound on γ and is harmless. □

Proposition 5.44 (Higher derivatives, and the condition K-025 (M1))**.** *(A) In the coupling and threshold chain, with the variables restricted to the couplings, the thresholds, and the MA2 fields, no consumer needs derivatives of order greater than 2. Bałaban's field expansion goes to order 4, and the source derivatives of D2/D3 go to all orders; these are outside the claim. (B) The condition K-025 (M1), "M large", is a block-size choice. There are 26 lower bounds on M, for example $M \gg \kappa_1$, $M \ge M(\kappa)$, $M \ge cL$, $M \ge 48\kappa_1/\varepsilon$, $M/M_1 > 3(6 + c_X)$, together with the bounds of* `SOURCES_A3_SIGMA.md` *5.1 and 5.3. There is no upper bound, and the order of constants puts M before α. So K-025 (M1) is the convention K-025: M is at least the largest of these lower bounds.*

Proof (sketch). We list every consumer of derivatives in the chain and every condition on M in the pages read. □

5.5 Step 6: thresholds and second history derivatives

Let $\Delta_j(c) = B_j(u(c); \bar{u}) - B_j(u(c); u(c))$ be the threshold mismatch, with the integer parameters frozen (convention L3).

Proposition 5.45 (Reduction of (H-thr), and Lemma HG (L-191))**.** *Assume the density form M2b of the block bound, the extraction cost M4, the history holomorphy M5, the history weights M3, and the frozen-integer convention L3 (ledgers R-039, K-001). Then*

$$\sup_c |\Delta_j'(c)| \le 3(1+3r)A\,\Phi(g_j^{-2}).$$

The sum of this bound along the trajectory is at most $C\beta^{-1}\gamma^{-2(1+a)}(\log\gamma^{-2})^P \exp(-(\log\gamma^{-2})^q)$*, with* $q = 2p_1 - (d+5)r_0 > p_0$*, uniformly in* K*. Lemma HG (L-191) then splits the budget:* $x \le \ln(6/5)$ *for (H-hist) (ledger row R-037) and* $x_{\mathrm{thr}} \le \frac{1}{4}$ *give* $\sum_j |B_j'| \le \frac{1}{2}$ *and* $\frac{1}{2} \le |v_k| \le \frac{3}{2}$*.*

Proof (sketch). The coarea formula writes Δ_j as the contribution of the shell between the two thresholds. The indicator χ stays inside the integral, so $\Delta_j \neq 0$ in general. This reduces Δ_j' to threshold derivatives and mixed threshold–weight derivatives of B_j. L-267 (Lemma S) bounds them. The fluctuation cutoff is $\delta_j' = g_j A_1 p_1(g_j)$ with $p_1 < p_0$. Bałaban bounds one bond only through the block quadratic form CMP 122 II (1.78), with loss $R_j^{-(d+5)}$. So a per-link density bound is not available, and we use the block form M2b. The mixed derivative needs the history holomorphy M5. The sum along the trajectory uses $u_i \ge \gamma^{-2} + \beta m$ and an integral comparison that keeps the factor γ^{-2}/β. Lemma HG is an additive Gronwall argument. □

Hypotheses. M2b, M3, M4, and M5 are not proved here. Proposition 5.47 replaces M2b by L-285 (Lemma T) and M5 by (H-hist2) (ledger row R-038). Section 5.12 gives a route to (H-thr) that does not use R-038.

Remark 5.46. CMP 122 II (1.77)–(1.78) prints p_1 where p_1^2 is meant. With p_1^2 the tail factor reads as in Proposition 5.47.

Proposition 5.47 (L-285 (Lemma T), and the summed bound for (H-thr))**.** *(i) (L-285 (Lemma T).) The shell indicator is at most the large-field function* $1-\chi$ *at the lower threshold. So Bałaban's printed tail factor of CMP 122 II (1.77)–(1.78), with* p_1^2*, bounds* Δ_j *itself with exponent at least* $\kappa_2(\log s)^q$*,* $\kappa_2 = \frac{1}{2}L^{-(d+5)}$*,* $q = 2p_1 - (d+5)r_0 > p_0$*. (ii) Assume (H-hist2) (ledger R-038), the history weights M3, the extraction cost M4 (with the units and the small-field analogue of CMP 122 II (1.73)), the transfer of Bałaban's printed large-field factor to the small-field shell terms, and the frozen-integer convention L3. Let* D *be a uniform bound on* Δ_j''*. Then*

$$\sup_c |\Delta_j'| \le 2(2AD)^{1/2}\Phi_T(u_{j-1})^{1/2} + (2A/r)\Phi_T(u_{j-1}).$$

The sum over j *is uniform in* K *under (HT1) (H-195)–H-198 (HT3), and Lemma HG of Proposition 5.45 applies unchanged.*

Proof (sketch). The zero-order bound on Δ_j uses L-285 (Lemma T), the history weights M3, and the extraction cost M4. M4 is a Cauchy estimate on the analyticity radius $\alpha_{1,j} = g_j C_1(\log g_j^{-2})^{q_1}$ of CMP 119 (2.28). So no history derivative enters. The derivative then comes from the interpolation inequality $\|\Delta_j'\| \le 2(\|\Delta_j\|\,\|\Delta_j''\|)^{1/2} + \|\Delta_j\|/r$ (Lemma 7.1(a) of `SOURCES_D2.md`). This replaces the history holomorphy M5 by the uniform second-derivative bound (H-hist2). □

Hypotheses. For β_j with running thresholds, (H-hist2) contains the mixed threshold–history derivative. So it is a new hypothesis of the (H-hist) kind; a Cauchy estimate in the history does not give it. For the frozen-threshold map, Proposition 5.48 reduces (H-hist2). The block is ledger R-039.

Proposition 5.48 ((H-hist2), second history derivatives)**.** *Assume Theorem SM (L-281) and the joint holomorphy in the relative u-disc and the input ball, and the bounds of the large-field part of the frozen map. Let $F = B_j(\cdot;\bar u)$ be the frozen-threshold map on $|w-\bar u|\le 3r$ in u-slots. Then*

$$\sum_{i,l} |\partial_{u_i}\partial_{u_l} F| \le \varepsilon_j'' = C_H(1-\lambda^{1/2})^{-2} u_{j-1}^{-2},$$

so $\sum_j \varepsilon_j'' \le C_H(1-\lambda^{1/2})^{-2}\gamma^2/\beta$, uniformly in j, K, and volume, with product decay $\lambda^{m/2}$. If in addition the real uniform (H-hist) holds, the first-derivative half is $\varepsilon_j' \le \frac{1}{2}C_1(1-\lambda)^{-1}u_{j-1}^{-3/2}$.

Proof (sketch). Route 2(a) differentiates the real chain twice. The past slots stay real. The input variable is handled by Cauchy estimates (Theorem SM (L-281)). Only the own slot is complexified, on one relative u-disc (`SOURCES_SM.md` §4.2). The Jacobians $\partial_{u_i}\partial_{u_l} = \frac{1}{4}g_i^3 g_l^3 \partial_{g_i}\partial_{g_l}$ turn the $1/x$ losses into g^4. This is summable because $u_{j-1} \ge \gamma^{-2} + \beta m$. The rate $\varepsilon'' \propto u^{-2}$ follows from the u-disc alone. The product decay is $\lambda^{m/2}$; the decay λ^m fails already for a toy recursion. □

Hypotheses. For β_j with running thresholds the (H-hist2) step needs only a uniform bound. Lemma HG still needs a summable first derivative of β_j, from the real (H-hist). For β_j the mixed threshold–history derivative through L-133 (Lemma D) and M3 is open (ledger R-038). The real uniform (H-hist) is ledger R-037; its open inputs are CMP 102 (189) (ledger R-005), the coarea Jacobian, and the link from F2 to the activities α.

Proposition 5.49 (Theorem SM, the step-map Lipschitz bound)**.** *Assume Fact A (L-050) for the lower-order bracket terms (ledger R-010, in the form R-011 of Theorem 5.82); Fact B (L-082), that is (W4) (ledger row R-044) on CMP 116 (C1)–(C5), with α_4, α_6 fixed before ε_1; (I) (ledger row R-036) with L3; and (G-gain) (H-111) in both the hop and the own-slot bound. The small-field step map factors as $G = \hat G \circ \mathcal{L}$ with $\mathcal{L}$ linear. Let $\theta_0 = 2K_H K_V \varepsilon_1$, and let the per-label exponent be $\alpha_* = \beta_H$, a free parameter (choice K-028). Then*

$$|DG(I_0)[W]| \le \theta_0 \sum_{l\le k} L^{-\alpha_*(k-l)} \|W^{(l)}\|\, e^{-\kappa' d_{k+1}(X)}.$$

With it L-190 (Lemma HH) holds at rate $\lambda = L^{-\alpha_*} + C_V\theta_0$ when $C_V\theta_0 < 1 - L^{-\alpha_*}$. The bound (Eps) of Proposition 5.16 is a corollary, with $\varepsilon' = \theta_0(1-L^{-\beta_H})^{-1}$.

Proof (sketch). The per-input bound CMP 116 (1.24) gives the linear stage a fluctuation factor $\alpha_2^{-1}\varepsilon_1$ and a label weight $L^{-(k-l)}$ (Fact A). The map $\hat G$ is holomorphic on a ball of radius $R_V = \alpha_4\alpha_6 e^{-5\kappa}/(4(L+2)^4 O(1))$, free of ε_1 and E_0, by CMP 116 (C1)–(C5) (Fact B). A Cauchy estimate on this ball gives the contraction. The own-slot constant C_β has the same $1/g$ gap as S_0, so (G-gain) is applied in both slots. □

Hypotheses. Theorem 5.82 proves Fact A in the form R-011 relative to its inputs; the per-label decay of the lower-order terms enters only through it. (G-gain) is given by Propositions 5.51 and 5.53 (ledger R-037).

5.6 Step 7: history bounds and the gain

Proposition 5.50 (L-189 (Lemma HH), (H-hist) in column-sum form)**.** *Assume CMP 109 (0.29); uniform own-slot bounds* $\|\partial_{g_{l-1}} V^{(l)}\|_l \le S_0$ *and* $|\partial\beta_l/\partial g_{l-1}| \le C_\beta$*; and a step-map Lipschitz bound (SM) with* $\theta < 1 - L^{-\alpha}$*. Let* $\lambda = L^{-\alpha} + \theta$*. Then L-190 (Lemma HH) gives* $|\partial\beta_j/\partial g_i| \le (C_\Pi/C_V)S_0\lambda^{j-1-i}$ *for* $i \le j-2$*. The column sum is at most* $C_\beta + (C_\Pi/C_V)S_0\lambda/(1-\lambda)$*, uniformly in* i*,* j*,* K*, and the volume.*

Proof (sketch). The past enters step $k+1$ only through the curly bracket of CMP 109 (2.12), and only through the irrelevant remainders $V^{(l)}$; the marginal parts cancel (p. 298). A step-l remainder costs $L^{-\alpha(k-l)}$ per π_k cube by (0.29)–(0.30). An induction on $s_l = \|\partial_{g_i} V^{(l)}\|_l$ with one hop per step gives the rate λ in place of the one-hop rate $L^{-\alpha m}$. The rate is sharp. The bound itself is not sharp: $s_{i+1+n} = (\theta/\lambda)\lambda^n S_0$, so it is loose by the factor θ/λ. □

Hypotheses. Bałaban gives no per-hop Lipschitz constant θ. The constant C_β is Bałaban's inductive assumption (ledger R-006). Propositions 5.51 and 5.49 give S_0, C_β, and (SM). The step bounds must hold off the trajectory; this is close to (I) (ledger row R-036). Proposition 5.53 treats the large-field contributions. The whole chain is ledger R-037.

Proposition 5.51 ((G-gain) (H-111), small field)**.** *Assume Lemma G3 (L-174) and the margin* (M_r)*. Then for every step* k *and* $g_k \in]0,\gamma]$ *the own-slot derivatives of the small-field step, including the vacuum subtraction, obey* $\|\partial_{g_k} V^{(k+1)}\| \le C_V S_0$ *and* $|\partial\beta_{k+1}/\partial g_k| \le C_\beta := C_\Pi S_0$ *(L-098). Here* S_0 *is free of* $1/g_k$ *and of* ε_1*; the sharp-cut surface constant is* $O(\gamma_2^{-(n_b+1)/2}\varepsilon_1^{-1})$*. With Theorem SM (L-281) the column sum of L-190 (Lemma HH) is uniform. In u-slots it is* $O(g_i^3)$*, which meets the threshold* $Cg_i^{2+\delta}$ *of Proposition 5.18 with* $\delta = 1$*.*

Proof (sketch). By CMP 116 (1.34) the vertex $V_k'(Y,\cdot,B)$ is an analytic function of $x = g_k B$ on a polydisc of g-free radius ε_1. So a pointwise Cauchy estimate in x costs $|B|_Y/\varepsilon_1$. The ε_1 cancels against the vertex size, and the Gaussian at strength α_4 absorbs the one polynomial insertion. The $P^{(k)}$ part is cubic in B' by (1.39)–(1.40), so $|\partial_g| \le C_P|B|^3$ there; this part also uses (M_r). In the hop the factor $g_k|B|$ of (1.22)–(1.24) gives $\theta_0 \to \theta_g g_k$. □

Remark 5.52. Bałaban assumes on CMP 109 p. 264 bounds on the derivatives of the step in the couplings. Proposition 5.51 proves the first-derivative part of this assumption, on $]0,\gamma]$, for the small-field step.

Hypotheses. Lemma G3 rests on (Q)/c_-; Theorem 5.91 proves it relative to listed inputs. The margin (M_r) shrinks the χ_k threshold (ledger K-009). The large-field half is Proposition 5.53. The endpoint $g = 0$ is ledger K-020. All lie inside ledger R-037.

Proposition 5.53 ((G-gain), large field)**.** *Assume (Q)/*c_-*, the margin* (M_r)*, CMP 102 (189) through (M-A) (H-143) (ledger R-005), and the (M-B) (H-145) bookkeeping (Proposition 5.37). Freeze the thresholds at Bałaban's trajectory* x^**. Then the own-slot derivatives of the large-field outputs are uniform in* $g_k \in]0,\gamma]$ *with* $|g_k - x_k^*| \le \rho_0 x_k^*/2$*, in* k*,* K*, and the volume (L-097). For example*

$$|\partial_{g_k} R'^{(k)}(X)| \le 20\gamma^{2A_0-1}e^{-\kappa d_k(X)}, \qquad |\partial_{g_k} T_k(Y_i)1| \le 20\gamma^{A_0-1}e^{-\frac{1}{2}p_0(x_k^*)}.$$

With Propositions 5.51 and 5.49 this reduces the real uniform (H-hist) in both halves.

Proof (sketch). On the disc $|g-x| < \rho_0 x$, $\rho_0 = 0.1$, of Proposition 5.32 the printed bounds hold with $\gamma_0 \to 0.826\gamma_0$. One Cauchy step gives $|\partial_g F| \le 20x^{-1}\sup|F|$. The large-field suppression absorbs the factor $1/x$. Indeed $p_0(x) = A_0(\log x^{-2})^{p_0}$ with p_0 a positive integer and A_0 large (CMP 119

p. 244), so $x^{-1}e^{-\theta p_0(x)} \le x^{2\theta A_0 - 1}$. The mismatch average sits at $u' = 2x^{-2} - u \in [0.892, 1.093]x^{-2}$, which gives the constant $2.48 \le 2.6$. The Gaussian normalization adds a factor $e^{0.01n}$ per unit cube inside (M-B). By CMP 119 p. 279, β comes from E-class terms only. □

Hypotheses. The statement does not cover (1.103) at complex u, C^1 at $g = 0$, or derivatives of order greater than 2.

Proposition 5.54 (Lemma G3, the χ_k surface term)**.** *Assume (Q), read on the joint (X, B) majorant with $a = 0$, with constants c_- and Λ. Let $n = N^2 - 1$ and $U(g) = A_n D_n \varepsilon_1^{-1} e^{-\varepsilon_1^2/(8\Lambda g^2)}$. Then for every k and $g_k \in]0, \gamma]$ the g_k-derivative acting on the sharp cut $\chi_{k,Y_0}\chi^c_{k,P}$ of CMP 116 (2.3), in a term of (2.14), is at most $c_1' U(g_k) e^{\delta\kappa d_k(Z_0)} T_{\mathrm{maj}}$. The cut adds $S_\chi = c_2 A_n D_n \varepsilon_1^{-1} e^{-\varepsilon_1^2/(8\Lambda\gamma^2)}$ to S_0. This is uniform in g_k, X, k, and the volume, and it is $O(\gamma^p)$ for every p.*

Proof (sketch). Route 2 keeps Bałaban's cut, a round sphere $|B(b)| = \varepsilon_1/g_k$ per bond. It computes the surface term in polar coordinates. There the Gaussian marginal turns $g^{-1}(\varepsilon_1/g)^n$ into $\varepsilon_1^{-1} t^{n+1} e^{-ct^2}$, $t = \varepsilon_1/g$, with no $1/g$ left. The bond count is $c_d|Y_0| + |P|$, and the bond and polymer sums use (2.30). The surface term is real in the fixed-B computation of Proposition 5.51; a finite-difference test agrees to 10 digits at $n = 8$. The smooth cut $\hat\chi$ (Route 3, $\varepsilon_1 \to \varepsilon_1/2$, $\gamma \to \gamma/2$) is an alternative. Proposition 5.56 settles whether the cut acts on B or on B'. □

Remark 5.55. The constant γ_k printed in CMP 116 at this point does not give the bound under either parse of the formula. Proposition 5.54 does not use it.

Hypotheses. (Q)/c_- is row B-S.Q (ledger R-037). For the small field, Theorem 5.91 proves it relative to listed inputs. Numerical values of c_- and Λ are not computed. Higher derivatives are not treated.

Proposition 5.56 (Input (Q), the covariance lower bound $c_- > 0$)**.** *Assume the kernel bound CMP 99 (3.132) (printed on CMP 99 p. 422), (E_U), and the hypotheses of Lemma LOC (Proposition 5.57). Assume the residue of ledger R-001 (Proposition 5.60) and the two bridges $Q(U) = Q_{\mathrm{free}}(1 + O(\alpha_0))$ and $\|C(U)\|^2 \le L^{d+1}(1 + O(\alpha_0))$ (ledger R-018). Let $\alpha_5(1 + \gamma_0^{-1}) \le \frac{1}{2}$. Then the $B(b)$-block of the covariance of Bałaban's joint Gaussian majorant CMP 116 (2.23) is bounded below by $c_- > 0$ and above by Λ, uniformly in k, the volume, and regular U. After the localization of Proposition 5.57, $c_- = (\gamma_1 + \varepsilon_M)^{-1}$. In the free case $c_- \ge (a\kappa_d L^{d+1})^{-1}$, where κ_d depends on a and L unless $a = 1$. The cut of CMP 109 (2.9) is $|B(b)| < \varepsilon_1/g_k$ on every bond except b_0, a round sphere with no Jacobian.*

Proof (sketch). The precision of (2.23) is $Q = Q_0 - \alpha_5 \operatorname{diag}(Z, Z_0) \le Q_0$. Inversion reverses order, so the $B(b)$-block of Q^{-1} dominates that of Q_0^{-1}. Since Q_0 is coupled, the block relation is an inequality: the block is $\ge C_0 + C_0\Gamma\Gamma^* C_0 \ge C_0$, with $C_0 = C^{(k)}(Z_0, 0)$ the $\sigma = 0$ localized operator. The norm input $\|C^*\Delta_k C\| \le \gamma_1$, which CMP 116 (2.7) itself uses, follows from CMP 99 (3.132). The free-case constants come from CMP 95 (1.100) and CMP 96 (2.125). The coupling of Q_0 also enters Λ: G_Γ enters the B-block of Q_0^{-1}, and the value of Λ is the one of Theorem 5.91. □

Hypotheses. (E_U) is reduced to k-uniform kernel moments that Bałaban does not print (Section 5.14). Theorem 5.91 gives a second proof of (Q), from R-008 and R-009.

Proposition 5.57 (Lemma LOC (L-213), the localized covariance $C^{(k)}(Z_0, 0)$)**.** *Assume $\gamma_0 \le C^*\Delta_k C \le \gamma_1$ and $C_0^{-1} = Z_0 C^* \Delta_k C Z_0 - R$ with $\|R\| \le \varepsilon_M \le \gamma_0/2$, where ε_M is given by the R_2 bound CMP 116 (2.16). Then the localized covariance $C_0 = C^{(k)}(Z_0, 0)$ of CMP 116 (2.23) has $C_0(b, b) \ge c_- I$ on every bond $b \subset Z_0$ and $\|C_0\| \le 2/\gamma_0$. Here $c_- = (\gamma_1 + \varepsilon_M)^{-1}$, with no*

symmetry assumption on the truncated walk sum, and $\varepsilon_M = S_d(O(1)e^{-\delta_0 M/3} + O(\alpha_0 + \alpha_1))$ *with* $S_d \le d\coth(\delta_0/(4\sqrt{d}))^d$. *So* $\varepsilon_M \to 0$ *needs* $\alpha_0, \alpha_1 \to 0$ *as well as* $M \to \infty$.

Proof (sketch). The printed (2.5) is conditioning on Z_0^c. So $C^{(k)}(Z_0)$ is the inverse of the compression of $C^*\Delta_k C$ and inherits its bounds. The operator $C^{(k)}(Z_0, 0)$ is the walk sum with the walks that meet σ_0 deleted; it is not a compression. Its distance to the compression is Bałaban's bound (2.16) applied to R_2 at $\sigma \equiv 1$. The Schur test turns this into the norm bound ε_M. The replacement point is the real configuration U. □

Hypotheses. Bałaban states the R_2 bound (2.16) with "similarly" (ledger R-003); Proposition 5.70 and Theorem 5.78 reduce it to (W^c), R-008, and two readings. The upper bound γ_1 comes from CMP 99 (3.132) and (E_U). Bałaban states γ_0 in the prose after CMP 99 (3.158) and proves it there only at $U = 1$. For general U, Proposition 5.72 proves $\gamma_0 \ge m_L/(48d^2L^d)$ relative to the residue R-Δ2 (ledger R-004). The constants α_0, α_1 are small after M is fixed.

5.7 Step 8: Bałaban-level propagator inputs

This step treats the propagator bounds that Bałaban uses inside Theorem 1 but prints only in part: the kernel bound CMP 99 (3.132) (ledger R-001), its walk-expansion form (R-002), the remainders R_2, R_3 of CMP 116 (2.16) (R-003), the lower bound γ_0 (R-004), and CMP 102 (189) (R-005). Two readings recur in this step. We fix them first.

Convention 5.58 (the two unprinted conventions of R-001)**.** The background link $U(c)$ that enters the averaging is the parallel transport along the straight segment, and the axial contours are translation covariant. CMP 99 does not print either choice.

Convention 5.59 (reading of "intersect" in CMP 116 p. 16)**.** A walk "intersects" the set σ_0 when one of its domains shares an M_1-cube (equivalently, a set with interior) with a σ_0 cube. Boundary contact does not count. The competing reading counts boundary contact; it makes the leading walk X_0 depend on s, so that $C^{(k)}(Z_0, 0)$ degenerates and the printed (2.16) fails.

The kernel-decay form of R-001. CMP 99 p. 422 states (3.132) without proof ("we will not repeat these considerations here"). We prove it from two inputs.

Proposition 5.60 (two-scale kernel bound for $(QGQ^*)^{-1}$)**.** *Assume the kernel decay (K) of* QGQ^*, *from CMP 99 Theorems 3.3 and 3.12 for* G, *and the lower bound (C) (H-057),* $\hat{N} = S^{-1/2}(QGQ^*)S^{-1/2} \ge c_-$ *on* $\ell^2(\mathfrak{B}, \mu)$. *Then for* $y \in \Lambda_j$, $y' \in \Lambda_{j'}$,

$$|(QGQ^*)^{-1}(y, y')| \le 2K_{\text{ext}} L^{d/2+1}(L^j\eta)^{-2}(L^{j'}\eta)^{-d}e^{-\delta_1 d(y,y')},$$

uniformly in j, k, η, *the volume, and regular* U. *The same bound holds for* $(QG_1Q^*)^{-1}$ *plus a diagonal a-term. The rate comes from* $\varepsilon = \min\{\delta/2,\, c_-/(2c_+\Sigma(\delta/2))\}$; *it is much smaller than Bałaban's rate.*

Proof (sketch). L-125 (Lemma CT) is a Combes–Thomas argument for $\hat{N}$ on $\ell^2(\mathfrak{B}, \mu)$. It turns the kernel decay (K) and the lower bound (C) into decay of the inverse. It uses no Fourier analysis and no part of R-004. The G_1 clause follows in a non-circular order after the clause for G. □

Two other routes fail: a global perturbation from $U = 1$, and the Demko–Moss–Smith route. The lower bound (C) does not follow from face-midpoint bump extensions: these are not feasible, because $Q_j(U)$ is supported on both whole blocks (CMP 98 (14), CMP 99 (3.102)). We obtain (C)

instead from one feasible extension in the variational form $\langle B, N^{-1}B\rangle = \min\{\langle A, G^{-1}A\rangle : QA = B\}$; Propositions 5.61, 5.62, and 5.63 supply its inputs, and Proposition 5.126 reduces (C).

Hypotheses. Beyond (K) and (C), the bound uses the readings of Convention 5.58 and the choice $a = 1$ in CMP 99 Theorem 3.1. CMP 99 p. 399 allows this choice at no cost, so we take it as a convention. The torus side must be at least $c \cdot RM$ coarsest blocks. The consumers of R-001 need only a fixed rate that is set before M, so the small rate δ_1 is admissible.

The inputs (KE2) and ($\mathbf{H}_Q$) of (C).

Proposition 5.61 ((KE2) (H-136) and (H_Q) (H-123))**.** *(KE2): Bałaban's H of CMP 96 Cor. 2.8 satisfies $Q_{98}(1)H = I$ on all of $\mathfrak{B}$, interfaces and Λ_0 included, and H and ∇H have the decay of CMP 96 (2.151). (H_Q): on $B^j(c_-) \cup B^j(c_+)$,*

$$|K_j(U; c, b) - q_j(c, b)\,\mathrm{Ad}\,U(\Gamma_{c_- \leftarrow b})| \le c_H \alpha_0 \ell_j^2 \eta_j^d, \qquad c_H = 2C_1' + c_T(d)L,$$

uniformly in $j \le k$, under Convention 5.58. The constants $C_1'(d, L)$ and $c_{\mathrm{cone}}(d)$ are not valued.

Proof (sketch). (KE2) is a citation. CMP 96 uses the straight-line average CMP 95 (1.11)/(1.18); the invariance group $Q_j'\lambda = 0$ of CMP 96 (2.7) fits only this average; and CMP 98 p. 27 states that the propagators hold for the average (1.11). See also CMP 95 (1.12), CMP 96 (2.20), (2.35). For (H_Q) we apply Bałaban's induction (144)–(145) to the difference between $Q_j(U)$ and the product of the main terms. A cone-area bound compares the averaged background links with fine straight-line transports and fixed axial contours. Step (iii) uses a ribbon of at most $(d-1)L^2$ plaquettes and a lasso decomposition; its error is at most $9.6(d-1)\alpha_0\ell_i^2$ for $\alpha_0 \le \min(c_2, 1/(8(d-1)))$. The citation (112) is not needed. The corridor fallback $H + d\psi$ is exact within layers but not across interfaces, and we do not use it. □

Theorem QE, the input (C) of Lemma CT.

Proposition 5.62 (Theorem QE (L-252))**.** *Assume (KE2), (H_Q), the form bound S2 for G, and a torus side of at least $c \cdot RM$ coarsest blocks. Let $\mathcal{E}_U$ be the free multiscale minimizer dressed with axial transports. Then, in the S-weighted norm on $\mathfrak{B}$,*

$$\|Q(U)\mathcal{E}_U - I\|_S \le c_Q M \alpha_0 + c_{\mathrm{far}} e^{-\delta_H RM/(16L^2)},$$

uniformly in j, k, the volume, $\{\Omega_j\}$, and regular U. Hence $K_Q \le 2$, and $A = \mathcal{E}_U(Q(U)\mathcal{E}_U)^{-1}B$ is an exact extension: $Q(U)A = B$ and $A = 0$ on Λ_0. It obeys $\langle A, G^{-1}A\rangle \le K_{\mathrm{ext}}\|B\|_S^2$ with $K_{\mathrm{ext}} = 4(1 + O(M\alpha_0))K_{\mathcal{E}} + a$. This gives (C) with $c_- = 1/K_{\mathrm{ext}}$.

Proof (sketch). We write the defect of $Q(U)\mathcal{E}_U$ exactly as a $q|E|$-weighted sum of loop holonomies minus one. A cone-area lemma bounds each holonomy linearly in the distance. A Schur test with the decay (KE) and a near/far split across layers, with far terms e^{-cRM}, bound the defect. The error ε'' is of order $M\alpha_0$, and $\alpha_F = 2 \cdot O(1)M\alpha_0$ with $O(1) \ge 10$. So the Neumann series converges for $O(1)M\alpha_0 \lesssim 0.08$, which gives $K_Q \le 2$. A crude absolute-value Schur bound gives the energy. □

Numerically, the defect is linear in the curvature, with slope 0.010 at $d = 2$ and 0.019 at $d = 4$. The rigorous Schur constant is 41–76 times the true slope.

Hypotheses. (KE2) and (H_Q) hold by Proposition 5.61, and S2 for G holds by Proposition 5.63. The torus-side hypothesis and Convention 5.58 remain as hypotheses of R-001.

The form bound S2 for G.

Proposition 5.63 (S2 for G)**.** *Assume (KE2), the choice $a = 1$ in CMP 99 Theorem 3.1, and the block geometry of CMP 99. Then for every bond field A,*

$$|\langle A, \Delta'_\pi A\rangle| \le C'_\Pi M\alpha_0\big(\|DA\|^2 + \|D^*A\|^2 + \|\ell^{-1}A\|^2\big), \qquad C'_\Pi = C_\Pi + 2c_R(1+\kappa^2 a_0^2).$$

Hence $G^{-1} \le (1 + O(M\alpha_0))(D^*D + DD^*) + O(M\alpha_0)\ell^{-2} + Q^*aQ$ and $\langle \mathcal{E}_U X, G^{-1}\mathcal{E}_U X\rangle \le K(d, L, a)\|X\|_S^2$, uniformly in j, η, the volume, $\{\Omega_j\}$, and regular U. The constant c_- of (C) changes only by a factor $1 - O(M\alpha_0)$.

Proof (sketch). The route through CMP 99 (3.131) does not work: (3.131) is printed for one-block supports, and a tiling then forces sharp cut-offs that cost η^{-1} on the sup side. We use instead the exact identity (3.119), $\langle A, \Delta_\pi A\rangle = \langle \tilde A, \Delta \tilde A\rangle$ with $\tilde A = A - D\lambda$ and $\lambda = G'RD^*A$. The curl of the gradient is $\eta^{-2}(\mathrm{Ad}\, U(\partial p) - 1)\lambda$. The block L^2 bounds (3.46) of Theorem 3.1, CMP 96 Lemma 2.1, and a Schur test give $\|\ell^{-2}\lambda\|, \|\ell^{-1}D\lambda\| \le B_0 c_1 \|D^*A\|$. The operator Δ' of (3.10) has a second term $\eta^d \,\mathrm{tr}(DA)^2(\mathrm{Re}\, U(\partial p) - 1)$, which is at most $O(M\alpha_0)\|DA\|^2$; this term gives the summand $2c_R(1+\kappa^2 a_0^2)$ of C'_Π. CMP 99 (3.16) fixes the normalization of Q^*aQ with one constant a. □

Remark 5.64. In CMP 99 (3.120) the sign of the correction term must be a minus sign.

S2 for G_1, and the remainder R_2.

Proposition 5.65 (S2 for G_1; the bound on R_2)**.** *(A) Assume the hypotheses of Proposition 5.63. Then*

$$|\langle B, \Delta^{(2)}B\rangle| \le c_2 M\alpha_0\|\ell^{-1}B\|^2, \quad c_2 = \dim \mathfrak{g}\cdot C_{137}, \qquad |\langle A, \Delta_\pi^{(2)}A\rangle| \le 2c_2 M\alpha_0\big(\|\ell^{-1}A\|^2 + B_0^2 c_1^2\|D^*A\|^2\big),$$

so $\langle \mathcal{E}_U X, G_1^{-1}\mathcal{E}_U X\rangle \le K_1\|X\|_S^2$ with $K_1 = K(1 + O(M\alpha_0))$. (B) Assume the walk bounds (W^c) on the complex α'-space ($\alpha' \gg \alpha$), the real decay of $C^*\Delta_k C$ (R-008), Convention 5.59, and Convention 5.66 below. Then for $M \ge \max\{M_*, 96\kappa_1/\delta_0\}$ and $\alpha_0 + \alpha_1 \le \alpha_*$,

$$|R_2(b, b')| \le \big(O(1)e^{-\delta_0 M/3} + O(\alpha_0 + \alpha_1)\big)e^{-(1-\theta)\delta_A r}.$$

Proof (sketch). (A) By (3.135), $\Delta_\pi^{(2)}$ is the sandwich $(I - DG'RD^*)^*\Delta^{(2)}(I - DG'RD^*)$, so the form is taken at the $\tilde A$ of (3.119). The kernel of $\Delta^{(2)}$ is local by (3.137) and symmetric as a Hessian by (3.136). The symmetry is needed for the column sums; with it, a Schur test with AM–GM applies. (B) Write $C(\sigma; U_c) = A_Z(U)^{-1} - W'$ with $W' = W_\sigma(U_c) + [C(1; U_c) - C(1; U)]$. A one-variable Cauchy estimate on the complex walk bound bounds the bracket; this needs (W^c) on the α'-space, and not only on the α-space that W_σ needs. A Neumann series around the real $A_Z(U)$ lets $A_Z(U)$ cancel in R_2, so decay of A is needed at real U only. A two-constant theorem weakens the complex input to a sup bound but does not remove it. Cubes of σ_0 that touch the domain occur generically; Convention 5.59 decides how they count. The two covariances do not coincide on a gauge-fixed subspace, and the proof does not use such a coincidence. □

Convention 5.66 (distance of Z_0 to the boundary)**.** We read the construction of CMP 116 so that $\mathrm{dist}(Z_0, \Lambda_k^c) > RM$.

Hypotheses. (W^c) on the α'-space is proved in the exponential gauge, real and complex, without the transfer of CMP 99 Thm 3.15 (`SOURCES_WC_EXPGAUGE.md`, `SOURCES_WC_RESIDUE.md`); Theorem 5.78 states it, relative to its inputs. Part (A) uses the choice $a = 1$ and the block geometry. The lower bound (A3.5) for G_1 is not treated here.

The walk-expansion form R-002 of R-001. CMP 99 p. 422 says that $(QGQ^*)^{-1}$ "can be analyzed in the same way" and does not repeat the analysis. We state the walk expansion and prove it from (W_G) and (C).

Proposition 5.67 (walk expansion of $(QGQ^*)^{-1}$, L-063)**.** *Assume (W_G) (Proposition 5.68), the lower bound (C), CMP 96 Lemma 2.1, the Lipschitz cut-offs H-002, the block-locality of Q, and that κ_1 does not depend on M. Then for regular U, $(QGQ^*)^{-1} = \sum_\omega T_\omega$, with factors localized in $X \times X$, U-dependence only on $\tilde{X}_0^5 \cup \cdots \cup \tilde{X}_n^5$, and*

$$|T_\omega(y, y')| \le (4/c_-) L^{d/2+1} (L^j \eta)^{-1-d/2} (L^{j'} \eta)^{-1-d/2} (K_W/M_1)^{|\omega|} e^{-\frac{1}{2}\varepsilon d(\omega, y, y')}.$$

The sum converges absolutely. This is the (3.108) form at rate $\delta_W = \frac{1}{2}\varepsilon \ll \frac{1}{2}\delta_0$. Each factor kernel vanishes within about M_1 of ∂X (property (L1)). The same holds for $(QG_1Q^*)^{-1}$ with trees. The constant order contains the condition CMP 96 (2.59) on RM at the walk rate. Theorem 5.76 states the result in the rescaled form for $\hat{N}$, with explicit constants.

Proof (sketch). The local operator on a cube is the walk sum of G truncated to $\square^+$. Its positivity $\frac{1}{2}c_-$ comes from the global (C) by restriction, not from R-004. An accretive form of L-125 (Lemma CT) gives $\|C_\square\|_\varepsilon \le 4/c_-$, as in Bałaban's own local step (CMP 96 p. 237). The parametrix $P = \sum h_\square C_\square h_\square$ satisfies $\hat{N}P = I + \sum F_\square$. The three remainders (separation, truncation, commutator) are each at most K_F/M_1, and the Neumann series is the walk series. CMP 96 (2.84) has the factor $e^{-\frac{1}{8}\delta_0 M}$, and (2.82) has three error sums. The citation for $\breve{G}_2$ is CMP 99 p. 432. □

The Z_0-compressed variant (O3) and the variant on the complex α'-space (O4) hold by `SOURCES_WC_EXPGAUGE.md` and `SOURCES_WC_RESIDUE.md`; O4 holds relative to CMP 99 Thm 3.4 and the complex walk form of G, which rests on the printed sketch of CMP 99 Thm 3.10.

The walk expansions ($\mathbf{W}_G$) of G and G_1.

Proposition 5.68 ((W_G), reduction to the printed sketches)**.** *Assume the printed proof sketches of CMP 99 Theorems 3.7, 3.9, 3.10, the Neumann series (3.130), CMP 96 Lemma 2.1, and the readings (Q-locality, two-sided derivative bounds, locality of $C^{(2)}$, block counts) listed in `SOURCES_WG.md`. Let U be regular. Then for $M_1 \ge M_*(d, L)$ and $M\alpha_0 \le a_*(d, L)$, $G = \sum_\omega T_\omega$ is a walk expansion with bounded branching. For G_1 assume in addition the walk expansion A1-W of $(QGQ^*)^{-1}$ (Proposition 5.67); then G_1 has a tree expansion with rate $(1-\theta)\frac{1}{2}\varepsilon$. The rates, the M_1-cover, and the per-step form with a factor M_1^{-1} per step, also at the Δ'_π junctions, are as in Theorem 5.77. The per-step form needs the condition WG-1, $\alpha_0 \le c' M^{-1} M_1^{-10}$ for G and $\alpha_0 \le c'' M^{-1} M_1^{-12}$ for G_1. WG-1 is admissible in Bałaban's order of constants, but $M\alpha_0 \le a_0$ does not imply it.*

Proof (sketch). We expand the identity $G^{-1} = G_0^{-1} - \Delta'_\pi$ and the Neumann series (3.130), $G = \sum_n G_0(\Delta'_\pi G_0)^n$, with the printed walk expansions of G_0, G', and $(Q'G'^2Q'^*)^{-1}$ (CMP 99 Thms 3.10, 3.7, 3.9). Each Δ'_π carries one local J-insertion of size $O(M\alpha_0)$. A concatenation lemma splits the exponent at each junction, and CMP 96 Lemma 2.1 sums the junction points. A Δ'_π insertion has up to $N_V = 9$ walk blocks; this count gives the exponent of WG-1. For G_1 the same scheme runs on $G_1 = \sum_n G(\Delta_\pi^{(2)} G)^n$ with $\Delta_\pi^{(2)} = (I - DRG'D^*)\Delta^{(2)}(I - DG'RD^*)$. Its side branches use A1-W. The order $G_0 \to G \to (QGQ^*)^{-1} \to H \to G_1$ has no loop. □

Remark 5.69. Bałaban himself writes "assuming (3.132)" for the G_1 clause (CMP 99 p. 426). The phrase "after adjusting a definition of δ_0" is on CMP 99 p. 427.

The remainders R_2 and R_3 of CMP 116 (2.16) (R-003). CMP 116 p. 16 prints (2.16) for R_1 only and states it for R_2 and R_3 with "similarly".

Proposition 5.70 (R_2 and R_3)**.** *Assume the walk bounds (W^c) on the complex α'-space, the decay of $C^*\Delta_k C$ at real U (R-008), Convention 5.59, and Convention 5.66. Then the precision difference $R_2 := C_0^{-1} - \operatorname{Re} C^{(k)}(Z_0,\sigma)^{-1}$ obeys*

$$|R_2(b,b')| \le \big(O(1)e^{-\delta_0 M/3} + O(\alpha_0+\alpha_1)\big)e^{-\rho|b-b'|} \qquad \text{for } |\sigma(\Delta)| \le e^{\kappa_1}.$$

The rate is the printed $\rho = \frac{1}{2}\delta_0$ when the decay rate δ_A of $C^*\Delta_k C$ exceeds $\frac{1}{2}\delta_0$, and a smaller rate $\rho < \delta_A$ otherwise; (2.21) and Lemma LOC accept the smaller rate. R_3 holds in two weaker forms (R3a) (H-242) and (R3b) (H-243), which suffice for its only printed use (2.21). The input (E) of Lemma LOC follows at $\sigma = 0$ with $\varepsilon_M = O(e^{-\delta_0 M/2})$.

Proof (sketch). We use a Neumann series around the compression $C(1)^{-1} = Z_0 A Z_0$, $A = C^*\Delta_k C$ (case (a), (2.5)). The perturbation is $W_\sigma = C(1) - C(\sigma)$, the sum over the walks that meet σ_0. Each such walk makes a round trip to distance $> \frac{2}{3}M$ from Z_0. This gives $O(e^{-\delta_0 M/3})$ once the weight $e^{16\kappa_1}$ is absorbed, for $M \ge 96\kappa_1/\delta_0$. The walk bounds (W) are CMP 99 (3.107)–(3.108). For $C^{(k)}(Z_0)$ they contain the walk expansion of $(QGQ^*)^{-1}$, which CMP 99 does not print (R-002). The $O(\alpha_0+\alpha_1)$ part comes from a one-variable Cauchy estimate on the analyticity space; the Cauchy radius respects the plaquette condition. The expansion runs around the real $A_Z(U)$ as in Proposition 5.65(B), so L-228 (Lemma N) and the Cauchy disc need decay of A at real U only. □

Hypotheses. (W^c) is Theorem 5.78, relative to its inputs: the G_1 tree expansion (CMP 99 Thm 3.10, printed sketch), the lower bound (C) of A1-W, the items R2 and R-Δ2 of R-004, CMP 109 p. 254 (Proposition 5.141), CMP 99 Thm 3.4, and the readings listed there. The real decay R-008 is Theorem 5.79.

The lower bound γ_0 (R-004). CMP 99 p. 428 states, after (3.158), that the proof of CMP 96 Lemma 2.4 for $U = 1$ extends to $C^*\Delta_k C$ with an arbitrary configuration U. It gives no proof.

Proposition 5.71 (the Kato route fails; the two-step mechanism)**.** *The lattice Kato (diamagnetic) route to $C^*\Delta_k C \ge \gamma_0$ fails. Bałaban's two-step mechanism at $U = 1$, made gauge covariant, gives $\gamma_0 \ge (48d^2)^{-1}L^{-d-1}$, independent of k, the volume, and M, given the following inputs: R1, that is CMP 99 Thms 3.1, 3.3, 3.11, 3.12 with a uniform bound on the oblique projection $\kappa_\pi = \|I - DG'RD^*\|^2$; R2, the k-step averaging comparison (a part of R-018); the small items R3–R5; and, for the σ-localized clause, the reading of the walk expansion in Proposition 5.74. For $L \ge 10$ the constant carries the factor Lm_L of Proposition 5.72.*

Proof (sketch). The Kato route fails because of an unfixed mode $\lambda \in \operatorname{Ran}(\Delta^{-2}Q'^*)$: once $\|RD^*A\|^2$ is included, the comparison gives 0 against 4.62. The example $A = d\delta_x$ does not refute the route, because the term $\|RD^*A\|^2$ changes it. Step (i) bounds the minimal-extension energy below by the unit-lattice curl energy: at $U = 1$, $\langle B, \Delta_k B\rangle \ge \|\partial_1 B\|^2$, by exact lattice Stokes and Cauchy–Schwarz, with $c_* \le 1$. Step (ii) is CMP 96 Lemma 2.4, a unit-lattice Poincaré inequality in axial gauge. The free constant is $(12d^2)^{-1}L^{-d-1}$. Both steps are made covariant in a local regular gauge on each region. The gauge-fixing part is removed through (3.115), and the relative errors are $O(M\alpha_0 + L^2\alpha_0)c(d,L)$ for $M \ge cL$. We give this argument only as a sketch; Proposition 5.72 writes it out in full. □

Proposition 5.72 (the lower bound γ_0, L-069)**.** *Assume R-$\Delta 2$ and the items R3–R5 of R-004. Let U obey CMP 99 (3.35)–(3.36) with $M\alpha_0$ small. Then*

$$\langle \tilde{B}, C^* \Delta_k C \tilde{B} \rangle \geq \gamma_0 \|\tilde{B}\|^2, \qquad \gamma_0 = \frac{m_L}{48 d^2 L^d}, \quad m_L = \min\bigl(2(1-\cos(\pi/L)), 1/L\bigr),$$

uniformly in k, the volume, and regular U. For $L \leq 9$ this is $(48d^2)^{-1}L^{-d-1}$; at $L = 13$ it is 0.7555 times that value.

Proof (sketch). Lemma 5 shows that the G_1-minimizer satisfies $RD^*A_* = 0$: the map $A \mapsto A - DG'RD^*A$ keeps QA and lowers $\langle A, G_1^{-1}A\rangle$ by exactly $\|RD^*A\|^2$. So the gauge-fixing part drops out, and no bound on κ_π is needed. Lemma 5 is formalized in Lean from (F1), (F3), (F4) as hypotheses (`GaugeMin.lean`). Step (i) of Proposition 5.71 holds with $c_* = 1$ and a covariant relative error $2(1+\delta^{-1})\varepsilon_2^2$. Step (ii) is CMP 96 Lemma 2.4, reproved with m_L in place of $1/L$ and made covariant in one regular gauge per corridor. The assembly gives $\gamma_0 = \gamma_L/4$ with $\gamma_L = m_L/(12d^2L^d)$. Fact (E) needs a factor $\sqrt{N}$. R-$\Delta 2$ follows from the form (3.136) and a block-average bound on $C^{(2)}$; the Schur step of R1 and R-$\Delta 2$ follow from the printed (3.137) (CMP 99 p. 423) and (3.46). The proof uses CMP 99 Thms 3.11–3.12 and (H_Q) (Proposition 5.61). □

Remark 5.73. The proof step inside CMP 96 (2.126)–(2.127) is false for $L \geq 10$: it loses a factor L. The lowest eigenvalue of the slab problem is $L^{-d}\min(2(1-\cos(\pi/L)), 1/L)$. In $d = 2$, (2.127) fails from $L = 10$ with free block bonds and from $L = 15$ in axial gauge. The statement of CMP 96 Lemma 2.4 is not refuted; it holds with the constant m_L of Proposition 5.72. The failure of (2.127) for every $L \geq 10$ is checked in Lean (`SlabCounterexample.lean`, Appendix A).

Hypotheses. R3, R4, R5, R-Δ', and R2b remain inputs of R-004. The σ-localized clause is Proposition 5.74.

Proposition 5.74 (the σ-localized clause of R-004)**.** *Assume (W): the expansion of $C^{(k)}(Z_0)$ of CMP 116 p. 13, with the large domain X_0, obeys CMP 99 (3.107)–(3.108), with the counting of Cor. 3.8, uniformly in Z_0. Assume (N), $\gamma_0 \leq A \leq \gamma_1$, and Convention 5.59. Then for real $s \in [0,1]^{\sigma_0}$ the s-localized covariance $C_s = C^{(k)}(Z_0, s)$ obeys*

$$\|C_s - C_1\| \leq \eta_M = S_d B_W e^{-\frac{1}{3}\delta_0 M}, \qquad \langle v, C_s^{-1} v\rangle \geq (\gamma_0 - 2\gamma_1^2\eta_M)\|v\|^2 \geq \tfrac{1}{2}\gamma_0\|v\|^2$$

for $\eta_M \leq \gamma_0/(4\gamma_1^2)$, uniformly in s, Z, Z_0, k, the volume, and real regular U, with no α-condition and no symmetry assumption. The resolvent bound $\|C_1^{-1} - C_s^{-1}\| \leq 2\gamma_1^2\eta_M$ also gives the input (E) of Lemma LOC in norm at real points, without (2.16).

Proof (sketch). The walk sum of CMP 116 p. 3 expands the covariance, $C_s = \sum_\omega s^\omega T_\omega$. So $C_s - C_1 = -\sum_{\omega \in W_\sigma}(1 - s^\omega)T_\omega$, and one s-free absolute majorant over the walks that meet σ_0 dominates it termwise. Each such walk reaches distance $> \frac{2}{3}M$ from Z_0; for the zero-length walk this uses Convention 5.59. A Schur bound gives η_M; the uniformity of B_W in Z_0 uses part of the exponential decay. The resolvent identity with $\gamma_0 \leq Z_0 A Z_0 \leq \gamma_1$ turns η_M into the lower bound on the precision. Under the symmetry (S_s) the bound is $\frac{2}{3}\gamma_0$; we do not need (S_s). □

Hypotheses. (W) is Bałaban's assertion on CMP 116 p. 13 (R-002 enters here); in the exponential gauge this form is Theorem 5.78(ii), relative to its inputs. In (N), γ_0 is Proposition 5.72 and γ_1 comes through R-008. Complex σ stays with R-003.

CMP 102 (189) (R-005). CMP 102 p. 308 states (189) without proof ("we do not perform these calculations here").

Proposition 5.75 (the Hessian kernel bound CMP 102 (189))**.** *Assume R-001 in its kernel-decay form (Proposition 5.60), the published statements of CMP 102, R-018, and the second-derivative bookkeeping of the V_0 terms (90)–(97). Then the two-point Hessian kernel of the nonlinear action part V at fixed A' obeys*

$$|V''(A';b,b')| \le c_V \varepsilon_3 e^{-\frac{1}{2}(1-\alpha)^5\delta_0 d(b,b')},$$

with rate at least $\frac{1}{4}\delta_0$ for $\alpha \le 0.129$, free of the volume, and on the complex domain (77). The bound (190) follows through the Neumann series (188) at rate $\frac{1}{8}\delta_0$ for $\alpha \le 0.159$.

Proof (sketch). Every operator that CMP 102 inverts has the form I plus a small decaying kernel. So a weighted Schur Neumann series (CMP 96 Lemma 2.1) suffices, without Demko–Moss–Smith and without a Hessian lower bound. Lemma D2 (L-139) shows that the second derivative $\mathfrak{D}_2$ of the linearizing map solves $(I+\mathfrak{N})\mathfrak{D}_2 = C''[(I-H\mathfrak{D})\cdot,(I-H\mathfrak{D})\cdot]$, with the operator of (68), and has two-sided decay at rate $\frac{1}{2}(1-\alpha)^4\delta_0$. (Our $\mathfrak{D}_2$ is not Bałaban's $\mathfrak{D}_2$, which is the first derivative minus the linear term.) Composition then gives V'' at rate $\frac{1}{2}(1-\alpha)^5\delta_0$. V'' uses (3.132) directly through (87)–(88), and the decay of $\tilde{G}$ depends on R-001 through (3.137) and (3.133). The CMP 102 rates are fractions of the common $\delta_0 \le \delta_1$. □

Hypotheses. R-001 is Proposition 5.60. R-018 is Proposition 5.108. The ∇, Hölder, and Δ versions of (190) are readings of CMP 102 that this proposition does not prove.

5.8 Step 9: the propagator expansions written out in full

The three theorems of this step write out in full the constructions to which Propositions 5.67, 5.68, and 5.70 reduce. They do not remove an input. They replace the reduction to a printed sketch by a written proof that names its inputs. The inputs common to all three are the lower bound (C) (H-057), Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, and 3.10, and a short list of geometry and convention items. Theorems 5.145 and 5.143 write out CMP 99 Thms 3.7 and 3.9.

Notation. M_1 is the cube size of the walks (CMP 116 p. 3). M is the regularity parameter of CMP 99 (3.36) and the cube size of the s-rule, and $R_1 M_1 \le M$. A walk expansion is a sum $T = \sum_\omega T_\omega$ over walks of domains with the following properties: (L1) each factor kernel vanishes within $\frac{1}{3}M_1$ of the boundary of its domain (the band of CMP 116 p. 3); (L2) each factor depends on U only on $\tilde{X}^5$; (WB1) a per-walk bound $B\rho^{|\omega|}e^{-\delta d(\omega,y,y')}$ with the path distance (3.93); (WB2) bounded branching.

Theorem 5.76 (walk expansion of $\hat{N}^{-1}$, L-017)**.** *Let $\hat{N} = S^{-1/2}QGQ^*S^{-1/2}$. Assume (H1) (H-204): $\hat{N}$ has a walk expansion with (L1), (L2), $|\hat{n}_\omega(y,y')| \le b_G \rho_G^{|\omega|} m(y,y') e^{-\delta_G d(\omega,y,y')}$, $\rho_G = c_G/M_1$, at most $C_{\mathcal{D}}$ domains meeting a set of diameter $\le D_+ M_1$, and $\operatorname{diam}\tilde{X}^5 \le c_X M_1$. Assume (H2): the lower bound (C), $\operatorname{Re}\langle u, \hat{N}u\rangle \ge c_-\|u\|^2$, uniform. Assume (H3): the cube geometry of Bałaban's cover. Assume (H4) (H-206): the constant order $(d,L,a,a_0) \to \theta \to c_\pm, \delta_G, \Sigma, b_G, c_G, C_{\mathcal{D}}, c_X \to \varepsilon \to M_1 \to M \to \alpha_0$, with WG-1 in the M_1 form $\alpha_0 \le c' M^{-1} M_1^{-10}$ for G and $\alpha_0 \le c'' M^{-1} M_1^{-12}$ for G_1. Put $c_+ = C_{\mathcal{D}} b_G$ and $\varepsilon = \min\{\frac{1}{2}\delta_G,\ c_-/(8c_+\Sigma(\frac{1}{2}\delta_G))\}$. Then for $M_1 \ge M_W(d,L,a,a_0)$: (i) $\hat{N}^{-1} = \sum_\omega T_\omega$ is a walk expansion; (ii) $|T_\omega(y,y')| \le (4/c_-)(c_1\rho)^{|\omega|} m(y,y') e^{-\frac{1}{2}\varepsilon d(\omega,y,y')}$ with $\rho \le K_W/M_1$; (iii) $\sum_\omega |T_\omega(y,y')| \le (8C_{\mathcal{D}}/c_-) m(y,y') e^{-\frac{1}{2}\varepsilon d(y,y')}$; (iv) $\|\hat{N}^{-1}\|_{\frac{1}{2}\varepsilon} \le 8/c_-$. All constants are uniform in U, j, k, η, $\{\Omega_j\}$, and the volume.*

The proof uses, besides (H1)–(H4), the following inputs: (W_G) in the form of Theorem 5.77 (H-296); the lower bound (C) (H-298), reduced in Proposition 5.126; the readings H-303–H-307 (the cube geometry, the Lipschitz partition H-002, Q-locality, the WG-1 condition, and that κ_1 does not depend on M); and H-313, the walk expansion of G at complex points, from CMP 99 Thms 3.4 and 3.10.

There are two variants. The Z_0-compressed variant: let A have a walk expansion with $\operatorname{Re} A \geq \gamma$. Then the inverse of $A_Z = 1_{Z_0} A 1_{Z_0}$ has a walk expansion with prefactor $4/\gamma$. For $q = M/M_1 > 10$ and $q \geq 3(7 + c_X)$ it regroups into the form of CMP 116 p. 13: one domain X_0 with $\tilde{X}_0^5 = \tilde{Z}_0$, all other domains at distance $> \frac{2}{3}M$ from Z_0, the band kept, and a merged-slot cost $8C_W/\gamma$ per walk. For $(x + A_Z)^{-1}$, $x \geq 0$, the prefactor is $4/(\gamma + x)$. The regrouping uses the bound $\sum_\square \varphi_\square^2 \geq 1$ on Z_0; a tent family on a grid of spacing $M_1 L^k$ proves it. The complex variant: Lemmas CP and AW and a four-level chain of complex domains give the expansion on the α'-space. The chain constants $\Gamma_3 = 2C_{\mathcal{D}} B_1''' \Sigma(\delta_1''')$ and $\Gamma_1'' = 2C_{\mathcal{D}} B_A'' \Sigma(\delta_A'')$ carry the factor $C_{\mathcal{D}}$, so the constants k_1, k_2 are smaller by at most this factor. The G_1 clause uses the tree form of (H1), stated in `A1_WALK_FULL.md` 6.9.

Theorem 5.77 (Theorem WG: walk expansions of G and G_1)**.** *Let U satisfy CMP 99 (3.35)–(3.36) and fix $\theta \in (0,1)$. Let $\delta_{WG} = \min\{\delta_{310} - \theta_L,\ (1-\theta_{37})\delta_0,\ \frac{1}{2}\varepsilon_{39} - |2 - d/2|\theta_L\}$ be the minimum of the rates of the walk expansions of G_0, G', and $(Q'G'^2Q'^*)^{-1}$ (CMP 99 Thms 3.10, 3.7, 3.9, in the forms of `THM310_FULL.md`, Theorem 5.145, and Theorem 5.143), with $\theta_L = \log L/(RM)$. (G) (Theorem WG-G (L-316).) There are M_* and $a_* > 0$, depending on (d, L, a), θ, and the constants of these three expansions, such that for $M_1 \geq M_*$, $M\alpha_0 \leq a_*$, and RM with CMP 96 (2.59) at rate $\theta\delta_{WG}$, $G = \sum_\lambda T_\lambda$ is a walk expansion on the M_1-cover with constants $(B_0', \rho_*, (1-\theta)\delta_{WG})$, $\rho_* = \max\{c_0 M_1^{-1}, (K_V M\alpha_0)^{1/10}\}$. The bound holds also with one covariant derivative on either side. If in addition $\alpha_0 \leq c' M^{-1} M_1^{-10}$ (WG-1) (H-179), then $\rho_* \leq c_0 M_1^{-1}$, the CMP 116 (1.7) form. (G1) (Theorem WG-G1 (L-317).) Assume in addition Theorem 5.76 for $(QGQ^*)^{-1}$. There are M_W and $a_{**} > 0$ such that for $M_1 \geq \max\{M_*', M_W\}$, $M\alpha_0 \leq a_{**}$, and $RM \geq C(d, L, \theta, \varepsilon)$, $G_1 = \sum_\tau T_\tau$ is a tree expansion with rate $(1-\theta)\frac{1}{2}\varepsilon$. The per-step form needs $\alpha_0 \leq c'' M^{-1} M_1^{-12}$. All constants are uniform in U with (3.35)–(3.36). The rate δ_{WG} is smaller than the printed rate $\frac{1}{2}\delta_0$ of CMP 99 (3.108); every consumer uses only a fixed rate that is set before M.*

Proof (sketch). We expand $G = \sum_n G_0(-VG_0)^n$, $V = -\Delta_\pi'$, with the walk expansions of G_0, G', and $(Q'G'^2Q'^*)^{-1}$. L-228 (Lemma N) shows that the Neumann series gives the inverse in finite volume, so no separate existence of G is needed. V splits into 24 patterns with at most $N_V = 9$ walk blocks and one local J-insertion of size $O(M\alpha_0)$. L-115 (Lemma C) concatenates walks: it splits the exponent at each junction and sums the junction point by CMP 96 Lemma 2.1, so the rate factor $1-\theta$ is lost once. The 10 junction steps of a V-insertion give the exponent of WG-1. To obtain the band (L1), every domain is enlarged by one layer of M_1-cubes. This changes the branching count to $C_{\mathcal{D}} = 5^d c(d) C_{\mathcal{D}}^{\mathrm{in}}$ and c_X to $c_X + 2$, and adds $M_1 \geq 3c_{\mathrm{loc}}$ to M_*. The output satisfies (H1) of Theorem 5.76. Every consumer uses the M_1 form of WG-1. □

The proof uses the following inputs. CMP 99 Thm 3.10 and Thm 3.9 (printed proof sketches); Thm 3.7 with Cor. 3.8 (proof by reference to CMP 95 Prop. 1.2 and CMP 96 Prop. 2.2); CMP 96 Lemma 2.1 under (2.59), in the form $\frac{1}{4}\alpha\delta_0 RM > 2d \log c_0(\frac{1}{2}\alpha) + 1$ (CMP 96 p. 233); the bound $|J| \leq O(1) M\alpha_0 (L^j \eta)^{-3}$ (CMP 99 pp. 419, 422); for (G1) also Theorem 5.76, hence (C). It also uses these readings: the scale bookkeeping C0 (written out for the vector $\hat{J} = S^{3/2}J$); the locality of Q, Q', D, and the parallel transports; the assignment of ι-domains; the locality of $C^{(2)}$; and the M_1-cover, which CMP 116 p. 3 asserts.

Theorem 5.78 (Theorem WC: the covariance $C^{(k)}(Z_0)$ in the exponential gauge)**.** *Fix $d \geq 2$ and odd $L \geq 3$. There are M_1, M with $M/M_1 > 3(6 + c_X)$ and $M/M_1 > 10$, a smallness α_*, a*

four-level chain of complex domains $\alpha' \subset \alpha'' \subset \alpha''' \subset$ the a_1-domain of CMP 99 Thm 3.4, and constants $\gamma_{\exp} = \frac{1}{4}\gamma_e(d,L) > 0$, $\varepsilon'' > 0$, $\rho > 0$, such that for every real regular U and every Z_0: (i) $\langle \tilde{B}, A_Z \tilde{B}\rangle \geq \gamma_{\exp}\|\tilde{B}\|^2$ on the reduced space of Z_0; (ii) $C^{(k)}(Z_0) = \sum_\omega T_\omega$ with (L1), (L2), $|T_\omega(b,b')| \leq (4/\gamma_{\exp})(c_1\rho)^{|\omega|}e^{-\frac{1}{2}\varepsilon d(\omega,b,b')}$, absolutely convergent, and after regrouping in the form of CMP 116 p. 13; (iii) at every U_c in the α'-space, $\mathrm{Re}\langle u, A_Z(U_c)u\rangle \geq \frac{1}{2}\gamma_{\exp}\|u\|^2$, $C^{(k)}(Z_0;\cdot)$ is analytic, and (ii) holds with $4/\gamma_{\exp} \to 8/\gamma_{\exp}$ and $\varepsilon \to \varepsilon''$; (iv) (i)–(iii) hold for $(x + A_Z)^{-1}$, $x \geq 0$, uniformly in x, with the per-walk prefactor $4/(\gamma_{\exp} + x)$ (real) and $8/(\gamma_{\exp} + x)$ (complex), so the $x^{-1/2}$ integral for $(C^{(k)})^{1/2}$ converges walk by walk. The constants are uniform in k, η, Z_0, $M \geq M_1$, and the background. The proof does not use CMP 99 Thm 3.15 or any transfer from the axial gauge.

Proof (sketch). The lower bound comes from L-153 (Lemma E), $S_y \leq K_b(P_y + G_y)$ on each block, with $K_b = 8dL^d(1 + 16dc_{\mathcal{G}}) \leq 64d^3L^{2d+1}$ under $16d^3L^d c_{\mathcal{G}}\tau_p^2 \leq 1$, $\tau_p = 16\sqrt{N}c_4 M\alpha_0$. In the exponential gauge the chain terms of the average do not vanish. Each carries a per-bond weight at most $1 + L^{-d}$, which gives the factor $\kappa_{\mathrm{ch}} = 2(1 + dL^{d+1})$ in the side sums. Then $\gamma_e^{-1} = K_b(1 + 6d\kappa_{\mathrm{ch}}/m_L) + 6/m_L$ and $\gamma_{\exp} = \frac{1}{2}\min(\frac{1}{2}, c_G)\gamma_e$ with $c_G = 1$. For example, $\gamma_{\exp} \approx 7.6 \cdot 10^{-10}$ at $d = 2$, $L = 3$. For Federbush's mean (CMP 109 (0.10)) the exact weights give $c_M \leq 4\sqrt{N}$, and $\kappa \leq \sqrt{2}$ for $w \leq 0.054/\sqrt{N}$. The walk expansion is the Z_0-compressed variant of Theorem 5.76, applied to the tree expansion of $A_{\exp}$; this tree expansion rests on Theorem 5.77(G1). The Combes–Thomas Schur sums carry the factor of $|e^x - 1| \leq |x|e^{|x|}$: $\Sigma(\frac{1}{2}\delta) \leq (1 + 4/(e\delta))S(\frac{1}{4}\delta)$, which changes only ε, ε^a, and ε''. □

The proof uses the following inputs. Theorem 5.77(G1), which rests on printed sketches; the complex walk form of G (CMP 99 Thms 3.4 and 3.10); the inputs of R-004; CMP 109 p. 254 for the results of CMP 98 (Proposition 5.141); and CMP 109 p. 253 for the operators G_1, H_1, $(QG_1Q^*)^{-1}$, Δ_J, $\tilde{Q}$ of CMP 99 (ledger R-022). It is conditional on (C) and on $c_{-,1} \geq c_-(1 - O(M\alpha_0))$; Lemma 5.129 gives $c_{-,1} = c_-/(1 + \delta_1)$ relative to the inputs of (C). It assumes the readings I1–I9 of `WC_FULL.md` §9: among them, the term match of CMP 109 (2.11) with CMP 99 (3.156), the locality of $\tilde{C}^{(2)}$ and Q, the identification $\tilde{Q} = Q_1(U_1)$, Convention 5.66, and the tilde convention.

5.9 Step 10: the remaining Group A rows and Lemma G3

This step treats the Group A rows R-008, R-009, R-010, R-012, and R-015. It also proves Lemma G3 (L-174), the residue item N2 of the reduction of R-006. Each result holds relative to the inputs that it names. These inputs are not proved here, so the rows stay in the dependency table. Nothing in this step makes Theorem A or Theorem B unconditional.

The operator of R-008. Bałaban's C in $C^*\Delta_k C$ is the constraint map $I - hL\tilde{Q}$. It is not a square root of the covariance. The operator $C^*\Delta_k C$ is the *precision*: CMP 116 (2.7) reads $(C^{(k)})^{1/2} = (C^*\Delta_k C)^{-1/2}$. So the lower bound $A_Z \geq \frac{1}{4}\gamma_e$ of Theorem 5.78(i) is the γ_0 side (row R-004), and it cannot give the upper bound γ_1. The row R-018 enters the upper bound through A3-R4 and the item I3 of `WC_FULL.md`.

Theorem 5.79 (Upper bound and decay of the precision $C^*\Delta_k C$ (row R-008))**.** *Let U be real and regular, $s \equiv 1$, and let Λ be as in `WC_FULL.md` 0.8.*

(N) (L-226 (Lemma N).) Assume C2 of `WC_FULL.md` (the bound (C) (H-057) for G_1, which rests on the torus-side hypothesis, two conventions, and one inference step for G_1), A3-R4, I1, and I3. Then $\|C^*\Delta_k C\| \leq \gamma_1^N$ with $\gamma_1^N = \|C\|^2 \max\{K_{\mathrm{ext},1} - a + c_J + 2c_{\mathcal{G}},\, a + c_J\}$, uniformly in k, the volume, and U.

(D) (Theorem D (L-138).) Assume the walk expansion W_{G_1} of Theorem 5.77(G1), Theorem 5.76, I2, and I9. Then $|(C^*\Delta_k C)(b,b')| \le 2B_A e^{-\delta_A|b-b'|}$ with $\delta_A = (1-\theta)\min\{\frac{1}{2}\varepsilon, \delta_1\}$, and the series (2.7) for $(C^{(k)})^{1/2}$ converges walk by walk for $\gamma_1 := 4B_A S(\theta'\delta_A)$.

Both bounds hold for real $s \in [0,1]$ with the same constants.

The proof of (N) can also use a two-sided route (§2.3 of `SOURCES_A7_CDC.md`) in place of L-208 (Theorem L). The clause for complex s with $|s| \le e^{\kappa_1}$ and complex U on $\mathcal{U}(\alpha'')$ is only sketched; it needs M_1 large after κ_1. The value $\gamma_1^N \approx 243(K_E + 648)$ at $d = 4$, $L = 3$ rests on unprinted constants and is not proved.

The positivity device of R-009. Only $H(K) = \operatorname{sym}\operatorname{Re} K$ enters the modulus of the Gaussian measure, where $K = C^{(k)}(Z_0, \sigma)^{-1}$. Bałaban states (without proof) that $C^{(k)}(Z_0, 0)$ is symmetric. We do not prove this claim: the parametrix walk series is not reversal-symmetric, and the truncation need not keep symmetry. The claim is also not needed. The majorant uses $H_0 = \operatorname{sym} C_0^{-1} \ge \frac{3}{4}\gamma_{\exp}$ in its place.

Theorem 5.80 (Positivity of the localized Gaussian measures (row R-009))**.** *Assume Theorem 5.78(i)–(iii), with its residue (row R-003), and the five readings I-a–I-e of `SOURCES_A8_POS.md`: quadratic forms in place of symmetric operators; the σ-weighted walk sum by the rule of CMP 116 p. 3; three geometry facts; $W < \infty$ at rate $\varepsilon''/8$; and "$\operatorname{Re} C^{-1}$" as the symmetric real part. Let M satisfy the M-large conditions of the order of constants (K-025). Then:*

(R) (Real case.) At $(U, 0)$ with $s \equiv 1$, $K = A_Z$ is real symmetric (L-266 (Lemma S)) with $\gamma_{\exp} \le A_Z \le \Gamma_1$, so $d\mu_{C^{(k)}(Z_0)}$ is a positive Gaussian measure.

(D) (Lemma D.) For $|\sigma| \le e^{\kappa_1}$ the σ-truncation obeys $\|C^{(k)}(Z_0,\sigma) - A_Z^{-1}\| \le \eta_M = (16/\gamma_{\exp})e^{2\cdot 3^d \kappa_1} W e^{-\varepsilon'' M/8}$.

(P) (Lemma P (L-242).) The complex Gaussian exists. Its modulus is $r_\sigma\, d\mu_{\Gamma_\sigma}$ with $1/(2\Gamma_1'') \le \Gamma_\sigma \le 4/\gamma_{\exp}$, which is Bałaban's (2.15).

(M) (Majorant.) With $H_0 = \operatorname{sym} C_0^{-1} \ge \frac{3}{4}\gamma_{\exp}$ the majorant (2.23)–(2.25) holds at a cost $\exp(O(\eta_M^2)|Z_0|)$. The extra term $O(\eta_M^2)\|ZX\|^2$ in (2.24) is a cost, and (2.25) absorbs it.

Lemma G3 (Theorem 5.91) reads R_2 of CMP 116 (2.16) as $H_0 - H_\sigma$.

Remark 5.81 (CMP 116 p. 8). Bałaban writes that the lower-order bracket terms are "bounded similarly as in (1.24)" (CMP 116 p. 8). For the raw Taylor terms of order $n \le 4$ this is false as a bound: the cube count $(6L)^4(L^j\eta)^{-4}$ leaves a sum over j that diverges in k. The sentence holds for the terms after the transformations of CMP 109 Sect. 4. Theorem 5.82 states this form, which we call R-011.

Theorem 5.82 (Lower-order bracket terms (row R-011))**.** *Assume the residue R1–R4: R1, the unprinted constants are finite and fixed before ε_1; R2, the R- and C-terms obey the same mechanism as the F-terms; R3, Bałaban's bounds of CMP 109 Sects. 4–5 as printed; R4, the reading of the tail constant of (4.5). Then, after the transformations of CMP 109 Sect. 4: the order-0 term is removed by the $t_\square$-derivative; the order-1 term is zero by (4.14); the marginal parts of (4.34) and (5.43) cancel exactly against the counterterm, because β_j is defined as the coefficient β in (5.43) (p. 298). Every other term obeys the (1.24) form with the factor $8B_0C_1e^{16\kappa_1}\alpha_2^{-1}g_k|B|$ and field power 5, or $4 + \beta_H$. The other terms satisfy (1.29): the F-terms with exponent $(1-2\delta)\kappa$, and the R- and C-terms by the same mechanism.*

Here (1.29) reads $O(M^q)$ and $(1/16)\kappa_1$. The R- and C-terms (item R2) are not written out. The per-label exponent of Fact A (L-050) in Theorem SM (L-281) is the Hölder exponent β_H of CMP 109 (3.31). Bałaban does not print β_H, so it needs its own symbol. The θ of L-186 (Lemma HH) has no numerical value, so $\theta < 1 - L^{-\beta_H}$ holds by ε_1 small.

Proposition 5.83 (The free exponent β_H)**.** *Every consumer of Theorem 5.82 needs only $\beta_H > 0$, chosen before ε_1 and γ. Each of its conditions is an ε_1- or γ-smallness condition. So $\beta_H \in (0,1)$ with $\beta_H \le \beta_0 < 1$ is a free parameter, first in the order of constants (ledger K-028). The consumers take the following forms: Fact A has $\alpha_* = \beta_H$; (Eps) reads $2\varepsilon' = 2\theta_0(1 - L^{-\beta_H})^{-1}$; L-189 (Lemma HH) needs $\theta < 1 - L^{-\beta_H}$, with column sum $\le C_\beta + 2C_\Pi\theta_0 S_0(1 - L^{-\beta_H})^{-1}$; and $\lambda_g = L^{-\beta_H/2} + \theta_g\gamma$ in §4.3 of `SOURCES_SM.md`.*

The cost is the factors $(1 - L^{-\beta_H})^{-1}$, a larger $B_4(\beta_0)$, and K_T fixed before ε_1. At $L = 13$ the (G-gain) (H-111) margin $1 - 13^{-\beta_H/2}$ is 0.120 at $\beta_H = 0.1$ and 0.685 at $\beta_H = 0.9$. The L-186 (Lemma HH) margin is 0.901 at $\beta_H = 0.9$.

Convention 5.84 (the exponent β_0 in CMP 102 Thm 1 (9))**.** We read "$\beta_0 = 1$" in CMP 102 Thm 1 (9) as a misprint, and we take $\beta_0 < 1$. The competing reading takes $\beta_0 = 1$ literally. This β_0 is not the β_0 of CMP 119 p. 255.

Proposition 5.85 (The other constants of CMP 119 p. 256 (row R-012))**.** *The "other constants" of CMP 119 p. 256 are further scale-indexed numbers $X_j = f(g_j)$: $\alpha_{0,j}$, $\alpha_{1,j}$ ((2.28)), $g_j^{\kappa_0}$ ((2.31)), $\delta_k = (A_1/A_0)\varepsilon_k$ (p. 265), $p_1(g_k)$ ((3.43)), and $g_j(x)$ ((2.24)). Every similar inequality for them follows from (2.6)–(2.9). The three unprinted cross-scale steps ($R_k \le LR_{k+1}$ on p. 268, the monotonicity of k_1 on p. 270, and $\varepsilon_{k+1} \le (1 + \beta_0)\varepsilon_k$) also follow from (2.9). The side conditions H-255–H-265 are admissible and acyclic.*

Proof (sketch). The list rests on 43 points of introduction on CMP 119 pp. 255–284. The step for ε_{k+1} uses monotonicity to control the factor $(1 + \beta_0)^2$. Three entries join the order of constants: K-029 ($M_1 < M_2 < M$, order $M_1 \to M_2 \to M$), K-030 ($M_2 \ge 1/\delta$), and K-031 ($A_1/A_0 \le 1/(80B_3)$, A_1 after A_0). The condition $r > 1$ holds by "$r \ge 2$" (CMP 119 p. 245). □

The list does not depend on the form of $p_1(g)$ inherited from CMP 122 I, on the CMP 109 restrictions outside CMP 119, or on a print defect of p. 278.

The whole-lattice coefficient (R-015). $\beta_{j+1}(g_j)$ is defined by CMP 109 (1.20)–(1.22), p. 264. It is the mixed second moment $\sum_x \Pi_{\mu\nu}(x)x_\mu x_\nu$ of the vacuum-polarization tensor at $U = 1$, taken as the limit of periodic-torus kernels $T \nearrow \mathbb{Z}^d$. Bałaban states (without proof) that this limit exists "by the localized representation (1.7)". (1.7) and (1.9) say only that each term depends on U restricted to X. They do not say that each term is the same function on every torus and on $\mathbb{Z}^d$. So the existence of the limit needs an extra hypothesis, (Loc) (H-141) below.

Remark 5.86 (CMP 119 p. 253). The citation "(1.41)–(1.43)" on CMP 119 p. 253 for the definition of β_j is a slip; the definition is CMP 109 (1.20)–(1.22).

Proposition 5.87 (Whole-lattice limit of β_j (Prop. A15-N))**.** *Fix j, $g \in [0, \gamma]$, and $U = 1$. Assume (Loc): for every non-wrapping polymer X the torus term equals the $\mathbb{Z}^d$ term of its lift; (Bd): CMP 109 (1.18); (H): the linearized background map, including the J channel, decays exponentially, at the same rate on the torus and on $\mathbb{Z}^d$; and (Cnt): the standard polymer count. Let N_j be the number*

of level-j cells per side and $m_j = \frac{1}{2}\min(\delta_H, \kappa - \kappa_0)$, up to a fixed factor. Then the limit in (1.22) exists, and for the torus coefficient $\beta_j^{(N)}$ in moment form

$$|\beta_j^{(N)}(g) - \beta_j(g)| \le C_j N_j^6 e^{-m_j N_j/2},$$

with C_j independent of N. The Fourier form adds $O(N_j^{-2})$.

Proof (sketch). The first-order term drops by gauge invariance. The non-wrapping terms are equal term by term under (Loc). The wrapping polymers and the image terms are bounded by (Bd), (H), and (Cnt); this gives the exponential error. The power N_j^6 is not sharp. □

Exact equality $\beta_j^{(N)} = \beta_j$ cannot be derived in general. Image terms and wrapping polymers are real error sources. No explicit N with $\beta_j^{(N)} \neq \beta_j$ is known, and Bałaban never claims coefficient equality. Equality holds term by term only for non-wrapping interior X under (Loc). The hypothesis (Loc) does not follow from (L2) of Theorem 5.77, which covers walk factors only. The case $g = 0$ is Proposition 5.6 (ledger R-030). Theorem 2w (L-001) uses only the definition of β_j (through R-029, R-030, and R-006), so it does not use R-015. Bałaban's assertion that the $\mathbb{Z}^d$ limit of (1.22) exists belongs to row R-006. The implication Theorem 2w ⇒ Theorem 1 and Theorem B use R-015 through R-032 and R-013: (Loc) for $g > 0$, the reading of $\partial\Lambda_{k+1}$ on CMP 119 p. 278, and the R-term part, which cannot hold literally for terms built from merged R′ inputs.

The reduction of R-006 and Lemma G3.

Proposition 5.88 (Reduction of R-006 (Proposition H0-red, L-183))**.** *The use of R-006 in Theorem 2w, item 7, reduces to a residue N1–N9 in which no constant depends on j or on the past. The residue item N2 is Lemma G3. The reduction does not use Bałaban's C^∞ and all-derivative claims.*

Lemma 5.89 (Bound on G_Γ)**.** *Let $\Gamma_k(Z_0, 0) = 1_{Z_0} A_{\exp}(0) 1_{Z_0^c} R(0)$ with $R(0) = (C^{(k)})^{1/2}(0)$, and assume Convention 5.90. Then $G_\Gamma = \|\Gamma_k(Z_0, 0)\| \le \bar{G} := \Gamma_1 K_{1/2}$ with $\Gamma_1 = 2B_A S(\delta_A)$ and $K_{1/2} = 8\gamma_{\exp}^{-1/2} S(\frac{1}{2}\varepsilon) + 4\gamma_1^{-1/2} S((1-\theta')\delta_A)$. The bound is uniform in M, the volume, j, Z_0, and regular U.*

Proof (sketch). A Schur test on the walk expansions of $A_{\exp}(0)$ and $R(0)$. □

Convention 5.90 (the s-rule on the resolvent expansion)**.** The s-rule acts on Bałaban's resolvent expansion (3.185) of [13] (CMP 116 p. 13). We identify $R(0)$ with the walk sum of Theorem 5.78(iv) with the σ_0-walks deleted (the reading I-b of Theorem 5.80).

Theorem 5.91 (Lemma G3 (N2))**.** *Assume the two-sided form of Theorem 5.79, Theorem 5.80, and Bałaban's chain CMP 116 (2.15)–(2.25) read as pointwise exponent bounds on the closed small-field set (including R-003). Assume Convention 5.90, the residue items N3 (the Kotecký–Preiss derivative step, the margin (M_r) of R-017 with K-009, dominated differentiation), N6 (Bałaban's sup induction), and N9 (R-015, definition only). Assume the conditions*

(Hα) $(\alpha_5 + G_\Gamma^2 \delta_S)\Lambda_0 \le \frac{1}{2}$, with $\delta_S = O(\eta_M^2)$ (ledger K-026);

(Hγ) $\frac{1}{20}\gamma_2 \varepsilon_1^2/\gamma^2 \ge 4\kappa + 1$, and $\frac{1}{20}x - 4M^4 e^{-x/20} \ge 0$ at $x = \gamma_2\varepsilon_1^2/g_k^2$ (ledger K-027).

Then (Q) holds in the joint (X, B) form with

$$c_- = (\gamma_1^N + \varepsilon_0)^{-1} \geq \tfrac{4}{5\gamma_1^N}, \qquad \Lambda = 2\Big(1 + \tfrac{4}{3\gamma_{\exp}} + \tfrac{16}{9\gamma_{\exp}^2} G_\Gamma^2\Big),$$

$\varepsilon_0 = 2\Gamma_1^2 \eta_M^0 \leq \frac{1}{4}\gamma_{\exp}$*, and Lemma G3 holds: the cut adds* S_χ *to* S_0*, and* $C_\beta = C_\Pi(S_0^{\mathrm{GGAIN}} + S_\chi)$*, uniformly in* j*, the past, the volume, and regular* U*.*

Proof (sketch). The B-block of Q_0^{-1} is $H_0^{-1} + H_0^{-1}\Gamma\Gamma^* H_0^{-1}$, so G_Γ enters Λ. The proof uses no symmetry of A_Z and no L-207 (Theorem L). The payment of $e^{|P|}$ comes from the fifth factor $e^{-(1/10)x|P|}$, through (Hγ); this is the reason for the second condition in (Hγ), which is used for (2.31). □

The one-bond-fixed form of (Q) is a corollary: fixing $B(b)$ integrates the compression exactly, with Schur complement $(\Gamma_{bb})^{-1}$ and no cost factor ($a = 0$). The conditions (Hα) and (Hγ) are smallness conditions of Bałaban's type, one unit stronger than his own. Numerically, 2000 random instances satisfy the bounds. By Lemma 5.89, (Hα) is the single block-size threshold

$$M \geq \max\{M_\Gamma, M_{\alpha 5}\}, \quad M_\Gamma = \tfrac{4}{\varepsilon''} \log\big[16 h^3 \Gamma_1^4 \bar{G}^2 \bar{\Lambda}_0 (16W/\gamma_{\exp})^2\big], \quad M_{\alpha 5} = \tfrac{3}{\delta_0} \log(8 c_5 \bar{\Lambda}_0),$$

$h = 4/(3\gamma_{\exp})$, together with Bałaban's smallness of $\alpha_0, \alpha_1, \alpha_4, \gamma_2$. The constant κ_1 does not enter M_Γ, and the order of constants has no cycle.

Hypotheses. With Theorem 5.91, the use of R-006 in Theorem 2w is a reduction with residue N1–N9, and N2 is proved relative to the inputs above. This gives the small-field input (Q)/c_- of ledger R-037. R-006 stays a row: N5 (continuity at $g = 0$, a reduction), N6, N8 (readings), and N9 are not proved here, and N7 is R-031. In N6 the past-uniformity of E_0 is reduced, not proved.

5.10 Step 11: locality, the boundary, the 2D gap, and the order of constants

This step treats the Bałaban-level residue of row R-015, the R-term row R-032, and the 2D gap of row R-028. It also fixes one acyclic order of constants. Each result holds relative to the inputs that it names. Nothing in this step makes Theorem A or Theorem B unconditional.

Locality of torus activities. Work at level k on $T_N = \mathbb{Z}^4/N\mathbb{Z}^4$ in level-k units, with projection π. A set is *liftable* if it lies in $\pi(B)$ for a box $B \subset \mathbb{Z}^4$ of side $< N/2$. A domain X is *non-wrapping* if its collar $X^\sharp$ (X plus one layer of width c_k, the largest cube side at level k) is liftable; $\operatorname{diam}_\infty X \leq N/2 - 3c_k$ suffices. $\hat{X}$, $\hat{U}$, $\hat{J}$ denote lifts.

Convention 5.92 (free boundary conditions in CMP 109 p. 290)**.** We read "H_j with free boundary conditions" (CMP 109 p. 290) as the operator on $\mathbb{Z}^4$.

Theorem 5.93 (Theorem LOC)**.** *Assume (FI) (H-103): each walk factor is given by the same local formula on* T_N *and on* $\mathbb{Z}^4$ *(CMP 99 Sect. C); (U): the solutions of (1.3)–(1.4) are unique (CMP 102); Convention 5.92; and the Group C rows K-033–K-037: (Cube) (K-033), the reading that* $U_j(\square_0, \cdot)$ *of [14] uses only data on* $\square_0$*; (Cases) (K-034), Bałaban's statement (CMP 116 p. 2, without proof) that the other term classes are localized "in exactly the same way" as the typical term; (Conv) (K-035), cube sides divide* N*, the partitions are* π*-images of the* $\mathbb{Z}^4$ *partitions, and* $M = KR_0M_0$ *with* K *a power of* L*; (Det) (K-036), a domain-assignment rule for the closed-walk terms of the determinant expansion of CMP 102. Then for every* k*, every coupling history in Bałaban's range, and every non-wrapping* $X \in \mathcal{D}_{k+1}$*,*

$$E_{T_N}^{(k+1)}(X, g, U, J) = E_{\mathbb{Z}^4}^{(k+1)}(\hat{X}, g, \hat{U}, \hat{J}) \quad \textit{for all } (U, J) \in \mathcal{U}_{k+1}^c(X, \alpha_0, \alpha_1).$$

There is no error term.

Proof (sketch). The mechanism is the s-rule of CMP 116 p. 3. At $s = 0$ off Y_0, every walk-expanded operator is a finite walk sum inside Y_0, and the background equations (1.3)–(1.4) decouple by components. A bijection between the surviving torus walks and the $\mathbb{Z}^4$ walks gives equal kernels (Lemma W (L-305)). Uniqueness gives equal background fields (L-224 (Lemma N)). The cluster assembly and the subtractions are cube geometry (Lemmas A, S). In L-167 (Lemma G)(b), CMP 116 (2.7) is split as follows: for large x the resolvent is a power series in $C^*\Delta_k C$, and the integral with $\tilde{G}_3(x)$ covers $x \in [0, \gamma_1]$. The determinant expansion of CMP 102 does not follow the s-rule. It is a trace of walk expansions, and a trace splits exactly by walk under (Det). The block-local terms (a Jacobian and $\log \sigma$) are intrinsic by the same argument. The image terms of a periodized kernel sit in winding walks, so they enter only wrapping activities. The induction on k uses that β_{k+1} is the same $\mathbb{Z}^d$ number in both volumes; a torus coefficient would break exactness at the next step. □

With Theorem 5.93, Proposition 5.87 holds relative to (Bd), (H), (Cnt), and the hypotheses of Theorem 5.93; (Loc) (H-141) is then not a hypothesis. Bałaban's assertion that the $\mathbb{Z}^d$ limit of CMP 109 (1.22) exists at $g > 0$ follows from Proposition 5.87. So row R-006 contains no assertion of Bałaban that is used without proof.

The boundary of the small-field region.

Convention 5.94 (walk domains of the s-rule (K-038))**.** (Enl) The s-rule localization domain of a walk is $\bigcup X_i^5$ as a whole-lattice set. Bałaban prints neither form. The competing convention takes $\bigcup X_i^5 \cap \Lambda$; under it, part (f) of Proposition 5.95 holds with a margin of at most one layer of M-cubes. The s-truncation of the Δ_k walks follows the same convention.

Proposition 5.95 (Proposition BDRY)**.** *Assume the hypotheses of Theorem 5.93, Convention 5.94, (L2$_\Lambda$) (H-139) through (L2) of Theorem 5.76, (Dom), the reading (Conv$_\Lambda$) (H-091), and the reading that the Λ_j are nested.*

(f) The small-field terms $E_0^{(k+1)}(\Lambda_{k+1}, X, z)$ of CMP 119 p. 278 with $X \subset \Lambda_{k+1}$, also those with X next to $\partial\Lambda_{k+1}$, equal the whole-lattice terms exactly.

(Thr) Let N_k be the number of level-k cells per side of the torus. If $N_k > 22M$, every wrapping or image term of the coefficient analysis at step k lies in Bałaban's irrelevant class $X \not\subset \tilde{\square}^2$ (CMP 109 (4.36), CMP 119 (3.48)), and the torus correction to β_j is at most $\varepsilon(M) = C_j(22M)^6 e^{-11 m_j M}$, valid for $m_j M \geq 1$.

Proof (sketch). On the image of CMP 119 (1.20), $S_1 \cap \Lambda_1 = \emptyset$. The change of variables (1.22) acts only on S_1^* bonds. So H, the Jacobians, and $V(S)$ reach only terms that meet Λ^c, and these go to the large-field part. Step $k+1$ (p. 279) keeps every $X \subset \Lambda_{k+1}$. Lemma W$_\Lambda$: with $s = 0$ off $Y_0 \subset X \subset \Lambda$, every surviving walk of $C(\Lambda)$ lies inside Y_0 and does not see $\partial\Lambda$. Here $C^*\Delta_k C$ is walk-expanded, not of finite range (CMP 116 (2.7), CMP 119 p. 275). For (Thr), the local set is $\tilde{\square}^5$, of side $11M$, which gives the threshold $N_k > 22M$. □

The R-term row R-032. A finite-volume R-term increment cannot be compared with an infinite-volume one: the whole lattice of CMP 109 has no R-terms, and no infinite-volume large-field construction exists. CMP 119 p. 279 prints instead that the R″ terms with $X \subset \Lambda$ "do not depend on Λ_{k+1}". Claim L$_G$ (L-209) proves this: the step output on X is one map $\mathcal{G}_X$ of the inputs on X, the same in every volume.

Proposition 5.96 (Row R-032)**.** *Assume the hypotheses of Theorem 5.93 and Proposition 5.95, the F2 inputs ((W4) (ledger row R-044), (Eps), (D); ledger R-044), Bałaban's (1.18), (3.48), (4.36), and (Vol). Then: (a) β_j has no R part, because it is subtracted from E_{pure} (CMP 119 p. 279); (b) E_{pure} on a non-wrapping $X \subset \Lambda$ equals the $\mathbb{Z}^4$ term exactly; (c) E_R is uniform in K through F2, with the threshold $N_k > 22M$.*

Row R-032 in this form is an input of the implication Theorem 2w (L-001) ⇒ Theorem 1. It is not an input of Theorem 2w.

The volume hypothesis (Vol). Proposition 5.95 needs $N_k > 22M$ at every step $k \le K$. N_k is smallest at $k = K$, where $N_K = \ell$. So the torus must have side $\ell > 22M$ final-scale cells. This is (Vol) of Section 2.1 (ledger K-032). It is a condition, not a choice, and it is admissible.

Convention 5.97 (the exponent m in CMP 109)**.** CMP 109 prints the torus exponent (p. 251, $L_\mu = L^m$) and the cube exponent (p. 257, $M = L^m$) with the same letter m. We read this as reuse of a symbol, and we fix the two exponents m_T and m_M independently. The competing reading takes them equal; then $\ell = 2M$, and (Vol) fails at the last steps.

(Vol) does not depend on K, the proof of Theorem 2w does not use it, and it excludes only tori with $\ell \le 22M$.

The 2D gap ($\mathbf{G_{2D}}$). Let $A' = H' + D'D'^\dagger$ be the charged block of the SU(2) Wilson Hessian at constant flux f on the coarse $\mathbb{Z}^2$ lattice, in Landau gauge, as in §5 of `SOURCES_T2_S123.md`. Write $\alpha = ef$, so $|\alpha| = 2|f|$.

Theorem 5.98 (The 2D gap ($\mathrm{G_{2D}}$))**.** *Let $0 \le c < 1$. For every real f with $0 < |f| \le ((1-c)/100)^2/2$, the operator A' has no spectrum in $(-2c|f|, 2c|f|)$, and neither has H' restricted to $\ker D'^\dagger$. For example, $c = \frac{1}{2}$ gives a gap $|f|$ for $0 < |f| \le 1.25 \cdot 10^{-5}$. The constants do not depend on j, L, or the averaging.*

Proof (sketch). A lattice ladder operator $\Pi = \nabla_0 + i\,\mathrm{sgn}(\alpha)\nabla_1$ gives $X(X - 2|\alpha|) \ge -157|\alpha|^3$ on low states. The lattice commutator is tracked exactly through $2|\sin(\alpha/2)|$. So the scalar magnetic Laplacian $\mathfrak{L}$ has no spectrum in $[(1+\theta)|\alpha|, (3-\theta)|\alpha|]$ for $\theta > 26|\alpha|^{1/2} + 157|\alpha|$. Then $A' = \mathfrak{L} \otimes 1 - 2\alpha\sigma_y + R$ with $\|R\psi\| \le 48|\alpha|^{3/2}$ near 0, and a contradiction argument gives the gap. No invariant subspace is needed. The $\ker D'^\dagger$ form follows from $A'D' = D'\mathfrak{L}$. No Harper-band or Helffer–Sjöstrand result is used. The identification of the midpoint transport is proved. □

Remark 5.99 (the proved range is narrow). The proof covers only $|ef| \le 10^{-4}$. Numerically, the gap holds on all of $\alpha \in (0, 3.02]$, with physical ratio ≥ 0.95: at 11 rational fluxes with denominators up to 2213, the smallest |eigenvalue| of A' is $|\alpha|(1 - 0.125|\alpha|)$, and a sparse shift-invert computation at $\alpha = 10^{-4}$ gives the ratio 1.0000125. So the proved range is far from sharp, and the numerics are not a proof. Numerically, the Nielsen–Olesen mode is the band at $-|ef|(1 + 0.12|ef|)$, not a zero mode. The upper edge $3|ef|$ of H' is not proved and is not needed. Prop. 5.6 of `SOURCES_T2_S123.md` uses ($\mathrm{G_{2D}}$) for $|f| \le \min(f_0, f_1(j))$, where $f_1(j)$ comes from its size condition $g_f \ge 4\eta_1 + 4C_M^2|f|^3$. This f-range need not fit a fixed window, because (M_j) is a limit $f \to 0$; the proved range suffices (Section 5.11).

With Theorem 5.98, Prop. 5.6 and (Z_j) of Proposition 5.7 hold given the hypotheses of that proposition. Theorem 5.102 uses Theorem 5.98 to prove (M_j) (row R-028).

The canonical order of constants.

Proposition 5.100 (Canonical order of constants (ledger K-005))**.** *There is one acyclic order of all constants of Section 5. The level constants (rates, prefactors, Γ_n, k_n, ε'') are bounds on domains capped by a_1, a_{**}, and $M\alpha_0 \le a_0$. So they are free of M and of the α's, and they come before M_1. The domain sizes α''', α'', α' are smallness choices, and they come after M. The order contains the entries $\alpha_0 \le \alpha_0'/(CM)$ for real α_0 with $C > c_J$, and $M \ge 1/c_J$. The block size M has 32 lower bounds and no upper bound (ledger K-025), so the admissible set of M is an up-set.*

Proof (sketch). The orders printed in `WC_FULL.md` 6.9 and `A1_WALK_FULL.md` 9.1 contain two nominal cycles. Cycle A, on the M_1 side: $M_1 \to M \to \alpha''' \to \alpha'' \to B_{A''}, \delta_{A''} \to M_1$. Cycle B, on the M side: $M \to \alpha''' \to \alpha'' \to \alpha' \to \varepsilon'' \to M$. The split of level constants and domain sizes in the statement removes both. The J scale counts M once. By Lemma 5.89, G_Γ is bounded uniformly in M, so (Hα) is one more lower bound on M and adds no cycle. □

The plaquette row of Lemma M1 (L-220) is nested by a reading only (amendment E6 of `SOURCES_ORDER_CANON.md`).

5.11 Step 12: the implicit Group A rows, (M_j), L5, and the order of constants

This step treats the implicit or unprinted Group A rows R-016–R-019 and R-020, the Group B rows R-028, R-031, R-034, and R-041, the certificate of (H2) (ledger row R-025) at more values of L, and the order of constants. Each result holds relative to the inputs that it names. Nothing in this step makes Theorem A or Theorem B unconditional.

The fixed-level input (M_j) (ledger R-028).

Convention 5.101 (Moyal coefficients)**.** The coefficients $b_{t,m}$ of `KAPPA_BOUNDED.md` (1.1) are the Moyal coefficients of §3.1 of `SOURCES_T2_MJ_REST.md` at $\sigma = 0$.

Theorem 5.102 (The fixed-level input (M_j))**.** *Fix $j \ge 1$ and odd $L \ge 3$. Assume (G_{2D}) (Theorem 5.98), (I3) (Proposition 5.9), the results of Proposition 5.7, Prop. 2.2 of `U3_LEMMA51.md`, and Convention 5.101. Then the one-loop increment of level j at constant abelian flux f satisfies $\delta\Gamma_j(f) - \delta\Gamma_j(0) = n_2 d_j f^2 + o(f^2)$ as $f \to 0$, with d_j the formal coefficient of `KAPPA_BOUNDED.md` (1.1). The charged-sector error is $O(|f|^{9/4}) + o(1)f^2$. The neutral sector contributes its formal coefficient exactly in the limit.*

Proof (sketch). Region 1 of Prop. 4.4 of `BETA_FP_M.md` is not an inequality. It is an asymptotic identity: the f^2 coefficient of a spectrally localized trace equals the per-momentum formal integral. We replace the region split by a split of $\log|x|$ in three pieces with a shifted logarithm. The gapped piece F_s has shift $s = |f|^a$, $a = \frac{3}{4} \in (\frac{2}{3}, 1)$, so $f/s \to 0$ and a Moyal expansion has a small parameter (L-056 (Lemma A), with an exact two-factor trace identity and an approximate Moyal inverse to order $M = 14$). The Nielsen–Olesen window piece is the term II of Prop. 5.5 of `SOURCES_T2_S123.md`. The localized remainder is $O(s^3 + |f|^3)$. The neutral sector (N2) follows by an induction on bordered Hessians: $\pm\partial^p \Delta_i \le C_p \Delta_i(0)$ for $|f| \le f_*$. On a finite torus the d constant 1-forms are zero modes. They are not gauge directions, and they form a null set of the per-cell trace, so the infinite-lattice statement does not see them. The numerics of Section 6.8 agree with the exponent count. □

Only the $o(f^2)$ form is proved. The rate $O(\varepsilon|f|^3\log(1/|f|))$ printed in Prop. 4.4 of `BETA_FP_M.md` is not proved, and no consumer uses it. The f-range of Theorem 5.98 suffices. (M_j) is a limit $f\to 0$ at fixed j, so the gap is needed only on some punctured neighbourhood $0<|f|\le f_{\lim}$, and its size plays no role. The size condition $g_f \ge 4\eta_1 + 4C_M^2|f|^3$ of Prop. 5.6 of `SOURCES_T2_S123.md` has a left side linear in $|f|$ and a right side $O(f^2)$ ($\eta_1 = C_\eta f^2$), so it holds for all small $|f|$. The usable range is $0<|f|\le \min(f_0(c), f_1(j,L), f_*(j,L))$. The numerical gap up to $\alpha\approx 3$ (Remark 5.99) is supporting evidence only. So row R-028 holds relative to the hypotheses of Theorem 5.102. The chain from Proposition 5.7 through (M_j) to Prop. 1.1 of `KAPPA_BOUNDED.md` is complete relative to these hypotheses.

History dependence L5 (ledger R-031).

Convention 5.103 (causality of the step)**.** Bałaban's step $k+1$ uses only $g_0,\dots,g_k$. This is our reading of the construction; CMP 109 p. 298 defines β_j only on the trajectory.

Theorem 5.104 (Lemmas DT-nest and DS-nest)**.** *Assume item 7 on trajectories, the continuity (C-tr) (H-089) of one step in its inputs, and Convention 5.103. Then the conclusions (a-tail), (b), (c-pair) of Lemma 3.6, with the same constants $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, $g_1 = (\gamma^{-2}+2R)^{-1/2}$, hold for true (unclamped) trajectories, with no clamp. Every evaluation of β lies on a true trajectory with all $u_i \ge \gamma^{-2}$, inside Bałaban's domain. The same holds for Lemma 3.4 with its constants.*

Proof (sketch). We use nested intervals. The envelopes $p_k, q_k = t + T_0(k,K) \mp (b/4)(K-k)$ subtract $\beta_{k+1}(0)$ exactly. The slack shrinks by $b/4$ per step, which equals the worst single-step error $C_1\gamma$, so negative increments are absorbed. At each step the intermediate-value theorem selects a closed subinterval of u_0 on which u_{k+1} sweeps the next target interval. The window bound (5.1) is used at every endpoint $n \le K$. □

The proof of (C-tr) uses the single CMP 116 step, Theorem SM (L-281), §2–§3 of `SOURCES_HHOL.md`, Proposition 5.2 or R-015, and the enlarged input list of N6. It evaluates the step map at one off-trajectory mixed point per step. The alternative argument by continuity on the clamped cube needs a γ condition with a constant that is not defined, and we do not use it. The history form of N3 needs that alternative; the one-argument form of N3 does not.

The cut and the analyticity domain (ledgers R-017, R-019).

Convention 5.105 (the map C on the remaining bonds)**.** The map C is the identity on the bonds that the cut χ_k of CMP 109 (2.9) covers, and the norm on Y is the per-bond sup of CMP 99 (3.41).

Proposition 5.106 (The cut lies in the analyticity domain (row R-017))**.** *Assume Convention 5.105. On every bond b that the cut χ_k of CMP 109 (2.9) covers, $B'(b) = g_k B(b)$ exactly. Hence* $\operatorname{supp}\chi_k \subset \{g_k|B|_Y < \varepsilon_1\}$*, the analyticity domain of CMP 116 (1.34), with margin $r=1$ and no slack.*

Proof. CMP 109 (2.9) takes the product over $T^{(k)}\setminus\{b_0(c)\}$, and $hB = 0$ off the bonds $b_0(c)$ (p. 267). CMP 116 (2.3) writes the cut directly as $|B(b)| \ge \varepsilon_1/g_k$ on the same bonds. The bound (1.20) of CMP 116 gives only the converse inclusion. □

The option K-010 keeps the cut at ε_1 and runs CMP 116 §1 with domain constant ε_1/r, under the condition $4B_0C_1e^{16\kappa_1}\varepsilon_1 \le \frac14\alpha_2$. At $r=1$ the option K-009 is not needed.

Proposition 5.107 (CMP 102 (190) at complex B (row R-019))**.** *Assume three readings: the complex clauses of CMP 102 (189) and (73), which Bałaban does not print, and the identification of the set (172) with the ball $|B| < 2C_1\varepsilon_1$. Then the bound (190) of CMP 102 holds at complex B on this ball. Without the ball reading, the two-constants theorem gives the bound on $|B| < \frac{1}{2}C_1\varepsilon_1$ with harmonic measure $\omega = (4/\pi)\arctan\frac{1}{3} = 0.4097$, so the decay rate is multiplied by* 0.59*.*

Four of the five named inputs carry a printed complex clause (Prop. 4, Prop. 6, (55), (80) of CMP 102); (130) is linear. The fallback still needs a complex sup bound on kernel entries, from (187) and (55). The Cauchy step of §4.4 of `SOURCES_MA2.md` reaches the boundary of the open ball; a limit $\rho \uparrow C_1\varepsilon_1$ keeps its constant c_N/C_1.

The regular class (ledger R-018).

Proposition 5.108 (The regular class (row R-018))**.** *Assume the printed results CMP 102 Thm 1, CMP 98 Prop. 2 and (139), CMP 99 (3.133), and CMP 109 p. 263; the readings I2 and I3; the reading that $U_{k,Z}$ obeys CMP 102 Thm 1; the reading that B and B_5 are fixed numbers of (d, L); and the conditions (A18-1) (H-063)–H-066 (A18-4) (ledger K-042), with $M \geq 5M_\square^{99}$ $(10 \mid C^{122}M)$. Then: (i) the background domains of CMP 109, 116, and 122 lie in the regular class of CMP 99 (3.35)–(3.36), with the unnamed cube constant $c_\square^{109}$ of CMP 109 as inclusion constant; (ii) Lemma Q (L-250): $Q_k(U) = Q_{\text{free}} + O(\alpha)$ in a per-row axial gauge, with sum $c\alpha L/(L-2)$, uniformly in k for $L \geq 3$, and $O(M\alpha_0)$ at operator level; (iii) L-109 (Lemma C): $\|C(U)\|^2 \leq c_C(1)(1 + O(\alpha))$; (iv) $c_J = O(\alpha_0)$, with no factor M; (v) σ-localized operators add nothing.*

Proof (sketch). Every explicit use of (3.36) in CMP 99 goes through the current J, so a bound on J replaces (3.36) up to $O((M\alpha_0)^2)$. Lemma Q telescopes $Q_k(U) - Q^k$ over levels on CMP 98 (124)–(125), (139)–(146); the term of level j carries a factor 2^{k-j-1}. L-109 (Lemma C) uses the printed definition of $b_0(c)$: each row of the linear one-step map contains exactly one eliminated variable. Part (i) needs two factors L in the inclusion step for (d). For the cube patching we use CMP 99 at cube scale $M_\square^{99} = CM/10$. □

The item (C2) of `SOURCES_MA2.md` is part of this row.

The large-field T-step rate loss (ledger R-020).

Proposition 5.109 (Rate loss of the large-field T step (row R-020))**.** *Assume the count R1 and Bałaban's statement (without proof) R3 that $H'(Z_0)$ obeys CMP 116 (2.38). Then in one large-field T step the decay rate κ loses at most the fraction 13δ: 10δ printed for the small-field step (CMP 116 (2.28)–(2.41)) and at most 3δ unprinted for the Z_0-marked cluster expansion of CMP 119 (3.44)–(3.47). No loss inside T is an additive constant in κ.*

Proof (sketch). "CMP 109 Sects. 6–7" is CMP 116 Sects. 1–2 in a preprint numbering, so the two deferral targets of row R-020 are one printed source. Each merging step costs a δ-fraction of the current rate: one δ for a sum over pieces, and one for a volume factor by (2.30). The tree bound is printed as CMP 116 (2.27). A finer count gives 2δ for the unprinted part, so 13δ is an upper bound. □

For ledger R-046 this gives the condition (K1_δ) (H-216) $\delta \leq \frac{1}{13}(1-2(1+4\beta)/(\theta_K L))$ with $\theta_K \geq \frac{1}{2}$ and $\kappa \geq 2m_{\text{tot}}$; it replaces (K1) (H-215). At $L = 13$, $\beta = \frac{1}{4}$ this is $\delta \leq 5/169 \approx 0.0296$, so we choose $\delta = 0.025$ (0.03 fails; Lean `K1_numeric_13` in `lean/YmLean/K1Prime.lean`). The conditions (K2) (H-217), (K3) (H-218), κ_{eff}, and $\kappa \geq 1536$ are unchanged. Proposition 5.19 uses this form.

The propagator replacement (ledger R-016).

Theorem 5.110 (The sourced covariance (row R-016))**.** *Assume Theorem 5.78 (with Proposition W_N and §6 of `WC_FULL.md`), Theorem 5.77 (L-115 (Lemma C), WG-G1, 5.2, 5.5), Theorem 5.76 for G_1 (which rests on (C) (H-057)), Theorem 5.80, Lemma 5.89, (2.a) of Proposition 5.12, §1.1 and §1.4 of Proposition 5.16, and the tree conditions K-041. Assume also the readings H-035 (M1), the completeness of the property list of CMP 116 pp. 12–17; (S_H) (H-166), the scale normalization of D_S, J_S; the volume convention of the Ω-setting of CMP 122; and the choice of the s-version for $(C_f)^{1/2}$. Then the sourced covariance $C_f = (A_{\exp} - g^2\mathcal{W}_S)^{-1}$, $\mathcal{W}_S = C^*\mathcal{V}_S C$, has the properties H-281–H-291 that CMP 116 uses for $C^{(k)}$, and it supports the uses (U-a) (H-292) whole-volume C_f and $(C_f)^{1/2}$, (U-b) (H-293) σ-differences of $(C_f)^{1/2}$ and Δ^f, and (U-c) (H-294) the tree form of Δ^f.*

Proof (sketch). The sentence of CMP 116 p. 3 licenses only the replacement in (1.3)–(1.4), so we do not cite it for C_f. We prove the properties directly. $\mathcal{W}_S$ is not of finite range, because $H_1 = G_1Q^*(QG_1Q^*)^{-1}$ is nonlocal. We write $\mathcal{W}_S$ as a tree sum of the type of Proposition W_N (L-312) of `WC_FULL.md`, on the complex chain of §6 there, with at most 12 junctions per term and $K_W^T = 2C_{\mathcal{D}}(C'_{\mathcal{D}})^{12}B_{\mathcal{W}}$. The conditions (A16-a^T) (H-060) and (A16-b^T) (H-062) are linear in g^2r. In (U-b), R_2 is anchored at both ends in Z_0, and the product form needs concatenation at the level of elementary blocks; this does not change the order of constants. □

The completeness item H-035 (M1) is treated in Section 5.13. Euclidean covariance of C_f is false for non-constant f, and no consumer may use it.

The large-field holomorphy chain (ledger R-041, chain item 5).

Proposition 5.111 (The items (C1), (C2) of (H−))**.** *(C1) Assume R-019 (Proposition 5.107) and R-018 (Proposition 5.108). Every condition on the ε_1 and C_1 of CMP 102 is an upper bound (and $C_1 \geq 1$), so ε_1 can be raised per scale to give room for the Cauchy step. The four cases K1–K4 of Proposition 5.107 give radii $\kappa_H \in \{2, \frac{1}{2}, \frac{1}{2}, \frac{1}{8}\}$ and constants c_N/C_1 to $16c'_N/C_1$, all free of C_1. The price is the condition (C1*) $c_m M_0\, g_k p_0(g_k) \leq \frac{1}{2}\kappa_H C_1^{102} a_*^{102}$ (ledger K-039), with M_0 placed in the order of constants before γ (K-040). (C2) Assume the readings that the averaging map $m = M^\bullet \circ Q_k^{s*}$ is local and analytic, so that V' always enters through m, and that $R_1 < L$ suffices for condition (1) of CMP 102. Then (C2) reduces to R-018.*

Here the map of (H−) is CMP 122 I (1.74), and "~ 4" means four layers of M-cubes. So chain item 5 reads: R-041 ⇐ (M-A) (H-143) ⇐ (H−) ⇐ R-005 + (K1–K4) (ledger row R-019) + R-018 + (C1*). R-041 is a reduction.

The sweep of CMP 119 (2.6)–(2.9) and N3 in the DT form (ledger R-034).

Proposition 5.112 (Row R-034)**.** *(i) Each of the 19 rows of the sweep of CMP 119 (2.6)–(2.9) holds on the pages. CMP 122 I–II Sect. 1 contains about ten further cross-scale uses and five exponent or threshold conditions, listed as H-271–H-280 and H-266–H-270 (ledger K-043). Each is a (P-up) (H-156) window, a (P-low) (H-155) window with a factor $(1+\beta_0)^a \operatorname{poly}(n-m)$, or a single-scale γ-smallness threshold. The form $p_1(g) = (\log g^{-2})^{p_1}$, $p_1 < p_0$, is printed on CMP 122 I p. 183.*

(ii) Assume (S1) (H-244)–H-247 (S3). Then Prop. 4.3 of `BETA_FP_M.md` holds for $j \geq j_1$. Below j_1 it needs the open bound (S3w) (H-248), which Theorem 2w (L-001) does not use.

(iii) The assembly of Prop. 3.6 of `BETA_FP_M.md` *holds.*

(iv) N3 in the DT form. Lemma HR (L-192) (real base) takes J as the component of 0 in $\{c \in [-2r, 2r] : u_k(c) > \gamma^{-2}\ \forall k \le K\}$ and gives $S_3 \le (D'/\beta + 1)\gamma^3 + 2\gamma/\beta$. N3-C (complex) splits the bracket at $\beta/2$, gives $\operatorname{Re} u_i \ge \gamma^{-2} + \max(0, (\beta/2)(K - i) - D')$, and has slack 6.7 at the radius of Proposition 5.30, with no extra $+\eta$.

The history form of N3 needs the clamped-cube argument after Theorem 5.104.

The order of constants, section F. Section F of `SOURCES_ORDER_CANON.md` adds 28 condition entries (lines N-260, N-300, N-320, N-380, N-460) to the order of Proposition 5.100. It renames clashing symbols as a_*^{102}, C_1^{102}, ε_1^{102}, κ_H, θ_H, θ_K, $M_\square^{99}$, and others. The extended order is acyclic. (K1$_\delta$) uses the bound $\theta_K \ge \frac{1}{2}$, so δ is a constant of the order (line N-060), and $\kappa \ge 2m_{\mathrm{tot}}$ is on line N-070. The cube cap of CMP 102 Thm 1 is a condition on ε_1^{102} and ε_0^{109}, not on M, so M has lower bounds only. Four constants are two-sided, and each has an existence condition: p_1 (exists under the N-410 condition on p_0); ε_0^{102} (nonempty by CMP 102 Thm 1); ε_1^{102} per scale (exists if and only if (C1*) holds and $\varepsilon_1^{(0)} \le a_*^{102}$); and N per scale (exists for γ small; $N = R_j$ is admissible). The budget α_5 is shared between (Hα) and the entries of line N-490. The constant C_1 of (K3) (called C_1^{FX}) and the bound $\kappa \ge 1536$ are not in this order.

The lattice log coefficient (ledger R-027).

Proposition 5.113 (Reduction of Step B of (F))**.** *Theorem 3.1 of* `BETA_FP_H.md` *uses (F) (ledger row R-027) only through (3.1). The relation (3.1) is equivalent to $G(\varphi) - G(\lambda\varphi) \to n_2 c_2 \ln(1/\lambda)$ as $\varphi \to 0$, for $G = F_W(\varphi)/\varphi^2$ and $\lambda = L^{-2k}$. Assume (L-LL) (H-137): the limit of $H_\varphi(\mu) = \varphi^{-2}[W(\varphi; 0) - W(\varphi; \mu|\varphi|)]$ as $\varphi \to 0$ exists (its value is not needed; it contains the Nielsen–Olesen zero). Assume also a gauge-covariant kernel bound (Lemma 1, a standard bound that we cite without proof) and the gauge-fixing part of Step C. Then (3.1) holds. The ghost part of Step C is proved.*

The continuum side is X-005. (L-LL) is Theorem 5.133 (Section 5.12), and (F) itself is Theorem 5.134.

Remark 5.114 (the single-step coefficient β_1). Numerically, the single-step values β_1 at $L = 5, 7, 9, 11$ do not follow the slope $c \log L$, $c = 0.3715$ (Section 6.1). The question is whether 0.3715 is compared with the right quantity (the single-step β_1 against the fixed-point β_∞), or whether the difference sits in the coarse bracket, in the dependence on the family, or in the $N_g = 8$ grid at $L \ge 9$. A possible approach: write $\beta_1(L) - \beta_1(L') = c \log(L/L') - [B_c(L) - B_c(L')]$, with $B_c(L)$ the constant of the one-step coarse bracket (Proposition 3.1 of `SOURCES_BETA1_F_TENSION.md`). In this form the data measure $B_c(L)$, and they neither refute nor support R-027 (Section 6.9). The convergence of $B_c(L)$ is not proved.

(H2) at $L = 15, 17, 21$ (ledger R-025). The certificate of Theorem 6.2 runs unchanged at $L = 15, 17, 21$ and certifies (H2) there (Section 6.8). The targets use only $L = 13$ (K-003).

5.12 Step 13: the sub-items of (H-thr), (W4) and (S), the condition (C), and the lemma (L-LL)

This step treats sub-items of rows R-039, R-044, R-045, and R-027, the page readings R-047, R-048, R-049, R-051, and the condition (C) (H-057). None of these results removes a hypothesis of Theorem A or Theorem B.

The threshold mismatch (H-thr) (ledger row R-039). The input list M2b, M4, M5 of Proposition 5.47 can be replaced as follows.

Proposition 5.115 (the inputs M2b, M4, M5 of (H-thr))**.** *Assume the frozen-threshold bounds ($R1_{\mathrm{f}}$) (H-241) of Proposition 5.116. Then: (i) M2b holds: the estimate of CMP 122 II p. 383 is pointwise in B, so the one-bond factor gives a Schur-complement density bound, after division by the bond-b Haar factor $c_H = O(1)$ and with the density $\sigma(g_k B)$ included; (ii) M4 holds with $P_D = np_1 + dr_0$, with the term $s^{-1/2}$ included; (iii) M5 is not needed: the mismatch splits into a weight part W_j and a threshold part T_j, and W_j is bounded by a directional Cauchy estimate at thresholds frozen at a real s', with an extra factor $30r\gamma^2$; the bound (MT2) holds for the split. The resulting bound on Δ'_j is summable along the trajectory, and the Gronwall step of Lemma HG applies. The hypothesis (H-hist2) (ledger R-038) is not an input of R-039.*

Here ($\mathrm{R1_f}$) consists of the sup and small-difference bounds of Proposition 5.32 at thresholds frozen at a real $s' \neq x^*$ within $3r$ of x^*.

Proposition 5.116 (frozen-threshold bounds ($\mathrm{R1_f}$), H-241)**.** *Assume (M-A) (H-143)–(M-C) (H-146), the bound (R2), L-285 (Lemma T) of Proposition 5.47, and Conventions 5.117 and 5.118. On the box $S = \{|s'_i - \bar{u}_i| \le 0.1\bar{u}_i\}$ the holomorphy and sup bounds of (H-hol) (ledger row R-041) in the large-field layer hold at thresholds frozen at s', with $\gamma_2, c'' \to 0.895\times$, $\gamma_0 \to 0.73\gamma_0$, $(C_0, \dots, C_4) \to 1.06\,(C_0, \dots, C_4)$, and one more lowering of γ. The difference between the objects frozen at s' and at x^* has the tail form $\le C_T \sum_i \lambda_C^{j-1-i} (\log \bar{u}_i)^{dr_0} e^{-\kappa_3 (\log \bar{u}_i)^q}$, with $\kappa_3 \ge 0.734\kappa_2$ on S and $\kappa_3 \ge 0.80\kappa_2$ on the $3r$-set of R-039.*

Proof (sketch). The threshold s' enters only through the cut radii $\delta(s') = s'^{-1/2} A (\log s')^p$ and the split that they induce. The vertex domains stay frozen at x^*. On S the cut ratios lie in $e^{\pm 1/18}$ and the large-field exponents in $e^{\pm 1/9}$. Besides the inclusion C3, three conditions of Proposition 5.43 compare a cut radius with a frozen domain radius. The joint scaling $(C_0, \dots, C_4) \to 1.06(C_0, \dots, C_4)$ restores each of them. The proofs of Propositions 5.31 and 5.32 then run uniformly on S. The tail form suffices for the weight part W_j of R-039. □

Convention 5.117 (the constants C_2–C_4 of CMP 119 (2.28))**.** We read the constants C_2, C_3, C_4 as free to be increased.

Convention 5.118 (the R-ratio cutoff)**.** The R-ratio cutoff in the numerator and the denominator of `SOURCES_HHOL_LF.md` 3.4 is frozen at x^*. The competing reading lets the cutoff move with s'; then the ratio telescopes only under a lower bound on the denominator, which we do not prove.

Remark 5.119. The inter-step cut-nesting slack is not checked on the full box S. On the $3r$-set of R-039 the thresholds move by $e^{\pm O(r\gamma^2)}$, so this point does not enter R-039. The hypotheses of R-039 are then: the hypotheses of Proposition 5.116; (R2)–(R4); the large-field coarea Jacobian (`SOURCES_G3_N2.md` 2.6); and the convention K-001 (integers and partitions frozen at x^*).

The R operation and the domains Λ_j (ledger R-044).

Proposition 5.120 (holomorphy of the R operation, (W4)-R, ledger row R-044)**.** *Assume W4-G (the holomorphy of $V(Y)$ in the inputs, for which Bałaban gives a sketch of the preparatory integrations in CMP 122 I Sect. 1) and the Ursell bound (7.13) (Convention 5.121). Make the changes $\varepsilon_1 \to \varepsilon_1/3$ in (C1)–(C4), $\alpha \to c'\alpha$, and $A_0 \to \sqrt{c'} A_0$. Then the output of Bałaban's R operation is holomorphic in the input activities on the set $3\times$(inputs), which contains the $2E_0$ ball, and it obeys $|R'(X)| \le O(1) c'_1 e^{-(1+\frac{1}{2}\beta)\kappa d}$. When $q_0 = p_0$, the constants C_0, C_1 increase with A_0.*

Proof (sketch). The denominators of CMP 122 II (1.1)–(1.2) and CMP 122 I (1.100) contain only the Wilson action. They do not depend on the inputs. The identity (1.10) turns the division into a product with an input-free exponential, and the final R operation (1.71)–(1.72), (1.98) divides by nothing. The inputs enter affinely. $T'_k(X)$ is holomorphic on the complex domain $\mathfrak{A}_c$, by domination under a positive input-free measure. This domination bound replaces Bałaban's positivity step (1.73), which does not apply at complex inputs. The bracket is a finite sum of products and real t-integrals, and the logarithm in (1.98) is the Ursell series, which converges under (1.97) with changed constants. □

Convention 5.121 (the Ursell bound)**.** We read the Ursell bound (7.13) of `SOURCES_B20_W4R.md` as the bound of CMP 116 (2.13).

Remark 5.122. The positivity display used in the line check of Proposition 5.16 is CMP 122 II (1.73) on p. 380. Its item L5 needs $A_0, A_1 \to 2A_0, 2A_1$. A fixed shrink of g cannot pay a factor 4 in $\gamma_0 p_0^2$, because $p_0(cg)/p_0(g) \to 1$.

Proposition 5.123 (the domains Λ_j are summation indices)**.** *In Bałaban's expansion, Λ_j is a summation index and not a domain that moves with the field. The expansion is a finite sum over region sequences outside the T operations (CMP 119 (1.25), (1.28); the sum in (1.25) runs over R_1, and the second factor sits on $R_1^c \cap \Lambda_1^c$), and each term has fixed regions. Hence (W4) (ledger row R-044) holds at fixed regions on a fixed finite slot space, and (P8) (H-240) holds for each Λ with $p_0(g_j) \geq 16/(\gamma_0 A_1^2)$. The term $E_R(X, 1; f)$ on $X \subset \Lambda_m$ is booked into e_k^f.*

Proof (sketch). The proof reads CMP 119 pp. 252–253 and CMP 122 I pp. 182–183. Several cutoff factors sit inside the integral. They are sharp and free of inputs and sources, so they do not change the bounds. □

So the hypotheses that R-044 adds beyond printed results are W4-G and the Ursell bound of Convention 5.121.

The (E14-c) (H-098) clause (ledger R-045).

Proposition 5.124 (the (E14-c) clause reduces to (S))**.** *The proof of Theorem B uses no interpolation lemma and no source derivative at $t \neq 0$. The only source-derivative object is $\kappa_n(f_1, \dots, f_n) = \partial_{t_1} \cdots \partial_{t_n} \log Z_\varepsilon[\sum_i t_i f_i]|_{t=0}$, with pairwise distinct, separated, real arguments (`ASSEMBLY_N1.md` §4). Hence the real-source form of (E14-c) that the proof uses is R-045 (S) at the real point.*

In the bound (S), the factor $n!$ comes from the polymer-size sum of L-111 (Lemma C) Step 4, and the factor $(n!)^\eta$ from Step 8. So the clause rests on the hypotheses of R-045 (S): S1, (FX) (ledger row R-043) with R-042, the choice K-014, the coarsest-scale (FX2) (H-191) radius, and N2/N3 of Q-060. S1 and R-042 are open.

The page readings R-047, R-048, R-049, R-051.

Remark 5.125 (four page readings). We use the following readings of Bałaban's pages. R-047 (CMP 98 pp. 36, 39–40): the constant C'_1 is not printed, and it cannot be derived from the printed lines. R-048 (CMP 119 p. 255): β_0 can be taken arbitrarily small, and it is fixed before γ. R-049: $R_{K+1} \geq L$ for $\gamma < e^{-1/2}$; the printed lines support that C_c does not depend on g; the $O(1)$ constants are not printed. R-051: every clause on CMP 109 pp. 251, 259, 291–292 and CMP 119 pp. 256, 258, 261. The T-step rate loss is at most 13δ (R-020, Proposition 5.109). The rate printed in CMP 99 (3.108) is $e^{-\frac{1}{2}\delta_0 d}$. `WC_FULL.md` uses a fixed rate $\delta_A < \frac{1}{2}\delta_0$, and no consumer needs the printed rate.

The condition (C) (ledgers R-001, R-002, R-003, R-008, R-009, R-016). Notation. $\mathfrak{B} = \bigcup_{j=0}^{k} \Lambda_j$ is the multiscale coarse lattice, $\mathcal{H} = \ell^2(\mathfrak{B}, \mu)$ with $\mu(y) = \ell_y^d$ and $\ell_y = L^j \eta$ for $y \in \Lambda_j$, and S is multiplication by ℓ_y^2. $Q = Q(U)$ is the multiscale averaging, $G = (\Delta_\pi + DRD^* + Q^* a Q)^{-1}$ with $a = 1$ is the fluctuation propagator of CMP 99 Thm 3.3, and $G_1 = (G^{-1} - \Delta_\pi^{(2)})^{-1}$ (CMP 99 (3.128), (3.135), (3.138)). Put $\hat{N} = S^{-1/2} Q G Q^* S^{-1/2}$ and $\hat{N}_1 = S^{-1/2} Q G_1 Q^* S^{-1/2}$.

Proposition 5.126 (the lower bound (C), H-057)**.** *Assume the following inputs: (KE2) (H-136), from CMP 96 Cor. 2.8 by citation, with C_H, δ_H uniform in the torus side (Convention 5.127); CMP 99 Thm 3.1 at $a = 1$, Thm 3.3 (3.46), Thm 3.11, and the constant C_{137} of (3.137) (ledger R-021); CMP 96 Lemma 2.1; (H_Q) (H-123), Theorem QE (L-252) (Proposition 5.62), and S2 (Propositions 5.63 and 5.65); the readings [I-a1] and [I-geom] of* `SOURCES_C_CONDITION.md` *and Convention 5.128; the conventions (Cv1), (Cv2) (K-044). Then there are $c_- > 0$ and $c_{-,1} > 0$ such that*

$$\langle B, \hat{N} B \rangle_{\mathcal{H}} \geq c_- \|B\|_{\mathcal{H}}^2, \qquad \langle B, \hat{N}_1 B \rangle_{\mathcal{H}} \geq c_{-,1} \|B\|_{\mathcal{H}}^2 \qquad \text{for every } B \text{ on } \mathfrak{B},$$

uniformly in k, η, the sequence $\{\Omega_j\}$, the volume, and every real U with (3.35)–(3.36), $M \geq M_1$, $M\alpha_0 \leq a_0$. Equivalently, $\min\{\langle A, G^{-1} A\rangle : QA = B\} \leq c_-^{-1} \sum_j \sum_{y \in \Lambda_j} \ell_y^{d-2} |B(y)|^2$ for every B, and the same holds for G_1. One can take $c_- = 1/(4K^D)$ and $c_{-,1} = c_-/(1+\delta_1)$, under $M\alpha_0 \leq a_0(d, L)$, $RM \geq \max(4L^2 \tilde{D}_0(d, L), c(d, L))$, $RM > 2dL$ (implied by $R > 18dL$), $\theta = \log L/(RM) \leq \delta_0/2$, $\theta_S \leq \delta_H/8$, $M \geq \tilde{c}_T(d, L)/22$, and (Vol) (K-032) $\ell > 22M$. The constants K^D, $\tilde{D}_0$, $\tilde{c}_T$ are those of Lemma 5.130.

(C) is an upper bound on a constrained energy, the γ_1 direction. It is not the lower bound γ_0 of R-004: R-004 bounds a different operator, the constrained precision $C^* \Delta_k C$ on unit fields. The proof combines Lemma 5.129 and Lemma 5.130 with the extension of Proposition 5.62.

Convention 5.127 (torus uniformity of (KE2))**.** We read CMP 96 Cor. 2.8 as giving C_H, δ_H independent of the torus side.

Convention 5.128 (symmetry of $\Delta^{(2)}$)**.** We read the kernel of $\Delta^{(2)}$ in CMP 99 (3.136) as symmetric, as the Hessian of a real function.

Lemma 5.129 (Lemma R)**.** *Put $\varepsilon = M\alpha_0$ and $g(A) = \langle A, G^{-1} A\rangle$, and assume R1($G$): $\|\ell^{-1} A\|^2 \leq K_G g(A)$ for all A. If $2B_0^2 c_1^2 \kappa' \varepsilon \leq \frac{1}{2}$, then $|\langle A, \Delta_\pi^{(2)} A\rangle| \leq \delta_1 g(A)$ for every fine A, with $\delta_1 = 4c_2 \varepsilon (K_G + B_0^2 c_1^2)$. Hence $(1 - \delta_1) G^{-1} \leq G_1^{-1} \leq (1 + \delta_1) G^{-1}$; for $\delta_1 < 1$, $G_1 > 0$ and $c_{-,1} = c_-/(1 + \delta_1)$.*

Lemma 5.129 replaces a Schur step for G_1 in the proof of Theorem 5.79. Its input R1(G) follows from CMP 99 Thm 3.3 (3.46), Thm 3.11, and CMP 96 Lemma 2.1. As a consequence, the input R1 of R-004 for G_1 reduces to R1(G).

The periodic-lattice seam. The dressed extension of Proposition 5.62 transports along axial paths $y_- \to b$. On the periodic lattice every single-valued path rule has a seam. For adjacent bonds on opposite sides of the seam, the thin loop of `SOURCES_QE_INV.md` 5.3 winds once around the torus and carries a Polyakov loop. The regularity (3.35) does not bound this loop. Example: take constant abelian links in one direction; every plaquette is 1, and the holonomy around the torus is arbitrary. The energy defect at the seam is at most about $4C_H^2 N_T^{d-1} e^{-\delta_H N_T} L^k$, with leading term $c(\delta_H) N e^{-\delta_H N_T}$. It is not uniform in k for any fixed torus side, and a hypothesis "side $\geq c \cdot RM$" does not remove it. Numerically, the seam defect is present (Section 6.10). So the dressed kernel must be cut off.

Lemma 5.130 (Lemma T (L-284)). *Let χ be a cutoff function, $\psi_{y'}$ the block partition of unity, and $d_* \le d$ the closure-bond metric. Define the block-interpolated cutoff $\tilde\chi_y = \sum_{y'} \psi_{y'}\chi(d_*(y',y)/D)$, and cut off the dressed kernel by $\tilde\chi$. Assume $RM \ge 4L^2D$, $RM > 2dL$, and that the torus side in coarsest-block units exceeds $2L(D+3(d+1))+2$. Then: (a) every support embeds in a box, and no transport path crosses a seam; (b), (d), (e) the cutoff defect is controlled through $Q(1)E = I$; (c-rep) the gradient bound of the cutoff holds with the constants $c_{\mathrm{adj}} \le 2d + (d-1)\lfloor L/2\rfloor$, $C_\chi = N_{\mathrm{ov}} c_\psi c_{\mathrm{adj}}$ with $N_{\mathrm{ov}} = 3^{d+1}$, $\tilde c_T \le c_T + 6Lc_{\mathrm{adj}}$, $\tilde D_0 \le D_0 + 2c_{\mathrm{adj}}$, and $3/D_0 \to 3C_\chi/\tilde D_0$ in K^D; all of them are (d,L) numbers. The side condition follows from (Vol) $\ell > 22M$ with $M \ge \tilde c_T(d,L)/22$ (ledger K-045).*

Proof (sketch). The distance d of CMP 96 (2.46) is constant on each block at fine points. So a cutoff $\chi(d(b,y)/D)$ has no useful gradient at block faces, and the interpolation over blocks is needed for (c-rep). The bound on c_{adj} uses a contour argument. Each consumer of (c) holds with the new constants. □

So no torus-side hypothesis remains beyond (Vol). The two unprinted conventions ($U(c)$ the product over the straight segment; translation-covariant axial Γ) change only constants; among contour choices no other choice of $U(c)$ is possible.

Remark 5.131 (a false reading in two derivations). The reading [I-c] of `SOURCES_QE_INV.md` 5.3, that the thin loop has small holonomy, is false on the periodic lattice. `SOURCES_S2_FORM.md` 4.3 uses the same reading. Lemma 5.130 replaces both uses, and Propositions 5.62 and 5.63 hold with the cutoff kernel and the constants of Lemma 5.130.

The free case: numerically, $c_-^{\mathrm{free}} = 1/(1+K_0)$ (Section 6.9). The hypotheses of Proposition 5.126 that remain beyond the project's proofs are the printed CMP 99 material of row R-021: Thm 3.3 (3.46) for G and Thm 3.1 at $a = 1$ rest on Bałaban's walk sketches, and the torus-side uniformity of (KE2) is Convention 5.127.

The lattice log coefficient (ledger R-027).

Proposition 5.132 ((F) from Step B). *(F) (ledger row R-027) follows from: Lemma 2 of `SOURCES_T2_F_STEPB.md` (the massive log coefficient equals the continuum one, given Step C); (L-LL) (Theorem 5.133); Lemma 1 (MPT) (H-148), a gauge-covariant kernel bound from magnetic Weyl bounds; and the gauge-fixing part of Step C. The ghost part of Step C holds.*

Theorem 5.134 below proves (MPT) and the gauge-fixing part of Step C, and it gives (F) by a route that does not use Lemma 2.

Theorem 5.133 (the lattice log limit (L-LL), H-137). *Let $W(\varphi; m^2)$ be the per-cell charged bracket $-\frac12 \log|\det|(A_\varphi + m^2) + \log|\det|(\mathcal{G}_\varphi + m^2)$ minus its value at $\varphi = 0$, for the constant abelian background of plaquette angle φ, and $H_\varphi(\mu) = \varphi^{-2}[W(\varphi;0) - W(\varphi;\mu|\varphi|)]$. For every $\mu > 0$, as $\varphi \to 0$ through real values,*

$$H_\varphi(\mu) \to h(\mu) = -4\big[I_{\mathrm{heat}}(\mu/2) + g(1,\mu/2)/(8\pi^2)\big].$$

Here $h(3) = 0.116322$, $h(5) = 0.169938$, and $h(\mu) = \frac{11}{12\pi^2}\ln\mu + 0.023444 + o(1)$ as $\mu \to \infty$. The coefficient $11/(12\pi^2) = n_2c_2$ with $n_2 = \frac12$ is the one that Step B of `SOURCES_T2_F_STEPB.md` §2.6 needs.

Proof (sketch). We use proper time instead of band-by-band semiclassics. In the fine bracket the ghost cancels the (b_2, b_3) block exactly, so W reduces to the 2D operator A' integrated over $k_\perp$. The negative Nielsen–Olesen modes split off exactly; they give a term of order φ^2 with no $\log\varphi$. The positive part is controlled by heat traces: L-272 (the second moment of the lattice Lévy area is $\le Ct$); L-301 ($|\tau e^{-tA'_\alpha} - \tau e^{-tA'_0}| \le C\alpha^2 t$ for $t \le 1/\alpha$, by a Dyson expansion, S1, and charge-conjugation evenness); L-275 (the scaled heat trace converges to the Mehler kernel, through a Feynman–Kac representation and the invariance principle for random-walk bridges); L-304 (the scaled integrated density of states converges to the spin-split Landau measure, through Lemma 4.1 of `SOURCES_T2_GAPRIME2.md`). The inputs are textbook facts (the invariance principle for random-walk bridges, the continuity theorem for Laplace transforms, counting monotonicity in the magnetic algebra, Bessel asymptotics) and the project lemmas 1.1 and 4.1 of `SOURCES_T2_GAPRIME2.md` and 5.1 and 5.2 of `SOURCES_T2_S123.md`; the identification step of Lemma 5.1 is proved in `SOURCES_T2_MPT_REST.md`. No Harper band-width theorem is used, and the gap theorem (Theorem 5.98) is not needed. □

No rate of convergence is proved, and Step B needs none. Numerically, an independent quadrature with a closed form for the Nielsen–Olesen constant gives $h(3) = 0.1163219$ and $h(5) = 0.1699377$; the lattice values are computed down to $\varphi = 0.00125$ (Sections 6.9 and 6.10). The numerical tension of the single-step β_1 (Section 6.9) concerns a different family and does not affect (F).

5.13 Step 14: a proof of (F), the bound (S_σ), the s-rule, the p. 254 transfer, and CMP 99 Theorems 3.7 and 3.9

This step gives a proof of (F), the bound (S_σ) at every scale, the s-rule convention, the p. 254 transfer, and CMP 99 Theorems 3.7 and 3.9 in full. None of these results removes a hypothesis of Theorem A or Theorem B.

A proof of (F) (ledger row R-027).

Theorem 5.134 ((F) by proper time, ledger row R-027)**.** *Let $G(\varphi) = F_W(\varphi)/\varphi^2$, where $F_W(\varphi)$ is the per-cell lattice bracket at the constant abelian background of plaquette angle φ. For every fixed $\lambda > 0$, $G(\lambda\varphi) - G(\varphi) \to n_2 c_2 \ln(1/\lambda)$ as $\varphi \to 0$, with $n_2 c_2 = 11/(12\pi^2)$. Hence the variant of (3.1) that `SOURCES_T2_F_STEPB.md` needs holds, and with it (F) at constant field.*

Proof (sketch). Theorem A (L-060) of `SOURCES_T2_MPT_STEPC.md` writes $W(\varphi; m^2)$ as a proper-time integral. It proves (MPT) (H-148), Lemma 1 of `SOURCES_T2_F_STEPB.md`, for $m^2 \ge 6|\alpha|$ with $\alpha = 2|\varphi|$: $W = \alpha^2 \tilde{w}_2(m^2) + R$ and $|R| \le C_A \alpha^4 m^{-4}$. The main step is a fourth-order Dyson bound, L-303: $|D_\alpha(t) - \alpha^2 c_A(t)| \le C\alpha^4 t(1+t)^2$ for $t \le 1/|\alpha|$. Charge-conjugation evenness removes the orders α and α^3. The proof uses no Weyl symbol classes and no Combes–Thomas bound. L-112 (Lemma C) gives $c_A(t)/t \to 11/(12\pi)$ from L-304 (proof of Theorem 5.133) and L-303. A Frullani-type argument with (L-LL) (Theorem 5.133) then gives the theorem. This route uses neither Lemma 2 of `SOURCES_T2_F_STEPB.md` nor Step C. The gauge-fixing part of Step C also holds: the midpoint-gauge symbols are exact in α, and all corrections are even in K.

The inputs are textbook facts (the invariance principle for random-walk bridges, Laplace continuity, counting monotonicity, Bessel bounds) and the project lemmas of the proof of Theorem 5.133, among them the identification step of Lemma 5.1 of `SOURCES_T2_S123.md` and the neutral sector of `SOURCES_T2_F_STEPB.md` §2.0, both proved in `SOURCES_T2_MPT_REST.md`. The midpoint convention of U3 Lemma 3.4 is the same as the one of Step C. □

Numerically (20 s), the finite-difference Hessian matches the predicted kernel to 10^{-15} at $\alpha = 0.3$, 1, 2; $c_A(t)/t \to 0.29178 = 11/(12\pi)$; and the fourth-order coefficient divided by t^3 tends to 0.0561, against the continuum value 0.05615. `SOURCES_T2_F_STEPB.md` §1 exchanges λ and $1/\lambda$; this is a change of notation only. The numerical tension of the single-step β_1 (Section 6.9) concerns a different family and does not affect Theorem 5.134.

R-045 (S) at every scale (ledgers R-045, R-043).

Proposition 5.135 (the bound (S_σ), H-168)**.** *Assume S1(b), (FX) (ledger row R-043) with constants independent of ℓ, and the ℓ-independent trajectory (Bałaban's construction with ℓ-free constants). For $\sigma \geq 16\varepsilon$ and minimal support diameter $D \leq \sigma$, with no upper limit on σ,*

$$|\kappa_n| \leq \big(C_{S'} \max(1, \sigma^{-a})\big)^n (n!)^{1+\eta} \prod_i N(f_i), \qquad C_{S'} = 16 C_A C_1 C_B C_2 C_\eta,$$

and $C_{S'}$ does not depend on ℓ. At $\sigma = 16\varepsilon$, (S_σ) (H-168) is R-045 (S).

Proof (sketch). The ℓ-dependence of R-045 (S) sits in three inequalities of `SOURCES_D4D5.md` §5.1. Two of them fail for $\sigma \gg 2M$, and their left sides are $O(1)$. Lemma CS (L-127) redoes Steps 6–7 of L-111 (Lemma C) with $\log_L^+$ and $\Lambda_1(\min(\sigma, 2M))$. □

Remark 5.136. L-111 (Lemma C) of Proposition 5.17 holds only for $\sigma < 2ML$, and it is not defined for $\sigma \geq \sigma_0$. Lemma CS covers all $\sigma \geq 16\varepsilon$, so the clause R-045 (S) of R-045 rests on Lemma CS. The bridge (B-N1) (ledger row R-045) uses the whole ladder σ_m, with top rung $d_B/4$ in N1-B, so it uses (S_σ).

Proposition 5.137 (the item (FX-D))**.** *The item (FX-D) (H-105) is §6.3(b) of Proposition 5.22, that is Q-060, which Proposition 5.25 proves. Assume R-042, N2, S1(c), Q-061, and K-014. Then the source radius is $c \cdot r$ at every scale. (FX-D) adds no hypothesis beyond R-042.*

The s-rule convention (ledger K-046).

Convention 5.138 (the s-rule, K-046)**.** The s-versions of the propagators in the cluster expansions are defined per distinct σ_0-cube met on the expansions of this paper: the walks of `WC_FULL.md` 5.3 and 5.5, the trees of $\mathcal{W}_N$, of Theorem 5.76, and of WG-G1, and the composites of R-016. The admissible walk set is the one covered by (FI) (H-103), and the localization set is $\bigcup_i \tilde{X}_i^5$. The competing reading is the form (3.185) of CMP 99.

Proposition 5.139 (admissibility of the s-rule)**.** *Assume (FI) (H-103) ("the same formula on T_N and $\mathbb{Z}^4$") for tree factors, WG-G1 (Theorem 5.77), the reading I-a, and K-035. Then Convention 5.138 is consistent with all twelve consumers of the s-versions. Each consumer uses only bounds: normalization at $s \equiv 1$, (L1)–(L2) with the band, per-walk bounds, analyticity on $|s| \leq e^{\kappa_1}$, the anchored difference, multi-affinity, and mixed partials up to order $|\sigma_0|$ (from joint analyticity). No consumer needs the form (3.185).*

The symmetry of $C(Z_0, 0)$ comes from Theorem 5.80, not from the convention.

The completeness item H-035 (M1) of R-016.

Proposition 5.140 (the uses of the fluctuation covariance)**.** *In CMP 119 Sect. 3 and CMP 122 II, every use of the fluctuation covariance falls in the classes H-281–H-291, (U-a) (H-292), (U-c) (H-294), Q-047, and Q-048. Here Q-047 is the gauge and Euclidean covariance of $C^{(k)}$: Bałaban uses only block translations and the hypercubic point group about block points, and C_f satisfies $O_r C_f(U; d_S) O_r^{-1} = C_f(rU; r \cdot d_S)$ jointly with the source d_S. Q-048 is the large-field Gaussian with covariance $C^* \Delta^{(j)}_{\Lambda^c} C$; its only use (CMP 122 II p. 383) follows from H-281 and the Schur complement bound $S \geq \frac{3}{4}\gamma_{\exp}$, relative to $Q1^{\Omega}$.*

The list depends on Q-049 (row R-024, a reduction) and K-035 (Q-047(c)). Two items lie outside H-035 (M1). First, CMP 119 pp. 275–276 use the walk expansions in the multiscale Ω-geometry, while `WC_FULL.md` and Theorem 5.110 are single-scale (row R-024). Second, CMP 122 II uses a lower bound on the R-operation Hessian, which needs a smallness condition if the source enters the R-operation Wilson term; this is Q-050, proved in `SOURCES_D3_NEW4.md`.

The transfer of CMP 98 results (ledger R-003, input R4; rows R-022, R-023).

Proposition 5.141 (the p. 254 transfer)**.** *Assume the proofs of CMP 98, R-018, and the analyticity axiom for the mean M. Then the sentence of CMP 109 p. 254 that "all results of [12]" (CMP 98) stay valid for the new averages holds for Propositions 1–2 of CMP 98 and for their other uses U3–U9.*

Proof (sketch). The gauge-covariance reading of CMP 98 (64)–(108) holds, so U3, U4, U6 are proved. Bałaban's own average already carries axial chain terms, so $\tilde{Q}$ and Q_1 differ in chain layout only. The factor S_b obeys $S_b \leq 2(1 + dL^{d+1})$, and it lowers $\gamma_{\exp}(d, L)$. □

Remark 5.142. `WC_FULL.md` §3.1(2) applies A3-4.3 in the exponential gauge, where the axial condition fails. So that step needs a condition inside the blocks. The repair is the same (d, L) constant change, a smaller $\gamma_{\exp}$, and Theorem 5.78 holds with it.

The p. 254 sentence does not cover the CMP 99 operators G_1, H_1, $(QG_1Q^*)^{-1}$, and Δ_J. They rest on Bałaban's assertion on CMP 109 p. 253 that the results are "valid universally" (row R-022). The assertion on CMP 109 p. 275 that the new backgrounds have "approximately the same regularity properties" is row R-023; R-018 does not cover it. Both rows are reduced, with open items (Section 7.7).

CMP 99 Theorems 3.7 and 3.9 in full (ledger R-002). Theorems 5.76 and 5.77 rest on Bałaban's printed sketches of CMP 99 Theorems 3.7, 3.9, and 3.10. The next two theorems give full proofs of the first two, and the remarks after them record errors of the print.

Theorem 5.143 (CMP 99 Theorem 3.9 in walk-expansion form)**.** *Assume the block-diagonality of Q'_j (CMP 98 (77)–(80)) and the small-field bound on averaged links (AL-r) (H-072). Then CMP 99 Theorem 3.9 holds in walk-expansion form for $\hat{N}' = S^{-1}Q'G'^2Q'^*S^{-1}$. The expansion is built from the walk expansion of G' (Theorem 5.145) with step factor c'/M_1, $b_G = B'^2 L^{2+d/2} c_1(\frac{1}{4}\delta')$, and rate $\delta_G \geq \frac{7}{16}\delta'$. Lemma LB (L-210): $Q'G'^2Q'^* \geq \gamma_N^2 S$ for every real U with (3.35), with $\gamma_N(d, L, a) > 0$; this bound needs no further condition. The (3.99) rate is $\frac{1}{2}\varepsilon - |2 - \frac{d}{2}|\theta_L$ ($= \frac{1}{2}\varepsilon$ at $d = 4$).*

Proof (sketch). Theorem 5.76 is applied to $\hat{N}'$. Lemma LB uses a covariant bump trial function and Cauchy–Schwarz, and it needs only $|A|$ and $|\nabla A|$ on single blocks. The same argument does not give (C) (H-057) for QGQ^*: the corridor average has no diagonal dominance at $p = \pi$. The domain-change estimate, which the print cites to CMP 89, becomes a walk-truncation bound. □

(AL-r) is proved relative to CMP 98 Prop. 2 and one reading (`THM39_FULL.md`); its complex form (AL-c) (H-071) is proved as well. As a consequence, the common rate (I3) (H-209) of Theorem 5.77 is at most $\frac{1}{2}\varepsilon - |2 - \frac{d}{2}|\theta_L$, and M_*, a_* are computed with this rate; this creates no circularity.

Remark 5.144 (errors in the print of CMP 99 Theorem 3.9). CMP 99 has three misprints here: (3.96) drops a square; p. 411 prints the reversed inequality $e^{-x} \geq 3!x^{-3}$; p. 412 prints Q' for Q'^*. It also has four gaps: the smallness of $M\alpha_0$ is not stated; the domain-change estimate is not written; (3.99) is not proved; the local inverses rest on Cors. 3.5–3.6 and the $U = 1$ Fourier bound. The printed rate $\frac{1}{2}\delta_0$ in (3.99) is replaced by the rate of Theorem 5.143.

Theorem 5.145 (CMP 99 Theorem 3.7 and Corollary 3.8)**.** *Assume the multi-size cube class (H1) of `THM37_FULL.md`, which extends the printed class (3.35) and R-018, and Convention 5.147. Then CMP 99 Theorem 3.7 and Corollary 3.8 hold with rate* $(1-\theta)\delta_0$ *for every* $\theta \in (0,1)$*, with explicit constants* $K_1 = 2(3d+1)c_h L^2 B_0 e^{c\delta_0}$ *and* $M_{37} = \max\{M_{\mathrm{th}}, 3L+1, 2C_{\mathcal{D}}^{\mathrm{in}} K_1 c_1(\theta)\}$*,* $C_{\mathcal{D}}^{\mathrm{in}} \leq 3(2L+2)^d$*.* G' *exists by a finite-dimensional Neumann series. The theorem includes an* M_1 *version and a complex form on the (3.37) domain, with* $K_1^c = 8K_1$ *and a radius that carries a factor* $1/L^2$*.*

Corollary 3.8 ("similar estimates for other norms") holds in three forms. The rate cost $\log L/(RM)$ of the scale shift is not printed. CMP 99 Theorem 3.10 is also written out in full, relative to named inputs (`THM310_FULL.md`).

Remark 5.146 (the partition of unity (X-PU), H-180). The printed partition of unity (CMP 96 (2.36) with CMP 95 (1.118), on the cover of CMP 96 p. 229) gives $\sum h^2 = 2$ in a band near level interfaces. So the agreement step (Lemma AG (L-073)) fails where a cube crosses an interface. This is an error in the print. The repair $c_h \to L\, c_h$ (a factor L in K_1) holds on flat faces, and the constant K_1 of Theorem 5.145 includes it. The same constant change applies in CMP 99 Theorems 3.9, 3.10 and in Theorems 5.76, 5.77, 5.78, whose constants are to be read with c_h replaced by $L\, c_h$.

Convention 5.147 (corners of the cover)**.** At corners of level interfaces we read the repaired partition of unity as satisfying the same bound as on flat faces.

Order of constants and a new condition (ledgers K-005, K-047).

Remark 5.148 (an unprinted side condition). Add to the canonical order of constants the constants of R-018, (C1*), the R-012 addendum, R-016, R-020, and Lemma G3 (L-174). The order stays acyclic. CMP 119 (2.9), line 2, needs the side condition $\gamma^2\beta' \leq 1/L$ (K-047), which is not printed. The Lean file ScaleChain (`eq29_line2_full`) contains it. It is an upper bound on γ after L and β', of Bałaban's type, and it adds no cycle.

5.14 Step 15: the alternative ($\mathbf{E}_U$) route

This route is an alternative to R-001 for the γ_1 input. No result of this paper depends on it.

Uniform block extension ($\mathbf{E}_U$). The condition (E_U) asks for a fine field A with $Q(U)A = B$ and covariant energy $\leq K\|B\|^2$ for every block field B, uniformly in k.

Proposition 5.149 (the free constant and the dressed extension)**.** *(i) Numerically, in the free case* $K_0(4) = 103.25,\ 146.73,\ 185.64,\ 195.19$ *at* $N = 3, 5, 13, \infty$ *(*$= \kappa_4 - 1$ *at* $a = 1$*), for the straight-line average* Q_{98}*; the* $N = \infty$ *tail is* $\propto 1/M^3$*.* K_0 *is an upper bound. (ii) Assume the k-uniform moments of the free kernel* E_0 *(Proposition 5.150), the constant* C_1' *of R-047, and three readings: the lattice-sum factors, the boundary cutoff at* $\partial\Lambda$*, and the gauge-fixing term. Then for regular* U *the dressed extension* $A = \mathcal{E}_U(Q(U)\mathcal{E}_U)^{-1}B$ *gives* $K \leq K_0(1 + c_{2/3}(M\alpha_0)^{2/3} + c_1 M\alpha_0)$*. The k-step remainder constant is* $2C_1'$ *(CMP 98 (143)).*

Proof (sketch). We sketch the argument; it is not written out in full. K_0 is computed in momentum space (`numerics/k0_d.py`) from the CMP 98 (124)–(125) form of $Q(V_0)$, an equal-weight average over L^d straight lines with an $O(L^2\alpha_0)$ remainder, and is checked in real space at $d = 2$. For regular U the free extension kernel is dressed by parallel transport. The constraint error $\|Q(U)\mathcal{E}_U - I\|$ is bounded linearly in the plaquette bound $\alpha_F \le 20M\alpha_0$ by lattice Stokes. IMS patching over flat frames of radius $R \propto \alpha_F^{-1/3}$ gives the power 2/3. The relation (139) is on journal p. 39. □

Transport and back do not give $Q(U)A = B$ exactly; the inverse $(Q(U)\mathcal{E}_U)^{-1}$ is needed. The linear form $K \le K_0(1 + cM\alpha_0)$ is not proved, and $K \to K_0$ is not proved.

Proposition 5.150 (k-uniform moments of E_0, ledger R-050)**.** *The free minimal-extension kernel E_0 of the straight-line average $Q_{98}(1)$ (composition of k one-step averages, $N = L^k$) obeys $|E_0(y,c)|, |\nabla_\nu E_0(y,c)| \le Ce^{-\delta|y-c|}$, with C, δ depending on d only. All moments ν_j, γ_j, the block sums, and Γ_j are bounded uniformly in k.*

Proof (sketch). The proof uses symbol identities in momentum space, lower bounds on the symbols, a weighted L^2 Combes–Thomas bound uniform in η, and pointwise regularity from the symbol. It uses no Bałaban estimate. □

This gives the first hypothesis of Proposition 5.149(ii). The other hypotheses of that proposition remain.

The norm bound γ_1 (ledger R-008).

Proposition 5.151 (γ_1 from (E_U))**.** *Assume CMP 99 (3.132) (ledger R-001), or in its place (E_U) with the hypotheses of Proposition 5.149(ii); R-018 for the J-term and $\|C(U)\|$; and the multiscale correction $e^{-\delta RM}$ to the free constant (a reading). Then the norm bound $\|C^*\Delta_k C\| \le \gamma_1$ holds uniformly in k and regular U, with $\gamma_1 = L^{d+1}[(1+\varepsilon)K + \varepsilon a + c_J]$, where $\varepsilon, c_J = O(M\alpha_0)c(d,L)$ and K is the constant of (E_U). For single-scale $U = 1$ the free bound $K \le a(\kappa - 1)$ holds, which is 195.19 at $d = 4$, $a = 1$.*

Proof (sketch). Comparing (3.155) with (3.156) identifies $\langle B, (QG_1Q^*)^{-1}B\rangle$ as the minimum of $\langle A, G_1^{-1}A\rangle$ over $QA = B$; this needs only $G_1 > 0$ and Q onto. By Woodbury, $(QG_1Q^*)^{-1} = N_1^{-1} + a$ exactly, from the split $G_1^{-1} = M_1 + aQ^*Q$ of (3.128). So the term $-a\langle B, B\rangle$ of CMP 99 (3.156) cancels. The remaining minimal energy is bounded by a trial extension of B (the condition (E_U)), the Δ'_π and J parts by $O(M\alpha_0)$ perturbation (J is a second derivative of A), and the free constant by the CMP 95 Fourier formula (1.99)–(1.101). We use $\|C\|^2 \le L^{d+1}$. □

A local difference operator for which unitary U suffices does not give the bound: the leading term $(QG_1Q^*)^{-1}$ is an inverse whose naive coefficients scale as L^{2k}, so the regularity (3.35)–(3.36) is needed for the upper bound. The s-localized operator $C^{(k)}(Z_0, 0)$ that Lemma G3 (L-174) uses is handled by LOC (R-003, R-004). Theorem 5.79 proves R-008 relative to the inputs of Theorem 5.78, without the kernel form of R-001, so Proposition 5.151 is not used.

6 Numerical results and the computer-assisted proof of (H2)

This section collects the numerical results that bear on the hypotheses of Section 3. Sections 6.1–6.3 give the one-loop increments and the per-level quantities of the Route K proof. Sections 6.4–6.6 give the values of (H2) (R-025), the one-step values, and one rigorous point enclosure. Section 6.7 proves (H2) at $L = 13$ (Theorem 6.2). Section 6.8 extends the proof to $L = 15, 17, 21$ and gives the

numerics of the 2D gap and of the neutral sector. Section 6.9 treats the single-step β_1, a second certificate at $L = 13$ on part of the Brillouin zone, and the numerics of (L-LL) (H-137) and of the free constant of (C) (H-057). Section 6.10 gives the seam test of (C), (L-LL) at smaller flux, and the converged-grid values of β_1. Section 6.11 gives the design of the cell certificate.

All values are for SU(2), in infinite volume, in Bałaban's normalized-trace units. Except for Sections 6.6, 6.7, the first paragraph of Section 6.8, and the cell certificate of Section 6.9, all values are floating-point values without rounding control; they enter no proof. Only odd L is inside the proofs. The values at $L = 2, 4$ use corner blocks and are outside them.

6.1 First increment $\beta_1(0)$

Sources: `NUMERICS` §6–§7; `UMK_BETA1.md`.

L	$\beta_1(0)$	method	$b = c \log L$	$\beta_1 - b$	notes
2	0.630	$N_g = 12$, $p \to 0$ fit	0.2575	+0.372	a second code gives 0.6296
3	0.664	$N_g = 8$, $p \to 0$ fit (0.6625–0.6661)	0.4082	+0.256	two codes agree; N_g $6 \to 8$ shift ≈ 0.015
4	0.757	$N_g = 6$, single $p = 0.2$; $O((Lp)^2)$ bias ≈ 0.01	0.5150	+0.242	—
5	0.749 ($N_g = 12$, $p = 0.1$); fit 0.743	umklapp code	0.5979	+0.145	—
7	0.803	umklapp code, $N_g = 8$, $Lp = 0.5$	0.7229	+0.080	—
9	0.833 ($Lp = 0.5$); 0.846 ($Lp = 0.9$), 0.851 ($Lp = 1.0$)	umklapp code, $N_g = 8$	0.8163	$\approx +0.02$	grid not converged; see Section 6.9
11	0.850 ($Lp = 0.5$); 0.868 ($Lp = 1.0$)	umklapp code, $N_g = 8$	0.8908	≈ -0.04	grid not converged; see Section 6.9
13	$0.857293 \pm 1.5 \cdot 10^{-5}$ ($Lp = 0.5$); $0.8757528 \pm 3.1 \cdot 10^{-6}$ ($Lp = 1.0$)	ball-arithmetic port `numerics/umk_beta1_arb.py`, $N_g = 8$	0.9529	−0.096	the ball covers the rounding of the $N_g = 8$ sum only

The first-step constant $\beta_1 - c \log L$ is +0.37, +0.26, +0.24, +0.15, +0.08, +0.02, −0.04 for $L = 2, 3, 4, 5, 7, 9, 11$, and −0.096 at $L = 13$. It decreases with L and changes sign between $L = 9$ and $L = 11$. $\beta_1(0)$ itself is positive at every L computed. The umklapp code agrees with `ym_momentum` to 10^{-12} per K. The decrease does not level off, and the slope of β_1 in $\log L$ is well below c. Section 6.9 gives the converged-grid values and discusses this.

6.2 Later increments ($L = 2$ only)

quantity	value	notes
$\beta_1 + \beta_2$ (composed, $N_g = 4/6/8$)	1.148 / 0.967 / 0.953	single $p = 0.2$; Lp bias
per-level $\beta_2(0)$	≈ 0.32 ($0.953 - 0.630$). Per-level recursion 0.317–0.341 ($N = 8$–12, single $p = 0.2$). Symmetric Nyquist: 0.33334 ($N = 8$, $p_1 = 0.2$), 0.31679 ($N = 12$, $p_1 = 0.2$), 0.32283 ($N = 12$, $p_1 = 0.4$). Level-1 values carry N-aliasing of about $\pm 5\%$.	two codes agree on the per-level recursion
$\beta_3(0)$	0.40 ($N = 8$), 0.286 ($N = 12$), 0.261 (symmetric Nyquist, $N = 8$)	not converged; not usable
abelian sector S_k^{ab}	$(1 - L^{-4k})/4$, matched to all digits	exact; the counting part is formalized in Lean (`AbelianCount.lean`)
(U) test	$\lvert\beta_{j+1}(p) - \beta_{j+1}(0)\rvert \le 0.2p^2$ at every computed point ($L = 2$ levels 0–1, $L = 3$ level 0, $p \le 0.8$)	uniformity in j cannot be tested with two levels

All computed increments are positive.

6.3 Quantities of the Route K proof

quantity	value	notes
$d_1 = b - \beta_1$, $L = 3$	-0.256 ± 0.01 (fits $-0.254 \cdots - 0.258$)	two codes agree
d_1, κ_1, $L = 2$	-0.372, 1.114	two codes agree
d_2, d_3, $L = 2$	≈ -0.44, ≈ -0.45 (d_3 uses the unconverged β_3)	bounded in appearance; not a test
$\rho_1 = \max_K \lVert H_j(K)\rVert / L$, $L = 3$, $j = 1, 2, 3, \infty$	1.652, 1.736, 1.757, 1.760	(HS) fails; $h^2/L^4 \approx 0.34 < 1$; one-step C^1 factor $\rho_1^2/L > 1$ for $j \ge 2$
ρ_1, $L = 5$, $j = 1, 2, \infty$	1.860, 1.9565, 1.963; $\rho_1^2/L = 0.771$	—
ρ_2 ((H2)), $L = 3, 5, 7, 9$	see Section 6.4: 2.0634, 2.108, $\ge$ 2.1131, $\ge$ 2.1142 at $j = \infty$ against thresholds 3, 5, 7, 9	two codes agree at $L = 3, 5$
a_*, $L = 5$	0.0107 (binding mode near $K = 0$, gauge/ghost sector)	—
(H_A), $n = 2$, $L = 3$	$2.056 < 2.828$	coarse grid, one level pair
$\sigma_j^{(1)}(0)$ J-part, $j = 0, 1, 2$ ($d = 2$ proxy)	0.750000, 0.75015, 0.75051 against 0.75	agrees with L-308 (Theorem W)
$\sigma_j^{(2)}(K)$ scan	not run: no $d = 4$ constant-field per-momentum code	—

6.4 Values of (H2)

Code: `numerics/h2_scan.py` (umklapp Schur pieces, one colour, Gram form; `H2_LARGE_L` §1). The one-colour reduction is exact at $U = 1$. The H and Δ_{out} blocks and the Gram identity are exact, and so is the separable Prop. A form. The sign in `H2_LARGE_L` §4 is $H = +[\mathfrak{B}^{-1}]_{f,c}$.

L	j	ρ_2 (scan max)	threshold L	notes
3	2	2.00857	3	two codes agree
3	3	2.05601	3	spot value 2.05599 at the argmax
3	∞	2.0634 (2.06332–2.06377 over l_{max} 2–5)	3	margin 31%; $q_1 \le 0.473$, $q_2 \le 0.053$
5	2, 3, 4, ∞	2.098, 2.108, 2.108, 2.108	5	an independent dense code gives 2.098283 at $j = 2$; a second search gives 2.107921 at $j = \infty$, same argmax; margin 58%
7	∞	$\ge$ 2.1131 (face diagonal only)	7	spot value 2.11309
9	∞	$\ge$ 2.1142 (one point, $(2.6, 2.6, \pi, \pi)$)	9	the $j = 2$ point of `H2_LARGE_L` gives ≥ 2.1104

- Shape. The argmax lies on the face $K_3 = K_4 = \pi$, on the diagonal $K_1 = K_2$: $\approx 2.22 \to 2.40$ ($L = 3$, $j = 2 \to \infty$), ≈ 2.54 ($L = 5$). Off-face points are lower: the best off-face values are 1.995 ($L = 3$, $j = 2$), 2.004 ($L = 3$, $j = \infty$), 2.088 ($L = 5$, $j = \infty$). The corner (π, π, π, π) stays below the face peak, but the gap shrinks: 0.0070, 0.0028, ≥ 0.0025, ≥ 0.0023 ($L = 3, 5, 7, 9$). The corner is the obvious place for a competing maximum at large L.

- Level dependence. At $L = 5$, $\rho_2(\infty) - \rho_2(j) = 9.64 \cdot 10^{-3}$, $4.0 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ ($j = 2, 3, 4$), with ratios 24.1 and 25.0, against $L^2 = 25$. At $L = 3$ the ratios are 7.5 and 8.7. Levels $j = 4, \dots, \infty - 1$ are not sampled at $L = 3$.

- L-trend. At $j = \infty$: 2.0633, 2.1079, ≥ 2.1131, ≥ 2.1142 ($L = 3, 5, 7, 9$). This is consistent with a bounded ρ_2 near 2.11–2.14. The values at $L = 7, 9$ are lower bounds, so they do not give a rate in L. The statement that (H2) becomes easier with L is a measured trend for $L \le 9$, not a result for $L > 11$.

- All values are lower estimates of a supremum (grid plus local search).

6.5 One-step values

By Section 3.7.4, $\rho_1^2 < L$ implies (H2). The measured one-step values are as follows.

- $L = 3$: $\rho_1^2 = 3.10 > 3$. The one-step route fails (Section 3.7.2); the two-step map is needed.

- $L = 5$: $\rho_1 = 1.860$ ($j = 1$, Wilson fine kernel, dense), 1.9565 ($j = 2$), 1.963 ($j = \infty$). So $\rho_1^2 \le 3.853 < 5$ and $\rho_1^2/L = 0.771$, as floating-point values.

- $L = 7$: $\rho_1 \ge 2.032$ (diagonal only), $\rho_1^2/L \approx 0.59$.

- $L = 13$: measured $\sup \rho_1^2 = 4.37$; Theorem 6.2 gives $\sup_{j,K} \rho_1^2 \le 10 < 13$.

So at $L \geq 5$ the one-step map suffices for L-011, as far as measured. For a computer-assisted proof this replaces L^8 blocks per K (two-step map) by L^4 blocks per K (625 times fewer at $L = 5$).

6.6 A rigorous point enclosure at $L = 3$

The program `numerics/h2cert_proto.py` (`H2_CERTIFY.md`) uses ball arithmetic (python-flint). At $L = 3$, $j = \infty$, and the face point $K^* = (2.4035, 2.4035, \pi, \pi)$ near the argmax of Section 6.4, it gives

$$\rho_2(\infty, K^*) \in [1.9993,\ 2.1273] < 3.$$

The width comes from the umklapp tail and scales as N^{-4} in the truncation N. At $N = 12$ the program gives [1.5475, 2.5792], which contains the floating-point value 2.063361, in agreement with the N^{-4} scaling. At $N = 6$ the program stops with an assertion instead of returning a wrong interval. The enclosure holds under three premises P1–P3 on the input data. P1 states that the loop-family coefficients of the (0.4) row are exact rationals $k/648$, k even; it is proved there. P3 is the fidelity of the loop family to Bałaban's paper, that is, Convention 6.1. Further results:

- One-step factorization: $\rho_2(j, K) \leq \rho_1(j, K) \cdot \max_l \rho_1(j-1, k_l)$, exactly. So two-step work is needed only where the right side is $\geq L$, near the face diagonal.
- The Oishi–Rump bound gives $\|\mathfrak{B}^{-1}\| \in [42.7,\ 43.6]$ at K^*. The values 177 and 195 of Section 3.9 use the full Δ instead of $\frac{1}{4}\Delta$; this factor reconciles them. No $\|\mathfrak{B}^{-1}\|$ enters ρ_2.
- The enclosure of the umklapp tail holds for $M = \infty$. For finite M its lower bound is 0, with class-wise factors $(\pi/2)^{12}$ for two or more large components.
- A uniform Lipschitz grid is not a practical route: $\Lambda_K \approx 2.8 \cdot 10^4$ would need about 10^{13} cells. The replacement is a per-cell invertibility test with a mean-value enclosure.

This is one point at one level. It does not bound the supremum over the Brillouin zone or over j. Section 6.7 does this at $L = 13$.

6.7 (H2) at $L = 13$

Here $\rho_1(j, K) = \|H_j(K)\|/L$, with $H_j(K)$ the one-step gauge-fixed minimizer map of Section 3.7. The map depends on Bałaban's averaging operator, which we fix as follows.

Convention 6.1 (reading of the averaging operator (0.4) of CMP 109)**.** The averaging operator is defined by the path family of [8, (0.4)] as read in `NUMERICS` §5, with the gauge choice fixed there. A different reading of the path family in (0.4) would change H_j. Theorem 6.2 and Proposition 6.4 are statements about H_j defined by this convention.

Theorem 6.2 ((H2) at $L = 13$)**.** *Assume Convention 6.1. For $L = 13$, every level $j \geq 1$ including $j = \infty$, and every coarse momentum $K \in \mathrm{BZ}$,*

$$\rho_1(j, K)^2 \leq 10 < 13.$$

Hence $\rho_2 \leq \sup_{j,K} \rho_1(j, K)^2 \leq 10 < L$ by the factorization $\rho_2 \leq \rho_1(j) \max_l \rho_1(j-1, k_l)$ (Section 3.7.4). That is, (H2) (R-025) holds at $L = 13$.

Proof (sketch; the full argument is in `H1_ANALYTIC.md` §0–§8). Write $s = L^{-2}$, $k_l = (K + 2\pi l)/L$, $M = L^{j-1}$, d^M for the Prop. A symbol at level M, $P_\perp$ for the projection onto $\hat{K}^\perp$, and T, T_2, σ, $Z = (P_\perp T P_\perp)^+$ as in `H1_ANALYTIC` §1.

1. Exact split. $\rho_1(j,K)^2 L^2 = \lambda_{\max}\big(ZT_2Z + cc^\dagger/(\sigma|\hat{K}|^4)\big)$, with $c^\dagger = \hat{K}^\dagger(1-TZ)$. The first term is the transverse part, the second the orthogonal pure-gauge part.

2. Three exact lemmas. L-264 (Lemma S): the averaging symbol is $E_q(k_l) = E_l^0 - \hat{K} g_l^\dagger$, with E_l^0 diagonal (block average times straight path), $|E^0_{l,\mu\mu}|$ built from $a_{l,\rho} = \sin(K_\rho/2)/(L\sin(k_{l,\rho}/2))$, and $\sum_l \prod_\rho a_{l,\rho}^2 = 1$ (Fejér). Tower identity: $d_\mu^{ML}(K) = L^{-2}\sum_l w_{l,\mu} d_\mu^M(k_l)$. Hence $P_\perp T P_\perp = P_\perp \operatorname{diag}(d^{ML}(K)) P_\perp$, and Z is the next-level kernel of Prop. A. L-166 (Lemma G): a closed form of the gauge row g_l from the path average, so the averaging operator is explicit.

3. Majorant. L-283 (Lemma T) (transverse): $\lambda_{\max}(ZT_2Z)/L^2 \le \gamma_L F(K)$, with $F(K) = 1/(|K|^2 \underline{d}(K))$ and $\gamma_L = \operatorname{sinc}(\pi/2L)^{-2}$. This follows from $T_2 \le d_{\max} T$, $ZTZ = Z$, $d_{\max} \le L^2\gamma_L/|K|^2$, and $\|Z\| \le 1/\underline{d}$. Lemma Gauge (L-180): the gauge part is $\le (1+\tau)^2/(|\hat{K}|^2 S_L)$, with explicit sums V, U and $\tau = (V + |\hat{K}|U)/\underline{d}$. So $\rho_1(j,K)^2 \le B_L(K) := \gamma_L F(K) + (1+\tau)^2/(|\hat{K}|^2 S_L)$. By Cauchy–Schwarz $F \le F_{CS}$, a closed form made of one-dimensional factors in each K_ρ.

4. Uniformity in j. B_L does not depend on j. For every $M \in \{1, L, L^2, \dots, \infty\}$ the fine symbol satisfies $d^M \le 1/|\tilde{q}_0^M|^2 \le 1/|\hat{k}|^2$, because the Fejér weights sum to 1. The coarse symbol d^{ML} dominates the truncated $M = \infty$ sum termwise. The proof uses no monotonicity in j and no O5 rate.

5. Branch-and-bound. B_L is invariant under the 384 signed permutations of K by construction. So the supremum over the sorted chamber $0 \le K_1 \le \cdots \le K_4 \le \pi$ is the supremum over BZ. The program `numerics/h1analytic/bb_arb.py` encloses $\gamma_L F_{CS} + \bar{G}_L$ on boxes $K = \pi u$ with exact dyadic u. It evaluates every one-dimensional value in arb (python-flint), and it rounds outward by one ulp after every floating-point operation, with Higham bounds on every sum. Result: $\sup_{\text{chamber}} B_{13} \le 9.9995277$, on 25,169 boxes with 8,075 gauge evaluations and no failed leaf. Boxes that contain $K = 0$ are enclosed directly; $\rho_1(j,0) = 1$. □

Margins. The measured $\sup \rho_1^2$ is 4.37. The majorant is tight to 2.6% at $|K| \approx 1.5$: $\min B_{13}/\rho_1^2 = 1.026$ over 896 cases and a local search. The program refines boxes only until each is ≤ 10, so its margin to 10 is $5 \cdot 10^{-5}$; its margin to the threshold 13 is 23%.

Consistency checks. The split, Lemmas S and G, the tower identity, L-283 (Lemma T), and Lemma Gauge agree with the production objects to $\le 5.3 \cdot 10^{-13}$, and $B_{13} \ge \rho_1^2$ holds at 896 (j,K) cases. A second construction of $H_j(K)$ from the definitions (position-space (0.4) paths, sparse KKT, ghost-domain gauge) reproduces the production ρ_1 to $5.8 \cdot 10^{-9}$ (10^{-14} with the same d symbol). The two parts of the split agree to $3 \cdot 10^{-8}$, and the conventions of L-264 (Lemma S) agree to $1.2 \cdot 10^{-13}$. The invariance of ρ_1 under the 384 signed permutations (P5a, P5b) holds; the analytic proof does not need it, the cell sweep does.

Lean. `lean/YmLean/Tower.lean` formalizes the algebraic core of the majorant: the tower identity in weighted form, the Rayleigh and rank-one bounds, `majorant_core`, and `lemmaT_core`. `BlockNorm.rho2_le` is the one-step to two-step factorization.

Remark 6.3 (inputs of the proof). Besides Convention 6.1, the proof uses the following.

- Prop. A at $M \ge 27$ and $M = \infty$. Two implementations of the same formula agree, and an independent recursion agrees at $M = 3, 9$.

- L-166 (Lemma G) and the chain of bounds, proved in `H1_ANALYTIC` and checked numerically at points. The Fourier and umklapp derivation, Lemmas S and G, and the identification of the split are not formalized in Lean.

- The program `bb_arb.py`, read line by line and compared with a floating-point program on 30 boxes. It is not formally verified. It relies on python-flint arb and on IEEE correctly rounded $+, -, \times, \div, \sqrt{\cdot}$.

The cell certificate of Section 6.9 gives an independent bound at $j = 1$ outside $|K|_\infty \le 0.32$.

6.8 Other values of L, the 2D gap, and the neutral sector

Proposition 6.4 ((H2) at $L = 15, 17, 21$)**.** *Assume Convention 6.1. For $L \in \{15, 17, 21\}$, every level $j \ge 1$ including $j = \infty$, and every $K \in \mathrm{BZ}$, $\rho_1(j, K)^2$ is bounded by the value in the table below, which is less than $L - 1$. Hence $\rho_2 \le \sup \rho_1^2 < L$, and (H2) holds at $L = 15, 17, 21$.*

Proof. The lemmas behind the majorant B_L of Theorem 6.2 are stated for odd $L \ge 3$, and B_L does not depend on j. The program `numerics/h1analytic/bb_arb.py` was run with target $L-1$ on the initial grid $n_0 = 16$ of the sorted chamber (3,876 boxes, no refinement). It covered all boxes with no failed leaf (`numerics/H1_OTHER_L.md`). The factorization `rho2_le` gives the bound on ρ_2. □

L	$\sup_{j,K} \rho_1^2 \le$	target	$F_{\max}$	$G_{\max}$
15	13.737723	14	10.020	4.005
17	12.826432	16	10.020	3.061
21	11.776985	20	10.020	1.951

The proof has the same inputs as that of Theorem 6.2. At each L an independent run reproduces the worst cell, $F_{\max}$, and $G_{\max}$ to the printed digits. At 5 points an independent construction of H_j gives $B_L \ge \rho_1^2$ with ratio ≥ 1.63. These values of L are not used by the two targets (K-003).

The 2D gap ($\mathbf{G}_{2\mathrm{D}}$). The proof of Theorem 5.98 covers only $|ef| \le 10^{-4}$. Numerics (`numerics/t2_gaprime2_check.py`, 141 values) show the gap on all of $\alpha = ef \in (0, 3.02]$. At 11 rational fluxes with denominators up to 2213, the smallest |eigenvalue| of A' is $|\alpha|(1 - 0.125|\alpha|)$, so the ratio to $|\alpha|$ is ≥ 0.95. A sparse shift-invert run at $\alpha = 10^{-4}$ gives the ratio 1.0000125. The input (M_j) needs the gap only on a punctured neighbourhood of $f = 0$ (`SOURCES_T2_FRANGE.md`). So these values are not inputs of any proof.

The neutral sector of (M_j). The model is the 2D $N \times N$ torus of `SOURCES_T2_S123.md` §6, in Landau gauge, with flux $\varphi = 2\pi m/N$ per plaquette, and the operator $4\,\mathrm{Hess} + D_r D_r^T$ restricted to the neutral (colour-3) block. Prediction: the spectrum is $\bigcup_K \{l(K), \cos\varphi\, l(K)\}$, with exactly $d = 2$ zero modes (the constant 1-forms). For $N = 12$–48 and $m = 0$–2 (`numerics/t2_mjrest_neutral.py`) the maximal deviation from the prediction is $\le 4.9 \cdot 10^{-8}$ (the finite-difference floor). There are two zero modes at every flux, and the gap is $\cos\varphi \cdot 4\sin^2(\pi/N)$ to relative error $\le 8 \cdot 10^{-8}$. A second program (`numerics/t2_mjrest_verify_neutral.py`) reproduces these values and gives, at new sizes,

N	m	φ	zero modes	first positive	$\cos\varphi \cdot 4\sin^2(\pi/N)$
36	3	0.5236	2	0.02631374	0.02631374
20	3	0.9425	2	0.05753652	0.05753652
64	2	0.1963	2	0.00944548	0.00944550

The zero modes span the two constant 1-forms (residual $\le 5.6 \cdot 10^{-7}$), and the charged–neutral leak is 0. Limits: 2D, Wilson level ($t = 0$, no Bałaban averaging), rational fluxes. The level-j neutral symbol is not checked numerically. Theorem 5.102 uses only the bounds (N1), (N2).

6.9 The single-step β_1, the cell certificate, (L-LL), and the condition (C)

The single-step β_1 against $c \log L$.

- The quantity. The single-step $\beta_1(L)$ is one step from the unit lattice with block size L, for SU(2) at $p \to 0$. Each L is a different constraint family. This is not the fixed-L composed family of Theorem 3.1 of `BETA_FP_H.md`, which concerns $\beta_\infty = \lim_k [\mathcal{C}(L^{k+1}) - \mathcal{C}(L^k)]$ at fixed odd L.
- The comparison value. $c = 11N^2/(12\pi^2) = 0.3715$ at $N = 2$ is the theoretical coefficient, not a fit. No computation gives β_∞.
- The data. The converged values are in the table below. All local slopes are far below c, and they decrease with L. So the data do not show that the first-step constant $\beta_1 - c \log L$ tends to a limit.

Remark 6.5 (a possible explanation of the slopes). A possible approach (`SOURCES_BETA1_F_TENSION.md`, Proposition 3.1) splits the single-step increment as $\beta_1(L) - \beta_1(L') = c \log(L/L') - [B_c(L) - B_c(L')]$. Here $B_c(L)$ is the constant of the one-step coarse bracket, the constant-field one-loop term of the tree kernel $\Delta_1^{(L)}$. The input (F) (R-027) concerns only the fine bracket and gives the term $c \log L$; it says nothing about $B_c(L)$. In this approach $\beta_1(L) = c \log L + C + o(1)$ holds if and only if $B_c(L)$ converges, and the β_1 data neither contradict nor support (F). The local coefficient $dB_c/d\log L$ is 0.230, 0.257, 0.291, 0.330 for $L = 5 \to 7$, $7 \to 9$, $9 \to 11$, $11 \to 13$ (`SOURCES_BCL_TEST.md`), and it shows no saturation. The region $|K| \sim 1/L$ carries less than 1% of the increment of B_c. At $f = 0$ the scaled tree kernel tends to $\kappa^2 H_\infty$, relative to U3 Theorem 4.1. The open part is the bulk behaviour of the single-step soft vertices.

Open Problem 6.6. Decide whether the constant $B_c(L)$ of the one-step coarse bracket converges as $L \to \infty$ through odd values.

L	$\beta_1(L, 0)$	grid	local slope from the previous L
5	0.7429 ± 0.0002	$N_g \to \infty$ extrapolation	—
7	0.7903 ± 0.0012	$N_g \to \infty$ extrapolation	0.141
9	0.8192 ± 0.0002	$N_g = 24$; $Lp = 1$, 0.5, 0.25	0.115
11	0.8354 ± 0.0005	$N_g = 28$; $Lp = 1$, 0.5, 0.25	0.081
13	0.8423 ± 0.0014	$N_g \geq 24$; $Lp = 1$, 0.5, 0.25	0.041 ± 0.009

Values: `SOURCES_BETA1_CODE_CHECK.md` §5.3–§5.4 ($N_g \to \infty$ first, then $Lp \to 0$ at fixed L); `SOURCES_BETA1_F_TENSION.md` §13.1 gives 0.7905 at $L = 7$. At $L = 13$ the raw values are 0.8432 ($Lp = 0.25$), 0.8440 ($Lp = 0.5$), and 0.8621 ($Lp = 1$; N_g 24 $\to$ 28 changes β_1 by $4 \cdot 10^{-4}$). At $Lp = 1$ the $11 \to 13$ slope is 0.034. The formula $(4\beta(0.5) - \beta(1))/3$ is biased low by 0.003–0.004 at $L \geq 9$. The difference $\beta(Lp = 1) - \beta(Lp = 0.5) = 0.0185$–$0.0195$ does not depend on L, so fixed Lp is the right comparison. The anchored-ghost scale $2.54/L$ cancels between the ghost and border parts in the total; it sets only the grid requirement. Finer grids lower β_1 more at larger L, so the grid cannot restore c. The two codes agree to 10^{-12} relative at commensurate p at $L = 3$, 5, and the abelian and pure-gauge limits hold exactly. The data neither contradict nor support Theorem 3.1 of `BETA_FP_H.md`, because the family is different.

The cell certificate at $L = 13$, level $j = 1$. The program `numerics/h1cert_drive2.py` covers the sorted chamber at level $j = 1$ with initial half-width $h = 0.08$ and bisection depth at most 2 (`numerics/H1CERT2_M1_RESULT.md`). It has 8,855 top cells and 14,423 cell records: 13,872 succeed, 348 split, 200 fail, and 3 are near $K = 0$. The largest bound on a successful leaf is $\rho_1 \le 3.59685 < \sqrt{13} = 3.60555$, so $\rho_1(1, K)^2 \le 12.937$ there. The margin is 0.0087 in ρ_1 (0.24%); it reflects the loose depth-0 enclosure, not the true ρ_1. The 200 failed cells are cells without a bound, not violations. All have depth 2, half-width 0.02, no exception, and a finite bound between 3.621 and 9.936. All descend from two top cells next to $K = 0$ ($|K|_\infty \le 0.32$), where the Taylor and Oishi–Rump enclosures grow as the cell width approaches $|K|$. There the true value is small: the analytic majorant gives $\rho_1^2 \approx 2$ near $K = 0$, and $\rho_1(j, 0) = 1$. The two certificates are tightest at opposite ends of the chamber: the analytic one near the face diagonal, and the cell one near $K = 0$. Result: at $j = 1$, the cell certificate bounds $\rho_1^2 < 13$ on the sorted chamber outside $|K|_\infty \le 0.32$, independently of the analytic majorant, under the premises P1a, P2, P5b. Theorem 6.2 does not depend on it.

(L-LL) (H-137). The program `numerics/t2_lll_window.py` (`SOURCES_T2_LLL.md` §6) takes the operator A'_α from the builder of `numerics/t2_gaprime2_check.py`, at flux $\alpha = 2\pi/q$ and $\varphi = \pi/q$. It computes $H_\varphi(\mu)$ directly from its definition, not from the proof.

φ	0.03977	0.02001	0.01001	0.00500	limit h (Theorem 5.133)
$H_\varphi(3)$	0.124869	0.121263	0.119109	0.117872	0.116322
$H_\varphi(5)$	0.183646	0.177929	0.174470	0.172467	0.169938

Checks: the Mehler identity of L-275 holds to relative error $9 \cdot 10^{-16}$; the evenness $A'_{-\alpha} = CA'_\alpha C$ is exact; the Nielsen–Olesen count $\tau(P_-)/(\alpha/2\pi)$ is 1.00000 at every q; the k-grid is stable to all printed digits. The error behaves as $H - h \approx \varphi(0.07 + 0.046\ln(1/\varphi))$ at $\mu = 3$ and $\varphi(0.10 + 0.077\ln(1/\varphi))$ at $\mu = 5$. So the observed rate is $O(\varphi\ln(1/\varphi))$; we prove no rate. In the continuum, $h(\mu) - \frac{11}{12\pi^2}\ln\mu \to 0.023444$. The proof of (L-LL) does not use the gap theorem, so the (G_{2D}) values of Section 6.8 bear only on (M_j).

The free constant of (C) (H-057). For $U = 1$ and a single layer, $c_-^{\text{free}} = 1/(1 + K_0)$ equals $5.10 \cdot 10^{-3}$ ($d = 4$), $7.82 \cdot 10^{-3}$ ($d = 3$), and $1.37 \cdot 10^{-2}$ ($d = 2$), independent of L as $N \to \infty$ (`SOURCES_C_CONDITION.md` §6, `numerics/t2_ccond_free.py`). The script reproduces $K_0(4, N = 3) = 103.248$, with a real-space check 48.198346 against 48.198347. This value enters the reduction of (C) (Proposition 5.126).

6.10 The seam test of (C), (L-LL) at smaller φ, and the converged β_1

The seam test of (C) (H-057). A second program reproduces the free constant $c_-^{\text{free}} = 1/(1 + K_0) = 5.10 \cdot 10^{-3}$ at $d = 4$ (Section 6.9). The seam test (`numerics/c_verify_seam.py`) uses $d = 2$, a torus of $N_T = 6$ unit blocks with N fine steps per block, and flat SU(2) links $U_1 = \exp(i\theta\sigma_3/N)$ in direction 1. Every plaquette is 1, and the Polyakov angle is θ. The field is the dressed extension with model kernel $E = e^{-d_1}$ and the axial path rule in the fundamental domain.

N	$\theta = 1$: total energy	seam row	cut-off $\chi(d/2.5)$: total	seam row
3	2.075	0.032	2.500	0
27	2.187	0.196	2.483	0
81	2.565	0.574	2.485	0

243	3.699	1.709	2.486	0

At $\theta = 1$ the seam energy grows by a factor 2.98 per factor 3 in N, so it is linear in $N = L^k$. The predicted value at $N = 243$ is $N|\mathrm{Ad}P - 1|^2 e^{-\delta N_T} \int e^{-2|t|} dt = 1.70$; the measured value is 1.709. At $\theta = 0$ the seam row tends to 0 like $1/N$. With the cut-off of L-283 (Lemma T) inside the fundamental domain ($D < N_T/2$) the energy is uniform in N. A cut-off with $D = 4 > N_T/2$ still meets the seam, and its energy still grows. So the threshold (T2) is needed.

(L-LL) (H-137) at smaller φ. A second program (`numerics/t2_lll_verify_window.py`) reproduces every value of Section 6.9 and extends them to $\varphi = 0.00125$. The k-grids $n_k = 1$ and $n_k = 3$ agree to 6 digits at $q = 628$.

φ	0.03977	0.02001	0.01001	0.00500	0.002501	0.001251
$(H_\varphi(3) - h(3))/\varphi$	0.215	0.247	0.278	0.310	0.341	0.372
$(H_\varphi(5) - h(5))/\varphi$	0.345	0.399	0.453	0.506	0.558	0.609

$H_\varphi(3) = 0.117174$, 0.116787 and $H_\varphi(5) = 0.171333$, 0.170699 at $\varphi = 0.0025$, 0.00125. The step per halving is constant, 0.031 at $\mu = 3$ and 0.054 to 0.051 at $\mu = 5$. So $H - h \approx \varphi(a + b\ln(1/\varphi))$ holds down to $\varphi = 0.00125$, with $b \approx 0.045$ ($\mu = 3$) and $b \approx 0.075$ ($\mu = 5$). An independent continuum quadrature gives $h(3) = 0.1163219$, $h(5) = 0.1699377$, and $h(\mu) - \frac{11}{12\pi^2}\ln\mu = 0.0142853$, 0.0204567, 0.0234371, 0.0234442 at $\mu = 3$, 5, 10^2, 10^4. A test of the Nielsen–Olesen term without the logarithm shows that its limit is finite, with a correction of order $\alpha\ln(1/\alpha)$. This explains the observed rate $\varphi\ln(1/\varphi)$.

The converged-grid β_1. The converged grids at $L = 9$, 11, 13 are in `NUMERICS` §7(f), and the code is cross-checked in `SOURCES_BETA1_CODE_CHECK`. The jax port of the code agrees with the numpy path to $4.4 \cdot 10^{-16}$ on a GPU and $7.8 \cdot 10^{-16}$ on a laptop. The three $L = 9$ points ($N_g = 24$) give $\beta = 0.8201$, 0.8231, 0.8423 at $Lp = 0.25$, 0.5, 1.0. The $Lp \to 0$ values are 0.8192, 0.8354, 0.8423 ± 0.0014 at $L = 9$, 11, 13. The converged slopes still fall: 0.081 at $9 \to 11$ and 0.041 ± 0.009 at $11 \to 13$. In the possible approach of Section 6.9, these data measure $B_c(L)$.

6.11 Design of the cell certificate and the tail level j_0

At $L = 13$ the analytic proof of Section 6.7 needs no tail level j_0, no bound on $\|\mathfrak{B}^{-1}\|$, and no Lipschitz constant. The cell certificate below gives an independent bound at $L = 13$ on part of the chamber (Section 6.9). A proof of (H2) by cells is a computer-assisted proof of substantial size. `H2_LARGE_L` §4 gives the design. It has three parts.

(a) Tail $j \geq j_0$. From the resolvent identity, $\rho_2(j) \leq R_\infty + (\rho_1^2/a_*)(\delta_{j-2} + \delta_{j-1})$. Here $R_\infty = \rho_2$ at $j = \infty$, $a_* = \inf_K a_j(K)$ with $\|[\mathfrak{B}^{-1}]_{f,f}\| = 1/a_j(K)$ exactly, and $\delta_m = \sup_K \|\Delta_{L^m} - \Delta_\infty\|$ (O5 decay). The 2-norm supremum of δ_m is at the corner: $\delta = 179$, 92.0, 18.7, 2.39, 0.317 ($M = 1, 3, 9, 27, 81$). The value of a_* at $L = 5$ is 0.0107. With these inputs, $j_0 \approx 8$ ($L = 3$) and ≈ 6 ($L = 5$). With the O6 coercivity constant, $j_0 = 25$, which cannot be used in practice; with $\|\mathfrak{B}^{-1}\| \leq b(L) \sim 10^{19}$ one needs $j \gtrsim 45$.

(b) Levels $2 \leq j < j_0$ and $j = \infty$. One needs a certified supremum over the Brillouin zone of $\|H_{(2),j}(K)\|$ (at $L \geq 5$: of $\|H_j(K)\|$, Section 3.7.4). Plan: interval arithmetic with local mean-value enclosures on boxes in the sorted chamber (1/384 of the Brillouin zone), 4×4 and 5×5 verified inverses, and one 4×4 λ_{max} per K. At $L = 3$ this needs about $2.5 \cdot 10^3$ boxes of half-width 0.05,

and the analytic form of Δ near $K = 0$, where $\rho_2(0) = 1$. At $j = \infty$ one also needs a certified tail bound for the Prop. A sum (l_{max} truncation; observed effect $4 \cdot 10^{-4}$). No explicit global Lipschitz constant is known: the strip width of A0/O5 is existential, and the constant that can be made explicit is $\sim 10^{40}$. Monotonicity in j is not proved and is not used.

(c) Inputs to certify first: $\|\mathfrak{B}_j(K)^{-1}\| \le B_{num}$ for all j, K (the O6 constant $b(L)$ is too large to use), the O5 constant C_0 explicitly (or certified δ_m), and certified a_*, ρ_1, R_∞.

At $L = 13$ the cell certificate runs as a sweep (`numerics/h1cert_cell2.py`) over the sorted chamber, which is valid by P5.

7 Hypotheses and open problems

This section states the hypotheses on which the two main theorems rest, and the open problems. The hypotheses are grouped by kind: the uniform inequality (H2), the fixed-level and structural inputs of Theorem 2w (L-001), the inputs of the sources block, and the readings of Bałaban's text. The item labels H, 1–13, B1, BH, and B3–B7 are those of the dependency table (Section 4.6). Section 7.7 lists every statement that the two main theorems use and that this paper does not prove: Bałaban-level statements, external results, our hypotheses, and conventions.

7.1 The uniform inequality (H2)

Both routes need one uniform hypothesis.

Hypothesis 7.1 (item H; (H2), R-025)**.** $\rho_2 = \sup_{j\ge 2} \sup_{K \text{ real}} \|(\oplus_l H_{j-1}(k_l))H_j(K)\|/L^2 < L$.

At $L = 13$ this hypothesis is a theorem (Theorem 6.2): $\rho_1(j,K)^2 \le 10 < 13$ for all $j \ge 1$ (including ∞) and all K, hence $\rho_2 \le 10 < 13$ by `rho2_le`. The proof is computer-assisted (interval arithmetic with outward rounding). The same certificate gives (H2) at $L = 15, 17, 21$ (Section 6.8). At $L = 3, 5, 9$ we have numerical evidence only. At $L = 3$ one point is enclosed: $j = \infty$, $K^* = (2.4035, 2.4035, \pi, \pi)$, $\rho_2 \in [1.9993, 2.1273]$ (Section 6.6). The paper fixes $L = 13$ (K-003), so (H2) at other L is not needed.

H-res. The premises of Theorem 6.2 (Section 6.7) are: the reading of Bałaban's (0.4) and the gauge choice; Prop. A at $M \ge 27$ and $M = \infty$; L-166 (Lemma G) and the bound chain; the correctness of the interval code `bb_arb.py`. The gauge-fixing function is a convention: the minimum-norm gauge among the gauges that select one point per residual orbit.

Consequences. Route K: (H2) $\Rightarrow$ ($C1\sigma$) $\Rightarrow$ (B2) (Theorems 3.18, 3.19). Route B: (H2) and (T1) together give (U) by a possible approach (Section 3.6.3). The cell certificate at $L = 13$, mode $M = 1$, covers the chamber outside $|K|_\infty \le 0.32$ at $j = 1$; it is a partial cross-check (Section 6.9). Positivity of the early steps $\beta_1(0), \dots, \beta_{j_*}(0)$ at odd $L > 11$ is needed only for literal (0.31) at every K (Section 3.9.2).

7.2 Fixed-level and structural inputs of Theorem 2w

Route K. All of the following must hold.

1. (B2) $\sup_j \sup_K |\sigma_j^{(2)}| < \infty$, or (B2w) (H-184). (B2) follows from (H2) (item H). Two other implications hold: ($C1\sigma$) + (h) $\Rightarrow$ (B2) is proved, and (H_A) $\Rightarrow$ (B2) is a possible approach. The inequality (HS) is false.

2. (PI) (R-026), probe independence. It holds at each fixed level relative to listed inputs: (I1) (H-207) for the composed fine matrix at every j, (I2) (H-208), and (I3), the lattice-symmetry input, for the full hypercubic group. We claim no uniformity in j.

3. (F) (R-027): the lattice log coefficient at constant field equals the continuum one. It is proved (Theorem 5.134), by proper time, with (L-LL) (H-137) (Theorem 5.133); the proof needs neither Lemma 2 nor Step C. The continuum side is cited from Savvidy [22] (X-005). A separate numerical tension remains: the single-step β_1 at $L = 5$–13 does not follow $c \log L$ (Section 6.9). It concerns a different family.

4. (M_j) (R-028): the formal f^2 coefficient equals the true one at each fixed level. The inputs are (S1) (H-244) $\mathcal{K}_t(f) \geq -C_1|f|$, (S2) (H-245) density of states $\leq C_2|f|$ near 0, and (S3) (H-247) a relative bound of the level difference, all on the lattice. (S1), (S3), $(S2)|_{t=0}$, and (S2c) (H-246) hold at fixed j given the Peierls input (I3). (I3) holds at every step, with constants independent of the step, given the identification of the one-step objects with the linearized averaging. (G_{2D}) holds for $0 < |f| \leq ((1-c)/100)^2/2$, with gap $2c|f|$ (Theorem 5.98). (M_j) is proved at fixed j relative to listed inputs (Theorem 5.102); region 1 and the neutral sector are proved in the $o(f^2)$ form, and the proved f-range suffices.

5. V1, the order-g^0 completeness of CMP 109 (1.3)–(1.5): $\beta_{j+1}(0)$ is the Maxwell coefficient of the per-step Gaussian determinant. It is proved at fixed level, conditional on (G2) (R-030), on Bałaban's inductive bounds, and on the declaration that the averaging is (0.4). It is history-free.

6. (G2) (R-030), per-cell infinite-volume densities. At fixed j the torus densities converge exponentially, and the p^2- and f^2-coefficients commute with $N \to \infty$. This is proved at fixed j relative to listed inputs, including (Gap_f) and $\tilde{Q}\tilde{Q}^T > 0$. We do not prove a j-uniform version.

Route B. The following hypotheses replace items 1–5.

B1. (T1) $\sup_j \sup_{q \text{ real}} |c^{(1)}_{T,j}(q)| < \infty$, the plane-wave form of L-071. A possible approach is sketched; the Landau-to-symmetric gauge conversion and Steps 2–5 of Theorem A3 (L-071) for (2.1) are not written.

BH. (H2) (R-025), that is, item H.

B3. (DF) $\sup_k |\mathfrak{d}_F(\theta_*; L^k)| < \infty$, the gauge-fixing difference at the plane wave. With the exact split $\Gamma^{ex} = \gamma_{rec} + \mathfrak{d}_F$ there is no Y-defect; all non-stationarity sits in $\mathfrak{d}_F$ (Section 3.6.2).

B4. Lüscher–Weisz (5.7) [16].

B5. A6 (not R-007): the plane-wave increment equals Bałaban's β up to the expansion point, uniformly in k. This is a uniform-in-k input of the same kind as (DF).

B6. (H0) (H-202), that is, item 7 (Lemma DT (L-005) (i)).

B7. $\theta_* < 1$, so that $c_2 \ln(1/\theta_*^2) \geq 0$ in C_B. This is a trivial condition on the choice of θ_*.

Route B also needs item 6 ((G2)), item 7, and items 8 and 10–13. It does not need (PI), (M_j), V1; it needs (DF) and A6 instead. In Route B, (U) is a hypothesis; a possible approach derives it from (T1) and (H2).

Both routes.

7. The own-slot bound $|\beta_{k+1}(h,g)-\beta_{k+1}(0)| \le C_\beta g$ on $]0,\gamma]^k \times [0,\gamma]$, for every past h, monotone or not. It follows from (H0) or from (G-gain) (H-111). (H0) is Bałaban's smoothness in g with bounds uniform in j and in the history (CMP 109 inductive assumption at odd $L > 11$, R-006). "Uniform in j" is our reading of p. 264 and is not printed. Uniformity in the history is not printed. The shooting evaluates β on clamped, non-monotone pasts. (G-gain) gives C_β directly only on monotone pasts. The running maximum $G_k = \max_{i\le k} g_i$ removes this restriction with no change of constant: the only use of monotonicity is a comparison of bound sizes in L-164, and G_k closes the induction for every past. The proof uses the Lipschitz form on $]0,\gamma]$ only. The use of (H0) reduces to Proposition H0-red (L-183), with residue N1–N9. N2 is Lemma G3 (L-174), proved relative to listed inputs (Theorem 5.91). The rest of the residue (N5, N6 relative to the small-field step of R-013, N8 a reading, N9 a definition, N7 = R-031) is not proved, so R-006 is an input.

8. L5, history dependence. It is proved relative to listed inputs by Route 2, nested intervals on true trajectories with no clamp (Theorem 5.104).

   Remark 7.2. A clamp proof of Lemmas DS/DT would need $\beta_{k+1}(g_0,\dots,g_k)$ defined and continuous on the whole cube $]0,\gamma]^{k+1}$, and Bałaban does not supply this. Bałaban's β_{k+1} exists only on histories where his induction has run; that domain must carry (c-pair) + K-018 (S), not only $g_i \in]0,\gamma]$. Clamped histories can violate the growth side of (c-pair). True trajectories with all $u_i \ge \gamma^{-2}$ inherit (c-pair) from (a-tail). This is why Route 2 uses nested intervals on true trajectories. Lean covers the one-argument β (Lemmas DS, DT, D, E) and the history form (item 13).

9. K-independence of $\beta_j(g)$. It holds. Item 9 is a volume statement, not a history identity: K enters β_j only through the torus size. CMP 119 (2.24), p. 249, and p. 279 define β_j as the whole-lattice coefficient, so the β_j of the recursion is K-free by definition. Lemmas DS/DT run at one fixed K and never compare trajectories. The window bound needs only (K1) (history-free at zero own coupling), (K2) C_β (item 7), and (K3) the window bound (5.1) of Theorem K (L-009). The residual volume question (the finite-volume step equals the whole-lattice step, R-015, R-032) is an input of Theorem 1, not of the shooting. Given (Loc) (H-141), the torus coefficient is within $C_j N_j^6 e^{-m_j N_j/2}$ of the whole-lattice one (Proposition 5.87); exact equality cannot be derived in general. (Loc) is proved (Theorem 5.93), and the boundary reading and the replaced R-032 are proved under (Vol) (K-032) (Proposition 5.95).

11. Odd L. All one-loop proofs need odd $L \ge 3$ (centred blocks: U3 Lemma 1.2, O5/Prop. A, evenness, Lemma 4.1 of LEMMA43). Bałaban assumes L odd > 11 throughout CMP 109, so (H0) and $\beta_j(g)$ at $g > 0$ exist only there. Theorem 2w (L-001) is stated for odd $L > 11$, or with (H0) at the given L as a hypothesis. Numerics: $\beta_1(0)$ up to $L = 11$ (Section 6.1); (H2) at $L = 3, 5$ (scans), 7 (face diagonal), 9 (one point); ρ_1 at $L = 3,\dots,13$. Only $L = 13$ is inside Bałaban's range; there (H2) is a theorem (Theorem 6.2).

12. SU(2) only. The charged/neutral split of Lemma 5.1 and (W) are SU(2) statements. The value $c = 11N^2/(12\pi^2)$ is general, but the chain is not established for $N > 2$.

7.3 Inputs of the sources block

These inputs stand between the trajectory of Section 3 and the bound (H-U) (H-116) of Section 4.2.

- (B-S) (R-035). The sourced version of Bałaban's Theorem 1, (E14-a, b, c), is open as a whole. It is reduced to its parts R-036–R-044, R-046 and to the reading (W6*) (H-361); joint holomorphy holds given the bounds.

- (H-hist) (R-037), via (G-gain) (H-111). In the small-field step the pointwise Cauchy bound and the uniform C_β on $]0,\gamma]$ hold. Lemma G3 (L-174), the χ_k surface term, is proved relative to listed inputs, with (Q) assembled from R-008 and R-009 in the joint (X,B) form (Theorem 5.91); its conditions are K-026 and K-027, and G_Γ is bounded uniformly in M. The large-field half is reduced to one Cauchy step on the large-field disc, with the factor $1/g$ absorbed by the exponent of p_0 (CMP 119 p. 244). So the real uniform (H-hist) is reduced in both halves. Its open inputs are CMP 102 (189), the large-field coarea Jacobian, and the F2–α link (a reading). The margin (M_r) is exact at $r = 1$. The (M-B) (H-145) bookkeeping and the Kotecky–Preiss derivative step are proved relative to listed inputs. Lemma G4 (L-178) is not used. Second derivatives: (H-hist2) (R-038).

- The Bałaban-level core. The propagator and covariance inputs R-001–R-005 reduce to published theorems:

  – R-001, kernel-decay form: reduced via L-125 (Lemma CT) to the printed CMP 99 Theorems 3.1, 3.3, and the G part of 3.12, with (KE2) (H-136) by citation (CMP 95 (1.11)/(1.18), CMP 96 (2.35)), (H_Q) (H-123) proved, and S2 proved for G and G_1. The condition (C) (H-057) is reduced (Proposition 5.126). L-284 (Lemma T) discharges the torus-side hypothesis under (Vol) (K-032), and the two unprinted conventions are row K-044. The printed d is block-constant on fine points, so part (c) of L-284 is replaced by a version with a block-interpolated cut-off and a closure-bond metric, at a (d,L)-constant cost. Residue: the printed CMP 99 inputs of (C) (row R-021), the reading $a = 1$ of Theorem 3.1, and the path metric.

  – R-002, the walk-expansion form of R-001 (CMP 99 Theorem 3.15, "analyzed in the same way"): reduced and written out. Theorem 5.76 gives the walk expansion of $(QGQ^*)^{-1}$ with (3.108)-form bounds at rate $\frac{1}{2}\varepsilon$, the Z_0-compressed variant (O3), and the complex α'-space (O4). Its input (W_G) for the final propagators is Theorem 5.77: the walk expansion of G and the tree expansion of G_1, built from Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10, the Neumann series (3.130), and CMP 96 Lemma 2.1. Both theorems hold relative to their listed inputs. The inputs are: the lower bound (C) (the residue of R-001); the three printed sketches, written out as Theorems 5.145 and 5.143; (W_G) at complex points (reduced); the condition WG-1 in the M_1 form $\alpha_0 \le c' M^{-1} M_1^{-10}$ ($c'' M^{-1} M_1^{-12}$ for G_1), admissible in Bałaban's order; and geometry and convention items (cube geometry, Lipschitz partition, Q-locality, $C^{(2)}$-locality, the scale bookkeeping C0, the M_1-cover, κ_1 independent of M).

    Remark 7.3. The printed partition of unity of CMP 99 fails near level interfaces ((X-PU) (H-180)). The repair replaces c_h by Lc_h on interface cubes, a factor L in a constant. Every changed constant is fixed before M_1, and no conclusion changes.

  – R-003 (R_2, R_3 of CMP 116 (2.16)): reduced. R_2 follows by a Neumann series around the compression, with the complex-U repair V7, from the complex walk bounds (W^c) on the α'-space, which are Theorem 5.78. That theorem gives, in the exponential gauge: $A_Z \ge \frac{1}{4}\gamma_e$ at real U; the walk expansion of $C^{(k)}(Z_0)$ in the CMP 116 p. 13 form; $\operatorname{Re} A_Z \ge \frac{1}{2}\gamma_{\exp}$ on the α'-space; and the resolvent bound with per-walk prefactor $4/(\gamma_{\exp} + x)$. It does not use Thm 3.15. Its inputs are Theorem 5.77(G1), (C), the inputs of R-004, CMP 109 p. 254, readings, the decay of $C^* \Delta_k C$ at real U (R-008), the boundary-contact reading, and $\operatorname{dist}(Z_0, \Lambda_k^c) > RM$. R_3 holds in weaker forms that suffice for (2.21). CMP 109 p. 254 is proved for the CMP 98 results (Proposition 5.141); for the CMP 99 operators it rests on the p. 253 assertion (row R-022).

– R-004 ($C^*\Delta_k C \geq \gamma_0$): proved from the printed CMP 99 Theorems 3.11–3.12, (H_Q), R-Δ2 (written out from the printed (3.137) by a Schur test), and items R3–R5: $\gamma_0 \geq m_L/(48d^2L^d)$ with $m_L = \min(2(1-\cos(\pi/L)), 1/L)$, uniformly in k, the volume, and regular U, for $M \geq L/2$. This equals $(48d^2)^{-1}L^{-d-1}$ for $L \leq 9$ and is 0.7555 times it at $L = 13$. The σ-localized clause reduces to (W) + (N), without symmetry.

Remark 7.4. In CMP 96 (2.126)–(2.127) one step of the proof is false for $L \geq 10$: a factor L is lost. The statement of Lemma 2.4 stands with the corrected constant.

– R-005 (CMP 102 (189)): reduced to R-001, published CMP 102 statements, R-018, and one bookkeeping item. On-shell, (189) is proved relative to listed inputs (real U_0, complex A', rate $a < \frac{1}{2}\delta_0$). The off-shell literal reading is open.

– R-008 is proved relative to listed inputs (Theorem 5.79). Here C is the constraint map and $C^*\Delta_k C$ the precision. The norm bound follows by L-226 (Lemma N) from C2 of Theorem 5.78, A3-R4, I1, I3; the decay follows from W_{G_1}, Theorem 5.76, I2, I9. Complex s is treated at the level of a proof sketch.

– R-009 is proved in a corrected form (Theorem 5.80). Bałaban's symmetry claim is not proved and not needed. The majorant uses $H_0 = \operatorname{sym} C_0^{-1}$, and Lemma D holds with $e^{-\varepsilon'' M/8}$. Readings I-a–I-e.

– R-010 is proved in the corrected form R-011 (Theorem 5.82). The raw Taylor terms do not obey (1.24); the transformed terms do, after the exact marginal cancellation. Per-label exponent β_H. The off-shell Δ entry is reduced to seven items.

– R-012 holds as a list (Proposition 5.85).

– R-015 is reduced to conventions. (Loc) is exact for non-wrapping polymers relative to K-033–K-037 (Theorem 5.93). The $\partial\Lambda_{k+1}$ reading (f) is exact relative to K-038 (Proposition 5.95). Threshold $N_k > 22M$, $\varepsilon(M) = C_j(22M)^6 e^{-11 m_j M}$.

– R-006: the assertion that the $\mathbb{Z}^d$ limit exists follows from Proposition 5.87 and Theorem 5.93. Its use in Theorem 2w (L-001) is reduced, with N2 = Lemma G3 proved (Theorem 5.91).

– R-032 is ill-posed as stated; it is replaced by (a)–(c), which are proved relative to listed inputs and (Vol).

– R-016 is proved relative to listed inputs (Theorem 5.110). R-017 is proved relative to the printed text (Proposition 5.106). R-018 (Proposition 5.108), R-019 (Proposition 5.107), and R-020 (Proposition 5.109) are reduced. Conditions K-039–K-043.

– The condition (C) is stated exactly in row R-001. It is reduced (Proposition 5.126, Lemma 5.129), with residue R-021. The proof closes the Schur step for G_1. On the periodic lattice a seam defect appears; the proof repairs it and replaces the torus-side hypothesis by (Vol). K-009 is not used.

– (L-LL) (H-137) is proved; (F) (R-027) is proved; (S_σ) (H-168) is proved; the s-rule is convention K-046; H-035 (M1) holds as a list; the p. 254 transfer is proved (Proposition 5.141); CMP 99 Theorems 3.7 and 3.9 are written out, with the printed errors corrected.

– R-022 and R-023 are reduced to page readings. The former open items of R-022 (Lemma $\mathrm{LB}_{\tilde{Q}}$ (L-211) and the k-step bound $\tilde{Q}_k\tilde{Q}_k^* \geq c$) and the gluing step of R-023 follow from the Dressing Lemma. R-024 (Q-049) is reduced; its sourced extension is proved relative to its inputs. Its residue is (Sep) on the R-operation-modified $T^{(j)}$ (a strong reading)

and (S_H^Ω) (a proof sketch). (NEW-2) (Q-048) is proved relative to $Q1^\Omega$ and the Schur step; it lies inside R-016.

The Bałaban-level core is a reduction, not a proof. It rests on:

- published propagator theorems;
- Bałaban's printed proof sketches of CMP 99 Thms 3.7, 3.9, 3.10 (through Theorems 5.76, 5.77, 5.78);
- the printed inputs of (C) (row R-021), the page readings of R-022 and R-023, and the residue of the multiscale assertion of CMP 119 pp. 275–276 (row R-024);
- the inputs of Theorems 5.79, 5.80, 5.82; R-018; two readings of R-003;
- the geometry and convention items of the write-outs;
- the conventions K-044 and K-046, the locality conventions K-033–K-038, and (Vol) (K-032).

- (H-hist2), on which the threshold-shell bound (H-thr) (R-039) depends: reduced to the real uniform (H-hist), with no new input, together with the M3 history weights and the frozen-integer convention (L3). Its β_j part (the threshold-history mixed derivative) is open.

- (FX) (R-043). Q-059(b), Q-060, and R2 are proved; Q-061 is proved relative to N3 (R-042); Q-062 is avoided by Bałaban's main form (1.98). N3, the HBETA strip bound $|\delta w| \le \frac{3}{2}\tau$, is reduced to (H-hol) (R-040, R-041), the real (H-hist), and (H2-hist) (H-120); its condition (a) is proved. (H-hol) is reduced to (M-A) (H-143) (via (H−) and CMP 102 (189)), (M-B) (proved), and (M-C) (H-146). (M-C) is reduced to the printed conditions of CMP 116 §2 (B1–B23 and four more), Lemma 2, (1.36), and p. 12 (2.5)–(2.6), with the shrinks $\varepsilon_1/(4\lambda_V)$, $\gamma/(8\lambda_V)$. Its inputs (I-V0) (H-133), (I-ε) (for CMP 109/116 and for the large-field layer of CMP 119/122), and (I-ord) (H-131) are proved, with two readings: the unprinted scaling of α_2–α_4, and q_0 versus p_0 (each branch closes). K-025 (M1) is a convention. No consumer needs coupling or threshold derivatives of order greater than 2. The two minor inputs of R2 ($R_{K+1} \ge L$, C_c independent of g) are page readings. The coarsest-scale (FX2) (H-191) radius is not a separate input.

- (B-S.D2C, B-S.Hβ) The domain hypothesis (D), and the conditions (I) (R-036, proved relative to listed inputs) and (H-hist) of the two-loop bound.

- (B-S.D2, B-S.D3) The recovery of the decay loss $\kappa \to \kappa/2$ is reduced by $L/2$-rescaling, with $\kappa \ge 1536$ at $d = 4$ (R-046). It uses the unprinted large-field T-step rate loss R-020, which is reduced with loss $\le 13\delta$; so ($K1_\delta$) (H-216) holds with $\delta = 0.025$ at $L = 13$. The scale condition (U) (K-024) ($\rho_1 \le c_* r_0/M^a$ at the unit scale) is a choice of constants, compatible with (B-N1) (R-045). It does not appear in Theorem B.

- (B-N1) The bridge from the source form to (H-U) holds given R-045 (S) (Proposition 4.7). The uniformity of R-045 (S) in support size and in ε is reduced: every factor of $C^n(n!)^{1+\eta}$ is independent of h, given L-127 (Lemma CS), S1 at all scales, and (FX) in the main form. The (E14-c) (H-098) clause is reduced to R-045 (S) (Section 5.12). Given (FX), the step (S_{src}) $\Rightarrow$ (H-U) holds with $s = 6$, $D > 1$.

7.4 Readings of Bałaban's papers

10. The cross-scale sweep (Section 3.4.3) locates every use of the trajectory in Bałaban's papers by search and page images; it is not a line-by-line reading. It covers all 19 rows, with the CMP 122 additions H-271–H-280 and H-266–H-270. "Similar inequalities hold for other constants, which will be introduced later" (CMP 119 p. 256) holds as a list (Proposition 5.85, R-012). "β_0 can be chosen arbitrarily small" is on CMP 119 p. 255, and β_0 is fixed before γ, as used (former R-048). The readings of Π (4.37)/(5.1), CMP 119 (2.42), (2.12)/(2.23), and the "L odd" page hold, with corrections of misprints (former R-051). The residue in CMP 119 Sect. 3 is the unprinted T-step rate loss (R-020). Theorem 1 itself rests on Bałaban's published proof, which uses the Group A items of Section 7.7.

13. The history form of Lemmas DS/DT is formalized in Lean: LemmaDHist.lean, LemmaDHistNested.lean (`lemma_DPrime_hist_adm`, `lemma_DDoublePrime_hist_adm`). That Bałaban's β_j meets the continuity hypotheses is an input. Lemma DT (L-005) in the one-argument form is also formalized (Section 3.4.2).

Remark 7.5 (misprints in Bałaban's papers). The following misprints are corrected where they are used. CMP 99 p. 411: "$e^{-(1/4)\delta_0 M} \geq 3!(\frac{1}{4}\delta_0 M)^{-3}$" should read $\leq$. CMP 99 p. 412: "(3.25)" should read (3.35). CMP 99 p. 415 (Thm 3.10): "(3.95)" should read (3.35), and $\tilde{\Box}_0^5$ and $\tilde{\Box}^8$ do not agree (16 against 18); we use 18. CMP 122 II p. 385: "$R_{m+1} = LR_m$" should read $R_{m+1} = L^{-1}R_m$. CMP 122 II (1.85), p. 386: a factor $\frac{1}{2}$ is dropped. CMP 119 p. 267: "[16]" should read [15]. CMP 119 p. 268: the factor $L^2(1+\beta_0)$ is dropped. CMP 109 (4.38): U_j should read U_k. CMP 122 II (1.77)–(1.78): p_1 should read p_1^2. CMP 109 (1.3)–(1.5): $E^{(j+k)}$ should read $E^{(j+1)}$. CMP 122 I p. 192 assumes $L_0^2 \leq L/3$, which conflicts with p. 182; this is not a misprint, and the order of constants records it.

Remark 7.6 (CMP 122 II p. 387). In CMP 122 II p. 387, Bałaban's proof of Case 2 of the large-field induction is invalid as written: one step is unproved and one step is false. Lemma G replaces it and gives the induction (1.80)–(1.89) with the improved bound in volume form. Bałaban's conclusions stand. Every use of (1.80), (1.89), or the improved bound in this paper needs only the form that Lemma G gives.

7.5 Open problems

The following problems are not needed for Theorem 1 or Theorem B. They are open for stronger statements.

Open Problem 7.7 (convergence of the running coupling)**.** Prove (A) in full and Lemma 5.1 for $m = 2$ (convergence of $\beta_j(0)$, with a rate). Prove C^2 and C^3 bounds on $\sigma^{(1)}$ under (H2) (R-025). Prove or refute (N).

Open Problem 7.8 (literal (0.31))**.** Prove $\beta_1(0), \ldots, \beta_{j_*}(0) > 0$ at an odd $L > 11$, with an explicit j_*. This gives literal (0.31) at every K.

Open Problem 7.9 (nontriviality)**.** Prove (NT) (H-101) at every layer, or the four-point hypothesis (NT4) (H-339). The order-g^4 tree route does not work: the connected four-point function has no order-g^4 term, and its leading term is the free order-g^8 ring. A nonzero connected four-point function proves only that the limit is not Gaussian, not that it is interacting. The single-distance form $(\mathrm{NT4})_{d_0}$ is reduced to a locality hypothesis (NT-loc) (H-341), and (NT-loc) is open.

Open Problem 7.10 (uniqueness)**.** Prove E13: convergence along the full sequence $\varepsilon \to 0$ (comparison of two trajectories). It is not needed for Theorem B.

7.6 Why the mass gap is not on this route

Write β_W for the coefficient of $\sum_p(1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p)$ in the Wilson action. Bałaban's action is $g^{-2}\sum_p(1 - \operatorname{Re}\operatorname{tr} U_p)$ with the normalized trace (CMP 109 (0.2)), so $\beta_W = 1/g_B^2$. Shen, Zhu, and Zhu [24] prove a mass gap for SU(N) lattice Yang–Mills with action $N\beta \operatorname{Re}\sum_p \operatorname{Tr} Q_p$ (their (1.2)) when $|\beta| < 1/(16(d-1))$ (their (1.3)); with $\beta_W = N^2\beta$, $N = 2$, and $d = 4$, this is $\beta_W < 1/12$. Bałaban's construction requires $-2(1+\beta_0)^{-1}p_0(g_k) + 1 + 2\kappa(100R_k)^d \le -\frac{3}{2}p_0(g_k)$ (CMP 122 II p. 389). With $d = 4$ and $R_K \ge 1$, this gives $p_0(g_K) > 4\kappa \cdot 10^8 + 2$ for every $\beta_0 > 0$ (twice this at Bałaban's $\beta_0 = 1/7$). The unit-scale threshold $g_K p_0(g_K)$ must be below 2, the largest value of $\|U - 1\|$ on SU(2); with $\kappa \ge 1$ this gives $\beta_W = 1/g_K^2 > p_0(g_K)^2/4 > 4 \times 10^{16}$ at the unit scale. So the two regimes are separated by at least seventeen orders of magnitude in β_W. No choice of L narrows this, because L does not enter the condition. A proof of a mass gap needs a mechanism outside the renormalization-group construction, and Bałaban's papers do not address it.

7.7 The inputs of the main theorems

The three tables below list every statement that the two main theorems use and that this paper does not prove. Group A holds Bałaban-level statements and external results (X). Group B holds our own hypotheses. Group C holds conventions and choices. The IDs are those of the trust ledger `TRUST_LEDGER.md`. Rows R-025–R-046 are not the Route B items B1–B7, and R-007 is not the item A6 of Route B. Sources are cited by paper and page. The last column states the form in which the paper uses the item, in the words of Section 4.6; "reading" marks an interpretation of Bałaban's text that we adopt.

Group A: Bałaban-level statements and external results

ID	statement	source	form used here
R-001	Kernel bounds on $(QGQ^*)^{-1}$, $(QG_1Q^*)^{-1}$, uniform in k and regular U	CMP 99 p. 422 (3.132)	asserted, "not repeated". The kernel-decay form is reduced via L-125 (Lemma CT) to CMP 99 Thms 3.1, 3.3, the G part of 3.12, H-136 (KE2), and S2. Residue: the condition H-057 (C): $\hat{N} = S^{-1/2}QGQ^*S^{-1/2} \geq c_-$ and $\hat{N}_1 = S^{-1/2}QG_1Q^*S^{-1/2} \geq c_{-,1} \geq \frac{1}{2}c_-$ on $\ell^2(\mathfrak{B}, \mu)$, uniformly in regular U, j, k, η, $\{\Omega_j\}$, and the volume; the γ_0 of R-004 does not imply it. (C) is reduced (Proposition 5.126), with $c_- = 1/(4K^D)$ and $c_{-,1} = c_-/(1 + \delta_1)$ by L-039 (Lemma R). L-284 (Lemma T) discharges the torus-side hypothesis under K-032 (Vol), with $M \geq \tilde{c}_T/22$ (K-045); a seam defect of the periodic lattice is repaired. The printed d is block-constant on fine points, so part (c) of L-284 is replaced by (c-rep), with the cut-off $\tilde{\chi}_y = \sum_{y'} \psi_{y'}\chi(d_*(y', y)/D)$ and the closure-bond metric $d_* \leq d$. New constants: $\tilde{c}_T \leq c_T + 6Lc_{\mathrm{adj}}$, $\tilde{D}_0 \leq D_0 + 2c_{\mathrm{adj}}$, $c_{\mathrm{adj}} \leq 2d + (d-1)\lfloor L/2 \rfloor$, $c_- \to \tilde{c}_- = 1/(4\tilde{K}^D)$, and the condition $RM > 2dL$ (implied by $R > 18dL$). Residue: the printed inputs R-021, the reading $a = 1$, the path metric, the conventions K-044
R-002	Walk-expansion form of R-001 (the walk expansion of $C^{(k)}(Z_0)$, bounds (3.107)–(3.108))	CMP 99 Thm 3.15, p. 416	asserted, "analyzed in the same way". Reduced and written out as Theorems 5.76 and 5.77. Inputs: H-057 (C); CMP 99 Thms 3.7, 3.9, 3.10 (printed sketches, written out as Theorems 5.145 and 5.143); CMP 96 Lemma 2.1; WG-1 ($\alpha_0 \leq c'M^{-1}M_1^{-10}$); geometry and convention readings. O3 and O4 are proved. L-210 (Lemma LB) holds without conditions. The rate in (3.99) is $\frac{1}{2}\varepsilon - |2 - \frac{d}{2}|\theta_L$. H-180 (X-PU) costs a factor L in a constant; the H-209 (I3) rate is corrected

Group A: Bałaban-level statements and external results

ID	statement	source	form used here
R-003	R_2 and R_3 obey (2.16)	CMP 116 p. 16	asserted, "similarly". Reduced (V7 repair) to (W^c) = Theorem 5.78 (no Thm 3.15 transfer), R-008 at real U, the boundary-contact reading, and the distance RM (a reading). R_3 holds in weaker forms. The input R4 (CMP 109 p. 254) is proved for the CMP 98 results (Proposition 5.141); the CMP 99 operators rest on R-022. One step of Theorem 5.78 costs a (d, L) constant in $\gamma_{\exp}$
R-004	$C^*\Delta_k C \geq \gamma_0 > 0$, uniform in k and U	CMP 99 p. 428	asserted, "we can prove". Proved from CMP 99 Thms 3.11–3.12, H-123 (H_Q), and R-Δ2 (written out from the printed (3.137), CMP 99 p. 423, by a Schur test); $\gamma_0 \geq m_L/(48d^2L^d)$; the σ-localized clause reduces to (W) + (N). The proof step in CMP 96 (2.126)–(2.127) is false for $L \geq 10$. The R1 Schur step gives $K_1 = B_0 c_1(\frac{1}{2})$. Residue: the index-range reading; $G_1 > 0$ (Bałaban)
R-005	Kernel decay (189) of $\delta^2 V/\delta A'^2$ and its consequence (190)	CMP 102 p. 308, Prop. 9	announced. Reduced to R-001, published CMP 102 statements, R-018, and one bookkeeping item. On-shell, (189) is proved relative to listed inputs (real U_0, complex A', rate $a < \frac{1}{2}\delta_0$), and so are the derivative entries of (190). Residue: the complex form of CMP 98 Prop. 5 (a reading of p. 43) and two sketches. Off-shell, the literal reading is open
R-006	H-202 (H0): $E^{(j)}$, β_{j+1} smooth in g with bounded derivatives; includes the assertion that the whole-lattice limit of (1.22) exists	CMP 109 pp. 263–264	inductive assumption. L-001 (Theorem 2w) uses only item 7 (Lipschitz form on $]0, \gamma]$, every past), which also follows from H-111 (G-gain) with the running-maximum lift. That use is reduced (H0-red), with residue N1–N9; N2 = L-174 (Lemma G3) is proved relative to listed inputs (Theorem 5.91); N6 rests on the small-field step of R-013. The $\mathbb{Z}^d$-limit assertion follows from Proposition 5.87 and Theorem 5.93
R-007	$E^{(j)}$, β_j analytic in the couplings	CMP 109 pp. 264, 266	asserted; not used

Group A: Bałaban-level statements and external results

ID	statement	source	form used here
R-008	$\|C^*\Delta_k C\| \le \gamma_1$ and uniform decay of $C^*\Delta_k C$	CMP 116 p. 13 (2.7); CMP 99 p. 428	premise. Proved relative to listed inputs (Theorem 5.79): C = constraint map, $C^*\Delta_k C$ = precision; inputs C2, A3-R4, I1, I3 (norm), W_{G_1}, A1-W, I2, I9 (decay); complex s at the level of a sketch
R-009	Positive Gaussian measure for the real pair $(U, 0)$; complex case "perturbative"	CMP 116 p. 15	asserted. Proved in corrected form relative to Theorem 5.78 and the readings I-a–I-e (Theorem 5.80). The symmetry of $C^{(k)}(Z_0, 0)$ is not proved and not needed ($H_0 = \operatorname{sym} C_0^{-1}$); Lemma D at $e^{-\varepsilon'' M/8}$
R-010	Lower-order bracket terms "bounded similarly as in (1.24)"	CMP 116 pp. 8–9	asserted, "similarly". Proved in the corrected form R-011 relative to CMP 109 Sect. 3–5 (Theorem 5.82). The raw Taylor terms do not obey (1.24) (the j-sum diverges); the transformed terms do, after the marginal cancellation. Exponent β_H (K-028). The (4.5) prefactor is printed, and $K_{\text{tail}} \le 6.32(32 c_B B_3/\alpha_2)^4/(\delta_0 M)$ for $e\delta_0 M \ge 4$. The clause (C-c) is false and is replaced by a revised clause of Lemma C1. The off-shell Δ entry (H-358) is reduced, by the extension F, to seven items; among them are the literal (189) item, the decay rate, and the conversion $c_J = L^3\alpha_0$
R-012	"Similar inequalities hold for other constants"	CMP 119 p. 256	asserted. Holds as a list (Proposition 5.85); all follow from (2.6)–(2.9); order entries K-029–K-031. The p. 277 condition holds in its scale-indexed form (factor $L^{j-1}\eta$ of (3.37)) by CMP 119 (2.6)–(2.7)
R-013	Theorem 1	CMP 122 II p. 355; CMP 116, 119, 122 I–II	proved by Bałaban, conditional on $g_k \in \,]0, \gamma]$. Uses R-001–R-005, R-008, R-009, R-010, R-014, R-015, R-017, R-018. The large-field induction CMP 122 II (1.80)–(1.89) holds by Lemma G, with the improved bound in volume form (Remark 7.6)
R-014	Induction closure recovers E_0	CMP 109 p. 280	asserted, constant unspecified; used only inside R-013; it carries the off-shell items of R-010

Group A: Bałaban-level statements and external results

ID	statement	source	form used here
R-015	Small-field terms and β_j are whole-lattice quantities, independent of Λ_{k+1}	CMP 119 p. 278, (2.24), p. 249; CMP 109 (4.35)–(4.37)	asserted, small-field part only. Reduced to conventions: H-141 (Loc) is exact relative to K-033–K-037 (Theorem 5.93), so Proposition 5.87 needs only (Bd), (H), (Cnt); (f) is exact relative to K-038 (Proposition 5.95); threshold $N_k > 22M$, $\varepsilon(M) = C_j(22M)^6 e^{-11 m_j M}$. Exact equality holds only term by term for non-wrapping X. The R-term part is R-032. Not in the set of L-001 (Theorem 2w)
R-016	Propagators may be replaced by operators with the same bounds	CMP 116 p. 3	asserted for (1.3)–(1.4). For C_f: proved relative to listed inputs (Theorem 5.110; tree form of $\mathcal{W}_S$); conditions K-041. H-035 (M1) holds as a list; (S_H) is proved on the torus; the volume rule is printed; the s-version of $(C_f)^{1/2}$ is a convention; Q-047(c) follows from the averaging convention (Avg). Q-048 (NEW-2, the positivity of the Ω-sequence operator, CMP 122 II p. 383) is proved relative to $Q1^{\Omega}$ and a Schur step, for the CMP 119 (2.21)/(3.23) form and the CMP 122 I (1.25) form (the same operator). Residue: U-A, U-B, U-C, and p_S (readings); the dependencies Q-049 (R-024) and K-035 of H-035 (M1)
R-017	$\operatorname{supp}\chi_k$ lies in the analyticity domain of (1.34)	used in CMP 116 Lemma 1	implicit; inside R-013. Proved relative to the printed text (Proposition 5.106): $B' = g_k B$ on every cut bond, $r = 1$. That C is the identity on the cut bonds is printed (CMP 99 p. 428, CMP 109 pp. 267–268). K-009 is not needed
R-018	Backgrounds lie in the regular class of CMP 99/102; the C-maps agree	used in CMP 109, 116, 122 I–II	implicit. Reduced (Proposition 5.108): inclusions, L-250 (Lemma Q), L-109 (Lemma C), $c_J = O(\alpha_0)$; the cube patching is replaced, with $M \geq 5M_{99}$ (K-042). Residue: CMP 102 Thm 1, CMP 98, CMP 99 (3.133). The hypothesis match for CMP 102 Thm 1 is a convention ($M_1^{(119)}$ a multiple of $R_1 M_1^{(102)}$, with R_1 above $18dL$ and an M_1 split; acyclic). It contains the residue (ii) of the R-operation geometry and (C2) of the (M-A) chain

Group A: Bałaban-level statements and external results

ID	statement	source	form used here
R-019	(190) at complex B	CMP 102 Prop. 9 (silent)	implicit. Reduced (Proposition 5.107). (73) at complex A' holds by a proof sketch through CMP 98 Sect. D. (189) at complex A' is proved relative to the complex form of CMP 98 Prop. 5; that form (a reading of CMP 98 p. 43) is the residue. A fallback without this reading uses two constants on $\lvert B\rvert < \frac{1}{8}C_1\varepsilon_1$ ($\omega = 0.4097$, half ball, rate $\times 0.59$). Consumers: R-041 and, through K_U, R-040
R-050	Uniform-in-k moments of the free extension kernel E_0	not printed	proved (constants depend on d only); not an input; listed for reference
R-020	Large-field T-step rate loss	CMP 119 Sect. 3 (defers to CMP 109 Sects. 6–7 = CMP 116 Sects. 1–2)	not printed. Reduced (proof sketch, Proposition 5.109): loss $\leq 13\delta$. The R1 count gives an unprinted cost of 1δ to 3δ; 13δ and $\delta = 0.025$ are kept. Open: R3 ($H'(Z_0)$ obeys (2.38)), with the items (H′-def) and (O1). Used by R-046 through H-216 ($\mathrm{K1}_\delta$)
R-021	Printed CMP 99 inputs of H-057 (C): Thm 3.1 (3.46) at $a = 1$; Thm 3.11 ($G > 0$); the constant C_{137} of (3.137); the H-136 (KE2) constants uniform in the torus side	CMP 99 pp. 399, 416, 423; CMP 96 Cor. 2.8	printed statements, taken as inputs. Thm 3.1 at $a = 1$ and Thm 3.3 rest on walk sketches; Thm 3.11 has a printed proof
R-022	"Valid universally for all averages", applied to the CMP 99 operators G_1, H_1, $(QG_1Q^*)^{-1}$, Δ_J under $\tilde{Q}$	CMP 109 p. 253	asserted. Reduced to page readings. The literal content is the independence of the group average M among averages with analyticity and (0.5)–(0.9). CMP 99 needs only these properties and a uniform Lipschitz modulus of M near 1, with $U(\Gamma_{y,x}) \to U(y,x)$ substituted consistently. $\tilde{Q}$, (3.115), H_1, Δ_J transfer with (d, L) constant changes. The lower bound L-211 (Lemma $\mathrm{LB}_{\tilde{Q}}$) and the k-step bound $\tilde{Q}_k\tilde{Q}_k^* \geq c$ follow from the Dressing Lemma; the one-step bound $\tilde{Q}\tilde{Q}^* \geq L^{-(d+1)}$ is proved

Group A: Bałaban-level statements and external results

ID	statement	source	form used here
R-023	$U_{k+1}(\square_0, M'(U))$ has "approximately the same regularity properties"	CMP 109 p. 275	asserted; not covered by R-018. Reduced to page readings. The glued field is CMP 99-regular with constant $cB_3M\alpha_0$. A gluing bound from the averaging constraint alone fails on the fine lattice. The Λ_0 constraint is exact, with no residual gauge freedom, and $\bar{U}^0 = U$ is forced by the printed uses. The two M (of CMP 102 and of CMP 109) are compatible under the order of constants, with rounding $(18L+2)M$, relative to a naming reading of M_1. The gluing step follows from the Dressing Lemma
R-024	Q-049: the walk expansions of CMP 99 Thms 3.7–3.10 "in their full generality" in the multiscale Ω-sequence geometry, for $C^{(k)}(Z_0)$ and the sourced A^f; the CMP 122 I (1.25) operator	CMP 119 pp. 251, 275–276; CMP 122 I (1.25)	asserted. Reduced: WC (i)–(iv) and W_N carry over with (d, L) constant changes, $\gamma_{\exp}$ unchanged. The sourced extension ($A^f \geq \frac{3}{4}\gamma_{\exp}$, the $\mathcal{W}_S$ trees, H-288, H-293 (U-b)) is proved relative to its inputs. The (1.25) operator is the CMP 119 (2.21) Schur form. The regularity of the R-operation background is Theorem R. The R-operation geometry is reduced under the condition $M > 24dLM_1$, with one residue inside R-018. Residue: (Sep) on the R-operation-modified $T^{(j)}$ (a strong reading) and (S^{Ω}_H) (a proof sketch) cited; Route B only
X-001	Lüscher–Weisz (5.7) one-loop coefficient	Lüscher–Weisz [16]	cited
X-002	OS II growth condition and reconstruction from torus data	OS II [20]	cited
X-003	Reflection positivity of the Wilson action, link planes	Osterwalder–Seiler [21]	cited
X-004	Convergence criterion of the final cluster expansion	Cammarota, CMP 85 (1982)	cited
X-005	(F): continuum log coefficient at constant field	Savvidy [22]	cited

Group B: our own hypotheses

ID	statement	source	form used here
R-025	(H2) $\rho_2 < L$	none	Theorem 6.2 at $L = 13$; also proved at $L = 15, 17, 21$ (Section 6.8). Premises: the reading of (0.4) (CMP 109 pp. 252–255), the gauge choice, premise P4 (proved), the correctness of the interval code. The gauge-fixing function is a convention: the minimum-norm gauge among the gauges that select one point per residual orbit. Other L are not used (K-003)
R-026	(PI) probe independence	none	proved at each fixed level relative to listed inputs: (I1) for the composed matrix at every j, (I2), and (I3) for the full hypercubic group. KAPPA (1.1) is not an input at any L (only an existential κ_* is used)
R-027	(F) lattice coefficient = continuum coefficient	X-005	proved (Theorem 5.134) by proper time, relative to textbook facts and the proved lemma H-137 (L-LL) (Theorem 5.133), without Lemma 2 or Step C. Outside (F) there is a numerical tension: the single-step β_1 at $L = 5$–13 does not follow $c \log L$ (Section 6.9)
R-028	(M_j) formal = true f^2 coefficient, H-244 (S1)–H-247 (S3)	none	proved relative to listed inputs at fixed $j \geq 1$, odd $L \geq 3$ (Theorem 5.102): region 1 as an asymptotic identity, neutral sector (N2); only the $o(f^2)$ form; the proved (G_{2D}) range suffices; the coefficient convention is the Moyal coefficients at $\sigma = 0$
R-029	V1: $\beta_{j+1}(0)$ is the Maxwell coefficient	CMP 109 (1.3)–(1.5)	proved at fixed level, conditional on R-030, Bałaban's inductive bounds, and the averaging (0.4)
R-030	(G2) per-cell infinite-volume densities	none (definition of β_j only)	proved at fixed j relative to listed inputs: $\tilde{Q}\tilde{Q}^T > 0$ at each fixed j, (Gap_f), and the analyticity of the averaging (proof sketch). No uniformity in j (not needed)
R-031	L5: β_{k+1} on the clamped cube, or nested-interval L-005 (Lemma DT)	CMP 109 p. 298	proved relative to listed inputs by Route 2 (Theorem 5.104): nested intervals on true trajectories, no clamp. (C-tr) holds by Route C; the causality reading H-047 and the on-trajectory uniformity of E_0 hold relative to the small-field step of R-013. Residue: R-013, R-018, CMP 102 Sect. G, L-024 + H-141, R-015 (definition only)

Group B: our own hypotheses

ID	statement	source	form used here
R-032	K-independence of $\beta_j(g)$, R-term part. The original form is ill-posed (the whole lattice has no R-terms); it is replaced by (a) β_j has no R part, (b) E_{pure} on non-wrapping $X \subset \Lambda =$ the $\mathbb{Z}^4$ term, (c) E_R K-uniform through F2	→ R-015; CMP 119 p. 279	proved relative to listed inputs (Section 5.10): K-032–K-038, the F2 inputs (R-044), CMP 109 (1.18), (3.48), (4.36). The Λ_j nesting is printed; H-091 (Conv_Λ) is printed in two parts and is a convention (K-050) with $O(1)(d)$ junction constants in the third. An input of Theorem 1, not of the shooting
R-033	Route B inputs (T1), (DF), A6, and (U)	none	hypotheses; Route B only. For (U) a possible approach derives it from (T1) and (H2)
R-034	The cross-scale sweep, `BETA_FP_M` Prop. 4.3, the Prop. 3.6 assembly, N3 in DT form	→ R-012	proved (Section 5.11): all 19 sweep rows, with the CMP 122 additions H-271–H-280, H-266–H-270 (K-043); Prop. 4.3 for $j \geq j_1$ only; L-192 (Lemma HR), N3-C
R-035	(B-S): sourced Theorem 1, (E14-a, b, c)	→ R-013	open as a whole; parts R-036–R-044, R-046. Reduced: joint holomorphy holds given the bounds; one residue, the reading (W6*) (H-361)
R-036	(I): $B_j(c) = \beta_j(u(c))$	none	by construction (K-001, K-011). Proved relative to listed inputs as Lemma I36, relative to K-001, K-011, Lemma U of the constant-source family, and (Hβ). Complex c also needs the family (A) (reduced)
R-037	Real uniform (H-hist)	→ R-001–R-004, R-002, R-008, R-009, R-010, R-017, R-018	reduced, in both halves. Small-field (Q)/c_-: Theorem 5.91 (with K-026, K-027). L-281 (Theorem SM) rests on R-011 with β_H. The Kotecky–Preiss derivative step I8 is proved relative to the Kotecky–Preiss theorem and the tree geometry (Lemma KP-D; it needs $\kappa \geq 2(24 + s_0)/\beta$, with s_0 not valued). H-145 (M-B) is proved relative to listed inputs (it needs $\beta_0 < 1/3$ for CMP 122 II p. 389). (M_r) is exact at $r = 1$ (R-017). G4 is not used. Open: the coarea Jacobian, CMP 102 (189), the F2–α link (a reading)
R-038	(H-hist2)	none	reduced; the β_j part is open

Group B: our own hypotheses

ID	statement	source	form used here
R-039	(H-thr) threshold mismatch	CMP 122 II p. 383	reduced: M2b, M4 derived, M5 not needed. H-241 ($\mathrm{R1_f}$) is reduced (Proposition 5.116); its items (i) and (iii) are proved; item (ii) is a reading. Open: (R2)–(R4), the large-field coarea Jacobian. R-038 is not an input
R-040	(H-hol), small field	conditions (C1)–(C5)	reduced; $\lvert\beta_j\rvert \le M_\beta$ is reduced to Proposition A, K_U (that is, R-005 and R-019), and K_2
R-041	(H-hol), large field	→ R-005, R-018, R-019	reduced. H-146 (M-C) is reduced to the printed CMP 116 §2 conditions and Lemma 2; H-133 (I-V0), H-131 (I-ord), (I-ε) (both layers) are proved; K-025 (M1) is a convention. (H−) ⇐ R-005 + R-019 (K1–K4) + R-018 + (C1*); (C1) is proved given R-019, R-018; (C2) is part of R-018; (C2)(b) uses the M_1-split convention
R-042	N3 strip bound	none	proved given R-036–R-041. Q-061 is proved relative to R-042. The inputs R-037, R-040 route through Proposition A
R-043	(FX) final Mayer expansion	not in CMP 116–122	reduced; remaining inputs R-042 and R-044 (W4). H-105 (FX-D) is Q-060 (proved). H-168 (S_σ) uses R-043 (FX) with ℓ-free constants. The coarsest-scale radius is not a separate input (Lemma G-1)
R-044	Sourced step bookkeeping: R-044 (W4), H-240 (P8), F2, (D), Λ_j	→ R-005, R-010, R-016	reduced. R-044 (W4)-R is proved relative to its inputs (Proposition 5.120); the Λ_j sub-item is proved relative to its inputs. The Ursell bound (7.13) is CMP 116 (2.13), and the bound is proved by Lemma KP-D. Residue: W4-G (Bałaban), the Kotecky–Preiss theorem, the tree geometry (a reading)
R-045	(B-N1) bridge and R-045 (S)	none	the R-045 (S) clauses are reduced, h-independent; the H-098 (E14-c) clause is reduced to R-045 (S); ($\mathrm{S_{src}}$) ⇒ H-116 (H-U) is proved; compatible with K-024 (U). H-168 (S_σ) is proved (Proposition 5.135), ℓ-free, with no upper limit on σ. L-111 (Lemma C) is false for $\sigma \ge 2ML$; L-127 (Lemma CS) replaces it

Group B: our own hypotheses

ID	statement	source	form used here
R-046	Recovery of $\kappa \to \kappa/2$ in the sourced step	none	reduced by $L/2$-rescaling, $\kappa \geq 1536$; uses R-020. H-216 ($\mathrm{K1}_\delta$): $\delta \leq \frac{1}{13}(1-2(1+4\beta)/(\theta_K L))$, $\theta_K \geq \frac{1}{2}$, $\kappa \geq 2m_{\mathrm{tot}}$; $\delta = 0.025$ at $L = 13$ (0.03 fails; Lean `K1_numeric_13`)

Group C: conventions and choices

ID	choice	source	kind
K-001	Freeze thresholds and partitions at Bałaban's trajectory x^*	`SOURCES_D3`, `SOURCES_HTHR`	free; the derivative needs R-039
K-002	Option B margin, $\theta_W + \tau_C \leq 0.2$	`SOURCES_MB`	free; costs a smaller γ
K-003	L odd, $L > 11$; target $L = 13$	v4 §1.1	loses generality
K-004	$G = SU(2)$, $d = 4$	Theorem 2	loses generality
K-005	Order of constants: the canonical acyclic order of `SOURCES_ORDER_CANON` (amended E1–E8), β_H (K-028) first	`SOURCES_ORDER_CANON`, `SOURCES_N3A`	acyclic; two nominal cycles are resolved by the split (Section 5.10)
K-006	Holomorphy radius ρ_0	`SOURCES_HHOL`	free
K-007	Cut on B, not B'	CMP 109 (2.9), (2.12)	reading
K-008	Sharp χ_k; smooth fallback	`SOURCES_G3`	free
K-009	Shrunk χ_k threshold	`SOURCES_GGAIN`	not used: R-017 holds with $r = 1$
K-011	Gauge-fixing weight $1/u_k(c)$	`SOURCES_D2C`	changes the construction; R-013 must hold for it
K-012	$h \equiv 0$ booking; freezing set	`SOURCES_D2C`	free
K-013	Additive source normalization	`SOURCES_HBETA_N3`	free
K-014	Main R-step form (1.90)–(1.98)	`SOURCES_FX_O23`	free; removes Q-062
K-015	κ_{final} fixed before M	`SOURCES_FX_R2`	free; not sharp
K-016	$2p_1 - (d+5)r_0 > p_0$	CMP 122 II p. 383	condition (Bałaban's own)
K-017	$32\varepsilon' \leq 1$, $S \geq 1$, $\kappa_0 > 2$	`SOURCES_D3W8`	changes the construction; R-013 must hold for it
K-018	(S): $\gamma^{-2} \geq c_{sw} D'$	v4 §1.2	condition; needs β_0 fixed before γ (a page reading)
K-019	ℓ even; thermal state; plaquette observable; test space D_n; origin at a site; compactness form	`ASSEMBLY` §0–§1	loses generality
K-020	Coupling domain $]0, \gamma]$	`SOURCES_GGAIN`	free
K-021	Route K primary, Route B alternative	v4 §1.3	free

Group C: conventions and choices

ID	choice	source	kind
K-022	$\theta_* < 1$ in Route B	`BETA_FP_PW`	free; trivial
K-023	$g_0(K, g)$: any shooting solution	L-006 (Lemma D)	loses generality
K-024	(U): $\rho_1 \le c_* r_0 / M^a$ at $\lambda_K = 1$	`SOURCES_UN1`	choice of constants, not a hypothesis on test functions; compatible with B-N1; cost C^n per n-point bound
K-025	(M1): $M = L^m$ at least the largest of 32 lower bounds; no upper bound; real $\alpha_0 \le \alpha_0'/(CM)$ with $C > c_J$	`SOURCES_ORDER_CANON`, `SOURCES_HIGHER_M1`	convention, free choice; the admissible set of M is an up-set
K-026	(Hα): $(\alpha_5 + G_\Gamma^2 \delta_S)\Lambda_0 \le \frac{1}{2}$	`SOURCES_G3_N2`, `SOURCES_G3_GGAMMA`	condition, not choice; stronger than Bałaban's $\alpha_5 \|C^{(k)}\| < \frac{1}{2}$. $G_\Gamma \le \bar{G}$ uniformly in M, so (Hα) is the single threshold $M \ge \max\{M_\Gamma, M_{\alpha 5}\}$. Under the s-rule convention K-046, the s-rule acts on Bałaban's (3.185) resolvent expansion (CMP 116 p. 13)
K-027	(Hγ): $\frac{1}{20}\gamma_2 \varepsilon_1^2/\gamma^2 \ge 4\kappa + 1$, and the (2.31) condition $\frac{1}{20}x - 4M^4 e^{-x/20} \ge 0$ at $x = \gamma_2 \varepsilon_1^2 / g_k^2$	`SOURCES_G3_N2`	condition, not choice; one unit stronger than Bałaban's $\ge 4\kappa$; γ-smallness after M, κ, γ_2
K-028	$\beta_H \in (0,1)$, $\beta_H \le \beta_0 < 1$: the unprinted Hölder exponent of CMP 109 (3.31), first in the order	`SOURCES_A10_LOWER`; `SOURCES_GGAIN_BETAH`	free; every consumer needs only $\beta_H > 0$; cost: factors $(1 - L^{-\beta_H})^{-1}$, a larger $B_4(\beta_0)$
K-029	$M_1 < M_2 < M$, order $M_1 \to M_2 \to M$	CMP 119 p. 245; `SOURCES_A11_CONST`	free; printed order
K-030	$M_2 \ge 1/\delta$	`SOURCES_A11_CONST`; CMP 119 (3.7), (3.17)	free; unprinted lower bound
K-031	$A_1/A_0 \le 1/(80 B_3)$, A_1 after A_0	`SOURCES_A11_CONST`; CMP 119 p. 266	free; Bałaban: "sufficiently small"
K-032	(Vol): torus side $\ell > 22M$ final-scale cells; even form $\ell = 2L^{m_T}$, $m_T \ge m_M + 1$	`SOURCES_A15_BDRY`, `SOURCES_T2_VOL`	condition, admissible; K-free; not used by L-001 (Theorem 2w); used by R-032 and Theorem B
K-033	(Cube): $U_j(\square_0, \cdot)$ of [14] uses only data on $\square_0$	CMP 109 p. 276; `SOURCES_A15_LOC`	proved from CMP 99 p. 394 and CMP 102 (5)
K-034	(Cases): the other term classes are localized like the typical term	CMP 116 p. 2	Bałaban's printed sketch
K-035	(Conv): cube sides divide N; partitions are π-images of the $\mathbb{Z}^4$ partitions; $M = K R_0 M_0$, K a power of L	`SOURCES_A15_LOC`	convention on an unprinted part, admissible; it also contains Q-047(c), the averaging convention (Avg), at a cost of at most one M-cube layer

Group C: conventions and choices

ID	choice	source	kind
K-036	(Det): domain-assignment rule for the closed-walk terms of the determinant expansion	CMP 102 pp. 271–272; `SOURCES_A15_LOC`	convention on an unprinted part, admissible
K-037	"H_j with free boundary conditions" is the $\mathbb{Z}^4$ operator	CMP 109 p. 290	proof sketch from CMP 109 pp. 290–292
K-038	(Enl): the s-rule domain of a walk is $\bigcup X_i^5$ as a whole-lattice set	CMP 116 p. 3; `SOURCES_A15_BDRY`	convention on an unprinted part, admissible; the Λ-restricted form costs at most one M-cube layer
K-039	(C1*): $c_m M_0\, g_k p_0(g_k) \le \frac{1}{2}\kappa_H C_1^{102} a_*^{102}$, uniform in k	`SOURCES_T2_HMINUSC`; `SOURCES_ORDER_CANONF`	condition, not choice; γ-smallness; ε_1^{102} per scale exists iff (C1*) and $\varepsilon_1^{(0)} \le a_*^{102}$
K-040	Slot of M_0 and a_*^{102} (with the cube cap $\varepsilon_1^{102} \le R_1 M_1 a_1^{102}/M$) after M, before γ	`SOURCES_T2_HMINUSC`; `SOURCES_ORDER_CANONF`	free entry of the order of constants; the cap is booked on ε_1^{102}, so M stays one-sided
K-041	R-016 tree conditions H-060 (A16-a^T), H-062 (A16-b^T), H-126 (H_κ^f), (H_η^f), $\bar\Gamma_1^f \eta_M^f \le \frac{1}{2}$, $C_{\mathcal{D}}'^f$, the $O(\gamma^2 r)$ budget entries	`SOURCES_A16_REPLACE2`; `SOURCES_ORDER_CANON` F	conditions; linear in $g^2 r$ or M thresholds; the α_5 slack is shared with K-026
K-042	H-063 (A18-1)–H-066 (A18-4) and $M \ge 5M_{99}$ with $10 \mid C^{122}M$	`SOURCES_A18_REGULAR`; `SOURCES_ORDER_CANONF`	conditions, entries of the order of constants; acyclic; ε_0^{102} two-sided but nonempty
K-043	CMP 122 I–II threshold and exponent conditions H-266–H-270, with the cross-scale list H-271–H-280	`SOURCES_A11_CONST` addendum; `SOURCES_B10_SWEEP`	conditions, Bałaban's own (printed); exponents before γ; N two-sided per scale, exists for γ small
K-044	(Cv1) $U(c)$ = product of the level-$(i-1)$ links on the straight segment; K-044 (Cv2) translation-covariant axial Γ	CMP 98 (42); `SOURCES_C_CONDITION` §5	conventions on an unprinted part, admissible; enter only H-123 (H_Q)(iii) of H-057 (C); other contour choices change only constants
K-045	L-284 (Lemma T) cut-off $\tilde\chi_y = \sum_{y'} \psi_{y'} \chi(d_*(y',y)/\tilde D_0)$ (closure-bond metric d_*) with threshold (T2): torus side $> \tilde c_T(d,L)$ coarsest blocks; $\theta_S \le \delta_H/8$, $\theta \le \delta_0/2$, $RM \ge \max(4L^2(\tilde D_0 + c_{\mathrm{adj}}), 2dL, c(d,L))$, $M\alpha_0 \le a_0(d,L)$, $M \ge \tilde c_T(d,L)/22$	`SOURCES_C_CONDITION` §2; `SOURCES_LEMMA_T_REPAIR`	conditions, not choices; $M \ge \tilde c_T/22$ is a lower bound on M; $\tilde c_T \le c_T + 6Lc_{\mathrm{adj}}$, $\tilde D_0 \le D_0 + 2c_{\mathrm{adj}}$; $RM > 2dL$ is implied by $R > 18dL$
K-046	s-rule: s-versions per distinct σ_0-cube on the expansions of this paper; H-103 (FI) on the admissible walk set; localization set $\bigcup \tilde X_i^5$	`SOURCES_SRULE_CONVENTION`	convention on an unprinted part, admissible; replaces the reading H-022; consumers use only bounds
K-047	$\gamma^2\beta' \le 1/L$, side condition of CMP 119 (2.9) line 2	Lean `ScaleChain`; `SOURCES_ORDER_CANON`; `SOURCES_A11_CONST` addendum	condition, not choice; γ-smallness of Bałaban's type; no cycle

Page readings. Four statements are page readings and enter no set. R-047: the constant $C_1'(d, L)$ is unprinted and not derivable from the printed lines; its value is not used. R-048: β_0 "arbitrarily small" is on CMP 119 p. 255, and β_0 is fixed before γ. R-049: $R_{K+1} \geq L$ follows from CMP 119 (2.5), and C_c is free of g. R-051: Π (4.37)/(5.1), CMP 119 (2.42) (corrected), (2.12)/(2.23), and the "L odd" page; the residue in Sect. 3 is R-020.

The free parameter β_H. The per-label exponent of Fact A (L-050) (Theorem SM (L-281)) is the Hölder exponent β_H of CMP 109 (3.31), which Bałaban does not print (Theorem 5.82). It is a free parameter, $0 < \beta_H \leq \beta_0 < 1$, and it comes first in the order of constants: $\beta_H \to d, L \to \cdots \to \varepsilon_1 \to \gamma$ (K-005, K-025, K-028). Every consumer needs only $\beta_H > 0$, because each condition it adds is an ε_1- or γ-smallness condition. The cost is the factors $(1 - L^{-\beta_H})^{-1}$, a larger $B_4(\beta_0)$, and K_T fixed before ε_1. Margins at $L = 13$: the (G-gain) (H-111) margin $1 - 13^{-\beta_H/2}$ is 0.120 at $\beta_H = 0.1$ and 0.685 at $\beta_H = 0.9$; the L-186 (Lemma HH) margin is 0.901 at $\beta_H = 0.9$.

Binding sets. *Theorem 2w (L-001)* (Theorem A(i), Route K) rests on one Group A item, R-006, on R-025–R-031, and on X-005; Group C: K-003, K-004, K-018, K-023. R-025 holds at $L = 13$ (Theorem 6.2). R-026–R-031 are proved relative to listed inputs. R-006 has no bare assertion left (Proposition 5.87, Theorem 5.93); its use is reduced, with residue N1–N9. If R-006 is taken through this reduction, Theorem 2w also rests on R-008, R-009, R-017, the inputs of Bałaban's sup induction, K-026, K-027, and K-033–K-037. The residue of R-031 is R-013, R-018, CMP 102 Sect. G, and the causality reading H-047. Route B rests on R-006, R-025, R-030, R-031, R-033, and X-001.

Theorem 2w ⇒ Theorem 1 (Theorem A(ii)) rests on the Group A items R-006, R-012, R-013, R-015, on the Group B items R-025–R-032 and R-034, and on X-005; Group C adds K-029–K-038, K-043, and K-047. Inside the proof of Theorem 1 (R-013), Bałaban also uses R-001–R-005, R-008, R-009, R-010, R-014, R-017, R-018, R-021–R-024, and the conventions K-044–K-046.

Theorem B rests on the Group A items R-001–R-006, R-008, R-009, R-010, R-012, R-013, R-015–R-024, on the externals X-002–X-005, and on the Group B items R-025–R-032 and R-034–R-046; Group C adds K-026–K-047. Of the Group A items that Bałaban prints without a proof, R-002, R-003, R-004, R-008, R-009, and R-011 have proofs relative to listed inputs, R-012 holds as a list, and R-001, R-005, R-006, R-015 are reduced; R-001 includes the condition (C) (H-057). Of the implicit or unprinted items, R-016 and R-017 are proved relative to listed inputs, and R-018, R-019, and R-020 are reduced. R-021 is a set of printed CMP 99 statements, taken as inputs. R-022, R-023, and R-024 are reduced. R-014 (inside R-013 only) is asserted. This does not make Group A or Group B proved. Every row stays an input, because its residual inputs are not proved: Bałaban's printed sketches, conventions, readings, (Vol) (K-032), and K-039–K-047.

Open inputs of Theorem B beyond those of Theorem 2w: the off-shell item H-358 of R-010 and R-014 (seven reduced items; of the literal readings of (189), reading (i) holds and readings (ii), (iii) are open); the complex CMP 98 Prop. 5 reading of R-019; R-018, with the residue (ii) of the R-operation geometry; the residue of R-024 ((Sep) and (S_H^Ω)); the page readings of R-022 and R-023; the reading (W6*) (H-361) of R-035; the sub-items of R-039; the β_j part of R-038; the inputs of R-042 (N3); the residue of R-020 ((H′-def), (O1)); THM37's multi-size cube class (H1), a hypothesis beyond the printed cube class; and R-014, which is asserted.

Kinds of conventions. Of Group C, K-003, K-004, K-019, K-023 lose generality; K-024 and K-025 are choices of constants; K-011, K-017 change Bałaban's construction, and K-009 is not used; K-016, K-018, K-026, K-027, K-032, K-039, K-041, K-042, K-043, K-045, K-047 are conditions;

K-028–K-031 and K-040 are free entries of the order of constants; K-007 is a reading; K-033 and K-037 are proved from the printed pages; K-034 is a printed sketch of Bałaban; K-035, K-036, K-038, K-044, K-046 fix unprinted parts of the construction.

Lean support. Lean certifies finite or algebraic steps only. No input of the tables above becomes a theorem because of Lean. The Lean files are in `lean/YmLean/`: 80 files, with 51 canon lines and 180 edges in `CanonOrder.lean`. No file contains `sorry`. The axioms are propext, Classical.choice, and Quot.sound, except for the H2Box certificate: `H2Box.lean` and its 16 run files `H2BoxRun00-15` each add one `native_decide` axiom, and the `H2Box.lean` theorems inherit them. `CanonOrder` uses no axiom.

row	Lean files	what is formalized
R-001	CombesThomas, BlockSchur, Schur, KernelRow	L-125 (Lemma CT) in kernel form; the row sum for `Fin n` only, not the block-lattice sum
R-002	ParametrixCore, WGAlgebra, Concatenation, WalkAffine	the A1-W core (μ-weighted L-124 (Lemma CT), the parametrix identity, the Neumann step, entry decay); WG L-228 (Lemma N), the G_1 resolvent and its two-sided Loewner bound, the (3.135) four-term expansion, the WG-1 exponent counts 10 and 12, pattern counts; L-115 (Lemma C) concatenation. Not: the chain expansions as matrix identities, the walk distance
R-003, WC chain	WeightedNorm, Resolvent, BlockInequalityE, CauchyPerturbation, TheoremL, Concatenation	the weighted norm and L-225 (Lemma N); L-153 (Lemma E); L-122 (Lemma CP); L-207 (Theorem L) assembly, $\gamma_{\exp} = \gamma_e/4$, $K_b \le 64 d^3 L^{2d+1}$ (the compression step not formalized)
R-004	SlabCounterexample, GaugeMin, Interp, WalkAffine, Resolvent	the CMP 96 slab step fails for $L \ge 10$; Lemma 5; the interpolation and resolvent steps of the σ-localized clause
R-008, R-009	QConstants, TheoremL, ComplexGaussian	L-226 (Lemma N) for rectangular C; $\gamma_{\exp}$; L-242 (Lemma P) core and `lemmaD_core`
R-010 (=R-011)	ElemBounds	the raw j-sum diverges (`a10Sum_*`)
R-012, R-034, K-047	ScaleChain, ElemBounds	(2.6)–(2.9), (T1)–(T4), H-156 (P-up)/H-155 (P-low), `eq29_line2_full` with $\gamma^2\beta' \le 1/L$; monotone maps
R-006, R-037 (G3, (Q))	SchurMarginal, QConstants, ComplexGaussian	block inverse; the B-block $H_0^{-1} + H_0^{-1}\Gamma\Gamma^* H_0^{-1}$; Schur positivity; one-bond compression; `c_minus_ge`; Λ and c_- bounds; the Lemma S form. Not: the Gaussian integrals of (Q), $\Gamma_{bb} \ge H_0^{-1}$
R-025	BlockNorm, OishiRump, CertBridge, Tower, Divergence, H1Majorant	`rho2_le`; the certifier bridge; the majorant core; identity (D) of P4; L-283 (Lemma T) bounds, the F_{CS} core, L-166 (Lemma G), the gauge step, L-264 (Lemma S) / P5 invariance; H2Box*: `Bst_le_ten`, `rho1_sq_le_ten` by `native_decide`, with the hypotheses hmaj, hsym, hzero. Not: the end-to-end $F \le F_{CS}$ over $\mathbb{Z}^4$

row	Lean files	what is formalized
R-026	TheoremC1	the two-step bound (B2)
R-028	NeutralSpectrum, MagneticShift	the neutral sector (spectrum completeness not formalized); the magnetic-shift algebra on $\mathbb{Z}^2$ with exact commutators (the analytic norm forms not formalized)
R-031	NestEnvelope	`nest_DDoublePrime`, `nest_DPrime`; the induction needs the continuous family $u_0 \mapsto u_k$ and the IVT, with hypotheses on the Good-k set; `env_of_terminal` is the single-sequence form
DS, DT, K	LemmaDPrime, LemmaDE, LemmaDDoublePrime, TheoremK, Gronwall	the shooting lemmas (one-argument form) and L-009 (Theorem K)
R-042 (N3)	StripBound, LemmaH	the strip bootstrap; L-182 (Lemma H)
R-043 (FX_O1)	CubeCount	the cube count
R-046	K1Prime, Interp	`K1_numeric_13` (13δ form, $\delta = 0.025$); `K1_iff`
canon (K-005, K-025), R-012	CanonOrder, ElemBounds	51 canon lines, 180 edges, `id_eq_rank`, `canon_acyclic`, `canon_wf` (the transcription is acyclic; the mathematics is not certified); monotonicity of the constant maps
none	Telescoping, DetIdentities, AbelianCount, Envelope	numerics identities, U(1) count, H-081 (B-NT); not needed by the targets

Most of the content of Group A sits on Bałaban's walk expansions, polymer activities, and inductions. These stay in prose.

A Lean formalization

The shooting lemmas of Section 3.4, the arithmetic of L-009 (Theorem K) and L-011, and the bridge lemmas of the certifier are formalized in Lean 4 with Mathlib; seven files formalize lemmas of the sources block (Schur test, block-valued Schur test, Combes–Thomas, interpolation and rescaling, resolvent and weighted Neumann bounds, the gauge condition of the constrained minimizer, the slab counterexample to CMP 96 (2.127)). There are 80 Lean files, with no `sorry`. The axioms are propext, Classical.choice, and Quot.sound, except for the H2Box certificate: `H2Box.lean` and its 16 run files `H2BoxRun00-15` each add one `native_decide` axiom, and the `H2Box.lean` theorems inherit them. The Lean support by ledger row is the table at the end of Section 7.7, and `lean/README.md` lists every file. The list below gives the main formalized statements and the first 25 files.

Formalized statements.

- L-004 (Lemma DS) (a)–(d) (`YmLean.lemma_DPrime`), Step 8 (`lemma_DPrime_step8`), L-006 (Lemma D), L-007 (Lemma E) (Cg^2 form), L-006 (Lemma D)+E (`lean/YmLean/LemmaDPrime.lean`, `LemmaDE.lean`). One-argument β only.

- L-005 (Lemma DT) (`YmLean.lemma_DDoublePrime`, `lean/YmLean/LemmaDDoublePrime.lean`, 146 lines): (a-tail) bracket $(3b/4)(K-k) - 2R \le T(k,K) \le (5b/4)(K-k) + 2R$; (b) shooting for $g \in]0, g_1]$ with the clamp inactive; (c-pair) with $\beta = 3b/4$, $B' = 5b/4 + 2R$, $D' = 2R$, for every trajectory with couplings in $]0, \gamma]$; (d) for $n - m \ge 8R/b$, lower constant $b/2$. Hypotheses

weaker than the note (continuity, any admissible γ, $R \geq 0$ derived). One-argument β only; L5 not formalized.

- The first 25 files (file, main theorems, source note):

  1. `LemmaDPrime`: `lemma_DPrime`; `LEMMA_DPRIME.md` §1–§2.
  2. `LemmaDE`: `lemma_D`, `lemma_E`, `lemma_DE`, `lemma_DPrime_step8`; `LEMMAS.md`.
  3. `LemmaH`: `gronwall_telescope`, `lemma_H`; `SOURCES_HBETA.md` L-182 (Lemma H).
  4. `DetIdentities`: `det_bordered`, `det_restrict_add_gauge`; `numerics/NUMERICS.md` §1.
  5. `Telescoping`: `telescoping_det`, `telescoping_log`; `R2_TELESCOPING.md` Prop. 3.1.
  6. `AbelianCount`: `su2_count`, `abelian_share`; `U1_INFINITE_LATTICE.md` Prop. 9.1 (a cross-check only, Section 6.2).
  7. `LemmaDDoublePrime`: `lemma_DDoublePrime`; `KAPPA_BOUNDED.md` §5.3.
  8. `TheoremK`: `telescoping_bound`, `window_of_partial` (the window bound (5.1) from partial sums); `KAPPA_BOUNDED.md` §5.
  9. `Gronwall`: `discrete_gronwall`, `contraction_bound` ($N_{j+1} \leq qN_j + Z$); `KAPPA_BOUNDED.md` Prop. 4.1.
  10. `CubeCount`: `cube_count_sharp`, `cube_count_lower` (L-236 arithmetic); `SOURCES_FX_O1.md`.
  11. `BlockNorm`: `rho2_le`, the exact factorization $\rho_2 \leq \rho_1(j) \max_l \rho_1(j-1, k_l)$; `H2_CERTIFY.md`.
  12. `OishiRump`: `oishi_rump`, `oishi_rump_eta`, `frobenius_power_bound`; `H2_CERTIFY.md` §5.
  13. `Envelope`: `envelope_nontrivial`, `envelope_not_scale_invariant` (H-081 (B-NT) in log form); `ASSEMBLY.md` §2.9.
  14. `Divergence`: `profile_divergence`, `P4_divergence` (identity (D) of premise P4); `H2_CERTIFY_P4.md`.
  15. `TheoremC1`: `c1_two_step`, `c1_kappa_bound`, `c1_window`; `B2_TWOSTEP.md` §2.
  16. `StripBound`: `strip_bound_linear`, `strip_bound_quadratic` (the $\frac{3}{2}\tau$ bootstrap); `SOURCES_HBETA_N3.md` §4.
  17. `CertBridge`: `cell_invertible`, `cell_sup_bound`, `vertex_bound` (interval-to-norm bridge of the cell certifier); `numerics/h1cert_cell.py`, `h1cert_cell2.py`.
  18. `Tower`: `tower_bounds`, `I1_compress`, `majorant_core`, `lemmaT_core` (algebraic core of the one-step majorant, Theorem 6.2); `H1_ANALYTIC.md`.
  19. `Schur`: `schur_test_weighted`, `schur_test` (weighted Schur test for the spectral norm); `SOURCES_LOC.md` §3.3, `SOURCES_S2_FORM.md` §2, `SOURCES_A3_SIGMA.md` 2.6.
  20. `CombesThomas`: `norm_entry_le`, `coercive_inv`, `combes_thomas`, `combes_thomas_exp`, `combes_thomas_decay`, `combes_thomas_schur` (the proof structure of L-125 (Lemma CT), the kernel-decay reduction of ledger R-001); `SOURCES_A1_KERNEL.md` §7. The identification with the block operators, the kernel-to-row-sum step, (K)/H-057 (C), and the rescaling are not formalized; the block-valued Schur test is in `BlockSchur`.
  21. `Interp`: `multilinear_convex`, `monomial_affine` (`SOURCES_A3_SIGMA.md` 3.2); `window_perturb`, `window_perturb_partial` (`SOURCES_T2_KUNIF.md` 5.2, chained to L-009 (Theorem K)); `rescale_exp`, `rescale_rate`, `K1_iff`, `K1_numeric` (`SOURCES_T2_KAPPA2.md`); `xexp_le`, `xexp_absorb` (`SOURCES_GGAIN_LF.md` §2).
  22. `Resolvent`: `resolvent_identity`, `resolvent_bound`, `lower_transfer`, `neumann_perturb`, `resolvent_chain` (`SOURCES_A3_SIGMA.md` §4); `wnorm_mul_le`, `wnorm_neumann`, `pwnorm_mul_le`, `pwnorm_neumann` (weighted Neumann series, row half; `SOURCES_A2_R2.md` §2).
  23. `SlabCounterexample`: `slab_fails_10`–`slab_fails_13`, `slab_fails`, `lambda1_lt` (the proof step CMP 96 (2.127) fails for every $L \geq 10$); `SOURCES_A3_GAMMA0_FULL.md` §3.2–§3.3.

24. `BlockSchur`: `block_row_le`, `block_schur_test` (Schur test with block norms $\|K_{xy}\|_2$ in place of $|K|$); `SOURCES_A1_KERNEL.md` §7, `SOURCES_S2_FORM.md` §2 Π3.
25. `GaugeMin`: `variational`, `constrMin_existsUnique`, `lemma5`, `lemma5_full` ($RD^*A_* = 0$ for the constrained minimizer, from (F1), (F3), (F4) as hypotheses); `SOURCES_A3_GAMMA0_FULL.md` §5.

Not formalized: interval arithmetic in the arb-based certificate (the separate Lean check of the branch-and-bound, `H2Box.lean`, has its own rational interval evaluator, proved correct in Lean), the enclosure premises P1a–P4, and the identity $\rho_1^2 L^2 = \lambda_{\max}(G)$. The history form of the shooting lemmas is formalized: LemmaDHist.lean, LemmaDHistNested.lean (`lemma_DPrime_hist_adm`, `lemma_DDoublePrime_hist_adm`). That Bałaban's β_j meets the continuity hypotheses is an input.

B Provenance of the results

Every result of this paper was first derived in a project note and then checked by an independent reader in a separate note. All derivations and all checks were written by AI agents (Claude models) under the author's direction; no human expert has reviewed them. The table gives, for each result, the note with the derivation and the note with the check. The notes are in the `notes/` directory of the repository [14].

result	derivation	check
A	THEOREM2_PROOF_v4.md	THEOREM2_PROOF_v2_VERIFY.md
3.2	THEOREM2_PROOF_v4.md, KAPPA_BOUNDED.md	KAPPA_BOUNDED_VERIFY.md
3.4	LEMMA_DPRIME.md	LEMMA_DPRIME_VERIFY.md
3.5	LEMMA_DPRIME.md	LEMMA_DPRIME_VERIFY.md
3.6	KAPPA_BOUNDED.md	KAPPA_BOUNDED_VERIFY.md
3.4.3	LEMMA_DPRIME_SWEEP.md	LEMMA_DPRIME_VERIFY.md
3.10	LEMMAS.md	LEMMAS_DE_VERIFY.md
3.11	LEMMAS.md	LEMMAS_DE_VERIFY.md
3.4.5	SOURCES_T2_B7_CLAMP.md	SOURCES_T2_B7_CLAMP_VERIFY.md
3.5.1	BETA_FP.md	BETA_FP_VERIFY.md
3.12	BETA_FP_LEMMA43.md	BETA_FP_LEMMA43_VERIFY.md
3.5.3	R2_TELESCOPING.md	R2_VERIFY.md
3.5.4	KAPPA_BOUNDED.md	KAPPA_BOUNDED_VERIFY.md
3.5.5	U3_AW.md	U3_AW_VERIFY.md
3.14	KAPPA_BOUNDED.md	KAPPA_BOUNDED_VERIFY.md
3.15	BETA_FP_PW.md	BETA_FP_PW_VERIFY.md
3.18	B2_TWOSTEP.md	B2_TWOSTEP_VERIFY.md
3.19	B2_TWOSTEP.md	B2_TWOSTEP_VERIFY.md
3.9	U3_GAUSSIAN_RG.md	U3_VERIFY.md
6.6	H2_CERTIFY.md	H2_CERTIFY_VERIFY.md
6.2	H1_ANALYTIC.md	H1_ANALYTIC_VERIFY.md, H1_ANALYTIC_VERIFY2.md
6.4	numerics/H1_OTHER_L.md	H2_CERTIFY_L15_17_21_VERIFY.md
5.2	SOURCES_GGAIN_NONMONO.md	SOURCES_GGAIN_NONMONO_VERIFY.md
5.3	SOURCES_T2_KUNIF.md	SOURCES_T2_KUNIF_VERIFY.md
5.5	SOURCES_T2_V1.md	SOURCES_T2_V1_VERIFY.md
5.6	SOURCES_T2_G2.md	SOURCES_T2_G2_VERIFY.md
5.7	SOURCES_T2_S123.md	SOURCES_T2_S123_VERIFY.md

result	derivation	check
5.9	SOURCES_T2_I3.md	SOURCES_T2_I3_VERIFY.md
5.10	SOURCES.md	SOURCES_VERIFY.md
5.11	SOURCES_D1.md	SOURCES_D1_VERIFY.md
5.12	SOURCES_D2.md	SOURCES_D2_VERIFY.md
5.13	SOURCES_D2C.md	SOURCES_D2C_VERIFY.md
5.14	SOURCES_D3.md	SOURCES_D3_VERIFY.md
5.15	SOURCES_D3R.md	SOURCES_D3R_VERIFY.md
5.16	SOURCES_D3W8.md	SOURCES_D3W8_VERIFY.md
5.17	SOURCES_D4D5.md	SOURCES_D4D5_VERIFY.md
5.18	SOURCES_HBETA.md	SOURCES_HBETA_VERIFY.md
5.19	SOURCES_T2_KAPPA2.md	SOURCES_T2_KAPPA2_VERIFY.md
5.20	SOURCES_UN1.md	SOURCES_UN1_VERIFY.md
5.21	SOURCES_B21_S.md	SOURCES_B21_S_VERIFY.md
5.22	SOURCES_FX.md	SOURCES_FX_VERIFY.md
5.23	SOURCES_FX_O1.md	SOURCES_FX_O1_VERIFY.md
5.25	SOURCES_FX_O23.md	SOURCES_FX_O23_VERIFY.md
5.28	SOURCES_FX_R2.md	SOURCES_FX_R2_VERIFY.md
5.29	SOURCES_HBETA_N3.md	SOURCES_HBETA_N3_VERIFY.md
5.30	SOURCES_N3A.md	SOURCES_N3A_VERIFY.md
5.31	SOURCES_HHOL.md	SOURCES_HHOL_VERIFY.md
5.32	SOURCES_HHOL_LF.md	SOURCES_HHOL_LF_VERIFY.md
5.33	SOURCES_MA.md	SOURCES_MA_VERIFY.md
5.36	SOURCES_MA2.md	SOURCES_MA2_VERIFY.md
5.37	SOURCES_MB.md	SOURCES_MB_VERIFY.md
5.38	SOURCES_MC.md	SOURCES_MC_VERIFY.md
5.40	SOURCES_MC_IV0.md	SOURCES_MC_IV0_VERIFY.md
5.41	SOURCES_T2_IEPS.md	SOURCES_T2_IEPS_VERIFY.md
5.43	SOURCES_T2_IEPS_LF.md	SOURCES_T2_IEPS_LF_VERIFY.md
5.44	SOURCES_HIGHER_M1.md	SOURCES_HIGHER_M1_VERIFY.md
5.45	SOURCES_HTHR.md	SOURCES_HTHR_VERIFY.md
5.47	SOURCES_HTHR2.md	SOURCES_HTHR2_VERIFY.md
5.48	SOURCES_HHIST2.md	SOURCES_HHIST2_VERIFY.md
5.49	SOURCES_SM.md	SOURCES_SM_VERIFY.md
5.50	SOURCES_HHIST.md	SOURCES_HHIST_N1_VERIFY.md
5.51	SOURCES_GGAIN.md	SOURCES_GGAIN_VERIFY.md
5.53	SOURCES_GGAIN_LF.md	SOURCES_GGAIN_LF_VERIFY.md
5.54	SOURCES_G3.md	SOURCES_G3_VERIFY.md
5.56	SOURCES_Q.md	SOURCES_Q_VERIFY.md
5.57	SOURCES_LOC.md	SOURCES_LOC_VERIFY.md
5.60	SOURCES_A1_KERNEL.md	SOURCES_A1_A4_VERIFY.md
5.61	SOURCES_QE_KE_HQ.md	SOURCES_QE_KE_HQ_VERIFY.md
5.62	SOURCES_QE_INV.md	SOURCES_QE_INV_VERIFY.md
5.63	SOURCES_S2_FORM.md	SOURCES_S2_FORM_VERIFY.md
5.65	SOURCES_S2_G1_V7.md	SOURCES_S2_G1_V7_VERIFY.md
5.67	SOURCES_A1_WALK.md	SOURCES_A1_WALK_VERIFY.md
5.68	SOURCES_WG.md	SOURCES_WG_VERIFY.md
5.70	SOURCES_A2_R2.md	SOURCES_A2_R2_VERIFY.md
5.71	SOURCES_A3_GAMMA0.md	SOURCES_A3_GAMMA0_VERIFY.md
5.72	SOURCES_A3_GAMMA0_FULL.md	SOURCES_A3_GAMMA0_FULL_VERIFY.md, SOURCES_SWEEP_BC_VERIFY.md
5.74	SOURCES_A3_SIGMA.md	SOURCES_A3_SIGMA_VERIFY.md
5.75	SOURCES_A4_D189.md	SOURCES_A1_A4_VERIFY.md

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5.76	A1_WALK_FULL.md	A1_WALK_FULL_VERIFY.md
5.77	WG_FULL.md	WG_FULL_VERIFY.md, SOURCES_F19_WGRATE_VERIFY.md, SOURCES_F22_VERIFY.md
5.78	WC_FULL.md	WC_FULL_VERIFY.md, SOURCES_P254_TRANSFER_VERIFY.md, SOURCES_P253_P275_VERIFY.md
5.79	SOURCES_A7_CDC.md	SOURCES_A7_CDC_VERIFY.md
5.80	SOURCES_A8_POS.md	SOURCES_A8_POS_VERIFY.md
5.81	SOURCES_A10_LOWER.md	SOURCES_A10_LOWER_VERIFY.md
5.82	SOURCES_A10_LOWER.md	SOURCES_A10_LOWER_VERIFY.md
5.83	SOURCES_GGAIN_BETAH.md	SOURCES_SWEEP_H.md
5.85	SOURCES_A11_CONST.md	SOURCES_A11_CONST_VERIFY.md
5.87	SOURCES_A15_WHOLE.md	SOURCES_A15_WHOLE_VERIFY.md
5.88	SOURCES_T2_H0.md	SOURCES_T2_H0_VERIFY.md
5.89	SOURCES_G3_GGAMMA.md	SOURCES_G3_GGAMMA_VERIFY.md
5.91	SOURCES_G3_N2.md	SOURCES_G3_N2_VERIFY.md
5.93	SOURCES_A15_LOC.md	SOURCES_A15_LOC_VERIFY.md
5.95	SOURCES_A15_BDRY.md	SOURCES_A15_BDRY_VERIFY.md
5.96	SOURCES_A15_BDRY.md	SOURCES_A15_BDRY_VERIFY.md
5.98	SOURCES_T2_GAPRIME2.md	SOURCES_T2_GAPRIME2_VERIFY.md
5.100	SOURCES_ORDER_CANON.md, SOURCES_ORDER_CHECK.md	SOURCES_ORDER_CANON_VERIFY.md, SOURCES_ORDER_CANON_VERIFY2.md, SOURCES_ORDER_CANON_VERIFY3.md
5.102	SOURCES_T2_MJ_REST.md	SOURCES_T2_MJ_REST_VERIFY.md
5.104	SOURCES_T2_B7_CLAMP.md	SOURCES_T2_B7_CLAMP_VERIFY.md
5.106	SOURCES_A17_A19.md	SOURCES_A17_A19_VERIFY.md
5.107	SOURCES_A17_A19.md	SOURCES_A17_A19_VERIFY.md
5.108	SOURCES_A18_REGULAR.md	SOURCES_A18_REGULAR_VERIFY.md
5.109	SOURCES_A23_TSTEP.md	SOURCES_A23_TSTEP_VERIFY.md
5.110	SOURCES_A16_REPLACE2.md	SOURCES_A16_REPLACE2_VERIFY.md
5.111	SOURCES_T2_HMINUSC.md	SOURCES_T2_HMINUSC_VERIFY.md
5.112	SOURCES_B10_SWEEP.md	SOURCES_B10_SWEEP_VERIFY.md
5.113	SOURCES_T2_F_STEPB.md	SOURCES_T2_F_STEPB_VERIFY.md
5.115	SOURCES_B15_M245.md	SOURCES_B15_M245_VERIFY.md
5.116	SOURCES_HHOL_LF_SPRIME.md	SOURCES_HHOL_LF_SPRIME_VERIFY.md
5.120	SOURCES_B20_W4R.md	SOURCES_B20_W4R_VERIFY.md
5.123	SOURCES_B20_LAMBDAJ.md	SOURCES_B20_LAMBDAJ_VERIFY.md
5.124	SOURCES_B21_E14C.md	SOURCES_B21_E14C_VERIFY.md
5.125	SOURCES_BOOKKEEPING.md	SOURCES_BOOKKEEPING_VERIFY.md
5.126	SOURCES_C_CONDITION.md	SOURCES_C_CONDITION_VERIFY.md
5.129	SOURCES_C_CONDITION.md	SOURCES_C_CONDITION_VERIFY.md
5.130	SOURCES_C_CONDITION.md, SOURCES_LEMMA_T_REPAIR.md	SOURCES_C_CONDITION_VERIFY.md, SOURCES_SWEEP_DE_VERIFY.md, SOURCES_LEMMA_T_REPAIR_VERIFY.md
5.132	SOURCES_T2_F_STEPB.md	SOURCES_T2_F_STEPB_VERIFY.md
5.133	SOURCES_T2_LLL.md	SOURCES_T2_LLL_VERIFY.md

result	derivation	check
5.134	SOURCES_T2_MPT_STEPC.md, SOURCES_T2_MPT_REST.md	SOURCES_T2_MPT_STEPC_VERIFY.md, SOURCES_T2_MPT_REST_VERIFY.md
5.135	SOURCES_B21_S_PRIME.md	SOURCES_B21_S_PRIME_VERIFY.md
5.137	SOURCES_FX_D.md	SOURCES_FX_D_VERIFY.md
5.139	SOURCES_SRULE_CONVENTION.md	SOURCES_SRULE_CONVENTION_VERIFY.md
5.140	SOURCES_M1_SWEEP.md	SOURCES_M1_SWEEP_VERIFY.md
5.141	SOURCES_P254_TRANSFER.md, SOURCES_P253_P275.md	SOURCES_P254_TRANSFER_VERIFY.md, SOURCES_P253_P275_VERIFY.md
5.143	THM39_FULL.md	THM39_FULL_VERIFY.md
5.145	THM37_FULL.md, THM310_FULL.md	THM37_FULL_VERIFY.md, THM310_FULL_VERIFY.md
5.148	SOURCES_ORDER_CANON.md	SOURCES_ORDER_CANON_VERIFY.md, SOURCES_ORDER_CANON_VERIFY2.md, SOURCES_ORDER_CANON_VERIFY3.md
5.149	SOURCES_EU.md	SOURCES_EU_VERIFY.md
5.150	SOURCES_T2_EUMOM.md	SOURCES_T2_EUMOM_VERIFY.md
5.151	SOURCES_CDC.md	SOURCES_CDC_VERIFY.md
4.1, 4.2, 4.8, 4.11	ASSEMBLY.md	ASSEMBLY_VERIFY.md
4.5, Hypothesis (NT)	ASSEMBLY.md, SOURCES_L1_NT.md	ASSEMBLY_VERIFY.md, SOURCES_L1_NT_VERIFY.md
4.10	ASSEMBLY.md	ASSEMBLY_VERIFY.md
4.1, 4.7	ASSEMBLY_N1.md, SOURCES_D4D5.md	SOURCES_HHIST_N1_VERIFY.md, SOURCES_D4D5_VERIFY.md
4.5.3 (uniformity of (S))	SOURCES_B21_S.md	SOURCES_B21_S_VERIFY.md
4.5.1, 4.6 (B-T2)	THEOREM2_PROOF_v4.md, KAPPA_BOUNDED.md	THEOREM2_PROOF_v2_VERIFY.md, KAPPA_BOUNDED_VERIFY.md
4.6 (B-T2.a)	LEMMA_DPRIME.md, LEMMAS.md	LEMMA_DPRIME_VERIFY.md, LEMMAS_DE_VERIFY.md
4.6 (B-T2.b)	LEMMA_DPRIME_SWEEP.md	SOURCES_B10_SWEEP_VERIFY.md
4.6 (B-T2.1–3)	U3_GAUSSIAN_RG.md, U3_REPAIRS.md, U3_LEMMA51.md, U3_AW.md	U3_VERIFY.md, U3_REPAIRS_VERIFY.md, U3_LEMMA51_VERIFY.md, U3_AW_VERIFY.md
4.6 (B-T2.4)	BETA_FP_LEMMA43.md	BETA_FP_LEMMA43_VERIFY.md
4.6 (B-T2.5, item 1)	KAPPA_BOUNDED.md, B2_TWOSTEP.md	KAPPA_BOUNDED_VERIFY.md, B2_TWOSTEP_VERIFY.md
7.1, 4.6 (B-T2.H1, H2)	H1_ANALYTIC.md, H2_LARGE_L.md	H1_ANALYTIC_VERIFY.md, H1_ANALYTIC_VERIFY2.md, H2_CERTIFY_P5_VERIFY.md, H2_LARGE_L_VERIFY.md, H2_CERTIFY_VERIFY.md
4.6 (B-T2.B)	BETA_FP_PW.md	BETA_FP_PW_VERIFY.md
7.2 (item 2)	SOURCES_T2_PI.md, SOURCES_R026_PI.md	SOURCES_T2_PI_VERIFY.md, SOURCES_R026_PI_VERIFY.md
7.2 (item 3)	SOURCES_T2_MPT_STEPC.md, SOURCES_T2_MPT_REST.md	SOURCES_T2_MPT_STEPC_VERIFY.md, SOURCES_T2_MPT_REST_VERIFY.md

result	derivation	check
7.2 (item 4)	SOURCES_T2_S123.md, SOURCES_T2_I3.md, SOURCES_T2_MJ_REST.md	SOURCES_T2_S123_VERIFY.md, SOURCES_T2_I3_VERIFY.md, SOURCES_T2_MJ_REST_VERIFY.md
7.2 (items 5, 6)	SOURCES_T2_V1.md, SOURCES_T2_G2.md, SOURCES_R030_RESIDUE.md	SOURCES_T2_V1_VERIFY.md, SOURCES_T2_G2_VERIFY.md, SOURCES_R030_RESIDUE_VERIFY.md
7.2 (item 7)	SOURCES_GGAIN_NONMONO.md, SOURCES_T2_H0.md	SOURCES_GGAIN_NONMONO_VERIFY.md, SOURCES_T2_H0_VERIFY.md
7.2 (item 8)	SOURCES_T2_B7_CLAMP.md, SOURCES_CTR_RESIDUE.md	SOURCES_T2_B7_CLAMP_VERIFY.md, SOURCES_CTR_RESIDUE_VERIFY.md
7.2 (item 9)	SOURCES_T2_KUNIF.md	SOURCES_T2_KUNIF_VERIFY.md
7.4 (item 10)	SOURCES_BOOKKEEPING.md	SOURCES_BOOKKEEPING_VERIFY.md
7.5	SOURCES_ROP_RESIDUE.md, SOURCES_P387_REPAIR.md, SOURCES_R024_ISTRONG.md	SOURCES_ROP_RESIDUE_VERIFY.md, SOURCES_P387_REPAIR_VERIFY.md, SOURCES_R024_ISTRONG_VERIFY.md
7.6	SOURCES_P387_REPAIR.md	SOURCES_P387_REPAIR_VERIFY.md
7.5 (nontriviality)	SOURCES_L1_NT.md, SOURCES_NT4.md	SOURCES_L1_NT_VERIFY.md, SOURCES_NT4_VERIFY.md
7.7	TRUST_LEDGER.md	TRUST_LEDGER_VERIFY.md
7.3 (R-022, R-023)	SOURCES_P253_P275.md, SOURCES_L1_A.md	SOURCES_P253_P275_VERIFY.md, SOURCES_L1_A_VERIFY.md
7.3 (R-024)	SOURCES_MULTISCALE_OMEGA.md, SOURCES_MULTISCALE_OMEGA_SOURCED.md	SOURCES_MULTISCALE_OMEGA_VERIFY.md, SOURCES_MULTISCALE_OMEGA_SOURCED_VERIFY.md
7.3 (B-S)	SOURCES_L1_E.md, SOURCES_A_SIGMA_STAR.md	SOURCES_L1_E_VERIFY.md, SOURCES_A_SIGMA_STAR_VERIFY.md
4.14	INFINITE_VOLUME_FABLE.md	INFINITE_VOLUME_FABLE_VERIFY.md, INFINITE_VOLUME_FABLE_VERIFY2.md
4.14 (L-320, L-331, L-323)	SOURCES_L2_L3L14.md, FABLE_HARD.md	SOURCES_L2_L3L14_VERIFY.md, FABLE_HARD_VERIFY.md
4.15	SOURCES_FH_U1_REPAIR.md	FABLE_HARD_VERIFY.md

C Label map

The numbered environments of this paper carry the names used in the project files. The first table maps them. The second table maps the section numbers of THEOREM2_PROOF_v4.md ("v4") and ASSEMBLY.md to the sections of this paper; section references inside the project files use the old numbers.

this paper	name in the project files	source
Theorem A	Route K summary: L-001 (Theorem 2w) ⇒ Theorem 1	THEOREM2_PROOF_v4 §6, §10
Theorem B	Theorem 1.3 (summary)	ASSEMBLY §1.3
Theorem 3.2	L-001 (Theorem 2w)	THEOREM2_PROOF_v4 §1.3
Lemma 3.4	L-004 (Lemma DS)	LEMMA_DPRIME; v4 §2.1

this paper	name in the project files	source
Remark 3.5	Remark 4	`LEMMA_DPRIME`; v4 §2.1
Lemma 3.6	L-005 (Lemma DT)	`KAPPA_BOUNDED` §5.3; v4 §2.2
Lemma 3.10	L-006 (Lemma D)	`LEMMAS`; v4 §2.4
Lemma 3.11	L-007 (Lemma E)	`LEMMAS`; v4 §2.4
Proposition 3.12	Prop. 2.1 (anchor-free identity)	`BETA_FP_LEMMA43`; v4 §3.2
Theorem 3.14	L-009 (Theorem K)	`KAPPA_BOUNDED`; v4 §4.1
Theorem 3.15 (Theorem $\mathrm{B_{PW}}$)	L-088 (Theorem B) = Theorem 3.2	`BETA_FP_PW`; v4 §4.2
Theorem 3.18	L-011	`B2_TWOSTEP`; v4 §5.3
Theorem 3.19	L-012	`B2_TWOSTEP`; v4 §5.3
Definition 4.1	H-116 (H-U)	`ASSEMBLY` §1.2
Theorem 4.2	Theorem 1.3	`ASSEMBLY` §1.3
Theorem 4.5	Theorem 1.4	`ASSEMBLY` §1.4
Proposition 4.8	Proposition of H-074 (B-C)	`ASSEMBLY` §2.4
Lemma 4.11	finite-form lemma of H-083 (B-RP)	`ASSEMBLY` §2.7
H-200 (GC1)–H-201 (GC3)	(G1)–(G3) of the dependency table	`ASSEMBLY` §3
($\mathrm{S_{src}}$)	R-045 (S), final sourced bound	`SOURCES_D4D5`
items H, 1–13, B1–B8	items of the open list	`THEOREM2_PROOF_v4` §10
Theorem 5.79	R-008	`SOURCES_A7_CDC`
Theorem 5.80	R-009, corrected form	`SOURCES_A8_POS`
Theorem 5.82	R-011	`SOURCES_A10_LOWER`
Proposition 5.85	R-012 list	`SOURCES_A11_CONST`
Proposition 5.87	Prop. A15-N	`SOURCES_A15_WHOLE`
Theorem 5.91	L-174 (Lemma G3) = N2	`SOURCES_G3_N2`
Theorem 5.93	L-026 (Theorem LOC)	`SOURCES_A15_LOC` §7.5
Proposition 5.95	L-104 (Proposition BDRY)	`SOURCES_A15_BDRY`
Theorem 5.98	($\mathrm{G_{2D}}$), Thm 5.1 + Cor. 5.2	`SOURCES_T2_GAPRIME2`
Remark 5.99	range remark for ($\mathrm{G_{2D}}$)	`SOURCES_T2_GAPRIME2`, `SOURCES_T2_FRANGE`
Theorem 5.102	(M_j), Theorem 4.2 and §5	`SOURCES_T2_MJ_REST`
Theorem 5.104	Lemmas DT-nest, DS-nest, Route 2	`SOURCES_T2_B7_CLAMP`
Proposition 5.106	R-017 identity	`SOURCES_A17_A19`
Proposition 5.107	R-019 reduction	`SOURCES_A17_A19` §4–§5
Proposition 5.108	R-018 reduction	`SOURCES_A18_REGULAR`
Proposition 5.109	R-020 reduction	`SOURCES_A23_TSTEP`
Theorem 5.110	R-016 in tree form	`SOURCES_A16_REPLACE2`
Proposition 5.116	H-241 ($\mathrm{R1_f}$), frozen-threshold bounds	`SOURCES_HHOL_LF_SPRIME`
Proposition 5.120	R-044 (W4)-R	`SOURCES_B20_W4R`
Proposition 5.126	the H-057 (C) condition, §8 verdict	`SOURCES_C_CONDITION`
Lemma 5.129	L-039 (Lemma R)	`SOURCES_C_CONDITION` §3.2
Theorem 5.133	H-137 (L-LL), Theorem 5.1	`SOURCES_T2_LLL`
Theorem 5.134	H-148 (MPT) L-060 (Theorem A), L-112 (Lemma C), L-090 (Theorem B)	`SOURCES_T2_MPT_STEPC`
Proposition 5.135	H-168 (S_σ), L-127 (Lemma CS)	`SOURCES_B21_S_PRIME`
Proposition 5.141	p. 254 transfer	`SOURCES_P254_TRANSFER`
Theorem 5.143	L-044, L-210 (Lemma LB)	`THM39_FULL`
Theorem 5.145	L-045, Cor. 3.8 forms	`THM37_FULL`
Theorem 4.14	L-196 (Theorem IV), §7.1	`INFINITE_VOLUME_FABLE`
Remark 4.15	(Gap_U), §3.5	`SOURCES_FH_U1_REPAIR`

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v4 §1.2	Section 3.1
v4 §1.3	Section 3.2
v4 §1.4	Section 3.3
v4 §2 (notation)	Section 2.3
v4 §2.1	Section 3.4.1
v4 §2.2	Section 3.4.2
v4 §2.3	Section 3.4.3
v4 §2.4	Section 3.4.4
v4 §2.4 (L5)	Section 3.4.5
v4 §3.1	Section 3.5.1
v4 §3.2	Section 3.5.2
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v4 §3.4	Section 3.5.4
v4 §3.5	Section 3.5.5
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v4 §4.2a	Section 3.6.3
v4 §4.3	Section 3.6.4
v4 §5	Section 3.7
v4 §5.1	Section 3.7.1
v4 §5.2	Section 3.7.2
v4 §5.3	Section 3.7.3
v4 §5.4	Section 6.4
v4 §5.5 (arithmetic)	Section 3.7.4
v4 §5.5 (values)	Section 6.5
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`ASSEMBLY` §2.1	Section 4.5.1
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`ASSEMBLY` corrections	Remarks 4.10, 4.6; Section 4.7
`NUMERICS` §7 (rewritten)	Section 6.9

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